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Comparative Analysis of Physics-Based and AI-Based Molecular Docking Tools

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Abstract

Molecular docking predicts how small-molecule ligands bind their protein targets, yet physics-based and machine-learning tools embody very different trade-offs between geometric accuracy and physical realism. This thesis presents a systematic, provenance-documented comparison of a physics-based method (AutoDock Vina) against two artificial-intelligence (AI)-based methods, namely the diffusion-based generative model DiffDock-L (hereafter DiffDock) and the E(3)-equivariant regression model EquiBind, across two complementary settings. A calibration benchmark of 303 co-crystallised complexes enables ground-truth assessment of both pose accuracy (heavy-atom root-mean-square deviation, RMSD) and physical validity through the PoseBusters test battery. Because the dominant variant of each toolchain is both chosen and scored on this same benchmark, the reported rates are optimistic as estimates of performance on unseen targets. A crystal-free application to the calcium-release-activated channel Orai1, using molecular-dynamics receptor snapshots together with experimental and benchmark ligand sets, probes how the tools behave on a realistic membrane-protein target without a reference structure.

Across both settings, physical validity separates the tools most sharply in their raw output. On the calibration benchmark, adding the validity requirement to the near-native recovery criterion costs the refined pipelines at most 1.3 percentage points of complexes, so placement and pose selection rather than validity become the limiting factors on that endpoint. At the level of individual poses the gate remains costly even after minimisation, removing 15.5% of refined DiffDock poses and 48.2% of refined EquiBind poses. Pooled over every produced pose, raw AutoDock Vina is physically valid almost everywhere (98.2%), whereas the raw AI outputs are largely physically invalid (24.4% for DiffDock and 2.9% for unguided EquiBind), and AutoDock Vina followed by gnina rescoring reaches 99.5% on that same pooled basis. On the separate basis of the median per-complex validity rate, post-hoc minimisation is the dominant driver of usability, raising DiffDock from 13.3% to about 93% with smina and unguided EquiBind from 0% to about 57% with gnina, while it cannot recover an unsampled or grossly misplaced binding mode. Deepening the inspected ranking pool from rank-1 to the best of fifteen recovers valid, near-native poses for a further 21.1 percentage points of complexes with DiffDock and 20.5 with AutoDock Vina but only 5.0 with EquiBind, and extending it to thirty adds at most 0.7 points more, so the headroom is large, unequally distributed and almost exhausted by a top-15 pool. On Orai1 the three tools localise to markedly different sites, and there the combined screen for physical validity and admissible placement rather than validity alone decides which poses remain usable. Transmembrane-pore filtering, pose clustering and protein–ligand interaction-fingerprint analysis quantify their disagreement. Overall, the results argue for combined accuracy-and-validity criteria

and for geometry refinement as a default stage when AI-based docking tools of this generation are deployed, with the choice of refinement itself validated, since one relaxation stage tested here damaged the geometry it was meant to repair.

Keywords: Molecular docking; Physics-based versus AI-based methods; PoseBusters physical validity; Post-hoc pose refinement; Orai1 calcium channel

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1 Motivation

Molecular docking is vital in structure-based drug design, predicting the binding modes of small molecules to target proteins and estimating binding affinities. Computationally modelling protein-ligand interactions enables the screening of vast chemical libraries and thus significantly reduces drug development time and costs.

Computational molecular docking emerged in the 1980s from geometric shape-matching methods that predicted ligand binding modes from steric complementarity [1]. In 1990, Goodsell and Olson introduced an automated docking procedure based on simulated annealing [2], helping establish the AutoDock family. The combination of empirical interaction models with efficient volumetric energy evaluation enabled flexible-ligand conformational search and increasingly large virtual screens [3]. HADDOCK later extended data-guided docking to flexible, multistage modelling of biomolecular complexes [4], [5]. These classical approaches remain influential because their search and scoring procedures are explicit, reproducible and continuously maintained.

Artificial-intelligence (AI)-enabled protein-ligand docking methods have proliferated in recent years [6], [7]. DiffDock-L (hereafter DiffDock) and EquiBind are prominent examples. DiffDock formulates docking as equivariant diffusion [24], and the release used here is the larger retrained DiffDock-L [89], whereas EquiBind uses geometric deep learning for direct pose regression [10]. Despite reported gains in speed or accuracy, comparisons between classical and AI-based docking carry recurring limitations, namely evaluation from a selected output pose [12], [13], [34], search conditions that are not matched across the competing methods [11], [34], and the absence of post-docking optimisation together with explicit physical-validity screening [11]. The following research questions therefore guide this thesis.

RQ1 How do the quality and stability of poses predicted by AutoDock Vina, DiffDock and EquiBind differ?

RQ2 How does post-docking optimisation affect the quality and stability of predicted poses?

RQ3 How computationally efficient are physics-based and AI-based docking approaches?

RQ4 What opportunities and limitations do physics-based and AI-based docking tools present?

2 Literature Review

2.1 Molecular Docking Process

Molecular docking aims to predict geometry, bound pose and interactions of small-molecule ligands and target proteins. Ligands can be molecules, ions, peptides or proteins that bind to larger biomolecules to form complexes that serve biological functions. Ligands can be substrates that enzymes act upon, inhibitors that block enzyme activity, activators that enhance function, signalling molecules like neurotransmitters, or ions essential for structural or catalytic roles [16]. When docking, ligands often induce conformational changes in the protein, altering its functional state and triggering downstream biological effects [16].

Proteins are amino-acid sequences that fold into three-dimensional structures whose shapes support functions such as catalysis, structural organisation and cellular signalling. Small-molecule binding at protein pockets can modulate these functions, which makes proteins central targets in drug discovery [14], [15].

2.1.1 Physics-Based Molecular Docking

Physics-based protein-ligand docking is a computational process that predicts how ligands (small molecules) bind to proteins at the atomic level. This prediction is vital for understanding molecular recognition mechanisms. Next to predicting ligand conformation and position (sampling), binding affinities are assessed (scoring) in the docking process. The classical workflow proceeds in five stages and is described in detail in Figure 1.

AutoDock Vina

AutoDock Vina is a physics-based docking program that couples an efficient global search engine to an interchangeable empirical scoring function [22], [23]. It helps to keep the two components apart, because the engine that samples and refines candidate poses is largely independent of the scoring function that ranks them, and the same engine can be driven by more than one such function. Two are relevant here, namely the original Vina function and the later Vinardo function, both selectable within the same program [23]. After receptor and ligand preparation, the engine generates random poses inside a user-defined search box and refines them by iterated local search, namely random perturbation followed by gradient-based local optimisation with a BFGS-like quasi-Newton method over precomputed interaction grids. Similar poses are clustered by RMSD and the lowest-energy representatives are returned [23]. Because the chosen scoring

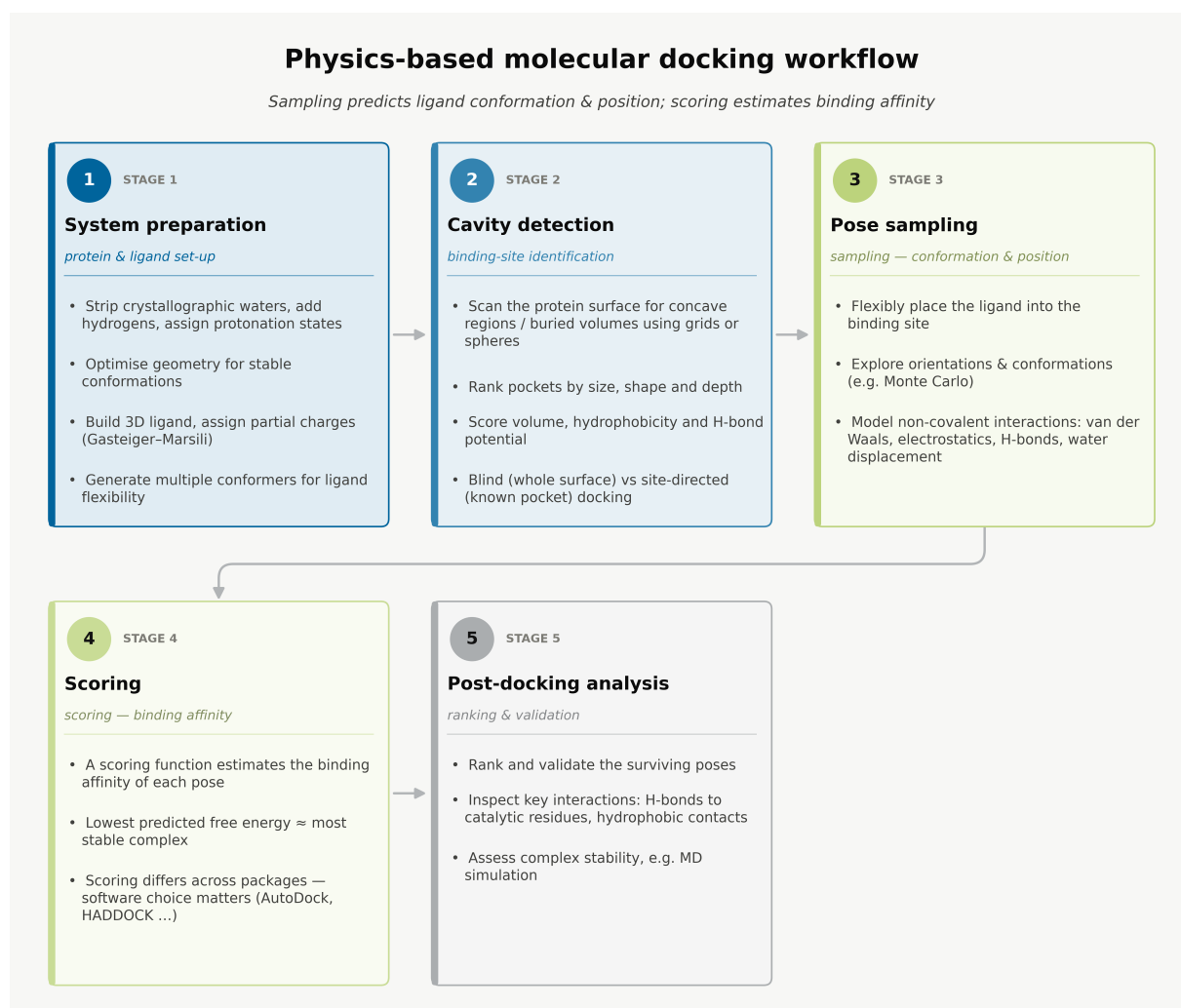


Figure 1: Physics-based Docking Flow Chart. The figure is an original synthesis drawn by the author from the workflows described in [15], [17], [18], [19], [20] and [21], and is neither reproduced nor adapted from any single source

function is evaluated together with its gradient, each pose is minimised against it during the search, so no separate post-docking optimisation is needed.

The default Vina scoring function expresses the predicted binding free energy as a weighted sum of pairwise atom-type terms over the surface distance between atoms, namely two attractive Gaussian steric terms, a repulsive parabola, a hydrophobic term and a hydrogen-bond term, divided by a factor growing with the number of rotatable bonds as a flexibility penalty. Its six weights were fixed once by tuning against the PDBbind set rather than derived from first principles, giving Vina a hybrid empirical and knowledge-based character [22]. The first Gaussian together with the repulsion term approximates the attractive well and steep wall of a dispersion interaction without its hard singularity, while the second, longer-range Gaussian adds a broader attraction further from contact, and this smooth steric model stays well suited to fast local optimisation.

Vinardo, short for Vina RaDii Optimized, is a reparametrised and deliberately simpler variant of this function [71]. It keeps the same families of terms but drops the second, longer-range Gaussian, which in Vina introduced a second steric minimum about 3Å beyond atomic contact and tended to seat ligands too deeply in the pocket. Vinardo also re-optimises the atomic radii, enlarging that of aliphatic carbon while shrinking those of nitrogen and oxygen, and re-fits the remaining term weights. These parameters were selected by screening candidate functions on the RMSD of crystallographic poses after local minimisation and then confirming the best by actual redocking, rather than by maximising correlation with measured affinities, on the finding that a good affinity fit does not by itself yield good docking behaviour [71]. The authors report that the resulting function improves scoring, docking and virtual-screening performance relative to Vina at a lower parameter count [71].

Switching from Vina to Vinardo therefore changes only the objective that the shared engine minimises and ranks, not the search machinery itself, which keeps the two directly comparable within AutoDock Vina. The full protocol, scoring equations and fitted weights for both functions are reproduced in the Appendix (Tools and Functions, AutoDock Vina).

2.1.2 AI-Based Molecular Docking

AI-based molecular docking tools employ diverse computational strategies. Common strategies include deep-learning neural networks that predict binding poses directly from protein and ligand structures, diffusion models like DiffDock [24] that treat docking as a generative sampling problem, geometric deep learning approaches such as EquiBind [9], [10], which predicts the bound pose directly with an SE(3)-equivariant network, and TankBind [25], [26], which encodes the protein with an equivariant network but predicts an inter-molecular distance map through a trigonometry module and then derives coordinates from it by optimisation, and regression-based models like KarmaDock [27] and GAABind [28] that predict binding affinities or poses with learned scoring functions. Other notable AI-based algorithms include SurfDock, DiffBindFR, DynamicBind, QuickBind, and hybrid methods like Interformer, which integrate traditional conformational

searches with AI-driven scoring functions. These tools also differ in the search space they require. DiffDock, EquiBind and TankBind need no user-specified binding site and localise the ligand themselves, whereas DeepDock and Uni-Mol must be given a pocket definition [12], [34]. Co-folding models likewise require no predefined site, and they predict protein and ligand structure jointly rather than placing a ligand into a fixed receptor structure [12], [29].

In contrast to traditional exhaustive conformational search algorithms and empirical scoring functions, AI-based tools learn complex interaction patterns from large datasets of protein-ligand complexes to directly predict optimal binding conformations. Moreover, AI-based methods tend to be computationally efficient, which improves scalability for large-scale virtual screening [49].

DiffDock

DiffDock is a diffusion generative model that treats pose prediction as a generative task rather than optimisation of a hand-crafted energy function [24]. Protein and ligand are represented as graphs, and the ligand pose is parameterised by translation, rotation and internal torsion angles. During training a forward process corrupts known poses by adding noise to these variables, and at inference the reverse process starts from random placements and denoises them over several steps, where an SE(3)-equivariant neural network predicts updates that move the pose toward a high-probability bound conformation. The equivariance keeps predictions consistent under rigid motions of the coordinates. Because sampling is stochastic, DiffDock returns several plausible poses, which suits multimodal docking landscapes. Its scoring has two learned components, namely a diffusion score model that steers the denoising trajectory and a confidence model that ranks poses by predicted correctness, that is whether a pose falls below a 2Å RMSD threshold rather than an affinity-like energy. These criteria live implicitly in the network weights, so performance depends heavily on the training data and confidence-model calibration [24]. Full details are given in the Appendix (Tools and Functions, DiffDock).

EquiBind

EquiBind is a direct geometric deep-learning approach to blind docking predicting the bound pose directly in a single forward pass of an SE(3)-equivariant neural network. Protein and ligand are encoded as graphs, and interaction-aware equivariant message passing between them lets ligand atoms attend to the most relevant protein regions, so the network infers both where the ligand binds and how it is oriented without a predefined site. Its central mechanism is the prediction of matching key points on the ligand and the protein surface, after which the best rigid-body transformation is computed by a singular-value-decomposition (Kabsch-style) step that places the ligand into the pocket, followed by a conformer-fitting stage that adjusts torsions. EquiBind uses no separate scoring function at inference, and is instead trained with losses that penalise the geometric error to the experimental reference, namely ligand RMSD-type, alignment and region losses, so the ranking rule is internalised during training and no affinity-like number is produced. Dropping both stochastic search and repeated denoising gives it a large

runtime advantage for high-throughput docking. Full details are given in the Appendix (Tools and Functions, EquiBind).

2.1.3 Optimisation Tools

smina

Unlike the tools above, smina and gnina are used here as lightweight refinement tools that re-score and locally optimise generated poses. smina is a fork of AutoDock Vina developed by Koes et al. [53]. It inherits Vina's empirical scoring function and search machinery unchanged but re-implements them so that individual energy terms can be reweighted or swapped and custom atom types defined. The energy it optimises is therefore the Vina objective itself, namely the weighted sum of two Gaussian steric terms, a steric-repulsion penalty, a hydrophobic contact term and a hydrogen-bond term, divided by a factor growing with the number of rotatable bonds, the same function described above. Beyond full docking, smina exposes a fast local-minimisation mode that relaxes a supplied pose into the nearest energy minimum by gradient-based optimisation without repeating the conformational search. This makes it well suited to post-docking optimisation, since a pose seated in roughly the correct pocket but clashing with the receptor can be minimised against the Vina function, removing the clash while leaving the placement essentially unchanged [55].

gnina

Because gnina is a fork of smina, both share the docking algorithm inherited from Vina [54], [65], [66]. Sampling uses Markov-chain Monte Carlo chains that apply random rigid-body and torsional perturbations, drive each candidate to a local minimum of the Vina energy by quasi-Newton minimisation, and accept or reject it by a Metropolis criterion, so the Vina energy is the quantity optimised during search. gnina differs in ranking, replacing the hand-crafted score with a learned one. An ensemble of convolutional neural networks rescores the poses, each reading a three-dimensional grid of Gaussian atom-type densities built with libmolgrid [74] and trained on two objectives, namely a cross-entropy classifier that separates poses within 2Å RMSD of the native mode from poses outside it (CNNscore) and a regression head that predicts a binding constant on a pK scale and is hinged for poses judged inaccurate (CNNaffinity) [73], [54], [65]. Their product CNN_VS is preferred for virtual screening [66]. gnina therefore couples a physics-based sampler to a deep-learning scoring function [56]. Full details appear in the Appendix (Tools and Functions, Optimisation Tools).

gnina Rescoring and gnina Refinement

gnina can be applied to an existing pose in two ways that are easy to conflate but are not the same operation, and this work uses both. The difference lies in what the learned score is

permitted to change. In the first mode the network only reorders poses, while in the second it also determines where their atoms end up [54], [56].

Rescoring is the mode gnina applies by default. The pose is first driven to a local minimum of the empirical function by the inherited quasi-Newton minimiser, and only once that minimisation has finished are the convolutional networks evaluated, after which the pose set is reordered by the learned score [54]. The network therefore changes the ranking of a set of poses and not the geometry of any one of them. Coordinates do shift relative to the input, because a local minimisation was performed, but that displacement is produced by the empirical objective in exactly the way smina would produce it. Rescoring is the cheapest way to use the model and is the configuration in which gnina has been most broadly validated, including the prospective CACHE screen in which it identified experimentally confirmed hits [66].

Refinement inverts this. The learned score becomes the objective of the final local minimisation, so the network decides the geometry. This is possible because the atom-density grid the network reads is constructed to be differentiable with respect to the input coordinates, which allows the score to be back-propagated into per-atom forces that then drive the same quasi-Newton minimiser [72], [74]. In this mode the empirical function is still evaluated and still reported, but it does not enter the objective unless it is explicitly mixed back in, which is not the default behaviour [56]. A further asymmetry is worth stating, because it is not obvious from the option name. The quantity that supplies the gradient is the pose-classification head, so refinement moves atoms toward higher CNNscore, whereas the ordering of the refined poses in this work is taken from CNNAffinity. The optimised criterion and the ranking criterion are therefore not the same number.

The published assessment of the two modes is asymmetric. The gnina authors report that refinement costs roughly an order of magnitude more computation than rescoring while performing no better, and they explicitly recommend against it, which is why rescoring remains the shipped default [54]. Earlier prospective work on convolutional-network minimisation found it useful on some targets without being uniformly better than minimisation against the empirical function [75]. The broader literature supplies a reason to expect this asymmetry. A learned score is fitted to discriminate poses rather than to reproduce a potential energy surface, and machine-learned scoring functions have been shown to degrade only gradually when a ligand is deliberately pushed into the protein instead of rising steeply as an explicit repulsion term does [76]. Frameworks that do optimise coordinates against a learned score generally recombine it with a traditional physics term for precisely that reason [83], [51], and deep-learning docking output is in general more prone to failing physical-plausibility checks than the output of classical methods [34], [13].

The consequence for this study is that the two modes answer different questions and are therefore never merged into a single gnina variant. Rescoring supplies the deep-learning pipelines with the geometric repair and the ranking they do not otherwise provide. Refinement is applied to the AutoDock variants as a direct test of the recommendation above, because poses that enter it are already almost entirely valid, which isolates what the network does to

geometry from what an empirical minimisation would have repaired anyway. The measured effect of each mode is reported in the Results, and the protocols are set out in the Appendix (Docking Protocols, AutoDock Vina).

2.1.4 Comparisons of Physics-Based and AI-Based Molecular Docking

The rapid emergence of AI-based molecular docking tools, which promise computationally efficient and reliable binding predictions, has drawn the attention of scholars and practitioners alike. New tools are usually released with a peer- or non-peer-reviewed paper setting out their advantages and disadvantages, and independent researchers have since compared different tools across diverse datasets. As Table 1 shows, these evaluations cover a wide range of tools while converging on a common set of comparison metrics.

Table 1: Physics-Based and AI-Based Docking Studies

Study (Year)	Docking Tools Compared	Datasets Used	Metrics for Comparison
Li et al. (2025) [13]	Glide SP, AutoDock Vina (physics-based); SurfDock, DiffBindFR, DynamicBind (generative diffusion); KarmaDock, GAABind, QuickBind (regression-based AI); Interformer (hybrid)	Astex Diverse Set, PoseBusters Benchmark, DockGen (pose/validity); DEKOIS2.0, DUD-E (virtual screening)	RMSD (pose accuracy), PB-valid (physical plausibility), Interaction Recovery, ROC-AUC, EF0.5%, BEDROC (VS), Generalisation (protein/ligand/pocket similarity)
Jiang et al. (2026) [12]	Glide, AutoDock Vina, MOE, Discovery Studio, gnina (physics-based); Uni-Mol Docking V2, DiffDock, SurfDock, Interformer, EquiBind, FABind, DeepDock, TankBind (AI); AlphaFold3, Boltz-1, Chai-1 (AI co-folding)	PoseX-SD (self-docking), PoseX-CD (cross-docking), Astex	RMSD, Success Rate, Physical Validity, Generalisation, Leaderboard Ranking
Gu et al. (2025) [32]	Glide, Surflex, rDock, LeDock (physics-based); CarsiDock, KarmaDock, DiffDock, FlexPose (AI-powered docking); RTMScore, EquiScore (AI rescoring)	VSDS-vd benchmark (TrueDecoy, TrueDecoy _{gap} , RandomDecoy, MassiveDecoy)	RMSD (pose accuracy), PB-valid (physical plausibility), combined RMSD-and-PB-valid criterion, EF0.5%, EF1%, EF5% (VS enrichment)

Table 1: Physics-Based and AI-Based Docking Studies (continued)

Study (Year)	Docking Tools Compared	Datasets Used	Metrics for Comparison
Schaller et al. (2024) [33]	Fred (physics-based docking); Hybrid, Posit (guided by a co-crystallised ligand), all OpenEye Toolkits	589 KLIFS kinase structures with 423 ATP-competitive ligands (cross-docking)	Heavy-atom RMSD (pose accuracy), bootstrapped success rate at 2 Å, Chemgauss4 and Posit probability scores

Recent evaluations show that learned docking models remain sensitive to physical validity and structural novelty [13]. None of the cited studies evaluates Orai1, so AutoDock Vina is retained as an established baseline rather than assumed to be universally superior.

3 Methods

The methodological premise of this work is exploratory rather than confirmatory. Instead of committing to a single docking engine, it deliberately combines a physics-based method (AutoDock Vina) with two complementary AI-based methods, the diffusion model DiffDock and the regression model EquiBind, and couples each deep-learning method to a lightweight physics-based refinement step (smina/gnina). Mapping where each toolchain performs well, and testing whether their union covers more of the benchmark than any single tool, provides the ground for the docking of the reference-free Orai1 target.

The comparison rests on a small set of measures applied identically to every tool. Geometric accuracy is the symmetry-corrected heavy-atom RMSD of a docked pose to the crystal ligand, with the conventional success threshold of $\text{RMSD} \leq 2\text{\AA}$. This in-place RMSD is further resolved into a shape and a position component through the approximate decomposition $\text{in-place}^2 \approx \text{placement}^2 + \text{form}^2$. The form error is the best-fit (Kabsch) RMSD that remains once translation and rotation are removed, and the placement error is the residual diagnostic recovered from that relation rather than an independently measured component, with a pose judged form-correct when its form error is at most $1\text{\AA}$. The decomposition separates a distorted ligand conformation from a merely mis-positioned one, since only a conformationally faithful pose can present its pharmacophore correctly and reproduce the native interaction pattern.

Physical plausibility is further assessed with the PoseBusters battery, which flags chemically or sterically impossible poses (PB-valid). Because a pose can be geometrically close yet physically broken, the two are combined into a single valid-near-native criterion, namely $\text{RMSD} \leq 2\text{\AA}$ and PB-valid, which serves as the headline metric wherever a crystal ligand exists. These are further resolved by pose ranking (the ranked or scored top-1 pose versus the best or oracle pose that is closest placed to the crystal ligand) and, for the binding-site question, by cross-tool clustering of the docked poses against the crystallographic pocket.

3.1 Datasets

Validating the reliability of produced poses by AutoDock Vina, DiffDock and EquiBind required the careful selection of an experimentally validated dataset as ground truth. A reliable dataset provides protein and ligand complexes across a wide range of chemical scaffolds, sizes and structural classes.

3.1.1 Calibration Dataset

The calibration dataset is the PoseBusters benchmark [34], which in its published form comprises 308 high-quality protein-ligand crystal complexes released from 2021 onwards, all of them deposited entries retrieved from the Protein Data Bank [41]. That benchmark is used here in full, namely the entries enumerated in the identifier file `posebusters_pdb_ccd_ids.txt`, published by the PoseBusters authors through the project repository and obtained from it on 10 June 2026. The copy of the distribution held for this work also ships a larger list of 428 entries in `posebusters_benchmark_set_ids.txt`, which is the set as it stood in the earlier preprint release. The benchmark authors withdrew 120 of those entries because the ligand lies within 5.0 Å of a protein symmetry mate and therefore carries a crystal contact, and the journal article reports the remaining 308. That withdrawal is a correction by the benchmark authors rather than a restriction imposed here, so the 120 excluded entries are not failures, omissions or a sample this work selected. The shared comparative analysis retained 303 complexes after preparation and processing, while the timing comparison used the 302 cases with comparable timing records.¹ The post-2021 release criterion reduces direct temporal overlap with the published training data of early AI docking models, but it does not exclude homologous proteins, similar ligands or familiar binding pockets. For each retained entry, the cognate ligand is re-docked into its experimentally determined receptor and compared with the crystallographic binding mode. The source set spans a broad range of chemical and structural properties (Table 2). A detailed description of the dataset is provided in Appendix G.

Table 2: Key Characteristics of the Calibration Benchmark Set

Descriptor	Median	IQR (25–75 %)	Range (min–max)
Heavy atoms	24	16–32	6–64
Molecular weight (Da)	359	236–462	100–872
Rotatable bonds	5	2–7	0–20

¹ Five of the 308 source entries, namely 7B2C_TP7, 7D6O_MTE, 7FRX_O88, 7M31_TDR and 8F4J_PHO, produced no DiffDock output and were therefore dropped before the three-tool comparison, leaving the 303 complexes analysed throughout. Because that exclusion removes one tool's own production failures from the denominator of all three, a sensitivity check was run in which the five are retained and scored as failures for DiffDock. AutoDock Vina + gnina reaches a PoseBusters-valid pose within 2 Å on one of the five at rank-1 and on four of the five within its top-15 pool, so on that intention-to-treat basis its recovery becomes 90 of 308 at rank-1 and 155 of 308 at top-15, against 103 and 167 for DiffDock + smina. The DiffDock margin narrows from 4.6 to 4.2 percentage points at rank-1 and from 5.3 to 3.9 at top-15 without changing direction. Those five poses were screened against the receptor as distributed with the benchmark rather than through the per-tool validation receptors used elsewhere, so the check is a bound on the exclusion effect rather than a fifth set of table entries. One further complex, 7WCF_ACP, carries no usable wall-clock record in the whole-protein run and is excluded from the timing comparison alone, leaving 302 cases there. Under the superseded crystal-boxed run ten complexes lacked comparable timing records, namely 6XCT_478, 6YSP_PAL, 7C3U_AZG, 7LMO_NYO, 7PIH_7QW, 7Q25_8J9, 7SDD_4IP, 7WCF_ACP, 7WKL_CAQ and 8FO5_Y4U, which is why the earlier timing set numbered 293.

Table 2: Key Characteristics of the Calibration Benchmark Set (continued)

Descriptor	Median	IQR (25–75 %)	Range (min–max)
H-bond acceptors	6	3–9	0–21
H-bond donors	3	2–5	0–13
TPSA ($\text{\AA}^2$)	107	66–160	0–384
cLogP	0.9	–1.6–3.2	–8.4–14.0
Fraction sp^3 C	0.41	0.23–0.57	0.00–1.00
QED	0.43	0.29–0.60	0.04–0.91
Ring count	3	1–4	0–8

3.1.2 Experimental Dataset

The experimental dataset is the system that the comparison is ultimately intended to address. Unlike the calibration dataset, where a crystallographic pose, i.e. ground-truth, is available for every complex, this set targets the calcium release-activated calcium channel Orai1, a target of genuine pharmacological interest, for which experimentally resolved protein-ligand reference structures are not established, so that the binding mode must be predicted rather than re-docked. It pairs a set of Orai1 conformational snapshots with a chemically diverse selection of Orai1-binding compounds, allowing the three docking tools to be compared on a realistic prediction task once their behaviour has been characterised on the calibration dataset.

Table 3: Experimental Orai1 ligands. Proposed binding sites are model-derived (mutagenesis, docking, molecular dynamics) and filter-proximal rather than experimentally resolved ligand-bound poses. No cognate Orai1 complex is resolved for any compound, and the binding site of 2-APB (docked as the neutral form 2abp-NH2) is not experimentally resolved.

Ligand Name	2-APB	GSK-7975A	Synta-66
Molecular Formula	$\text{C}_{14}\text{H}_{16}\text{BNO}$	$\text{C}_{18}\text{H}_{12}\text{F}_5\text{N}_3\text{O}_2$	$\text{C}_{20}\text{H}_{17}\text{FN}_2\text{O}_3$
Molecular Weight (Da)	225.1	397.3	352.4
Heavy Atom Count	17	28	26
Rotatable Bonds	5	4	5
Aromatic Rings	2	3	3
Docked protonation	neutral	neutral	neutral
Reported IC_{50} (μM)	biphasic, no single inhibitory IC_{50}	4.1 (Orai1) [68]	1–3 (native CRAC current) [69]

Table 3: Experimental Orai1 Ligands (continued)

Ligand Name	2-APB	GSK-7975A	Synta-66
IC ₅₀ range (μ M)	potentiates near 5, inhibits above roughly 40 [68]	3.8–4.1 (Orai3, Orai1) [68]	1.0–3.0 [69]
pIC ₅₀	not defined	5.39–5.42	5.52–6.00
Orai1 IC ₅₀ (μ M)	not resolved	4.1 [68]	not reported [69]
Assay	CRAC current, biphasic [68]	whole-cell patch clamp [68]	native CRAC current, RBL cells [69]
Primary Mechanism	Bimodal modulator (pore, STIM1)	Channel blocker, filter-linked	Extracellular pore blocker
Proposed Site	Not experimentally resolved for 2-APB [38]	Putative selectivity-filter (E106), direct or allosteric [68]	Proposed extracellular TM1/TM3, loops 1/3 [69]
Site Evidence	Functional two-site evidence for native 2-APB, site not resolved [38]	Mutagenesis, pore-geometry dependent	Docking, MD, mutagenesis

Potencies are quoted with the assay that produced them. GSK-7975A values are whole-cell patch-clamp IC₅₀ measurements on overexpressed Orai1 (4.1 μ M) and Orai3 (3.8 μ M), the same source also reporting 0.8 μ M in RBL cells [68]. Synta-66 values are for the native CRAC current in RBL mast cells [69]. 2-APB is biphasic and assay-dependent, potentiating overexpressed Orai1 currents near 5 μ M and inhibiting them above roughly 40 μ M [68], so no single inhibitory IC₅₀ is quoted for it. pIC₅₀ is the negative decadic logarithm of the IC₅₀ in molar units.

Limited receptor flexibility was represented with four supplied Orai1 molecular-dynamics snapshots, namely Fr0, Fr300, Fr400 and Fr499 [37], [70]. No pH-based protonation was applied to them. The snapshots arrive with CHARMM hydrogens and CHARMM histidine states, receptor preparation strips every hydrogen and collapses HSD, HSE and HSP to plain HIS, and the PDBQT converter then re-adds polar hydrogens by its own geometric rules, so the receptor protonation reported here is neither inherited from the molecular-dynamics model nor titrated at a stated pH. The selected ligands were prepared for a nominal pH of 7.0. Both protonation states of GSK-7975A were prepared and docked, and the Orai1 analyses reported in this thesis use the neutral form, which is the species carried through the transmembrane filter, the validity report, the pose clustering and the interaction analysis. The descriptors above are those of that neutral species (C₁₈H₁₂F₅N₃O₂, 397.3 Da), and Table 20 tabulates the same species, so the two tables and the reported results now describe one consistent ligand. For reference the deprotonated

form is $C_{18}H_{11}F_5N_3O_2^-$ at 396.3 Da. The deprotonated input file was corrected during this work. Its earlier preparation had removed the phenol proton from the three-dimensional structure without assigning the compensating negative charge, so the file described a neutral radical with a dearomatised ring rather than the phenolate. It has been rebuilt as the phenolate with its heavy-atom geometry preserved exactly, and the deprotonated form does not enter any result reported here. 2-APB and Synta-66 were docked as neutral species.

The experimental ligand set contains three Orai-related chemotypes, namely 2-APB, docked in its neutral protonation state and labelled 2abp-NH2 in the per-compound results tables, together with GSK-7975A [39], [68] and Synta-66 [69]. The docked structure is native 2-APB (2-aminoethoxydiphenyl borate), so the reported 2-APB pharmacology [38] characterises the docked molecule directly. GSK-7975A was docked in its neutral form, which is the species whose descriptors appear in Table 3 and in Table 20 and which underlies every Orai1 result reported here. A deprotonated phenolate form was also prepared during ligand setup and is retained in the deposited input files under a name carrying the suffix deprot, but it enters no analysis in this thesis and its descriptors differ from the docked species, having one fewer hydrogen-bond donor and a molecular weight of 396.3 Da. No cognate Orai1 crystal pose is available for any of the docked compounds, so this set supports a reference-free case study rather than a conventional accuracy benchmark.

All three compounds are tool compounds for store-operated calcium entry, that is the STIM1 to Orai1 pathway that opens calcium release-activated calcium channels once the endoplasmic store is depleted, yet they act on it in three different ways, which is what makes the set informative for a docking comparison.

GSK-7975A and Synta-66 act at or near the extracellular pore mouth and the selectivity filter rather than in the conduction pathway, so for those two a chemically valid pose lodged inside the transmembrane slab is implausible. 2-APB is carried here as a compound whose site is not experimentally resolved and whose documented STIM1 arm is cytosolic, so the same expectation cannot be asserted for it on structural grounds. The placement criterion used in the results chapter therefore applies one operational study assumption uniformly to all three compounds rather than an established fact about each of them. Their differing scaffolds also load the three docking tools unevenly. The full pharmacology of each compound, namely GSK-7975A, Synta-66 and 2-APB (docked in its neutral form, 2abp-NH2), together with the sites, selectivity liabilities and mechanistic differences that make the biologically defensible region non-identical across the set, is set out in Appendix E.

Docking each of the four receptor snapshots with each of the three drug compounds yields 12 frame–ligand units. Each unit was modelled with every toolchain to allow comparison of the resulting poses.

3.2 Evaluation Metrics

Assessing generated poses requires three complementary questions. First, is the ligand placed in a receptor region capable of the interactions relevant to the pharmacological hypothesis? Second, does the ligand retain a credible internal conformation and fit within that region? Third, is the resulting protein-ligand complex physically and chemically admissible? The metrics below address placement, form and validity separately before combining them.

3.2.1 Near-Nativeness ($\text{RMSD} \leq 2\text{\AA}$)

The docking results are compared on three primary measures, applied identically to every tool. The first is geometric accuracy. It is the symmetry-corrected heavy-atom root-mean-square deviation (RMSD) between a docked pose and the crystallographic ligand, defined as

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^N d_i^2}{N}}$$

where N is the number of heavy atoms and d_i is the Euclidean distance between the i -th pair of corresponding atoms in their docked and crystallographic positions, computed in place without superposing the pose onto the crystal ligand. A pose counts as accurate at the conventional threshold of $\text{RMSD} \leq 2\text{\AA}$ [22], [48]. The symmetry correction matters because many benchmark ligands are wholly or partly symmetric, so a naïve atom mapping would overstate the error, and it is computed by graph-isomorphism atom mapping over the symmetry-equivalent assignments [42], [46]. Because a geometrically close pose can still be chemically or sterically impossible, accuracy is paired with physical plausibility assessed by the PoseBusters battery (PB-valid), described below [34].

The two are combined into a single valid-near-native criterion, namely $\text{RMSD} \leq 2\text{\AA}$ and PB-valid, which is the headline metric throughout the crystal-referenced calibration chapters and separates a pose that merely looks right from one that could be used. It is deliberately kept distinct in name from the operational usable-pose rule defined for Orai1 below, which is RMSD-free because that target has no crystal ligand to measure against. The $2\text{\AA}$ threshold is the established standard in the docking literature and the AutoDock Vina benchmark, keeping the results comparable with published work [22], [34], [48].

Because a tool can generate a near-native pose yet fail to rank it first, accuracy is reported under two selection regimes, namely the realistic top-1 pose and the oracle (best) pose, the most accurate pose anywhere in the tool's output. The gap between them isolates ranking quality from sampling quality and is developed through the per-rank success rate and the cumulative recovery within the first k ranked poses. A single top-1 view would conflate a weak confidence model with weak sampling and systematically understate the deep-learning tools.

Two secondary measures complete the comparison. Binding-site recovery asks the coarser question of whether a tool places the ligand in the correct pocket at all, assessed by clustering

the docked poses across tools and against the crystallographic pocket, which is the appropriate resolution for the reference-free Orai1 system. Computational cost is reported as wall-clock time and as core-seconds per qualifying pose, that is per pose that is PoseBusters-valid and within 2Å of the crystal ligand, since practical value for large-scale screening depends on qualifying poses returned per unit of compute.

3.2.2 Form (Kabsch RMSD $\leq 1\text{\AA}$)

Geometric accuracy scored as in-place RMSD, the quantity the $\leq 2\text{\AA}$ gate uses, conflates two independent errors, namely where the molecule sits and what shape it holds. The approximate decomposition $\text{in-place}^2 \approx \text{placement}^2 + \text{form}^2$ separates them, the form error being the best-fit (Kabsch) RMSD that remains once translation and rotation are removed and the placement error the residual diagnostic obtained from that relation. A pose is called form-correct when its form error is $\leq 1\text{\AA}$.

The heavy-atom RMSD remaining after this best fit is the form error, the component of the deviation that no repositioning can remove, that is the discrepancy in the ligand's own three-dimensional shape. With both ligands translated to a common centroid and R the optimal proper rotation obtained by the Kabsch singular-value-decomposition construction [60], it is

$$\text{RMSD}_{\text{form}} = \sqrt{\frac{\sum_{i=1}^N \|Rx_i - y_i\|^2}{N}}$$

where x_i and y_i are the centred coordinates of the i -th heavy-atom pair and N the number of heavy atoms. As with the in-place RMSD, the minimisation is taken over the ligand's symmetry-equivalent atom mappings so that chemically equivalent atoms are not spuriously penalised [42], and the value is read from the PoseBusters implementation [34].

For a fixed atom correspondence, rigid superposition gives the per-pose decomposition $\text{in-place}^2 = \text{placement}^2 + \text{form}^2$, and placement can be derived as $\sqrt{\text{in-place}^2 - \text{form}^2}$. That correspondence is not shared by the two terms as they are computed here, because the in-place and best-fit values are each minimised over the symmetry-equivalent atom mappings independently, so the two minimisations may settle on different mappings for the same pose. The decomposition is therefore approximate rather than an identity, and the placement value is reported as a residual diagnostic rather than as an independently measured component. Form recovery is reported for PoseBusters-valid outputs independently of the 2 Å in-place threshold, and a pose is form-correct at Kabsch RMSD $\leq 1\text{\AA}$.

This separation matters because binding is mediated by specific directional contacts, namely hydrogen bonds, hydrophobic burial, π -stacking and salt bridges, whose geometry is set by the ligand's bound conformation, so a pose reproduces the native interaction pattern only if its form is right. Isolating form therefore probes the physical prerequisite for the interactions the docking is meant to predict, and it localises the limiting stage, since a placement-limited pose (right shape, wrong position) implicates pocket targeting while a form-limited pose (wrong conformation)

implicates conformer generation or refinement. The tool-by-tool analysis is presented in the Results (Figures 3 to 5).

3.2.3 Physical and Chemical Validity (PoseBusters Checks)

Geometric and form closeness to the crystal do not by themselves guarantee that a pose is physically possible, since RMSD alone can be satisfied by chemically distorted or sterically clashing poses, the failure mode the original PoseBusters study attributes to AI-based methods [34]. Every pose was therefore run through the PoseBusters “dock” battery and is counted PB-valid only if it passes all twenty-two applicable PoseBusters checks (Table 15). The battery also defines four further checks that compare a pose against a co-crystallised reference ligand, namely the reference-dependent identity tests on molecular formula, bond graph, tetrahedral chirality and double-bond stereochemistry. No reference ligand was supplied to the checker in any run of this thesis, including the calibration benchmark, so those four checks were not applicable and were not executed, and their result columns are absent from every result file. PB-valid therefore denotes the twenty-two reference-free checks in all three datasets, which keeps the criterion identical for the crystal-referenced calibration set and the reference-free Orai1 case study.

The checks fall into three groups. The first covers chemical validity, namely that the pose file and the receptor file parse into RDKit molecules, that the ligand passes chemical sanitisation and converts to a standard InChI string, that all its atoms form a single connected component and that it carries no radical atoms. Because the four reference-dependent checks were not run, PB-valid does not certify that the molecular formula, the bond graph or the tetrahedral and double-bond stereochemistry of a docked pose still match the ligand that was submitted to the tool. The second covers intramolecular geometry, namely bond lengths and angles within distance-geometry tolerance, flat aromatic rings and double bonds, absence of internal steric clash and no gross conformational strain. The third covers intermolecular validity against the receptor, namely minimum distances to the protein, to organic and inorganic cofactors and to waters together with the corresponding volume overlaps, so that the ligand neither interpenetrates the receptor nor occupies space already taken. The full list of individual checks and their tolerances is given in the Appendix (Table 15).

3.2.4 Operational Placement (Transmembrane Exclusion)

The Orai1 target has no crystallographic ligand, so near-nativeness cannot be scored and the accuracy criterion is replaced by an operational placement rule. A pose counts as usable when it is PoseBusters-valid, placed on the receptor and outside the transmembrane exclusion slab defined below. This subsection states both conditions geometrically, because together they gate every Orai1 quantity reported in this thesis.

Placed on the receptor is a physical-contact gate rather than a pocket criterion, and it exists

because a generative model occasionally emits a rigid-body-translated pose that is internally intact yet sits hundreds or thousands of Å away from the protein. The test is applied to the ligand heavy atoms of the pose as written, with hydrogens ignored. A pose counts as placed when three conditions hold at once. Its coordinates must be readable and finite, with no atom further than 10^4 Å from the origin. The PoseBusters protein-ligand maximum-distance check, which requires the ligand to lie within the interaction range of the receptor, must pass, this verdict being authoritative wherever the column is present. And the heavy-atom centroid must fall inside a generous receptor envelope, namely a radial distance of at most 80 Å from the pore axis defined below and an axial fraction between -4 and 5 of the slab frame, where the axial fraction is the axial coordinate divided by the slab thickness so that 0 and 1 are the two ring planes. The envelope is deliberately far wider than the hexamer, whose outer protein shell reaches 32 to 36 Å from the axis, so it removes only translated artefacts and never a pose in contact with the protein. Poses failing the gate are recorded as invalid placement and enter no reported count.

Orai1 is a hexamer whose pore is lined by the TM1 helix of each of the six subunits. Along TM1 the pore runs from the cytosolic basic residue Arg91 to the extracellular selectivity filter Glu106, so the band of the structure lying between those two residue rings is the membrane-spanning region, which conducts ions rather than presenting a druggable surface pocket. The slab is built from that pair of rings. For a given receptor frame let p_{R91} be the centroid of the six Arg91 C α atoms, one per subunit, and p_{E106} the centroid of the six Glu106 C α atoms. These define the pore axis

$$\hat{a} = \frac{p_{E106} - p_{R91}}{\|p_{E106} - p_{R91}\|}, \quad L = \|p_{E106} - p_{R91}\|$$

and for any point x the axial coordinate is $t(x) = (x - p_{R91}) \cdot \hat{a}$, so that $t = 0$ is the Arg91 plane and $t = L$ the Glu106 plane. The exclusion slab is the region

$$\{x : 0 \leq t(x) \leq L\}$$

that is the set of points lying between the two ring planes, taken with no padding beyond either plane. A pose is flagged as transmembrane when the centroid of its ligand heavy atoms falls inside the slab, and it is removed from the usable count. The fraction of ligand atoms inside the slab and a majority-of-atoms flag are recorded alongside the centroid test so that a stricter atom-level rule can be applied without recomputing the geometry.

Two properties of this definition matter for interpretation. First, the slab is bounded only along the pore axis and is unbounded radially, so a ligand resting against the lipid-facing outer surface of the hexamer is excluded just as a ligand in the conduction pathway is. This is deliberate, since both positions sit within the membrane band and neither is a solvent-accessible surface pocket. The pore lumen itself is far narrower than the excluded region, with a mean backbone radius of 7.0 to 7.7 Å across the four frames against an outer protein shell reaching 32 to 36 Å from the axis in the same band. Second, because the four Orai1 snapshots are different conformations in different coordinate systems, the slab is rebuilt for each frame from the exact receptor file

that frame's poses were docked into, so poses and slab always share a coordinate system. The resulting axial thickness is stable across frames at 22.4 to 22.8Å.

The rule is an operational hypothesis rather than ground truth. It encodes the pharmacological premise that GSK-7975A and Synta-66 act at or near the extracellular pore mouth and the selectivity filter rather than inside the conduction pathway, and it extends the same expectation to 2-APB, whose site is not experimentally resolved. It cannot establish that an excluded pose is biologically wrong.

3.2.5 Clustering

Because a predicted pocket from fpocket or P2Rank carries no ligand atoms, the only quantity comparable across docked poses, predicted pockets and the crystallographic site is the three-dimensional centre, so clustering is performed on each pose's heavy-atom centroid rather than on full-atom RMSD, with RMSD retained as a complementary pose-quality measure. For every complex the centroids of all three tools are pooled into a pairwise Euclidean distance matrix and partitioned by complete-linkage agglomerative clustering cut at 8Å, the physical scale of a binding pocket. The number of clusters is discovered rather than fixed and may be one, so a complex on which all three tools genuinely converge is reported as a single site. Complete linkage guarantees that no cluster's internal spread exceeds 8Å and the cut is deterministic. This replaces the earlier silhouette-maximising K-medoids default, which could never return a single cluster and so split converged sites into spurious sub-pockets.

Candidate pockets within a complex are ranked by cross-tool agreement rather than by raw pose count, which stops a high-throughput tool from winning a site on pose count alone. That consensus rule is compared against four alternative prospective rules, including the legacy pose-count rule, in Figure 8.

For the reference-free Orai target the same 8Å centroid clustering is applied within each MD frame, but because the four snapshots occupy different coordinate systems poses cannot be clustered by absolute position across frames. Each pose is instead represented by the receptor residues within 4Å of the ligand, folded over Orai's six-fold symmetry, so that poses from different frames remain comparable without superposition. With no crystal to arbitrate, the consensus site is defined as the geometric median of the per-tool sites, one point per tool. Cross-tool agreement is then reported as the Euclidean distance between two tools' consensus sites within a frame-ligand unit, and a pair counts as agreeing when that distance is at most 5Å. The threshold is the same physical reasoning as the 8Å clustering cut, namely that two sites separated by less than the radius of a binding pocket are the same site, and it is deliberately more permissive than any pose-level RMSD criterion because it compares pocket centres rather than atom positions.

Cluster quality is assessed on three complementary axes. Silhouette, Calinski-Harabasz, Davies-Bouldin and nearest-cluster separation are defined only for complexes with at least two clusters and are therefore omitted for one-cluster cases. Compactness can be reported for a

single cluster, while bootstrap Jaccard overlap and adjusted Rand indices quantify reproducibility under resampling and against independent partitions. Crystal-site recovery is evaluated separately with the prespecified 4 Å centroid threshold. The cluster nearest the crystal is not automatically counted as recovered. Both the recovery rates under that threshold and the internal, stability and ranking-rule results are reported with Figures 7 and 8.

3.2.6 Statistical Analysis

Every inferential statement in the Results rests on the conventions set out here. All tests are two-sided and are read at a nominal α of 0.05. Inferential comparisons are accompanied by an effect size or by the underlying paired proportions or counts, as appropriate, so that the size of a difference can be separated from the confidence attached to it. The tests are distribution-free throughout, because per-complex yields, RMSD values and contact counts are bounded, skewed and frequently zero-inflated, so no normality assumption is made anywhere.

The inferential unit differs by design and is fixed before any test is run. For the calibration benchmark it is the complex, so the poses of one complex are first reduced to one value per complex per tool, either a per-complex yield, a per-complex best value or a per-complex binary recovery indicator. For the Orai1 system there is no crystal reference and each ligand is docked into four molecular-dynamics frames, so the unit is the frame-by-ligand pair. That unit removes the nesting of poses inside a docking run, but it does not make the units independent, because the four frames of a given ligand are snapshots of one receptor and remain correlated with one another. The effective number of independent experimental observations is therefore three, namely the three ligands. Orai1 cross-panel contrasts are for that reason never advanced as established findings. They are computed and corrected within the declared families below, and their adjusted p-values are reported so that the reader can see which would and would not survive, but the conclusion drawn from them is descriptive in every case, and every such contrast is re-tested at the ligand unit in a ligand-level sensitivity analysis reported with the accompanying results. Figures may display pose-level distributions for readability, but wherever a figure shows poses and the text quotes a test, that test was computed on the aggregated unit, and both values are then given side by side.

Most benchmark comparisons are paired, because the same complexes pass through every tool and every ranking depth. Paired binary outcomes such as recovery at a given depth, the existence of at least one PoseBusters-valid pose or crystal-cluster reach are tested with Cochran's Q as the omnibus across three or more conditions and with the exact McNemar test on the discordant pairs for each pair of conditions, with Wilson score 95% intervals on the underlying proportions. Paired continuous outcomes such as per-complex yields, RMSD medians and pose counts use the Friedman test with Kendall's W as its effect size for the omnibus and the Wilcoxon signed-rank test with the matched-pairs rank-biserial correlation for the pairs, with Hodges-Lehmann estimates for the median paired difference. Unpaired comparisons arise only where the two groups are different molecules or different ligand panels,

chiefly the Orai1 experimental ligands against the benchmark ligands docked into the same frames. These use the Mann-Whitney U test with Cliff's δ as the effect size, oriented explicitly in the text, or Fisher's exact test for pooled two-by-two counts, with the Kruskal-Wallis test and ϵ^2 for three-group comparisons and the Jonckheere-Terpstra test where an ordered trend is hypothesised. Monotone association between two per-complex quantities, for instance a tool's own rank against pose quality, is measured by Kendall's τ -b or Spearman's ρ per complex and then tested with a one-sample Wilcoxon signed-rank test against zero. Where a paired binary contrast returns no significant difference and that null result carries weight in the argument, the contrast is additionally reported as a difference of correlated proportions with a Newcombe hybrid score interval, and its resolution is characterised by the number of discordant pairs, the minimum detectable difference at 80% power and a pair of one-sided tests against a stated equivalence margin. This is applied once, to the calibration contrast between the two leading pipelines, and it is the only place in this thesis where an equivalence claim is made. Everywhere else a non-significant result is reported as an absence of detected difference and nothing more.

Multiplicity is corrected within explicitly declared families rather than across the thesis as a whole. Small families use Holm-Bonferroni step-down adjustment, and adjusted values are labelled p_{holm} or Holm p . They are the three tool-versus-tool contrasts within one metric, the depth contrasts within one tool, the variant contrasts within one family, the per-tool experimental-versus-control contrasts within one Orai1 metric, which is the three-test family behind the per-unit yield tests and behind the total-contacts test, the eighteen tool-by-metric cluster-quality cells, the four consensus-site contrasts, namely the three tool pairs and their pooled summary and the fifteen per-residue contact-frequency contrasts within one tool, which is the family behind the residue hot-spot test. Large screening families use the Benjamini-Hochberg false-discovery-rate procedure, and adjusted values are labelled q . They are the per-check PoseBusters sweep, the per-descriptor sweeps, the fifteen interaction-type contrasts within one tool, which is the family behind the interaction-type profile and which is corrected separately for each tool rather than pooled over the thirty cells of both mature tools, and the nine tool-by-metric rank-correlation cells. A condition that supplies too few units to test, such as EquiBind with a single usable experimental pose, is left out of its family rather than entered as a hypothesis, so the family shrinks accordingly. Ranking depths are nested cumulative pools rather than independent hypotheses, so in the body they are not treated as a further multiplicity factor and each depth is corrected within its own family, which is stated at the point of use. The appendix recovery chapter departs from that convention deliberately and says so, correcting its six prespecified between-variant comparisons jointly across all four depths as one 24-test family, which is the more conservative choice and is why an effect can be significant there at one depth and not at another. Uncorrected values are reported only where they are named as such.

Resampling is used where no closed-form test applies. Bootstrap confidence intervals for medians, effect sizes and cluster stability use 2,000 resamples with a fixed seed, resampling the inferential unit rather than individual poses so that within-complex correlation is preserved, and the bootstrap Jaccard index quantifying cluster reproducibility uses 25 resamples per

complex. Permutation tests, namely the paired permutation test and the three-way cross-tool co-occurrence test, use 10,000 permutations with a fixed seed and report the add-one-adjusted p-value, whose floor is therefore quoted as below $1e-4$ rather than as zero.

Missing and dropped cases are handled by listwise deletion on the analysed unit, and the surviving count is stated wherever it differs from the headline sample. A complex that produced no pose for one tool cannot enter a paired comparison involving that tool, a per-complex correlation is defined only when the complex contributes at least three poses with non-constant values, and a per-complex proportion is defined only when the complex produced at least one pose. Degenerate pairs, for instance two conditions whose paired differences are all zero, return an undefined p-value that is carried through the correction as undefined rather than being counted as a hypothesis.

Two caveats stand over the whole analysis. First, the four Orai1 frames of a given ligand are conformational snapshots of one receptor, so the twelve frame-by-ligand units are correlated rather than independent replicates, which is why Orai1 tests are treated as low-powered and an absent difference is reported as an absence of detected difference rather than as evidence of equivalence. Second, the calibration benchmark is used both to select the dominant variant of each toolchain and to report that variant's performance, so the reported calibration figures are optimistic relative to a fully held-out evaluation, and the point is made again where the variants are selected.

4 Results

4.1 Calibration Dataset

Of the 308 source benchmark complexes, 303 were retained in the common calibration analysis for AutoDock Vina, DiffDock and EquiBind. Every generated pose was screened with the PoseBusters suite and compared with the crystal reference by symmetry-corrected heavy-atom RMSD and Kabsch RMSD. Pocket-guided EquiBind variants based on fpocket and P2Rank did not improve the variant-selection endpoint defined below relative to the unguided configuration.

Both learned pipelines were evaluated before and after local optimisation with smina and gnina. The toolchain variants were chosen on one three-way criterion, called the variant-selection endpoint throughout this thesis, under which a complex counts only when one of its poses is at once PoseBusters-valid, within 2Å in place and within 1Å in Kabsch form. It is deliberately stricter than the headline metric defined in the Methods chapter, which requires validity and the 2Å in-place gate alone, and the two are never interchanged. Variant selection and the performance evaluation that follows were both carried out on this same calibration benchmark, with no held-out split between them, so the calibration figures reported below are optimistic

relative to a fully independent evaluation. For DiffDock the two refiners are separated by at most two complexes on the variant-selection endpoint, at 76 against 74 complexes at rank-1 and 133 against 132 at the top-15 depth, so the choice between them is immaterial on this criterion, although gnina yields the higher pooled physical validity at 87.1% against 84.5%. For unguided EquiBind, gnina produced the clearly higher validity and combined recovery. These are empirical configuration results. The study did not measure optimisation trajectories and therefore does not establish that one refiner is inherently gentler or more aggressive.

Accordingly, the main comparison uses DiffDock + smina and EquiBind + gnina as the primary refined variants. Tables and the first mention in each analysis use the full pipeline names. Subsequent prose may shorten them to the generator names for readability, while raw or alternative variants are always qualified explicitly. Appendix analyses that use raw or gnina-optimised DiffDock output identify the variant explicitly. Their values should not be interchanged with the main comparison.

The AutoDock arm was selected on the same variant-selection endpoint, across the six variants of Table 4. The scoring family is decided first and decisively. Vina reaches 71 complexes at rank-1 and 117 at the top-15 depth against 33 and 42 for the Vinardo reparametrisation on matched searches, an exact McNemar p of $2.8\text{e-}06$ and $7.0\text{e-}18$, so Vinardo is retained only as a scoring-function contrast, reported in Table 4 itself and otherwise described at protocol level in Appendix C, while Vina supplies every AutoDock result discussed in the remainder of the main text. Within the Vina family the three treatments are far closer, at 65, 71 and 69 complexes at rank-1 and 112, 117 and 111 at top-15 for the raw search, gnina rescoring and gnina refinement respectively. As with DiffDock's two refiners, no pair is separable on this endpoint, with gnina rescoring against the raw search at $p = 0.405$ and $p = 0.063$ and against gnina refinement at $p = 0.774$ and $p = 0.109$. The choice therefore rests on physical validity, where rescoring is never worse than the alternatives and gives the highest pooled rate at 99.5% against 98.2% for the raw search and 92.6% for refinement, the last of these losing validity not because it moves atoms, since rescoring moves them too and gains validity, but because the learned score rather than the empirical function drives its final geometry. AutoDock Vina + gnina is accordingly the dominant AutoDock variant and is used throughout the main Results, matching DiffDock + smina and EquiBind + gnina in being a single named variant per family.

One asymmetry in what that rescoring does is worth stating, because it is easy to read the three refinement columns as equivalent interventions. Vina already performs a local optimisation inside its own search, so a post-hoc gnina pass adds a learned ranking to a pipeline that is already complete, which is why it moves AutoDock only from 25.7% to 29.4% at rank-1 on the validity-aware near-native endpoint. For DiffDock and EquiBind the same pass repairs geometry that the generator never optimised, which is why it transforms their validity. The ordering of the three pipelines is unchanged by the choice of AutoDock variant, since DiffDock + smina leads at every depth on either arm, but the size of that lead is not, so raw and rescored AutoDock numbers are never compared with each other in what follows.

Table 4: Pose Production and Physical-Validity Summary of All Docking Variants

		PoseBusters				Near-Nativeness RMSD $\leq 2\text{\AA}$			Form Kabsch RMSD $\leq 1\text{\AA}$			PoseBusters Valid RMSD $\leq 2\text{\AA}$ Kabsch RMSD $\leq 1\text{\AA}$		
Docking variant		Poses	Comp.	Poses	%	Comp.	Poses	%	Comp.	Poses	%	Comp.	Poses	%
AutoDock (raw)	Vina	9,043	303	8,883	98.2%	151	215	2.4%	216	2,814	31.1%	116	151	1.7%
+ gnina rescoring		9,043	303	8,995	99.5%	151	221	2.4%	216	2,801	31.0%	117	152	1.7%
+ gnina refinement		9,043	303	8,372	92.6%	153	229	2.5%	215	2,773	30.7%	113	153	1.7%
AutoDock (raw)	Vinardo	9,062	303	8,839	97.5%	55	69	0.8%	182	2,706	29.9%	43	50	0.6%
+ gnina rescoring		9,062	303	8,984	99.1%	54	67	0.7%	181	2,689	29.7%	42	49	0.5%
+ gnina refinement		9,062	303	7,957	87.8%	55	70	0.8%	180	2,670	29.5%	42	51	0.6%
DiffDock (raw)		9,013	242	2,198	24.4%	152	1,683	18.7%	225	4,030	44.7%	83	722	8.0%
DiffDock + smina		8,993	300	7,597	84.5%	173	1,715	19.1%	237	3,915	43.5%	138	1,227	13.6%
DiffDock + gnina		8,993	300	7,831	87.1%	176	1,716	19.1%	235	3,885	43.2%	138	1,219	13.6%
EquiBind (raw)	unguided	9,089	35	263	2.9%	19	103	1.1%	185	2,479	27.3%	2	21	0.2%
EquiBind smina		9,082	218	3,565	39.3%	55	339	3.7%	184	2,521	27.8%	39	157	1.7%
EquiBind gnina		9,074	247	4,702	51.8%	65	373	4.1%	196	2,578	28.4%	42	187	2.1%
EquiBind (raw)	fpocket	9,090	13	27	0.3%	0	0	0.0%	180	2,449	26.9%	0	0	0.0%
EquiBind smina		9,090	197	1,381	15.2%	14	22	0.2%	182	2,407	26.5%	7	9	0.1%
EquiBind gnina		9,090	261	2,580	28.4%	24	35	0.4%	189	2,443	26.9%	14	19	0.2%
EquiBind (raw)	P2Rank	8,520	12	47	0.6%	0	0	0.0%	181	2,336	27.4%	0	0	0.0%
EquiBind smina		8,520	223	1,765	20.7%	18	31	0.4%	181	2,323	27.3%	5	11	0.1%
EquiBind gnina		8,519	282	3,242	38.1%	35	59	0.7%	190	2,380	27.9%	18	35	0.4%
Comp. counts the complexes of the 303 for which the variant produced at least one pose meeting that block's criterion, whereas Poses and % count individual poses and their share of the poses the variant produced.														

4.1.1 PoseBusters Validity

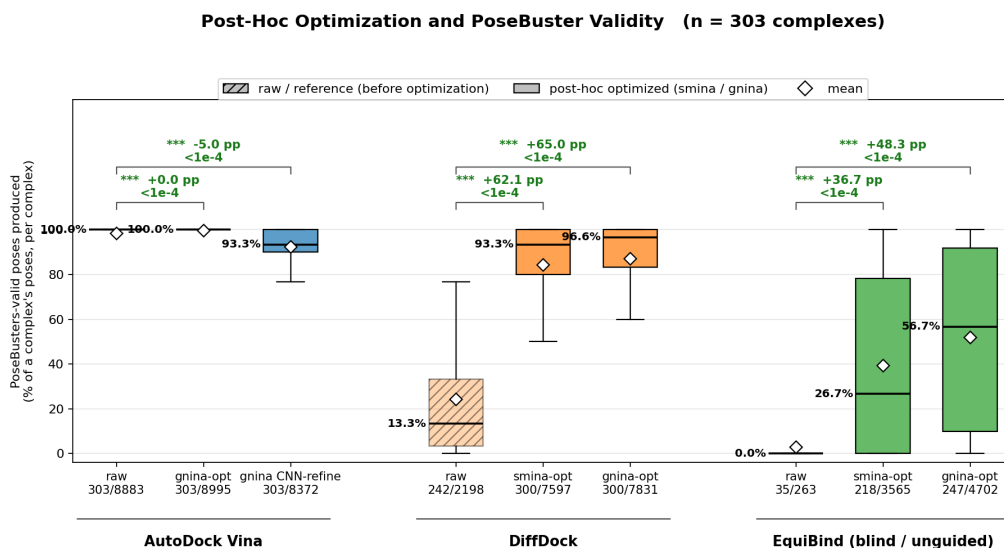


Figure 2: Post-hoc optimisation and PoseBusters validity. Each tool is shown raw beside its optimised variants, with the dominant variant of each family being AutoDock Vina + gnina, DiffDock + smina and EquiBind + gnina. The panel token `gnina-opt` denotes the gnina rescoring arm called AutoDock Vina + gnina throughout this thesis, and `gnina CNN-refine` denotes the separate gnina refinement arm, which is not the reported pipeline. Significance brackets carry Holm-corrected p-values, where *** $p < 0.001$, ** $p < 0.01$ and * $p < 0.05$

Figure 2 shows per-complex PoseBusters-validity yield across the 303 benchmark complexes for raw and refined outputs, with each variant's complex and pose counts matching Table 4. Post-hoc minimisation is the dominant driver of validity for the learned tools. The DiffDock median rises from 13.3% to 93.3% with smina and 96.6% with gnina, and unguided EquiBind rises from 0% to 26.7% with smina and 56.7% with gnina. The gains are near-unanimous across complexes and remain statistically detectable after Holm correction, with matched-pairs rank-biserial effects of 0.994 to 1.000 and adjusted p-values below 10^{-35} throughout. The Hodges-Lehmann median gain for DiffDock is +62.1 percentage points with smina and +65.0 with gnina. Within-tool omnibus tests confirm the effect, at Friedman $\chi^2(2) = 506.4$ with Kendall's $W = 0.84$ for DiffDock and $\chi^2(2) = 416.2$ with $W = 0.69$ for EquiBind. gnina is clearly stronger than smina for EquiBind, while the DiffDock refiners are much closer.

AutoDock behaves differently, and the three AutoDock boxes in the panel are what motivate the choice of its dominant variant. The raw search is already at the ceiling with a median per-complex yield of 100%, a pooled validity of 98.2% across all produced poses and at least one PoseBusters-valid pose for every one of the 303 complexes, that is full coverage with no ligand-preparation dropouts. Rescoring with gnina therefore leaves the median unchanged and instead lifts pooled validity to 99.5%, a shift carried by the minority of complexes that still had invalid poses to recover, at Friedman $\chi^2(2) = 322.8$ with Kendall's $W = 0.53$ over all 303 complexes and a rescoring-against-raw rank-biserial of 0.84 with $p_{\text{holm}} = 1.7\text{e-}07$ over the 49

complexes whose valid-pose count differs between the two arms. Refinement moves the median the other way, down to 93.3% for a Hodges-Lehmann loss of 5.0 percentage points, because minimising the coordinates of a pose that is already physically admissible has little left to repair and tends to move it off that admissible geometry, although this comparison does not isolate the mechanism. This is the empirical basis for taking rescoring rather than refinement as the dominant AutoDock treatment, and it is the mirror image of the learned tools, where the same minimisation is what makes their output usable at all. Physical validity is therefore where the physics-based engine retains its clearest advantage, and it is largely unaffected by removing the search box, because a pose minimised against an objective containing an explicit repulsion term is admissible wherever in the receptor it happens to sit.

Table 5 decomposes those pass rates over the three check groups defined in the Methods. Almost none of the discrimination comes from chemistry or from covalent geometry. Not one of the 81,373 poses in the nine variants of the table fails any of the six chemical-validity checks, so in every variant the pose file and the receptor file parse, and the ligand passes sanitisation, converts to a standard InChI string, remains a single connected component and carries no radical atoms. Covalent geometry is almost as clean, with 25 bond-length failures and 22 bond-angle failures across those same 81,373 poses, that is 0.03% on each check. Nearly all of the spread in Table 4 therefore sits in the intermolecular group, whose failure rate almost accounts for the invalid share on its own, at 75.2% failing against a 75.6% invalid share for raw DiffDock and 96.8% against 97.1% for raw EquiBind. Within that group a single check dominates, namely minimum distance to protein, which alone is failed by 73.8% of raw DiffDock poses and 96.5% of raw EquiBind poses against 0.2% for AutoDock Vina + gnina. The intramolecular group contributes materially for one tool only, since raw EquiBind fails it on 27.9% of its poses against at most 3.8% for any of the AutoDock or DiffDock variants in the table.

The per-check sweep that the analysis plan declares as a Benjamini-Hochberg family confirms that pattern. Comparing the three dominant variants complex by complex across all 303 complexes with a Friedman test on per-complex failure rates, one check at a time, the six chemical-validity checks are constant at zero and so supply no hypothesis, which shrinks the family to the sixteen checks that vary. Thirteen of those sixteen survive correction at $q < 0.05$, and minimum distance to protein carries by far the strongest signal at $\chi^2(2) = 393.3$ with $q = 6.2\text{e-}85$, followed by volume overlap with protein at $\chi^2(2) = 299.0$ with $q = 9.5\text{e-}65$. Bond lengths and bond angles separate the variants but only trivially, at $q = 8.3\text{e-}03$ on a mean per-complex failure rate of 0.10% for DiffDock + smina against zero for the other two variants. The three flatness checks do not survive, at $q = 0.155$ for aromatic ring flatness and $q = 0.368$ for both non-aromatic ring non-flatness and double-bond flatness.

Restricting the same decomposition to poses already within 2Å of the crystal ligand separates validity from placement, and most of the apparent gap turns out to be placement. Intermolecular failure for DiffDock + smina falls from 14.6% over all produced poses to 5.4% over its 1,715 near-native poses, and for EquiBind + gnina from 47.1% to 18.0% over 373 near-native poses. The pass-all rates converge accordingly, to 94.6% for DiffDock + smina and 81.8% for EquiBind

+ gnina against 98.2% for AutoDock Vina + gnina, so the 99.5% against 51.8% spread of Table 4 narrows to roughly sixteen percentage points once the pose is in the right place. A residual gap nevertheless survives that conditioning, because EquiBind + gnina still fails the intermolecular group on 67 of its 373 near-native poses against 4 of 221 for AutoDock Vina + gnina. Both of those subsets are small, so the residual difference is better read as a direction than as a calibrated effect size.

One qualification bears on how much independent corroboration this decomposition supplies. The group that does nearly all of the discriminating is also the group the search objective optimises against directly, because the Vina scoring function carries an explicit steric-repulsion term over ligand-receptor atom pairs, and minimum distance to protein together with volume overlap with protein are close to readings of that same term. A pose emitted by AutoDock, or minimised afterwards by smina or by gnina, has therefore already been driven down the coordinate that PoseBusters then scores it on, so part of the physics-based advantage is constitutive rather than an independent verdict. The refinement arm shows how tightly that coupling holds, and it does so by conceding the mechanism rather than by resisting it. gnina refinement is the one AutoDock treatment that replaces the empirical objective with the learned score, so it is the one arm whose final geometry is not driven by the explicit repulsion term, and it is also the only AutoDock Vina arm with a material intermolecular failure rate, at 5.4% against 0.1% for the raw search and 0.2% for rescoring. The intermolecular verdict therefore tracks the presence of that specific term rather than the fact of having been minimised at all, which is the strong form of the objection and is accepted here.

Two observations nevertheless keep the wider finding from dissolving into the circularity. The first is that raw EquiBind carries an intramolecular defect that no scoring convention explains, with 27.9% of its poses failing at least one intramolecular check, driven by internal steric clash at 22.1% and gross conformational strain at 23.3%, and it retains 10.7% intramolecular failure even on its near-native poses. That axis is not a reading of any docking objective and it orders the tools the same way. The second is quantitative and is set out in the next paragraph, namely that the validity requirement barely moves the recovery endpoint on which the tool comparison rests, so however much of the validity advantage is constitutive, it is not what produces the reported ordering.

There is a third and more direct check on how far that circularity reaches, namely to ask what the validity requirement actually costs the headline endpoint. Because the endpoint is a conjunction, the question has an exact answer, obtained by counting the complexes recovered under the 2 Å criterion alone and again under the same criterion with the PoseBusters requirement added. On the three variants carried through the Results the requirement removes at most four complexes of 303 at any depth. For AutoDock Vina + gnina it removes one complex at rank-1 and at top-5 and none at top-15 or top-30, for DiffDock + smina three to four complexes at every depth, and for unguided EquiBind + gnina one to three, which is a span of 0.00 to 1.32 percentage points across all three variants and all four depths. The recovery ordering, the paired differences and the intervals reported above are therefore set almost entirely by the 2

Å placement criterion, and they would barely move if the validity requirement were dropped altogether. Whatever advantage the physics-based engine draws from being scored on a quantity its own objective contains, that advantage is not what produces the recovery numbers, because the gate those numbers pass through is nearly transparent for every variant this chapter reports.

The requirement bites hard in exactly one place, namely the raw learned output that this work recommends against deploying. For raw DiffDock it removes 47 complexes at rank-1 and 55 to 58 at the deeper pools, a span of 15.5 to 19.1 percentage points, so the same gate that costs a refined pipeline one complex costs the unrefined one nearly a fifth of the set. That contrast is the useful form of the validity result. Physical validity discriminates decisively between raw and minimised learned output, and only marginally between the minimised pipelines, which is why refinement rather than tool choice is where it changes a practical decision.

Table 5: PoseBusters Failure Decomposition by Check Group

Docking variant	All Produced Poses						Near-Native Poses RMSD $\leq 2\text{\AA}$					
	Poses	Valid %	Chem. %	Intra. %	Inter. %	Min. dist. protein %	Poses	Valid %	Chem. %	Intra. %	Inter. %	Min. dist. protein %
AutoDock (raw)	9,043	98.2	0.0	1.6	0.1	0.1	215	97.7	0.0	0.9	1.4	1.4
AutoDock + gnina rescoring	9,043	99.5	0.0	0.4	0.2	0.2	221	98.2	0.0	0.0	1.8	1.8
AutoDock + gnina refinement	9,043	92.6	0.0	2.5	5.4	5.4	229	93.9	0.0	0.4	5.7	5.7
DiffDock (raw)	9,013	24.4	0.0	3.8	75.2	73.8	1,683	50.3	0.0	2.2	48.2	48.2
DiffDock + smina	8,993	84.5	0.0	1.5	14.6	13.4	1,715	94.6	0.0	0.1	5.4	5.4
DiffDock + gnina	8,993	87.1	0.0	1.4	12.0	10.7	1,716	94.5	0.0	0.2	5.3	5.2
EquiBind (raw)	9,089	2.9	0.0	27.9	96.8	96.5	103	20.4	0.0	10.7	78.6	78.6
EquiBind + smina	9,082	39.3	0.0	11.9	59.9	59.7	339	78.2	0.0	0.6	21.5	21.5
EquiBind + gnina	9,074	51.8	0.0	8.2	47.1	46.8	373	81.8	0.0	1.1	18.0	17.7

Valid % is the share of a variant's poses passing all twenty-two applied checks and reproduces the PoseBusters column of Table 4. Chem., Intra. and Inter. are the shares failing at least one check in the chemical-validity, intramolecular-geometry and intermolecular-validity groups of Table 15, and the three shares can sum past the invalid share because one pose may fail in more than one group. Min. dist. protein is the single minimum-distance-to-protein check, the dominant contributor throughout. Near-native poses are those within 2Å of the crystal ligand by symmetry-corrected heavy-atom RMSD without superposition, so their counts reproduce the near-nativeness block of Table 4, and the two halves of this table describe the same variants on all of their output and on their correctly placed output respectively.

Because each tool returns a ranked list rather than a single pose, the depth of the pool a user inspects changes how many physically valid poses are available. Three pools, rank-1, best of top-15 and best of top-30, are compared in Table 6. All depth gains reported in this chapter were assessed with exact two-sided McNemar tests on the discordant pairs, with Wilson score 95% confidence intervals and Holm-Bonferroni correction within the depth family, a complex counting as recovered whenever any pose within the inspected pool is both PoseBusters-valid and within 2Å. Because the pools are nested, a deeper pool can only add recovered complexes and never lose one, so for the depth steps every discordant pair runs in the same direction and the test asks only whether that one-sided count could have arisen by chance. The recovered share with its confidence interval is the more informative summary there, and the two-directional form of the test carries its usual meaning only in the between-tool comparisons.

In the pass-all analysis reported in Table 6, deepening the inspected pool from rank-1 to top-15 adds 62 recovered complexes for AutoDock Vina + gnina (+20.5 percentage points), 64 for DiffDock (+21.1 points) and 15 for EquiBind (+5.0 points). These are the ranking-depth view of the same PoseBusters-valid, within-2Å recovery tabulated across distance thresholds in Table 8, with which they agree at every reported depth.

Table 6: PoseBusters Pass-All Recovery by Ranking Depth

Method	Rank-1	Best of top-15	Best of top-30	top-1→15 gain	top-15→30 gain
AutoDock Vina + gnina	29.4% (89)	49.8% (151)	49.8% (151)	+62 (+20.5%) ***	+0 (+0.0%)
DiffDock + smina	34.0% (103)	55.1% (167)	55.8% (169)	+64 (+21.1%) ***	+2 (+0.7%)
EquiBind + gnina	15.2% (46)	20.1% (61)	20.5% (62)	+15 (+5.0%) ***	+1 (+0.3%)

Two-sided McNemar tests with Holm-Bonferroni correction, *** $p < 0.001$, ** $p < 0.01$ and * $p < 0.05$

A complex is recovered when *any* pose within the first d ranked poses is both PoseBusters-valid and within 2Å. This is the same PoseBusters-valid, within-2Å recovery reported across distance thresholds in the near-nativeness and form analysis (Table 8), and the two coincide at every reported depth (89/151/151 for AutoDock Vina + gnina at rank-1/top-15/top-30).

Almost all of these added complexes simply gain a near-native pose that was generated but under-ranked, so deeper pooling corrects mis-ranking rather than physical defects. PoseBusters validity has only a marginal net effect at this depth. It rescues a few complexes whose rank-1 near-native pose was invalid while excluding a few whose newly surfaced near-native pose still fails a check, so the valid recovery stays close to the raw near-native recovery.

Table 7: Decomposition of the Rank-1 to Best-of-Top-15 Pass-All Gain

Contributions top-15 gain	AutoDock Vina + gnina	DiffDock + smina	EquiBind + gnina
Complexes gaining a near-native pose (RMSD $\leq 2\text{\AA}$ gate)	+61	+63	+17
of which remain PoseBusters-invalid	0	-1	-2
Valid near-native rescued from a lower rank	+1	+2	0
Net valid complexes added	+62 ***	+64 ***	+15 ***
Two-sided McNemar test with Holm-Bonferroni correction, where *** $p < 0.001$			

Table 7 shows that deepening from rank-1 to the best of fifteen is highly significant for all three tools ($p < 0.001$), so a deeper pool taps a genuine reservoir of lower-ranked PoseBusters-valid poses, which for the deep-learning tools becomes accessible once gnina or smina minimisation has repaired their geometry.

4.1.2 Near-Nativeness and Form

Figures 3 and 4 and Table 8 summarise accuracy over the same 303 calibration complexes. The vertical axis reports the percentage of complexes for which a tool has at least one PoseBusters-valid pose within the distance threshold, evaluated over the best of its first d ranked poses.

Both figures are gated on PoseBusters validity and drawn at three ranking depths, with Figure 3 measuring near-nativeness against as-placed crystal RMSD and Figure 4 internal form against best-fit Kabsch RMSD.

Attention is drawn to the band between the top-1, top-15 and top-30 curves of each tool. Its width is the ranking headroom, that is the near-native valid poses that were generated but failed to rank first or within the top d , and it therefore separates sampling power from scoring power, a distinction the following paragraphs return to.

The three tools differ at every ranking depth on both metrics, with the Cochran's Q omnibus tests significant throughout (near-nativeness $p < 1e-7$, form $p < 1e-9$). At the $2\text{\AA}$ in-place RMSD threshold DiffDock has the highest point estimate at every depth, with rank-1 accuracies of 34.0% for DiffDock, 29.4% for AutoDock Vina + gnina and 15.2% for EquiBind. That lead is not statistically separable from AutoDock, however, since Holm-corrected exact McNemar tests leave the DiffDock advantage over the gnina-rescored arm short of significance at every depth ($p_{\text{holm}} = 0.219$ at rank-1, 0.188 at top-15 and 0.136 at top-30), while both tools remain decisively ahead of EquiBind throughout (all $p_{\text{holm}} \leq 3e-5$). Because a non-significant test is not evidence of equivalence, the contrast between AutoDock Vina + gnina and DiffDock + smina is reported as a

paired difference with its interval rather than as a failed test. Over the 303 paired complexes the DiffDock advantage on the valid-near-native endpoint is +4.6 percentage points at rank-1, with a 95% confidence interval of -2.2 to $+11.4$ points, +2.6 points at top-5 (-4.6 to $+9.8$), +5.3 points at top-15 (-2.1 to $+12.5$) and +5.9 points at top-30 (-1.4 to $+13.2$). Each is a Newcombe hybrid score interval for the difference of correlated proportions, taken over the 303 complex-level pairs and computed by `newcombe_paired_diff_ci` in the shared statistics module. Every interval contains zero, but the upper bounds run from $+9.8$ to $+13.2$ points while the lower bounds reach only -4.6 to -1.4 points, so a DiffDock advantage of practical size stays admissible at every depth whereas an advantage of the same magnitude for AutoDock Vina does not. The evidence is therefore inconclusive about the ordering of these two pipelines on near-nativeness. How much that leaves unresolved is itself quantifiable, and the paragraph below bounds it rather than leaving the contrast as a failed test.

That imprecision is a property of the design rather than a shortfall in execution, and it can be stated exactly. A paired binary test carries information only from the complexes on which the two pipelines disagree, because a complex both recover, or neither recovers, says nothing about which is better. At rank-1 both recover 40 of the 303 and neither recovers 151, so only the remaining 112 discordant complexes, 37.0% of the set, resolve the contrast. That discordant count rather than the 303 sets the resolution. It puts the minimum detectable difference at 9.8 percentage points for 80% power and leaves the observed 4.6-point difference tested at a power of 0.26, so a difference of the size actually present was unlikely to reach significance whatever the pipelines did. Detecting it reliably would require roughly 1,400 paired complexes, and even the 428 entries of the earlier preprint release of this benchmark would leave the minimum detectable difference near 8.2 points, so the ambiguity cannot be resolved by enlarging this benchmark. Two one-sided tests bound the same contrast from the other side. At a margin of 12 percentage points the two pipelines are equivalent at rank-1, at top-5 and at top-15, and at top-30 equivalence requires a slightly wider margin of 12.1 points, whereas no depth is equivalent within 5 points and only top-5 is within 10. The defensible statement is therefore that the data exclude a difference larger than about twelve percentage points in either direction while remaining unable to resolve one smaller than that, which is a bounded claim rather than an absence of one. On form at 1 Å the same ordering holds, at rank-1 accuracies of 52.1% for DiffDock, 46.9% for AutoDock Vina + gnina and 30.7% for EquiBind, and here the DiffDock advantage over AutoDock does reach significance at the top-15 depth ($p_{\text{holm}} = 0.023$) though not at rank-1 or top-30 ($p_{\text{holm}} = 0.109$ and 0.161). The defensible reading is therefore that DiffDock is ahead of the classical engine on point estimates at every depth and on both metrics, but that the two are separable only on form at intermediate depth, whereas the gap of both to EquiBind is unambiguous.

This point-estimate ordering is the reverse of the one obtained when AutoDock Vina was given a crystal-ligand-centred box, and the reversal motivates the matched blind comparison. Under blind conditions the diffusion model has higher in-place accuracy point estimates, without a statistically established advantage over AutoDock Vina, while the physics-based search retains a decisive advantage in physical validity, at a pooled 99.5% for AutoDock Vina + gnina against

84.5% for DiffDock + smina. Accuracy and admissibility are therefore separable properties, and the tool with the higher point estimate on one need not lead on the other.

The EquiBind deficit remains greater on in-place accuracy than on conformational form. Within each tool a Cochran's Q across the three depths confirms that ranking depth changes the outcome (Vina Q = 124, DiffDock Q = 128, EquiBind Q = 30, all $p < 1e-6$). The recovered near-native share from top-1 to top-15, with Wilson 95% intervals, is +20.5% (16.3 to 25.4) for AutoDock Vina + gnina, +21.1% (16.9 to 26.1) for DiffDock + smina and only +5.0% (3.0 to 8.0) for EquiBind + gnina. The further step from top-15 to top-30 adds at most 0.7 percentage points for any of the three reported variants, two further complexes for DiffDock + smina, so almost all recoverable headroom is already present in a top-15 pool.

Table 8: Near-Nativeness and Form Recovery at Different Ranking Depths

Metric	Tool	Ranking Depth	Threshold (Å)									
			0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5
RMSD	AutoDock Vina + gnina	1	0.0%	4.0%	9.9%	17.2%	21.1%	25.1%	27.7%	29.4%	30.0%	31.0%
	AutoDock Vina + gnina	15	0.0%	6.6%	14.5%	25.4%	33.3%	39.6%	44.2%	49.8%	50.8%	53.8%
	AutoDock Vina + gnina	30	0.0%	6.6%	14.5%	25.4%	33.3%	39.9%	44.6%	49.8%	51.5%	54.5%
	DiffDock + smina	1	0.3%	4.0%	10.9%	15.5%	21.5%	26.1%	31.4%	34.0%	35.3%	37.6%
	DiffDock + smina	15	1.0%	9.6%	20.5%	30.0%	40.6%	46.5%	51.8%	55.1%	58.1%	60.1%
	DiffDock + smina	30	1.0%	10.9%	21.8%	31.7%	41.6%	48.2%	53.1%	55.8%	59.4%	61.7%
	EquiBind + gnina	1	0.0%	1.3%	2.0%	5.0%	9.2%	11.2%	13.5%	15.2%	16.2%	16.8%
	EquiBind + gnina	15	0.0%	1.7%	4.0%	8.3%	12.2%	14.9%	17.5%	20.1%	21.5%	22.8%
	EquiBind + gnina	30	0.0%	1.7%	4.3%	8.3%	12.2%	14.9%	17.5%	20.5%	21.8%	23.1%
Kabsch RMSD	AutoDock Vina + gnina	1	10.2%	23.8%	33.7%	46.9%	56.4%	63.4%	70.3%	73.9%	79.9%	84.2%
	AutoDock Vina + gnina	15	17.8%	34.3%	49.2%	66.0%	80.2%	87.8%	91.7%	94.7%	97.4%	97.7%
	AutoDock Vina + gnina	30	18.5%	36.6%	52.1%	71.0%	83.8%	90.8%	95.4%	96.7%	98.3%	98.3%
	DiffDock + smina	1	11.9%	25.1%	38.6%	52.1%	61.7%	70.0%	76.2%	82.5%	85.8%	88.1%
	DiffDock + smina	15	21.1%	39.3%	57.4%	72.3%	80.2%	87.8%	94.1%	95.7%	96.0%	96.4%
	DiffDock + smina	30	22.4%	42.6%	60.4%	74.6%	84.5%	92.7%	96.0%	97.0%	97.7%	98.0%
	EquiBind + gnina	1	5.3%	11.2%	18.5%	30.7%	38.3%	48.2%	53.1%	60.7%	64.0%	68.0%
	EquiBind + gnina	15	13.5%	27.4%	37.3%	46.2%	56.4%	66.3%	69.0%	74.6%	75.6%	77.6%
	EquiBind + gnina	30	15.2%	30.0%	39.3%	49.8%	58.4%	67.7%	70.0%	74.9%	76.6%	78.2%

The band between the curves quantifies that separation [48], [50], since it shows how often a good pose is generated but not ranked first.

AutoDock Vina no longer has the highest point estimate on either near-nativeness or form once the crystal-centred box is withdrawn, and its ranking headroom remains substantial. Rank-1 valid-near-native recovery of 29.4% rises to 49.8% when the best of the top-15 poses is selected retrospectively, a gain of 62 complexes that is close to DiffDock's 64. Rescoring with gnina recovers part of that headroom prospectively rather than removing it, since the same figures for the unoptimised search are 25.7% and 47.9% with a gain of 67, so the learned rescorer converts a handful of retrospectively available poses into prospectively selected ones without closing the selection gap. The stronger conclusion is therefore that under a blind search the classical engine is no better at ranking its own output than the diffusion model is at ranking its own, and that both leave a large selection gap, consistent with the longstanding distinction between sampling and scoring [3], [48], [50].

DiffDock's rank-1 pose is near-native in 34.0% and form-correct in 52.1% of the calibration set, ahead of AutoDock Vina on both point estimates. Its top-30 pool contains a valid near-native pose for 55.8% of complexes and a form-correct pose for 74.6% at the primary 1 Å Kabsch threshold. Although DiffDock provides a confidence model [24], independent evaluations likewise show that generative docking can sample acceptable poses more reliably than it ranks them [13], [32], [34].

EquiBind's ranking depth improvement on near-nativeness is limited to +5.0% from rank-1 to top-15. The single-shot regression model barely samples distinct alternatives, so on this endpoint it is sampling-bound rather than ranking-bound and no rescorer can recover poses that were never generated. This is a genuinely different failure mode from DiffDock's, one that RMSD-only leaderboards conflate and that only the ranking-depth axis exposes.

Kabsch RMSD separates internal conformation from as-placed error. At the primary 1 Å threshold and the best-of-top-30 depth, form recovery is 71.0% for AutoDock Vina + gnina and 74.6% for DiffDock + smina. At the more permissive 2 Å threshold, the corresponding values are 96.7% and 97.0%. Both tools therefore score substantially higher on form than on in-place RMSD. For DiffDock, this pattern indicates that conformer generation is stronger than site localisation and ranking, consistent with its generative treatment of torsional, rotational and translational degrees of freedom [24].

EquiBind remains the weakest on form, reaching 30.7% at rank-1 and 49.8% at best-of-top-30 under the primary 1 Å threshold. This points to internal-geometry error in addition to misplacement and is consistent with current evaluations in which direct regression models trail diffusion and hybrid methods on combined geometry and physical validity [13], [34]. Its 19.1 percentage-point depth gain shows that the conformer-expanded pipeline sometimes contains an acceptable shape, even when that shape is not placed correctly.

Because the near-native valid poses DiffDock does produce already exist in a shallow top-15 pool, better pose selection is the intervention this study can show would pay off, which is the direction the field has converged on. Whether a larger or more diverse sampling budget would

Best-of-top-N pose accuracy vs distance threshold · n = 303 complexes

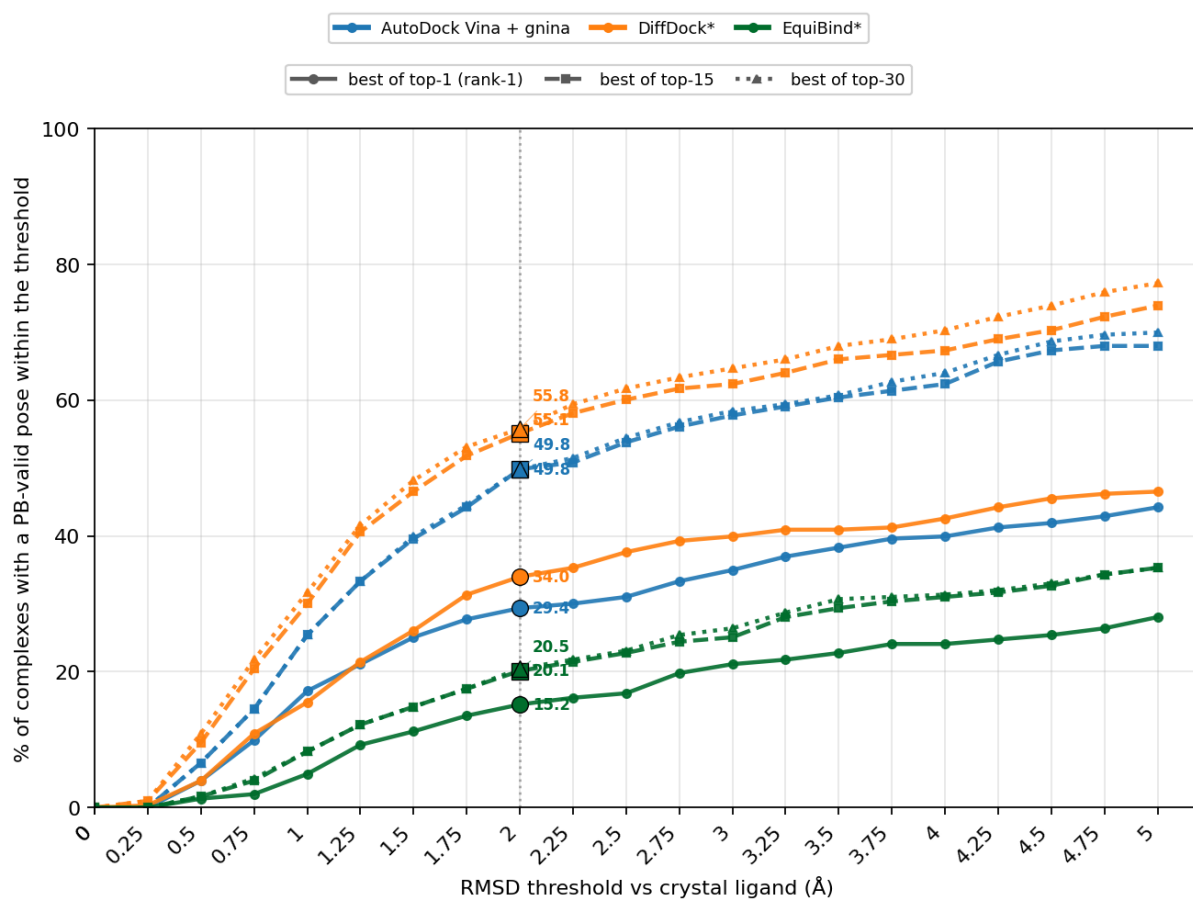


Figure 3: Best-of-top-N accuracy against the as-placed crystal RMSD, for AutoDock Vina + gnina, DiffDock + smina and EquiBind + gnina

help as well is untested here, since the thirty-pose pool was fixed and a top-30 view exhausts it [6], [11]. For EquiBind on near-nativeness the same logic does not apply, since no rescorer can recover poses that were never sampled.

Best-of-top-N pose accuracy vs distance threshold · n = 303 complexes

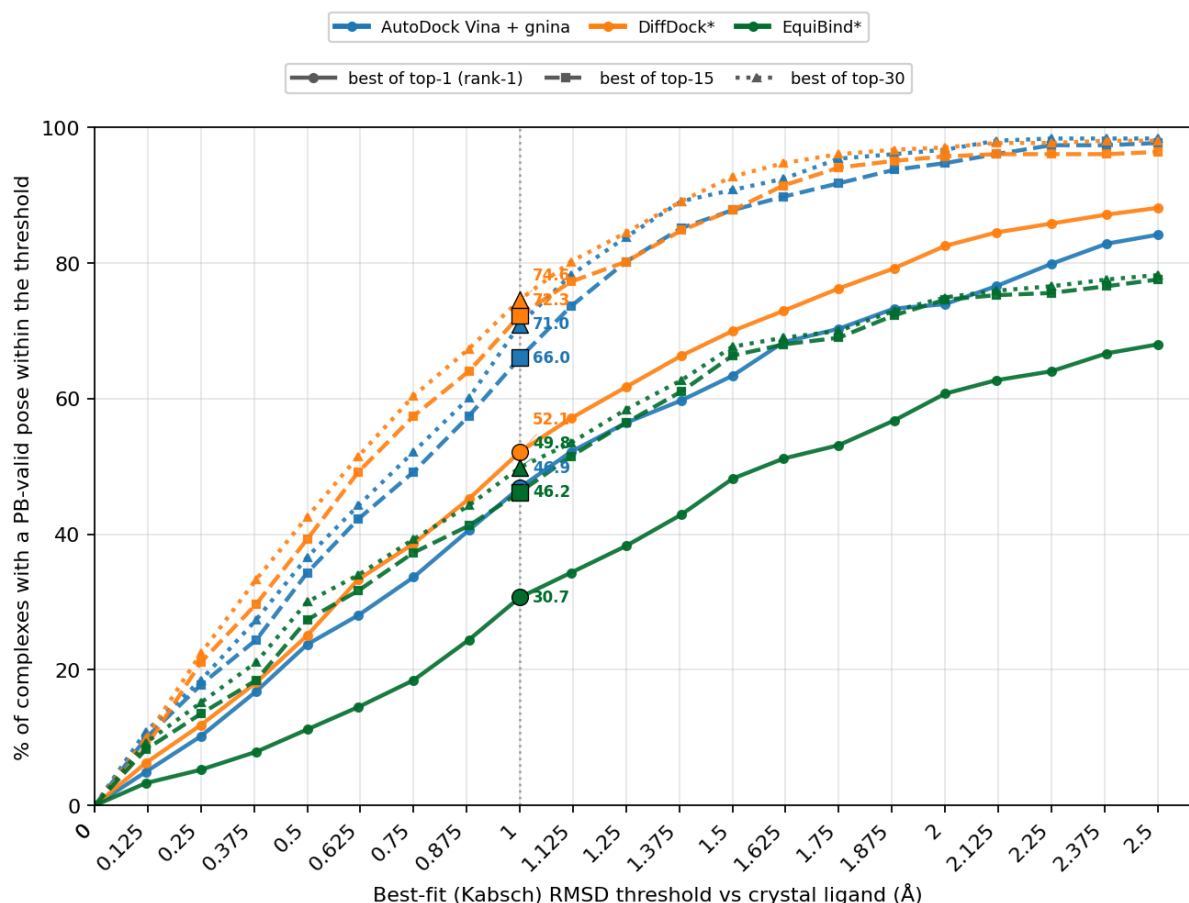


Figure 4: The same best-of-top-N accuracy against the best-fit (Kabsch) RMSD threshold, for the same three dominant variants

The depth-resolved curves separate generation from selection. DiffDock has the highest point estimate at every depth, and both it and AutoDock Vina retain material ranking headroom of comparable size. Both therefore carry a material selection component of comparable absolute size, 62 complexes for AutoDock Vina + gnina and 64 for DiffDock + smina, because valid near-native poses often appear below rank-1. That component is not the larger part of their residual loss. About seven in ten of their rank-1 failures hold no qualifying pose anywhere in the pool, so generation and localisation remain the greater constraint for both. EquiBind is limited almost entirely at generation and localisation because its best-of-pool recovery remains low [3], [13], [24], [34]. A top-15 pool captures almost all of the observed near-nativeness headroom, since extending the pool to 30 adds at most 0.7 additional percentage points. The form endpoint still moves on that extension, gaining 5.0 points for AutoDock Vina + gnina, from 66.0 to 71.0%,

2.3 points for DiffDock + smina, from 72.3 to 74.6%, and 3.6 points for EquiBind + gnina, from 46.2 to 49.8%, at the primary 1 Å Kabsch threshold of Table 8.

Figure 5 separates the two ways a valid pose can be wrong, plotting in-place RMSD against best-fit Kabsch RMSD so that placement-limited poses (correctly shaped but mislocated), form-limited poses (placed roughly right but folded wrong) and mixed cases fall in distinct regions, with the three columns widening the accepted pool from rank-1 to the best of the top-5 and top-15 and the arrow tracing each cohort median's drift from rank-1. Table 9 gives the underlying distributions.¹

AutoDock Vina + gnina returns a PoseBusters-valid pose for all 303 complexes, although only 58.4% of them still contribute once the cohort is trimmed to the crystal-site neighbourhood at rank-1, which is the coverage column of Table 9. Yet increasing ranking depth causes a decline in in-place accuracy. Its median in-place RMSD drifts from 1.9Å at rank-1 to 5.6Å at top-15, the largest movement of any tool, and a receptor-bootstrap interval on that +3.69Å drift spans +2.75 to +4.26Å, so it is comfortably separated from zero even though the two cohorts do not cover the same complexes. A separate within-complex test across disjoint rank bands also separates those bands on the 130 complexes that populate all three (Friedman $p \approx 6e-15$), which is an omnibus difference among the bands rather than a test of ordered trend. Meanwhile its placement-limited share swells from 51% to 81% and its form-limited share shrinks from 19% to 5%. Increasing ranking depth mostly adds correctly-shaped ligands scattered in the wrong place, consistent with the highest pose diversity of the three at a 5.7Å within-complex spread, so its ranking is informative only near the very top.

DiffDock is the depth-stable tool. Its rank-1 coverage is lower, but its cloud moves least, with an in-place drift of +0.46Å from 1.467 to 1.929Å against +3.69Å for AutoDock Vina and +1.21Å for EquiBind. Widening the pool fifteen-fold leaves the median near 1.5 to 1.9Å in in-place RMSD and about 0.9Å in form, from 0.885 to 0.918Å, and by top-5 and top-15 it is significantly the best tool on both axes. The form error moves by only +0.03Å across that fifteen-fold widening, and a receptor-bootstrap interval on that drift spans -0.05 to +0.12Å, so it cannot be distinguished from zero. A separate within-complex test across disjoint rank bands does separate those bands in form on the 137 complexes that populate all three (Friedman $p = 0.006$), again an omnibus difference rather than a test of ordered trend, so a difference across depth is real while its magnitude is negligible. Part of that stability is low pose diversity, namely a 2.34Å within-complex spread of near-duplicate poses against 5.74Å for AutoDock Vina, so its robustness to depth is

¹ Poses whose docked centroid lies more than 8Å from the crystal-ligand site are treated as outside the crystal-site neighbourhood and excluded before plotting, which removes 5,907 poses at the top-15 depth and leaves 5,549 PoseBusters-valid poses within that neighbourhood. The trim therefore discards a little over half of the 11,456 plotted candidates, so the panel shows the near-site poses only and is not a picture of the full output. Excluded poses are not off the receptor, since almost all of them remain in van der Waals contact with the protein, and they are set aside only because they sit away from the crystal site. Both axes are then clipped at 5Å to keep the dense near-native core legible, so a further 2,188 of those 5,549 poses lie beyond the frame and are not drawn. This is consistent with the in-place third quartiles reported in Table 9, which exceed 5Å at every depth for EquiBind and from the top-5 pool onwards for AutoDock Vina, while DiffDock stays below that mark throughout.

Form vs placement across ranking depth — top-1 net widened to top-5/15

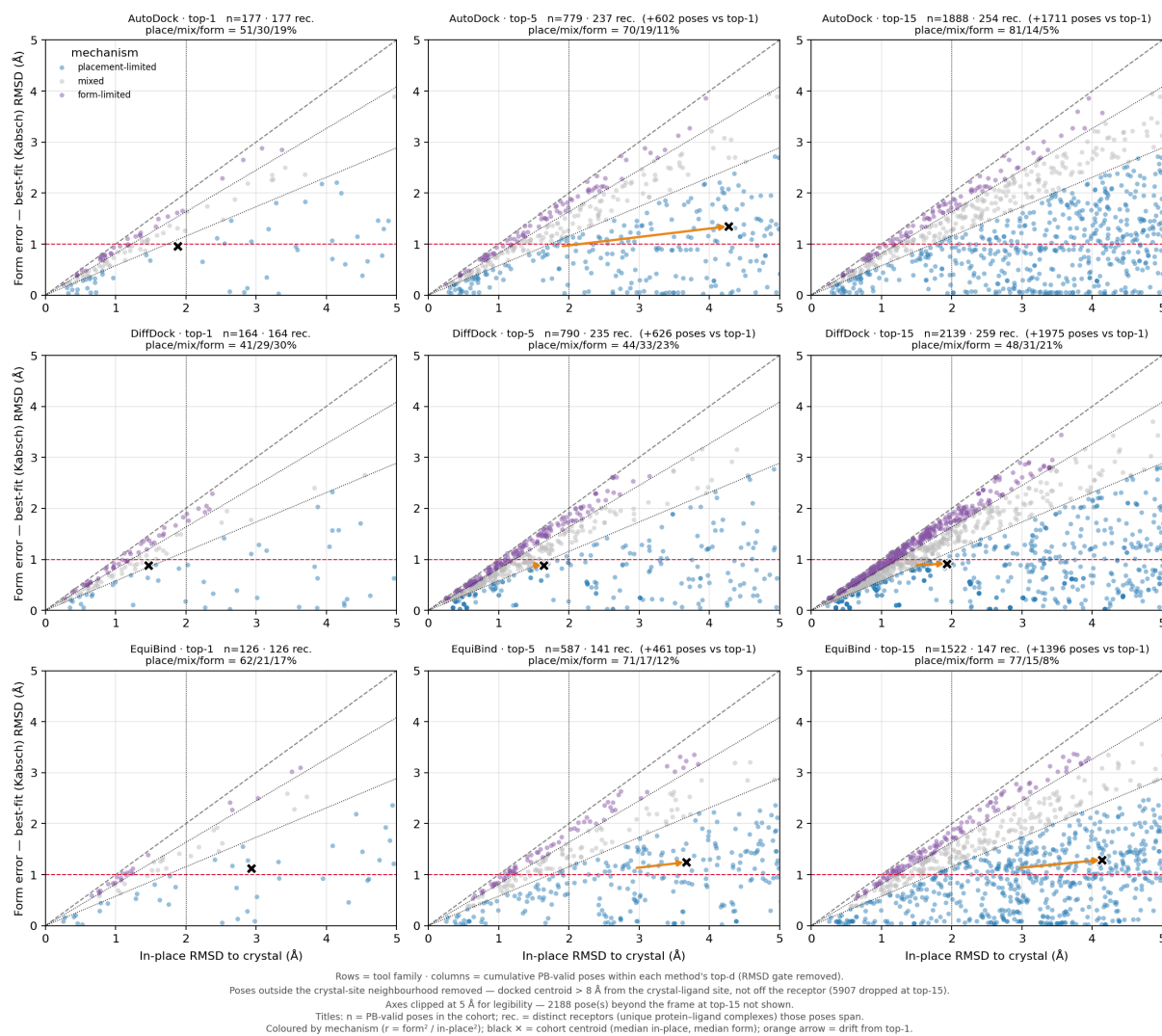


Figure 5: Form versus in-place RMSD across ranking depth. Rows are the dominant variant of each family, namely AutoDock Vina + gnina, DiffDock + smina and EquiBind + gnina

partly a reluctance to diversify.

EquiBind can generate PoseBusters-valid output, but validity does not translate into reliable in-place accuracy. At least one valid pose occurs somewhere in the full output for 247 of 303 complexes, yet valid-near-native recovery remains only 20.1% by the top-15 depth and 20.5% over the full thirty-pose pool. Its rank-1 median in-place RMSD of 2.9 Å already exceeds the 2 Å near-native threshold. gnina refinement can remove some local geometric defects, but it does not correct the pipeline's gross localisation failures.

Deeper ranks therefore add valid but mislocated poses for every tool, steeply for AutoDock and EquiBind and barely for DiffDock, so in-place RMSD rather than internal form is the axis that degrades with depth. DiffDock's more form-limited colouring is an artefact of its low in-place RMSD inflating the form-to-in-place ratio, since its absolute form RMSD is the lowest of the three.

Table 9: In-Place-RMSD-versus-Form Distributions by Tool and Ranking Depth

Tool (Ranking Depth)	n poses	Valid com- plexes	Coverage%	In-place median (Å)	In-place Q1	In-place Q3	Form median (Å)	Form Q1	Form Q3	Median r	% place-lim	% mixed	% form-lim
AutoDock Vina + gnina (1)	177	177	58.4%	1.882	0.823	4.885	0.955	0.482	1.537	0.306	50.8%	29.9%	19.2%
AutoDock Vina + gnina (5)	779	237	78.2%	4.273	2.142	6.845	1.349	0.742	2.101	0.140	70.1%	19.4%	10.5%
AutoDock Vina + gnina (15)	1888	254	83.8%	5.566	3.303	7.609	1.492	0.834	2.237	0.085	80.8%	13.9%	5.3%
DiffDock + smina (1)	164	164	54.1%	1.467	0.753	3.068	0.885	0.488	1.427	0.458	40.9%	28.7%	30.5%
DiffDock + smina (5)	790	235	77.6%	1.646	0.863	4.000	0.882	0.498	1.469	0.406	44.2%	32.5%	23.3%
DiffDock + smina (15)	2139	259	85.5%	1.929	0.921	4.893	0.918	0.517	1.569	0.363	47.9%	30.8%	21.3%
EquiBind + gnina (1)	126	126	41.6%	2.933	1.342	5.885	1.128	0.744	1.909	0.172	61.9%	20.6%	17.5%
EquiBind + gnina (5)	587	141	46.5%	3.672	2.008	6.313	1.245	0.762	1.979	0.137	71.2%	17.2%	11.6%
EquiBind + gnina (15)	1522	147	48.5%	4.138	2.565	6.582	1.292	0.782	2.019	0.106	76.7%	15.2%	8.0%

All rows are the blind whole-protein run with the dominant variant of each family, namely AutoDock Vina + gnina, DiffDock + smina and EquiBind + gnina. The cohort is each tool's PoseBusters-valid poses within the stated ranking depth, trimmed of poses whose centroid lies more than 8Å from the crystal-ligand site, that is outside the crystal-site neighbourhood rather than off the receptor, so the pose counts here are smaller than the produced-pose counts of Table 4 and the rows describe the near-site poses only. *Median r* is the median of $r = \text{form}^2 / \text{in-place}^2$, the share of a pose's squared deviation that is conformational rather than positional, and the three mechanism columns bin that same *r* at 1/3 and 2/3. The in-place axis of this table and of Figure 5 is the PoseBusters symmetry-corrected implementation, whereas the 2Å recovery gates of Table 8 and Figures 3 and 4 use this study's own symmetry-corrected RMSD. The two agree to about 0.1Å on these cohorts, at rank-1 medians of 1.882 against 1.918Å for AutoDock Vina + gnina and 1.467 against 1.566Å for DiffDock + smina, so the distributions here are not interchangeable with the counts gated there.

4.1.3 Cross-Tool Clustering

Accuracy and validity describe each tool separately, whereas clustering tests agreement about binding regions. For every complex, poses from all three tools were pooled without prefiltering on PoseBusters validity and clustered by ligand centroid. The cluster centre nearest the crystal ligand was treated as the recovered crystallographic region only when it fell within the prespecified 4 Å centroid threshold defined in the Methods, so a nearest cluster lying beyond that distance does not count as a recovery. Twenty-one of the 303 complexes have no cluster close enough to the crystal ligand to count as the site at all, so they carry no recovered region and contribute no pose to any composition statistic below, while remaining in the denominator of every rate reported here and counting as non-reaches. It is the same threshold that sets the retrospective oracle ceiling reported later in this subsection. All three arms are the whole-protein pipelines used throughout this chapter, with AutoDock Vina rescored and re-ranked by gnina, DiffDock refined by smina and EquiBind refined by gnina, so no arm carries a crystal-derived search box into the comparison. Figure 6 reports recovery within the top-N ranked poses for individual tools and combinations, with the corresponding rates in Table 10.

Under the whole-protein search no tool localises the pocket by construction, and the rank-1 field is correspondingly tight. DiffDock reaches the crystal cluster for 55% of complexes at rank-1, AutoDock Vina for 53% and EquiBind for 41%, and the AutoDock and DiffDock intervals overlap almost entirely, at 53 [48–59] against 55 [49–60]. The tools separate only with ranking depth. DiffDock gains 27.1 percentage points between rank-1 and top-15, from 54.8% to 81.9%, AutoDock gains 22.4 points, from 53.1% to 75.6%, and EquiBind is near flat at 7.6 points, from 40.9% to 48.5%, which Table 10 carries as the rounded 41% and 49%. Every pairing is most redundant at rank-1 and loosens once more than one pose is admitted, then recovers slightly and flattens beyond top-5, but none of them decays to independence. AutoDock and DiffDock loosen most, from $\phi = +0.29$ at rank-1 to $+0.18$ at top-5, and settle at $+0.25$ by top-15. The two learned tools are the most coupled pair from rank-1 through top-5, from $+0.43$ to $+0.29$, and by top-15 all three pairings have converged on one another, at $+0.26$ for AutoDock and EquiBind against $+0.25$ for the other two, with Fisher $p \leq 0.003$ throughout, so stacking EquiBind on DiffDock still adds little coverage.

Table 10: Crystal-Cluster Reach and Co-Reach by Ranking Depth

Statistic	top-1	top-5	top-10	top-15
AutoDock Vina + gnina reach [95% CI]	53 [48–59]	72 [67–77]	75 [70–79]	76 [70–80]
DiffDock reach [95% CI]	55 [49–60]	76 [71–80]	80 [75–84]	82 [77–86]
EquiBind reach [95% CI]	41 [36–47]	45 [40–51]	47 [42–53]	49 [43–54]
AutoDock + DiffDock co-reach ² ϕ	+0.29	+0.18	+0.23	+0.25

²Co-reach ϕ is the signed association between two tools' per-complex reach against independence, tested two-sided

Table 10: Crystal-Cluster Reach and Co-Reach by Ranking Depth (continued)

Statistic	top-1	top-5	top-10	top-15
AutoDock + EquiBind co-reach ϕ	+0.31	+0.23	+0.24	+0.26
DiffDock + EquiBind co-reach ϕ	+0.43	+0.29	+0.25	+0.25
All three reach, observed / expected	75 / 36	102 / 75	112 / 85	118 / 91

Figure 7 (A) shows that DiffDock now contributes most to the crystal cluster, with a median of six of its fifteen ranked poses inside it and 82% reach, ahead of AutoDock at a median of three poses and 76% reach, while EquiBind remains all-or-nothing at a median of zero and 49% reach. The ordering is significant on both pose count (Friedman $\chi^2 = 53.8$, Kendall W = 0.089) and reach (Cochran's Q = 107.4), and once the 4 Å gate is applied the two measures agree on which pairs are separable. On pose count DiffDock is clear of both rivals, at rank-biserial 0.49 against AutoDock and 0.37 against EquiBind with Holm-corrected Wilcoxon $p < 1e-4$ in both cases, and AutoDock Vina and EquiBind separate too, at a rank-biserial of -0.16 taken in AutoDock-first orientation (Holm $p = 0.032$), where the negative sign means that EquiBind contributes the larger number of poses, while on reach every pair separates as well, the AutoDock and DiffDock difference reaching Holm-corrected McNemar $p = 0.040$ and both tools remaining decisively ahead of EquiBind. As evident in Figure 7 (B) the correct pocket is usually a multi-tool basin, yet the consensus is polarised, namely complexes tend to be solved by all three tools at once (118 against 90.9 expected, permutation $p < 1e-4$) or by none of them (24 against 6.9 expected), with the intermediate states under-represented at 52 single-tool complexes against 59.1 expected and 109 two-tool complexes against 146.2. Where one tool does solve a complex alone it is usually DiffDock, in 31 of those 52 cases against 19 for AutoDock. Internal tightness in Figure 7 (C) no longer inverts the reach order, since DiffDock is now both the tool that reaches the crystal cluster most often and the one whose own poses sit tightest at a median 0.37 Å over the 232 complexes on which its spread is defined. What survives is the AutoDock anomaly. Its poses remain the loosest at 0.78 Å over 204 complexes, behind EquiBind at 0.59 Å over 138 (Kruskal–Wallis $H = 54.4$, $p < 1e-4$, $\varepsilon^2 = 0.092$). Those three groups are not the same set of complexes, because a tool contributes a spread only where it places at least two poses in the crystal cluster, so the missingness is informative rather than random and an unpaired test of them is only indicative. Restricting the comparison to the 102 complexes on which all three tools are measurable and treating them as paired observations still separates the three (Friedman $\chi^2 = 34.8$, $p < 1e-4$), so the difference is not an artefact of the differing composition. When AutoDock does find the pocket it therefore scatters its poses across it rather than converging on

by Fisher, where a positive value means the pair succeeds and fails together and a negative value means it is complementary. All three pairings are estimable in the whole-protein run, unlike the superseded boxed run in which AutoDock sat at a reach ceiling. Values in bold reach $p < 0.05$ on a two-sided Fisher test, which every cell of all three pairings does. Three-way agreement exceeds chance at every depth with a permutation p below $1e-4$, and the ranking depths are nested cumulative thresholds rather than an independent test family.

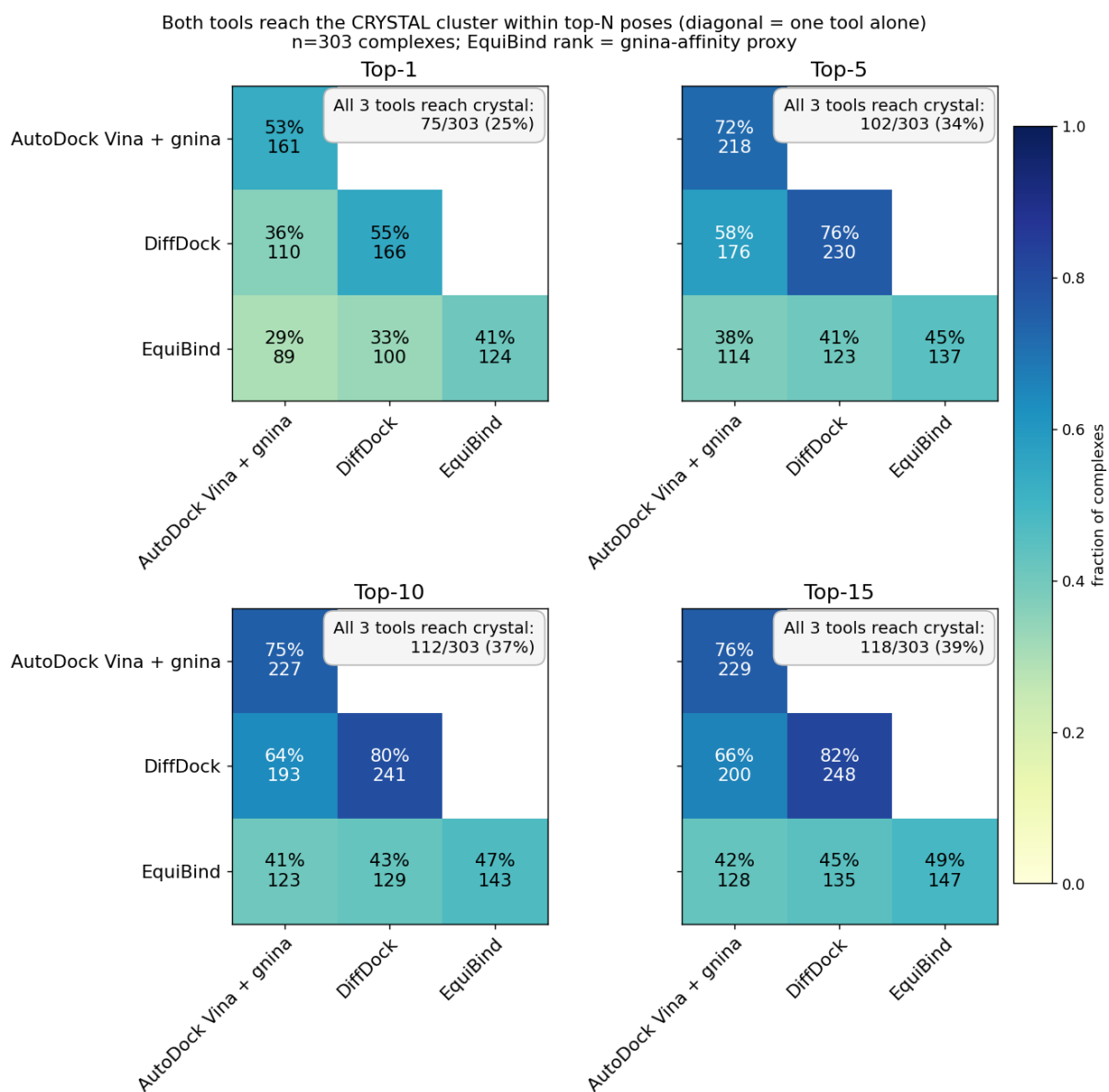


Figure 6: Top-N crystal-cluster co-recovery. Each lower-triangular panel is one ranking depth, with the AutoDock arm labelled AutoDock Vina + gnina

How homogeneous is the crystal-closest cluster? Composition, internal tightness and spatial spread

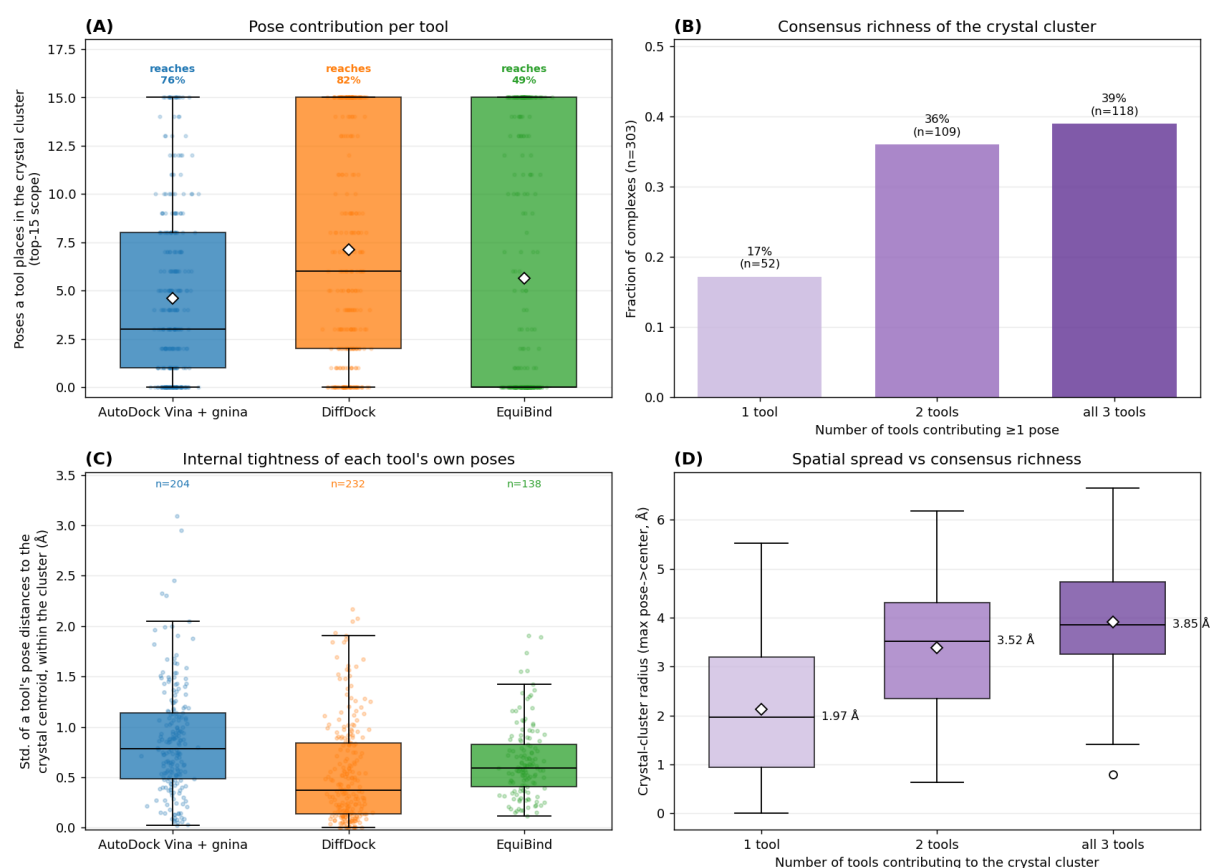


Figure 7: Composition and geometry of the crystal-closest cluster ($n = 303$, top-15 poses per tool, AutoDock arm labelled AutoDock Vina + gnina). Rates are taken over all 303 complexes, of which 24 contribute no pose to any panel, namely the 21 whose nearest cluster lies beyond the 4 Å crystal-site threshold and so carry no recovered region at all, together with three further complexes whose qualifying cluster holds only a pose ranked outside the top-15 pool shown here. Panel A gives the per-tool pose contribution and reach, panel B the consensus richness, panel C the internal tightness of each tool's own poses and panel D the spatial spread against consensus richness

one placement.

Figure 7 (D) shows cluster radius rising from a median 1.97 Å for single-tool clusters to 3.52 Å at two tools and 3.86 Å at all three (Jonckheere–Terpstra $z = 6.75$, $p < 1e-4$), but the trend disappears after controlling for pose count (partial $\rho = -0.04$, $p = 0.47$). The partition is reproducible under resampling (median bootstrap Jaccard 0.78). Internal indices are interpreted only for eligible multi-cluster complexes. Under the separate 4 Å crystal-site threshold, the retrospective oracle finds a recovered cluster in 93% of complexes, whereas the best inexpensive first-cluster rule reaches 55%. Consensus ranking is the strongest of the five prospective rules, tied with its medoid-centred variant at 55%, and it beats the legacy pose-count rule at 49% (Holm-corrected McNemar $p = 0.010$, omnibus Cochran's $Q = 171.4$). The 38-point shortfall against the oracle is nearly three times the 13-point shortfall seen under the superseded boxed search. This gap is a site-ranking problem. The crystal-compatible region is usually represented somewhere in the pooled pose cloud, but selecting it prospectively becomes markedly harder once the search is blind. Figure 8 summarises the internal validity indices, the resampling stability and the precision-at-one of all five ranking rules against that oracle ceiling.

In this benchmark, binding-site agreement is driven by DiffDock and AutoDock with EquiBind as the consensus bottleneck. The strongest redundancy is the one between the two learned tools at rank-1 and top-5, although by top-15 all three pairings have converged and none of them is strongest. The clustering is geometrically sound and the crystal-compatible region is present in the pooled ensemble for 93% of complexes, so for the pooled ensemble the residual loss is in ranking rather than search, even though EquiBind's own shortfall remains one of sampling. Pooling helps on both axes once the comparison is made like for like at the level of individual poses, taken over the full pose cloud rather than the top-15 pool the reach figures use. The best pose from the pooled three-tool ensemble lands within the 4 Å crystal-site threshold for 95% of complexes at a median oracle distance of 0.35 Å, against 84% and 0.56 Å for DiffDock alone and 78% and 0.73 Å for AutoDock alone. Almost all of that gain comes from the two stronger arms, since AutoDock and DiffDock together already reach 94.7% at 0.40 Å and adding EquiBind moves reach by 0.3 percentage points. The pooled cluster centre is necessarily coarser than any single pose, at a median 1.30 Å, which is a consequence of averaging over cluster members rather than a cost of pooling.

4.1.4 Interaction and Contact Analysis

The purpose of docking is to reproduce the protein–ligand interactions that drive binding. The interactions of the crystal ligand poses are compared with the interactions of PoseBusters-valid poses produced by the toolchains, with typed contacts detected by the PandaMap implementation [88] of the PLIP interaction-profiling rules [52], applied as heavy-atom distance and atom-type assignments, since the angular criteria are not evaluated on the code path used. This subsection was regenerated for the whole-protein search, so its AutoDock arm is AutoDock Vina + gnina throughout and is directly comparable with Table 4 and with the recovery analysis above. Because

Cluster-quality assessment — internal validity, stability, and ranking accuracy (n=303 complexes)

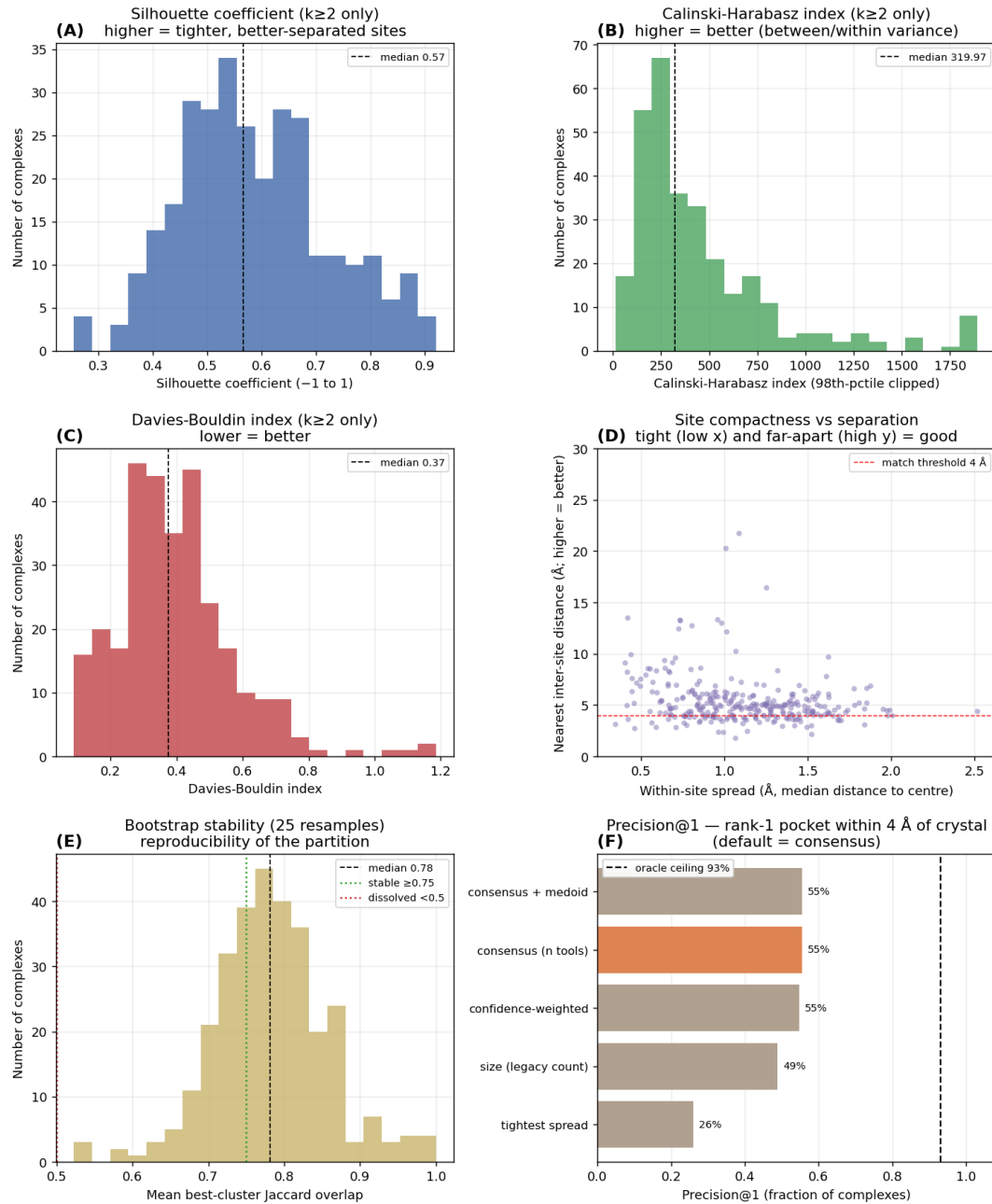


Figure 8: Cluster-quality assessment (n = 303, 8 Å centroid-clustering cut and 4 Å crystal-site recovery threshold). Panels A-C show internal-validity indices, panel D compactness against separation, panel E bootstrap stability and panel F precision-at-one of five ranking rules against the oracle ceiling

a typed fingerprint is keyed on residue identity, every pose and the crystal reference were profiled against one common author-numbered receptor per complex, verified heavy-atom coordinate-compatible with the receptor each pipeline actually docked into.³ Figure 9 reports the mean Jaccard overlap between typed interaction fingerprints, where each tool's repertoire for a complex is the union of its top-5 ranked poses and the overlap is averaged over all shared complexes.

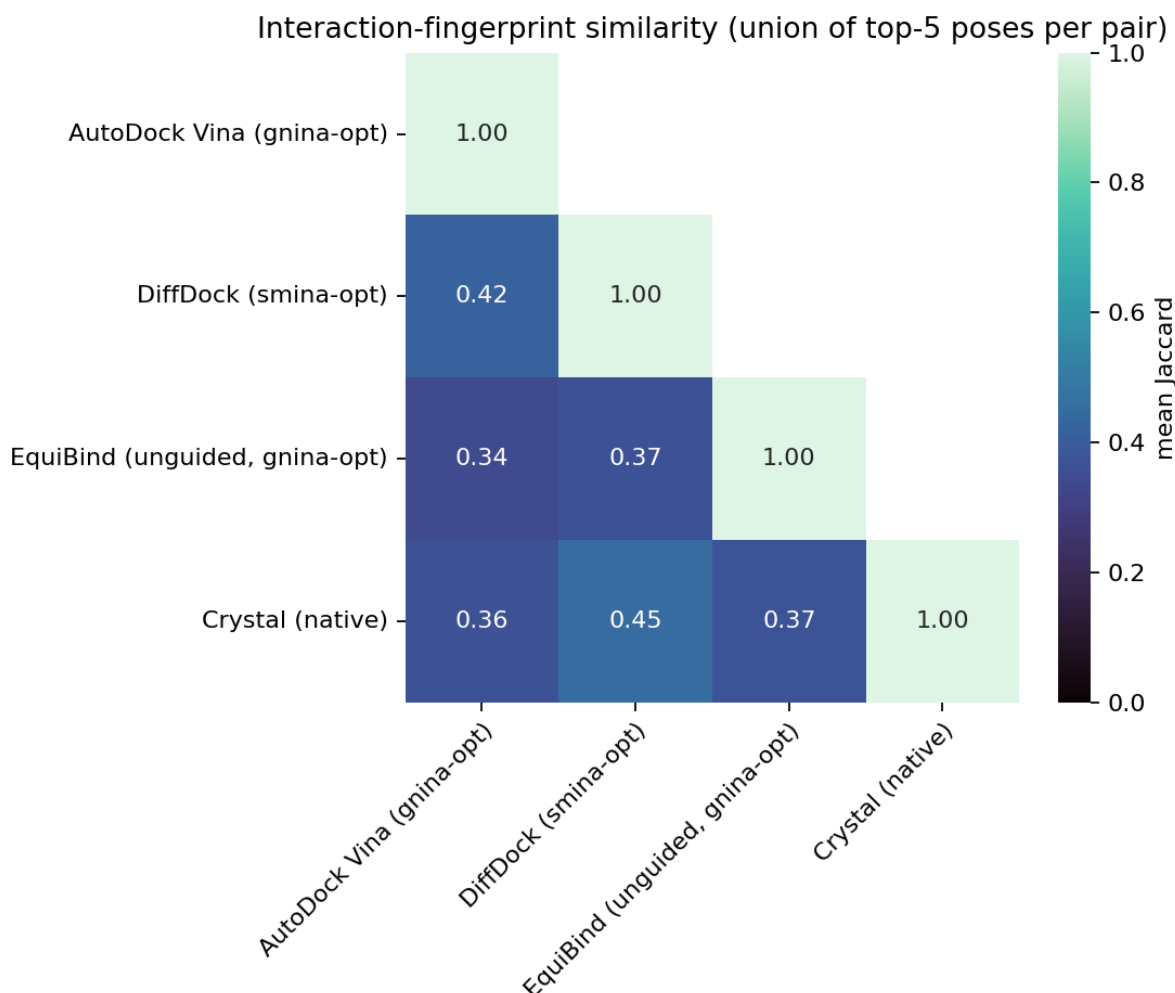


Figure 9: Mean Jaccard similarity of interaction fingerprints, whole-protein search

Under the blind whole-protein search DiffDock + smina has the highest mean typed-interaction Jaccard similarity to the crystal fingerprint (0.45), followed by EquiBind + gnina (0.37) and AutoDock Vina + gnina (0.36). The ordering is the reverse of the superseded crystal-ligand-

³The prepared receptors the toolchains emit do not share a residue-numbering scheme. The MGLTools receptor used by the whole-protein AutoDock search rennumbers residues sequentially from one, so its contacts cannot be matched against a crystal reference carrying author numbering. Profiling every arm against the common cleaned receptor relabels residues without moving any atom. That EquiBind, whose pipeline is untouched by the migration, returns a crystal Jaccard of 0.37 and an F1 of 0.44 both here and on the superseded boxed run is the check that the two runs remain on one scale.

boxed run, on which the site-restricted AutoDock search reached 0.67 against 0.44 for DiffDock and 0.37 for EquiBind. The whole-protein protocol therefore costs AutoDock roughly half of its measured fingerprint overlap, 0.67 to 0.36, while the two blind pipelines are essentially unmoved, EquiBind + gnina returning 0.37 in both runs and DiffDock + smina shifting only from 0.44 to 0.45, which locates the change in the AutoDock protocol rather than in the contact detection. That contrast does not isolate the search box itself, because the boxed AutoDock arm was raw AutoDock Vina while the blind one is AutoDock Vina + gnina, and receptor preparation also differs between the runs. Pairwise tool similarities are lower, at 0.34 to 0.42, showing that the methods often make different contact hypotheses. Because Jaccard combines shared and non-shared interactions, these values should not be interpreted as contact recall. They show overall fingerprint agreement, not the fraction of native interactions recovered.

Figure 10 resolves precision, recall and F1 against pose rank k , where k is each tool's own rank over all of its produced poses and only PoseBusters-valid poses are profiled. A complex therefore contributes no $k = 1$ point when its own top-ranked pose fails PoseBusters, which happens for 5 of the 303 complexes under AutoDock Vina + gnina and for 18 of the 300 complexes that retain a valid DiffDock + smina pose, whereas EquiBind's gnina-affinity rank is assigned over its retained valid poses and so always starts at $k = 1$.

At rank-1 the three pipelines are close, at an F1 of 0.53 for DiffDock + smina, 0.53 for AutoDock Vina + gnina and 0.46 for EquiBind + gnina, so on its own top-ranked pose the classical engine recovers the native contact pattern about as faithfully as the diffusion model. Those three means rest on different available-case cohorts, namely the 282, 298 and 247 complexes for which each pipeline has a PoseBusters-valid rank-1 pose to score, and no equivalence or non-inferiority test was applied to them. On the 278 complexes AutoDock Vina + gnina and DiffDock + smina share, the means are 0.5256 and 0.5266 with no detected paired difference (Wilcoxon $p = 0.69$). A sensitivity analysis that instead scores a missing valid rank-1 pose as zero across all 303 complexes reverses the order, at 0.517 for AutoDock Vina + gnina against 0.492 for DiffDock + smina, because DiffDock more often has no valid pose to nominate. The rank-1 comparison is therefore conditional on producing a valid top pose, and the ordering it gives is not robust to how that failure is counted. They separate with depth rather than at the top. AutoDock decays steadily to 0.36 by rank 5 while DiffDock stays near flat at 0.51 and EquiBind at 0.43, so DiffDock leads on the top-5 union of Figure 9 without leading on the pose a user would actually take. A per-complex Kendall τ between rank and score, tested with a one-sample Wilcoxon signed-rank against zero and BH-corrected across the nine tool-by-metric cells, shows why. AutoDock's own ranking puts better poses first ($\tau = -0.20, -0.32$ and -0.20 for precision, recall and F1, all BH $q < 0.001$), so its gnina re-ranking is informative about which of its poses to take, although only that arm was profiled here, so the effect of the re-ranking itself is not measured. EquiBind's gnina-affinity rank works for recall and F1 ($\tau = -0.26$ and -0.20 , both significant) and only marginally for precision ($q = 0.015$). DiffDock's native confidence shows no detected association with interaction recovery ($\tau \approx 0$ on all three, none significant), so nothing in these data lets its rank-1 pose be expected to recover more native contacts than its rank-5. That is an absence

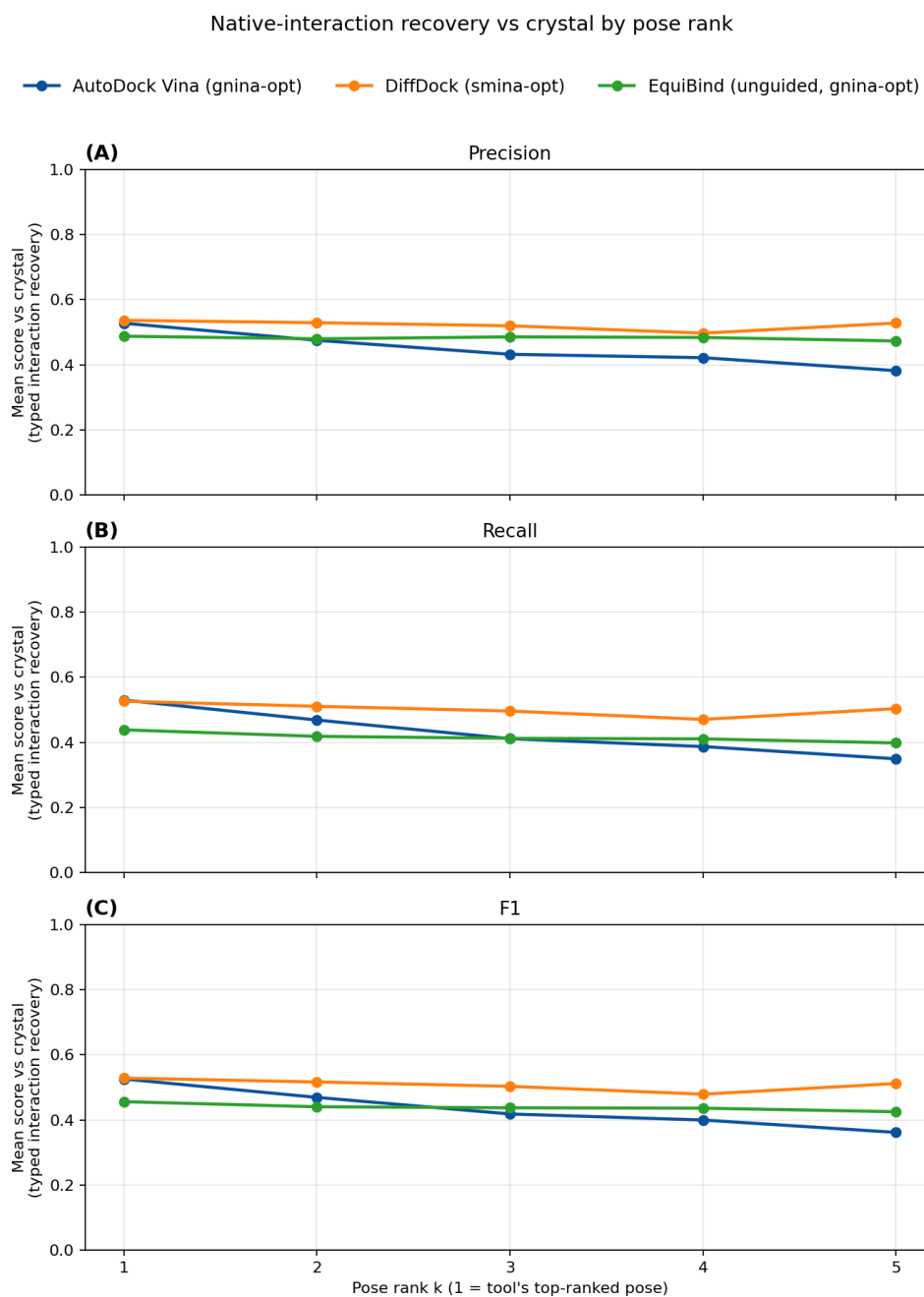


Figure 10: Native-interaction recovery versus the crystal by pose rank (1–5), whole-protein search

of detected ordering rather than a demonstration that the confidence carries no signal. That this rank-ordering pattern is reproduced unchanged from the boxed run, on an AutoDock arm whose absolute level has halved, indicates it is a property of the ranking signals rather than of the search box.

Figure 11 decomposes each pose's typed contacts into matched native contacts recovered, missed native contacts absent, and spurious non-native contacts invented.

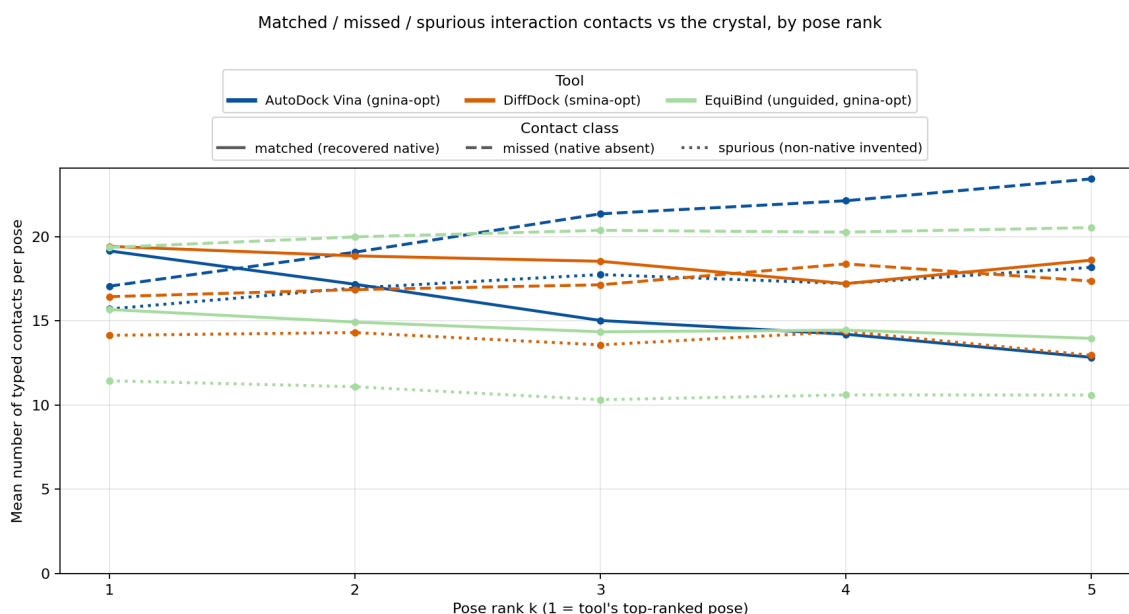


Figure 11: Matched, missed and spurious typed contacts versus the crystal, by pose rank, whole-protein search

At rank 1 AutoDock Vina + gnina and DiffDock + smina are almost indistinguishable, recovering 19.2 and 19.4 native contacts against 17.1 and 16.4 missed, with AutoDock inventing slightly more spurious contacts (15.7 against 14.1). The two diverge with depth. AutoDock falls to 12.8 matched against 23.4 missed by rank 5 and its spurious count climbs to 18.2, so it crosses from matched-dominated to missed-dominated within its own top five and is the only pipeline whose invented contacts increase significantly with rank, whereas DiffDock holds near 18.6 matched against 17.4 missed with a flat spurious count. EquiBind misses more native contacts than it recovers at every rank, 15.7 matched against 19.4 missed at rank 1, so it systematically under-reproduces the native network, though it invents the fewest non-native contacts (10.3 to 11.4). The same rank test orders AutoDock cleanly across all three classes ($\tau = -0.32$ matched, $+0.32$ missed, $+0.11$ spurious, all significant) and EquiBind on matched and missed (± 0.26), whereas DiffDock's counts are flat.

Taken together, the whole-protein interaction analysis mirrors the placement result. AutoDock Vina + gnina and DiffDock + smina show no detected difference in native interaction pattern on the pose each tool actually nominates, so neither can be preferred on rank-1 contact fidelity, although the data are inconclusive rather than positively equivalent. They differ in what lies

beneath that pose. AutoDock's series is steeply ordered, so its gnina ranking is informative but its lower-ranked poses are genuinely worse, while DiffDock's series is flat and unordered, so its confidence wastes a pool whose depth is uniformly good and a re-scorer stands to help it most. EquiBind is limited at source, recovering few native contacts regardless of rank. The earlier reading, on which AutoDock reproduced the native pattern most faithfully by a wide margin, does not survive the whole-protein protocol, although that contrast does not isolate the search box because the boxed AutoDock arm was raw AutoDock Vina while the blind one adds a gnina pass.

4.2 Experimental Dataset

The experimental evaluation asks whether the tool ordering established against crystallographic references survives on a target of genuine pharmacological interest, and whether the large benchmark library can stand in for the compounds that actually matter. That target is the human Orai1 channel, docked with the three functionally validated modulators 2-APB, GSK-7975A and Synta-66, and it removes the benchmark's most valuable asset, since no ligand-bound Orai structure exists and the receptor is an ensemble of molecular-dynamics snapshots rather than a crystal. Every accuracy measure that depends on a native pose therefore has to be replaced, and the chapter is organised around that replacement, opening with an end-to-end yield funnel, then moving from chemical validity, which needs no reference, through a biologically grounded placement criterion, on to cross-tool agreement and cluster quality, which are reference-free by construction, and finally to the interaction chemistry of the poses that survive both filters. The docking settings behind both panels, including the respects in which the panels differ, are set out in the Appendix (Docking Protocols, Orai1 Panels).

Two ligand panels pass through this pipeline. The experimental panel pairs three modulators with four Orai1 frames, giving twelve frame-ligand units. The control panel follows the established practice of screening a large, chemically diverse ligand library against the target to characterise a method's behaviour where no cognate reference exists [43], [44]. It yielded 1,215 successfully processed frame-ligand units. Of the 1,232 possible combinations of 308 calibration ligands and four frames, 17 produced no analysable pose in any tool and were dropped, spread across all four snapshots at six units for Fr300, five each for Fr400 and Fr499 and one for Fr0. The AutoDock arm covers 1,201 of those units, the shortfall being fourteen macrocyclic complexes whose gnina-rescored poses could not be reconstructed to a topology matching their input. Each tool is represented by one variant throughout both panels, namely AutoDock Vina + gnina, DiffDock + smina and EquiBind + gnina, matching the variants carried through the calibration benchmark. gnina rescoring was originally run only over the experimental panel and was extended to the full control panel for this work, so the two panels are now on the same AutoDock arm and the experimental-versus-control comparisons no longer have to fall back to the unoptimised search. Output is capped at the top ten ranked poses per unit, except in the interaction and contact analysis, which applies a tighter cap of the top three ranked poses per unit. That cap is an upper

bound rather than a guaranteed count, and the three tools do not meet it equally. AutoDock Vina and EquiBind returned the full ten poses on every unit they cover, giving 12,010 and 12,150 poses respectively, whereas DiffDock returned fewer than ten on 178 of its 1,215 units and 11,673 poses in all, with 89 of those units returning nine poses and ten of them returning a single pose. Every per-unit rate reported below is therefore taken over the poses that unit actually produced rather than over a fixed denominator of ten, and the pooled pose-level counts quoted alongside them carry the same tool-specific totals. Because poses within a frame-ligand unit are correlated and the twelve experimental units arise from only three chemotypes and one receptor trajectory, all comparisons are exploratory and are interpreted through effect sizes and uncertainty rather than p-values alone.

The evaluation opens with an end-to-end yield funnel. Table 11 reports, for each toolchain and modulator, how many produced poses are PoseBusters-valid and how many also satisfy the study's placement rule. The ligand must remain associated with the receptor and lie outside the R91-E106 transmembrane exclusion slab. The final usable fraction is therefore a combined physical-validity and operational-placement endpoint. Every tool is capped at forty poses per ligand across the four frames, making the resulting fractions directly comparable within this protocol.

Table 11: Physical-Validity and Operational-Placement Yield for the Orai1 Experimental Ligands

Tool	Ligand	Poses	PB-valid Poses	PB-valid %	PB-valid usable	PB-valid usable %
AutoDock Vina + gnina	2abp-NH2	40	40	100.0%	20	50.0%
AutoDock Vina + gnina	Synta-66	40	39	97.5%	24	60.0%
AutoDock Vina + gnina	GSK-7975A	40	40	100.0%	22	55.0%
DiffDock + smina	2abp-NH2	40	40	100.0%	24	60.0%
DiffDock + smina	Synta-66	40	37	92.5%	26	65.0%
DiffDock + smina	GSK-7975A	40	35	87.5%	32	80.0%
EquiBind + gnina	2abp-NH2	40	14	35.0%	1	2.5%
EquiBind + gnina	Synta-66	40	8	20.0%	0	0.0%
EquiBind + gnina	GSK-7975A	40	7	17.5%	0	0.0%
AutoDock Vina + gnina	all three	120	119	99.2%	66	55.0%
DiffDock + smina	all three	120	112	93.3%	82	68.3%

Table 11: Physical-Validity and Operational-Placement Yield for the Orai1 Experimental Ligands (continued)

Tool	Ligand	Poses	PB-valid Poses	PB-valid %	PB-valid usable	PB-valid usable %
EquiBind + gnina	all three	120	29	24.2%	1	0.8%

Each tool is capped at its top-10 ranked poses per frame-ligand unit, and on the experimental panel all three tools filled that cap on every unit, so every row rests on four frames $\times$ 10 poses = 40 produced poses. PB-valid usable counts poses that are PoseBusters-valid, placed on the receptor and outside the transmembrane exclusion slab defined in the Methods (Operational Placement). Percentages are taken over produced poses in both columns, so the two are directly comparable down a row.

AutoDock Vina produces the highest experimental physical validity but not the highest usable yield. Of its 120 generated poses 119 pass PoseBusters, and 50 to 60% per ligand also satisfy the placement rule, namely that a pose lies on the receptor and outside the transmembrane exclusion slab, giving 50% for 2abp-NH2, 60% for Synta-66 and 55% for GSK-7975A. That near-perfect validity is empirical rather than guaranteed by classical docking, as the calibration analysis still contains invalid Vina poses. In the Orai1 panel, the 53 excluded Vina poses fail the operational slab criterion.

DiffDock + smina is noisier on validity but places better. Validity runs from 87.5% to 100%, so the minimisation delivers mostly sane geometry, and 60% to 91.4% of the valid poses lie outside the transmembrane exclusion slab, ahead of AutoDock on all three ligands and most widely for GSK-7975A at 91.4% against 55%. On the re-docked pose set every complex delivers its full pose budget, including the boron-containing 2-APB (2abp-NH2), whose 40 of 40 poses are all physically valid but leak most at the placement stage, with only 60% of them outside the slab.

EquiBind performs poorly on both filters. Validity is only 17.5% to 35%, and the reason is a preparation step rather than a property of EquiBind itself. This panel is the one arm in the thesis that applied the in-protein UFF relaxation recorded in Table 14, and that step fixes every protein atom before minimising, so on a buried pose it cannot slide the ligand out of the clash and relieves the overlap by deforming the ligand instead. The resulting bond-length and bond-angle failures are then permanent, because gnina relieves clashes rather than correcting covalent geometry. That the cause is the relaxation and not the generator is visible in the calibration benchmark, which ran the same tool without that step and returns 25 bond-length and 22 bond-angle failures across all 81,373 poses of its nine variants, EquiBind included, as reported with the decomposition above. The intramolecular column of Table 5 is not a covalent-geometry column, and the 27.9% it records for raw EquiBind is internal clash and conformational strain rather than distorted bonds. The Orai1 experimental EquiBind yields below should therefore be read as a lower bound on what this configuration can do, and the control panel, which did not apply the relaxation, is the fairer picture of the tool. The few valid poses it does produce also sit almost entirely inside the transmembrane exclusion slab. Of its 29 valid poses across the

three ligands, only one survives the placement rule, so the usable yield is zero for Synta-66 and GSK-7975A and 2.5% for 2-APB. For this target EquiBind contributes essentially nothing usable.

The funnel shows that physical validity alone overstates utility on this target. AutoDock Vina and DiffDock both perform strongly on PoseBusters validity, but the combined screen for on-receptor placement outside the transmembrane exclusion slab separates them from EquiBind. The final usable column is therefore the relevant operational endpoint for this study. AutoDock Vina yields 50 to 60% usable poses, DiffDock is ahead of it on all three ligands at 60 to 80%, and EquiBind retains only one pose. Because GSK-7975A and Synta-66 act at or near the pore and selectivity filter, the exclusion slab is a modelling choice rather than proof that every excluded pose is biologically wrong [68], [69].

4.2.1 Orai Channel (MD Simulation)

The experimental target is the human Orai1 channel, the pore-forming subunit of the calcium release-activated calcium channel. It is a homohexamer arranged around a central ion-conduction pore [36]. The model spans residues 66 to 288 of each subunit and includes the four transmembrane helices and flanking regions, giving 1,338 residues and 10,362 heavy atoms without ligand, Ca²⁺ or other heteroatoms in the delivered model. Those figures describe the model as delivered and as used for validity screening. The AutoDock search receptor for the Fr0 snapshot alone was prepared with a permissive residue filter that dropped five residues carrying 41 heavy atoms, so that one search ran against 1,333 residues and 10,321 heavy atoms, as recorded in Appendix D. Signature residues R91, V102 and the E106 selectivity filter are represented. Two of the three modulators, GSK-7975A and Synta-66, are reported to act at or near the extracellular pore mouth and the selectivity filter rather than inside the conduction pathway, whereas the site of 2-APB is not experimentally resolved, and the mechanisms and preferred regions of the three are in any case not identical [38], [68], [69].

The receptor ensemble contains four molecular-dynamics snapshots of the same channel, namely an equilibrated starting structure (Fr0) and three later frames (Fr300, Fr400 and Fr499). The files follow a membrane-simulation preparation consistent with the structural and molecular-dynamics work of the contributing Orai group [37], [70]. Docking against all four frames provides limited ensemble sampling and partly mitigates, but does not remove, the rigid-receptor approximation [35].

Figure 12 shows the receptor used throughout this chapter, the Fr300 snapshot. The view is oblique and the six subunits are not colour-coded, so the hexameric arrangement around the central conduction pore is a property of the model described above rather than something the render itself resolves. The four frames are not evenly distributed. The starting structure lies approximately 7.3 to 7.5 Å from the three tightly clustered later snapshots before superposition, while the overall pore architecture remains similar. The same picture holds after alignment, since a Kabsch superposition on the 1,338 C α atoms still leaves the starting frame 5.90, 5.88 and 6.02 Å from Fr300, Fr400 and Fr499, whereas those three lie within 0.93 to 1.19 Å of one another, so

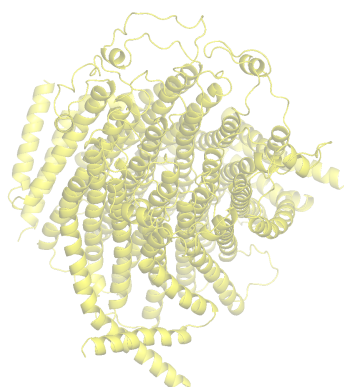


Figure 12: The docking receptor. Cartoon representation of the Fr300 molecular-dynamics snapshot (Orai1WT-MDSnap-Fr300), drawn in an oblique view. The image is a one-off illustrative render prepared in PyMOL for this figure and is not the output of any analysis script, so it carries no measurement and none of the quantities in the surrounding text is read from it

the reading of one outlier frame against a tight trio rests on an aligned number rather than on rigid-body offset alone. Orai is also atypical relative to the largely soluble calibration benchmark because its top-ranked cavity is the elongated, hydrated ion pore rather than a conventional buried hydrophobic pocket.

Crucially, the ensemble carries no crystallographic reference, so there is no RMSD-to-native ground truth for the Orai complexes, and the crystal-dependent analyses used for the calibration benchmark, namely near-nativeness, the form-versus-placement decomposition and native-interaction recovery, do not carry over. Operational admissibility is instead read from the ensemble itself, namely the placement of poses relative to the transmembrane exclusion slab, cross-tool and cross-frame clustering consensus, physical validity and the consistency of predicted interaction contacts, each reported in the subsections that follow. These are hypothesis-compatibility proxies rather than measures of accuracy, since nothing in the Orai1 arm can be scored against an experimental pose. A fuller account of the receptor model, its structural composition, molecular-dynamics provenance and pocket-detection profile, is given in the Appendix (Orai1 Receptor Model).

4.2.2 PoseBusters Validity

On Orai1, physical validity is the per-unit percentage of retained poses that pass all PoseBusters checks (Table 15). The denominator is every pose a unit contributes to the analysis after the top-ten cap, and no quality filter is applied within it, so nothing is removed for being invalid, off-receptor or inside the exclusion slab. The cap itself is a selection rather than a neutral truncation on the experimental panel, where thirty poses were written per unit and the best ten by each tool's own ranking are kept, that ranking being the gnina CNNaffinity re-ranking on the AutoDock arm, against ten written and ten kept on the control panel. The experimental panel therefore enters the cross-panel comparison with the more heavily selected pose set, which is

recorded with the other panel differences in Table 14. This endpoint measures chemical and geometric admissibility, not correct placement. The separate operational placement screen is applied in the next subsection.

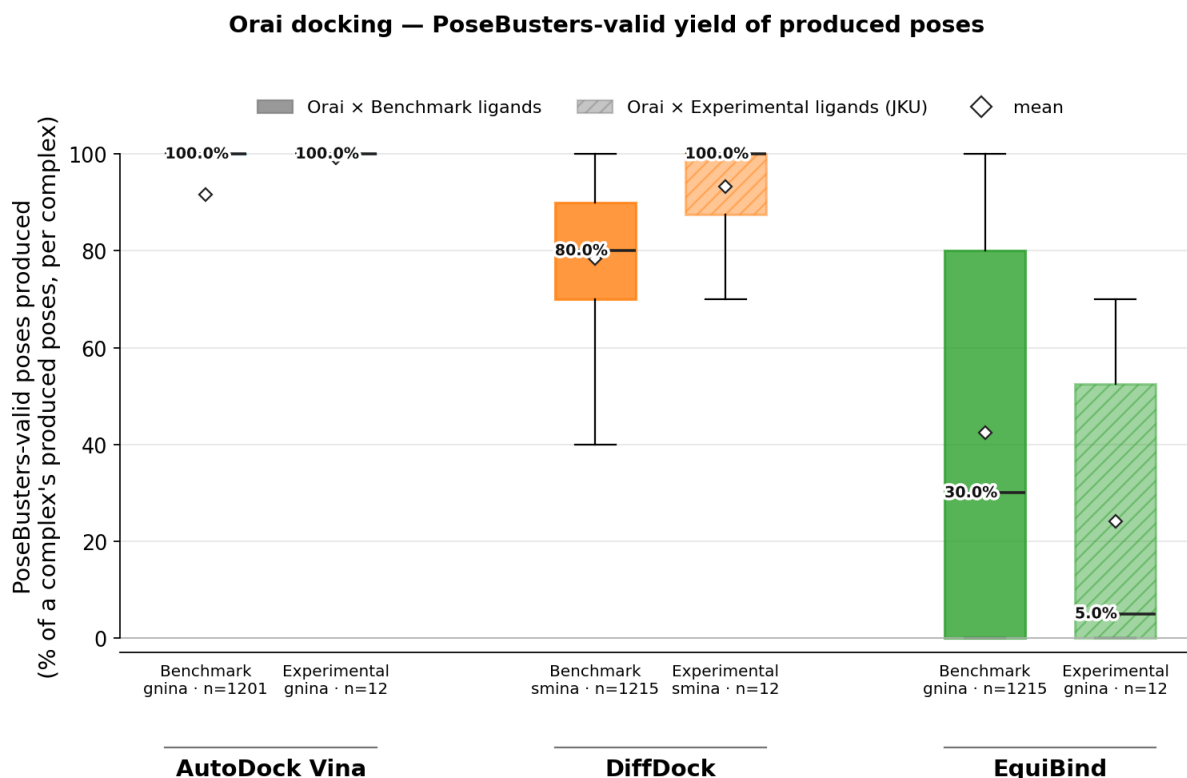


Figure 13: PoseBusters-valid yield of produced poses on Orai, benchmark against experimental ligands

In Figure 13 AutoDock Vina is the most stable of the three across the two ligand sets. Its median yield is 100% on both, and on the twelve experimental frame-ligand units it reaches a mean of 99.2% with an interquartile range pinned at 100%, against a benchmark mean of 91.6% with a thin failure tail in which 2% of units yield nothing. DiffDock is the tool that moves most between the sets, its benchmark median of 80% and mean of 78.5% rising to an experimental median of 100% and mean of 93.3% with an interquartile range of 88 to 100%. EquiBind is the weakest tool in both sets, its experimental median falling to 5.0% and its mean to 24.2%, with an interquartile range running from 0 to 53%.

The ceiling and floor rates separate the tools more sharply than the medians do. On the benchmark AutoDock returns a fully clean unit, in which every produced pose is valid, for 77.1% of units against 22.2% for DiffDock, a 3.5-fold gap, since on that panel the middle half of DiffDock units falls between 70 and 90% and rarely reaches a perfect unit. AutoDock and DiffDock are much closer on the experimental ligands, at 91.7% against 66.7%, so DiffDock is fully clean in two thirds of the experimental units. AutoDock and EquiBind are skewed in opposite directions, AutoDock being ceiling-bound with a mean below its median while EquiBind is floor-bound with a benchmark mean of 42.4% above its median of 30%, so no single central value describes both.

EquiBind is best read as all-or-nothing rather than mediocre, its lower benchmark quartile pinned at zero with 27% of units yielding nothing against 16% fully clean, and on the experimental ligands both tails move the wrong way, 50% of units yielding nothing and not one reaching a fully clean result, its best unit stopping at 70%.

Per tool the sets were compared with an unpaired two-sided Mann-Whitney U test on the per-unit yields and Cliff's δ , oriented so that a positive value means the benchmark is higher. DiffDock is the tool whose validity yield differs most between the sets, being cleaner on the experimental ligands with $\delta = -0.49$ at the frame-ligand unit and -0.68 once each ligand is collapsed to the median over its four frames, its experimental units sitting at a median of 100% valid against a benchmark median of 80%. That contrast does not survive multiplicity correction once the ligand rather than the frame-ligand pair is taken as the unit, at Holm $p = 0.131$ against the 0.009 obtained at the frame-ligand unit, so it is reported here as an effect size and a direction rather than as an established difference. AutoDock shows the smallest movement between the sets, $\delta = -0.16$ at the frame-ligand unit and -0.10 at the ligand unit with Holm $p = 0.195$ and 0.566, which is what its stability across the two sets would predict, with eleven fully clean units of twelve on the experimental side. EquiBind trends the other way, towards the benchmark, at $\delta = +0.30$ and $+0.36$ with Holm $p = 0.135$ and 0.556. The ligand-unit values are taken from the both-unit table of the ligand-level sensitivity analysis, which re-tests every Orai cross-panel contrast at both units.

The comparison is limited by the size and the chemistry of the experimental set, namely three ligands against 308, so the tests have little power and the non-significant results mean that no difference was detected rather than that none exists. Both EquiBind columns now use the same gnina refiner, so its cross-set contrast is no longer confounded by refiner choice. One earlier stage does still differ, because the experimental panel applied the in-protein UFF relaxation recorded in Table 14 and the control panel did not, and that relaxation deforms ligand geometry while relieving clash. The EquiBind cross-set contrast alone therefore carries a preparation difference as well as a ligand-set difference and is read here as descriptive only.

In summary, the ordering AutoDock Vina, DiffDock and EquiBind is preserved for physical validity in both Orai ligand panels. This is consistent with, but does not prove, transfer from the calibration benchmark because the experimental panel contains only three ligands. AutoDock output needs little geometric repair, DiffDock is mostly valid after smina refinement, and EquiBind requires strict validity screening even after gnina optimisation. On this metric no between-panel difference is established at the available power. The apparent DiffDock effect is the largest of the three and is reported descriptively, and with three experimental ligands none of the three contrasts survives multiplicity correction at the ligand unit.

4.2.3 Operational Placement and Usable Yield

The Orai1 target has no ligand-bound reference, so RMSD to a native pose cannot be computed. Accuracy is therefore replaced by an operational placement rule. Throughout this Orai1 section

the AutoDock arm is the gnina-rescored variant, that is the Vina search followed by gnina minimisation and CNN rescoring, which is the strongest AutoDock variant on the calibration benchmark. It is written simply as AutoDock Vina for readability. This now holds for the experimental-versus-benchmark comparison figures as well, both panels being pinned to the same gnina-rescored variant, so no comparison in this chapter is made across different engines. A pose is counted as usable only when it is PoseBusters-valid, remains associated with the receptor surface and lies outside the R91-E106 transmembrane exclusion slab. This rule focuses the analysis on an outer-pore hypothesis. It is not a near-native analogue or experimental ground truth.

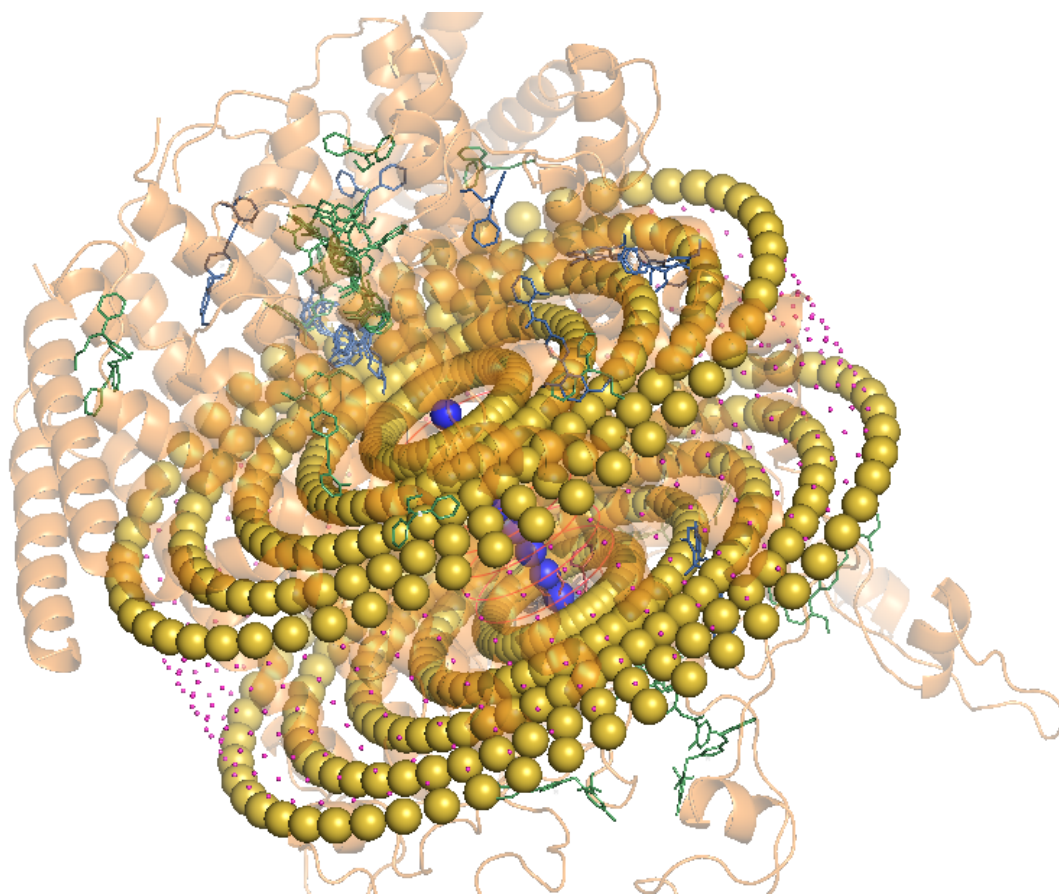


Figure 14: Orai Fr300 with docked ligands and transmembrane exclusion. The image is a one-off illustrative render prepared in PyMOL for this figure and is not the output of any analysis script, so it carries no measurement and none of the quantities in the surrounding text is read from it. The colour classes visible in it are a tan cartoon receptor, a dense layer of yellow spheres, a magenta dot surface, a small number of blue spheres and ligand sticks drawn in green and in blue. The render carries no key, so none of those classes is identified by the figure itself, and the exclusion slab, the pose sets and the residues that define them are specified in the Methods and in the surrounding text rather than labelled here

Figure 14 illustrates the rule on Fr300, showing the docked poses against the exclusion slab defined in the Methods, with the slab drawn out to the outer protein shell so that the picture

matches the radially unbounded test rather than the narrower pore lumen. Figure 15 then reports the percentage of produced poses that are PoseBusters-valid, remain on the receptor and lie outside the exclusion slab, for twelve experimental units and 1,215 control units (1,201 on the AutoDock arm, the fourteen missing units being the macrocyclic complexes noted above). The control-panel pose totals behind those units are 12,010 for AutoDock Vina, 12,150 for EquiBind and 11,673 for DiffDock, the last falling short of a full ten poses on 178 units.

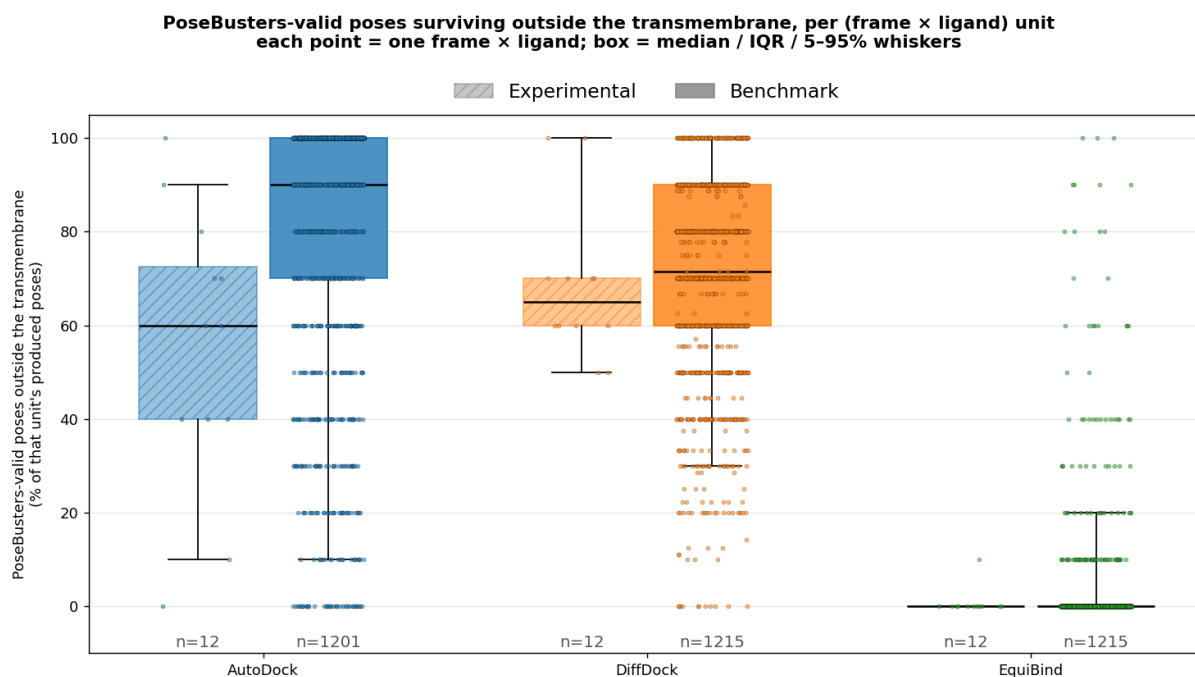


Figure 15: Usable pose yield per frame-ligand unit. Each point is one Orai frame paired with one ligand

Each tool is capped at its top ten ranked poses per unit and each unit contributes the yield of the poses it actually produced, so for any one tool the two ligand sets sit on one comparable axis. That denominator is not identical across tools, because DiffDock fell short of ten poses on 178 control units while the other two arms produced ten everywhere, which is why its benchmark usable median of 71.4% falls on a seven-pose unit rather than on a multiple of ten. AutoDock Vina gives the highest usable yield on the benchmark ligands but not on the experimental ones, a benchmark median of 90% of produced poses against an experimental median of 60%, with interquartile ranges of 70 to 100% and 40 to 72.5% and means of 79.2% and 55.0%. DiffDock sits below it on the benchmark at 71.4% and above it on the experimental ligands at 65.0%, with spreads of 60 to 90% and 60 to 70% and means of 70.9% and 68.3%. EquiBind is degenerate in both sets, its median and both quartiles pinned at zero and its mean only 2.9% on the benchmark and 0.8% on the experimental ligands.

Tested at the frame-ligand unit, which removes the nesting of poses inside a unit although the units of one ligand remain correlated with each other, with an unpaired two-sided Mann-Whitney U and Cliff's δ , one tool separates the two ligand sets. AutoDock returns less on the experimental ligands than on the benchmark ($U = 10803.0$, $\delta = +0.50$) and survives Holm correction across

the three tools ($p = 0.005$), and it is the one contrast in this chapter that survives at both units on all three experimental ligands, at $\delta = +0.90$ with a Holm-corrected p of 0.016 on ligand-collapsed medians of 60.0% against 95.0%, whereas DiffDock trends the same way without reaching significance ($U = 8462.0$, $\delta = +0.16$, $p = 0.667$) and EquiBind shows nothing ($\delta = +0.02$, $p = 0.832$). The pooled pose-level counterpart agrees. Its gaps are differences between pooled rates rather than between the per-unit means quoted above, which for DiffDock are not the same quantity because its per-unit pose counts vary. AutoDock's gap of 24.2 percentage points, from 79.17% on the benchmark ligands to 55.00% on the experimental ones, gives a Fisher p below $1e-4$ that survives correction, with DiffDock at 2.7 points (71.07% against 68.33%) and EquiBind at 2.0 points (2.86% against 0.83%) clearly non-significant. On this metric, therefore, AutoDock Vina is the only tool whose usable yield differs between the experimental ligands and the benchmark library, and the shortfall falls on the experimental set. With twelve correlated experimental units the tests have little power, so the two null results are an absence of detected difference rather than evidence that those sets behave equivalently.

Figure 16 describes the share of already PoseBusters-valid poses that lies inside the transmembrane slab. All rates in this paragraph are taken on the same top-ten capped basis as Table 11, with the denominator the tool's own PoseBusters-valid poses rather than a fixed pose count, so the differing control-panel totals of 12,010, 11,673 and 12,150 produced poses are already absorbed into it. AutoDock Vina loses 13.6% of valid control-panel poses (1,499 of 11,007) and 44.5% of experimental poses (53 of 119). DiffDock loses 9.6% (881 of 9,177) and 26.8% (30 of 112), respectively, while EquiBind loses 93.3% (4,808 of 5,156) and 96.6% (28 of 29 experimental valid poses). Because poses are clustered within frame-ligand units, these pose-level contrasts are descriptive and their nominal tests should not be treated as independent biological replication.

Figures 15 and 16 separate validity from conditional placement. DiffDock loses a smaller share of its already valid experimental poses to the transmembrane slab than AutoDock Vina (26.8% versus 44.5%), and the same ordering holds on the control panel (9.6% versus 13.6%). On the experimental panel that conditional advantage is large enough to give DiffDock the higher net usable yield as well, even though AutoDock Vina generates more valid poses. EquiBind fails the placement requirement outside the transmembrane exclusion slab almost completely. Of its 29 valid experimental poses, 28 lie in the slab and only one is usable. The combined placement screen, rather than physical validity alone, is therefore decisive for this Orai1 protocol.

Two limits bound these comparisons. The transmembrane-loss rates are pooled pose-level tests that treat correlated poses within a unit as independent, so they overstate significance and are descriptive only. Their per-unit counterpart, the usable-yield test above, is significant for AutoDock Vina alone, so AutoDock's 30.9-point benchmark-to-experimental loss gap is corroborated at the frame-ligand unit and again at the ligand unit, while the DiffDock and EquiBind gaps remain pooled descriptions. The cross-set differences are also reported as absolute differences rather than ratios, because the experimental denominator rests on three ligands and at most twelve units, which makes a ratio such as the apparent 0.31-fold reduction

Orai docking — cross-dataset difference in transmembrane loss of PoseBusters-valid poses, per tool

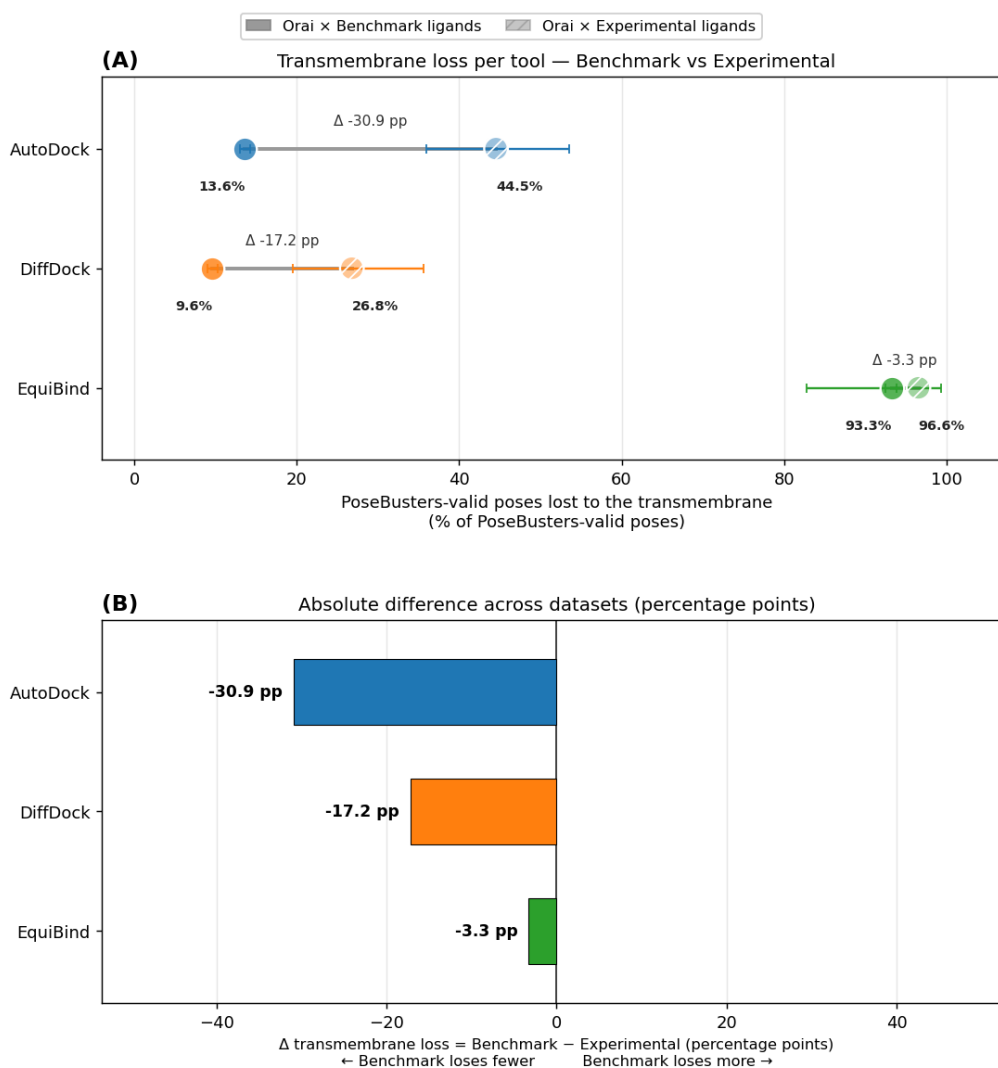


Figure 16: Transmembrane loss of PoseBusters-valid poses, benchmark against experimental ligands. Panel (A) loss rate per tool as pooled pose-level proportions with Wilson 95% intervals. Panel (B) the difference per tool as benchmark minus experimental, in percentage points

in loss unstable. Since the four frames are snapshots of one receptor, even the per-unit test is anti-conservative for the question of generalisation to new ligands.

4.2.4 Cross-Tool Clustering

The clustering approach is applied per MD frame and ligand pair, one pose cloud per frame, ligand and tool, to ask whether the three toolchains point to the same region of Orai and how well formed each tool's own cloud is. Since frames of the same ligand are snapshots of one receptor-ligand system rather than independent observations, the pooled medians below are mildly pseudoreplicated. A frame-robust summary that collapses each ligand to the median over its frames first gives 30.0 Å against the pooled 29.5 Å on the benchmark, so that figure is stable under the aggregation, whereas on the experimental ligands it rises to 42.2 Å against a pooled 30.1 Å. The experimental median is therefore not stable under frame-robust aggregation, which is what three ligands and eleven usable pairs would predict, so only the benchmark value should be read as a settled central estimate.

Figure 17 shows the distribution of the distance between each pair of tools' consensus sites for the two ligand sets. On the experimental ligands nothing at all falls left of the 5 Å agreement line, and on the benchmark panel the two EquiBind pairings are nearly empty there, but the AutoDock against DiffDock curve is not, its tallest bin sitting immediately left of that line even though the bulk of its mass lies far to the right. Distances run from a few ångströms to roughly 75 Å, since candidate sites scatter along the elongated Orai pore, and the per-pair medians sit at 26 to 31 Å on the benchmark ligands against 27 to 48 Å on the experimental ones. Agreement therefore has to be read from the continuous distance and the per-pair rate rather than from the binary all-pairs consensus flag, which demands that every tool pairing present in a unit fall within 5 Å at once and is met by 9.0% of the 1,204 multi-tool benchmark units, 108 of them, against none of the eleven multi-tool experimental units.

Figure 18 breaks the same agreement down by tool pair, giving the consensus-site distance in panel (A) and the percentage of pairs within the 5Å threshold in panel (B). One pairing separates the ligand sets more than the others. Effect sizes on this distance endpoint are oriented so that a positive Cliff's δ means the experimental ligands sit further apart, which inverts the benchmark-higher orientation used for the yield and cluster-quality endpoints elsewhere in this chapter. DiffDock and EquiBind show the largest gap, 47.6Å on the experimental ligands against 30.7Å on the benchmark with $\delta = +0.91$. It is nominally significant at both units, at Holm $p = 0.027$ over three usable experimental pairs and 0.006 once each ligand is collapsed to the median over its frames, but the collapse leaves only two experimental ligands, since no Synta-66 EquiBind pose survived the transmembrane exclusion and the top-ten cap that feed the consensus clustering. Two ligands cannot support an inferential claim, so this is an exploratory effect estimated from very little data rather than a corrected finding. AutoDock against DiffDock (30.1Å against 28.4Å, $\delta = +0.30$, $p = 0.275$) and AutoDock against EquiBind (27.2Å against 26.0Å, $\delta = +0.04$, $p = 0.900$) remain flat. The agreement rates tell the more important story. AutoDock and DiffDock

Cross-tool agreement on Orai — distance between tools' consensus sites, by ligand category

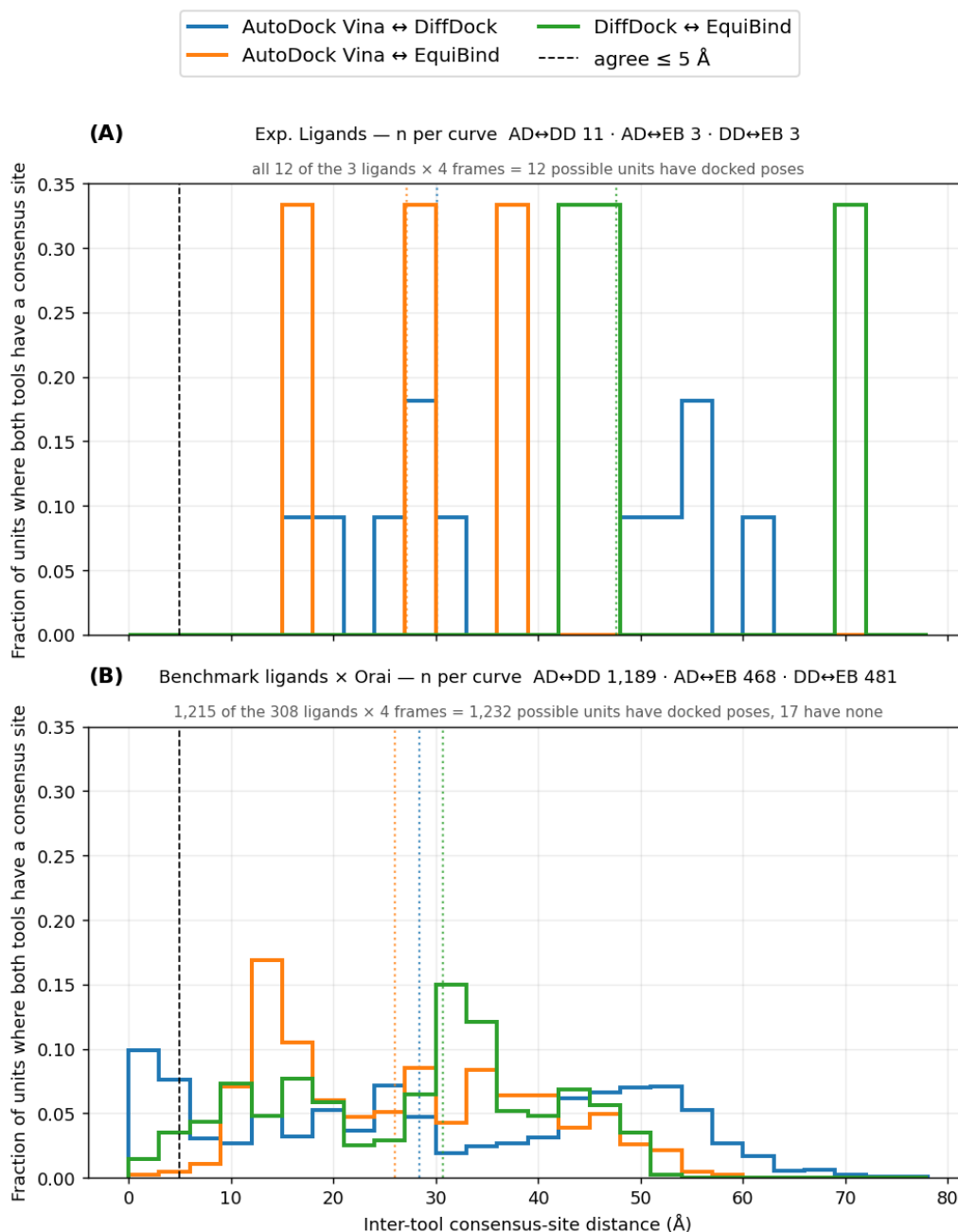


Figure 17: Distribution of inter-tool consensus-site distances on Orai. Step histograms per tool pair, with panel (A) the experimental ligands and panel (B) the benchmark ligands. The dashed vertical line marks the 5Å agreement threshold and the dotted lines the per-pair medians

Cross-tool agreement on Orai — Experimental vs Benchmark ligands

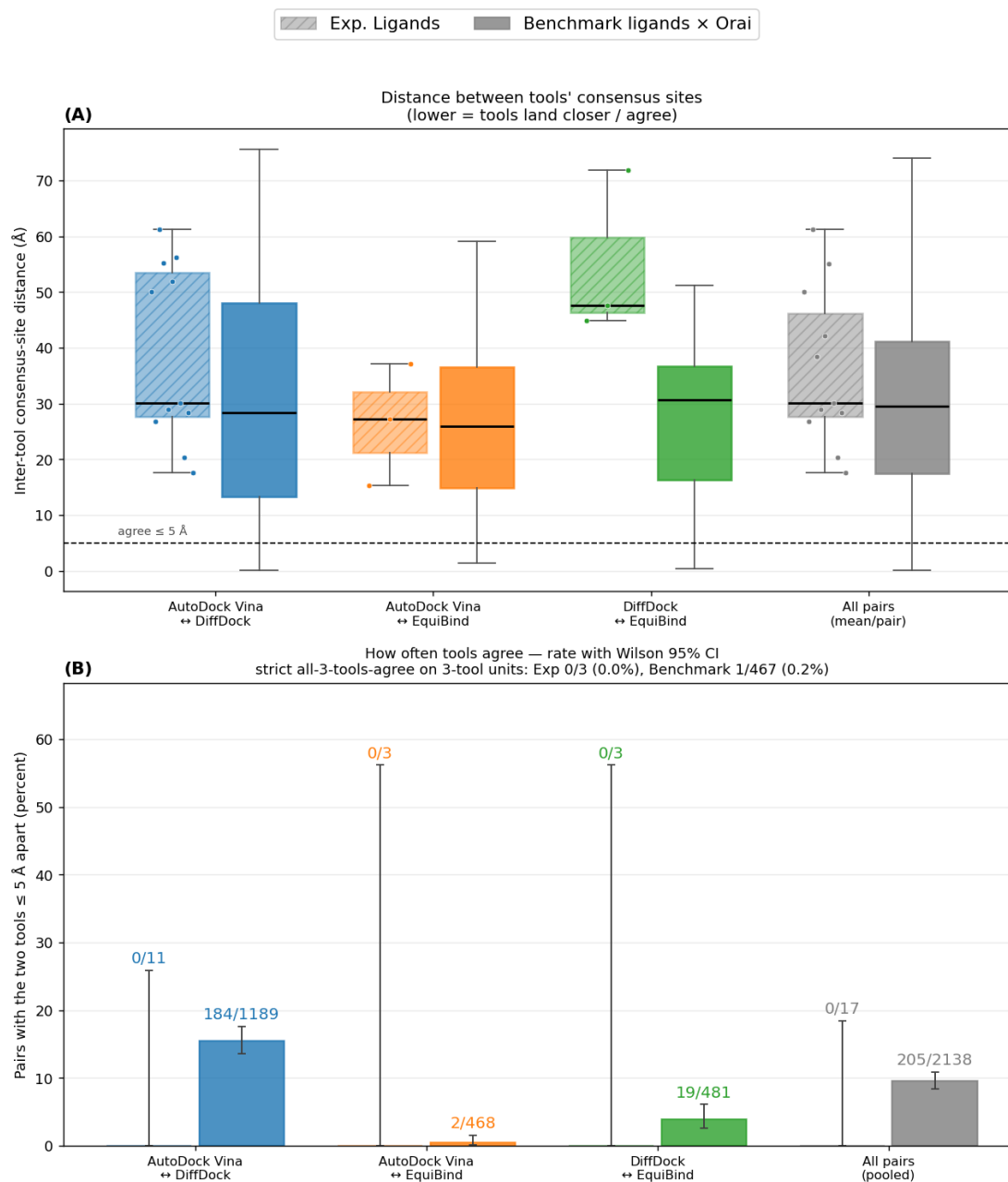


Figure 18: Cross-tool agreement on Orai. Panel (A) gives the inter-tool consensus-site distance per tool pair, hatched for the experimental ligands and solid for the benchmark set, and its fourth position summarises all pairs as the mean per pair. Panel (B) gives the percentage of pairs within the 5Å threshold per tool pair, with Wilson 95% intervals and the underlying counts, and its fourth position pools every pair into one rate rather than averaging the three per-pair rates

land within 5 Å in 15% of benchmark pairs, 184 of 1,189, and in none of the eleven usable experimental pairs, whereas every EquiBind pairing stays at or below 4%, namely 2 of 468 against AutoDock and 19 of 481 against DiffDock, with none at all on the experimental ligands, and those rate differences give a Fisher p of 1.000. Pooled over all three pairings, agreement within 5 Å is 10% on the benchmark, 205 of 2,138, and none of the seventeen experimental pairs. Three-way agreement, in which all three pairwise distances of a unit fall within 5 Å at once, is a far harder requirement than these per-pair rates, and it is reached in one of the 467 benchmark units that carry all three tools, 0.2%, and in none of the three experimental ones. The only real agreement channel is therefore AutoDock against DiffDock, and any consensus site on Orai is in practice a question about those two tools alone.

Figure 19 reports reference-free cluster quality per tool on Orai, the only form of cluster assessment available without a crystallographic site. AutoDock Vina forms the best-separated pose clouds in both ligand sets and the most compact ones on the benchmark, although on the experimental ligands DiffDock is the tighter of the two at a median within-cluster spread of 0.34 Å against 0.62 Å, lower being better. EquiBind forms the loosest clouds and often collapses into one diffuse mode. High bootstrap stability in such a degenerate case reflects reproducibility of the same broad cloud rather than accurate site resolution. The largest experimental-versus-control gap across the eighteen tool-by-metric cells is DiffDock's silhouette, at a median of 0.17 on the experimental ligands against 0.42 on the benchmark, so that on the benchmark-higher orientation used for these quality metrics, Cliff's $\delta = +0.53$ at the frame-ligand unit and $+0.72$ at the ligand unit. It reaches Holm $p = 0.031$ at the frame-ligand unit but falls to Holm $p = 0.516$ once each ligand is collapsed to the median over its four frames, so it is reported descriptively rather than as a corrected finding. The Calinski-Harabasz cell for the same tool moves the other way under that collapse, from Holm $p = 0.074$ to 0.031, which the ligand-level sensitivity analysis reported with these results flags as an artefact of removing the within-ligand spread rather than as new evidence, so it is not promoted here either. Everywhere else no set difference was detected at the available power, and the experimental panel contains only three ligands, so that absence does not establish equivalence or validate the control panel as a stand-in.

Figure 20 pairs the raw pose cloud against the filtered cloud within each frame-ligand unit of the benchmark panel, across the three quality axes. The medians quoted here are those of the paired subset, that is the units that carry both clouds, which for AutoDock Vina and DiffDock is close to the full panel but for EquiBind is only 49 units on spread and 37 on stability out of 1,215, so the EquiBind values sit below the wider raw boxes drawn in the figure. Filtering to poses that are PoseBusters-valid and outside the transmembrane exclusion slab does not tighten the clouds. Compactness is essentially unmoved for every tool, AutoDock going from a median 0.55 to 0.53 Å at a rank-biserial of -0.09 (Holm-corrected $p = 0.340$), DiffDock from 0.57 to 0.57 ($p = 1.000$) and EquiBind from 1.30 to 1.20 ($p = 1.000$). What filtering does improve is reproducibility rather than tightness. Bootstrap stability rises significantly for all three, AutoDock from 0.77 to 0.78 at a rank-biserial of $+0.25$, DiffDock from 0.70 to 0.72 at $+0.54$ and EquiBind from 0.83 to 1.00 at $+0.94$, each with a corrected p below $1e-3$, and DiffDock also gains on silhouette, from

Per-tool cluster QUALITY on Orai — Experimental vs Benchmark ligands ('raw' pose cloud; reference-free)

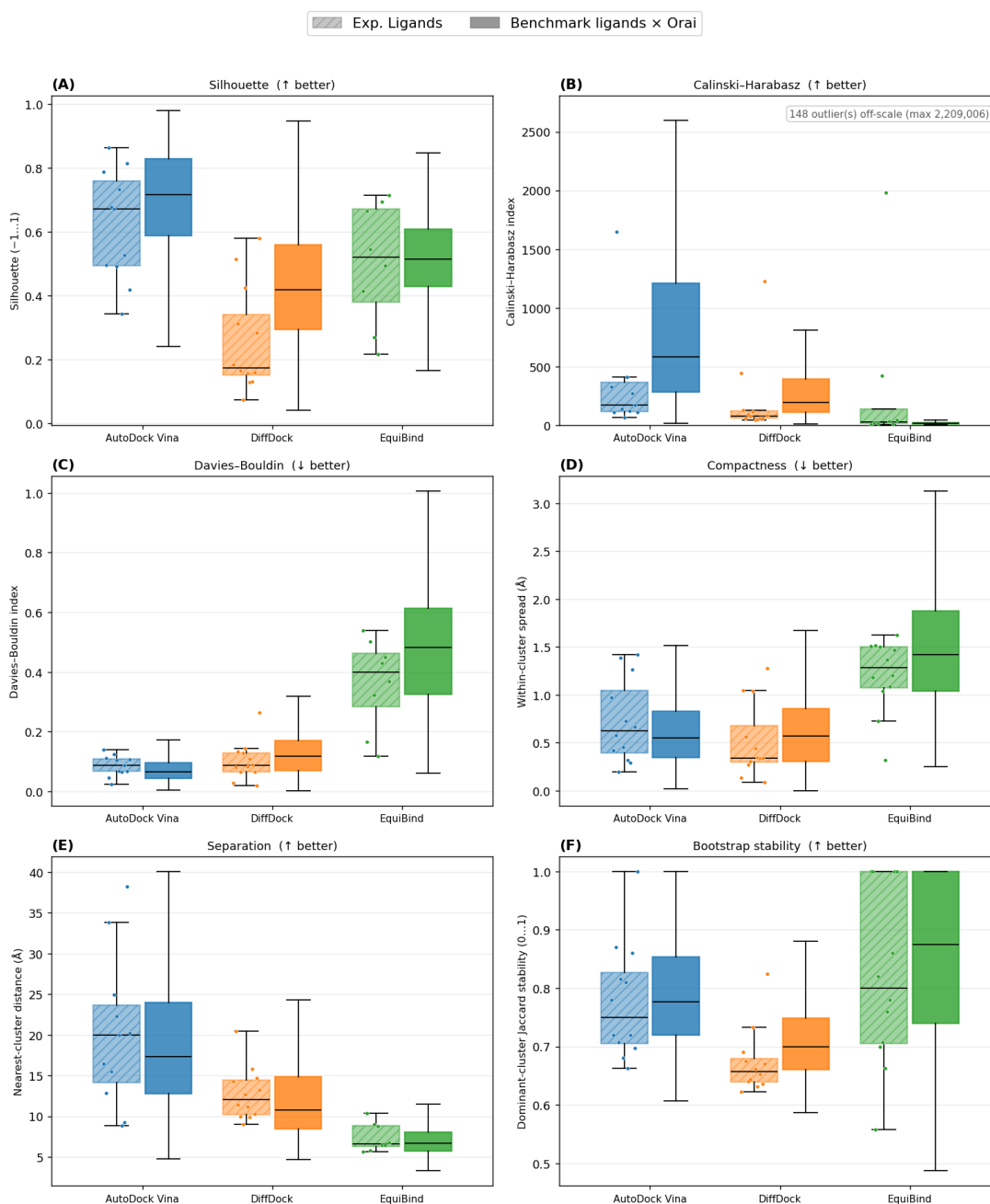


Figure 19: Reference-free cluster quality per tool on Orai, the experimental ligands against the benchmark control panel on the raw pose cloud. Panel (A) silhouette, panel (B) Calinski-Harabasz index, panel (C) Davies-Bouldin index, panel (D) compactness, panel (E) separation and panel (F) bootstrap stability. Panel (B) is truncated on the vertical axis, with 148 outliers off-scale up to a maximum of 2,209,006

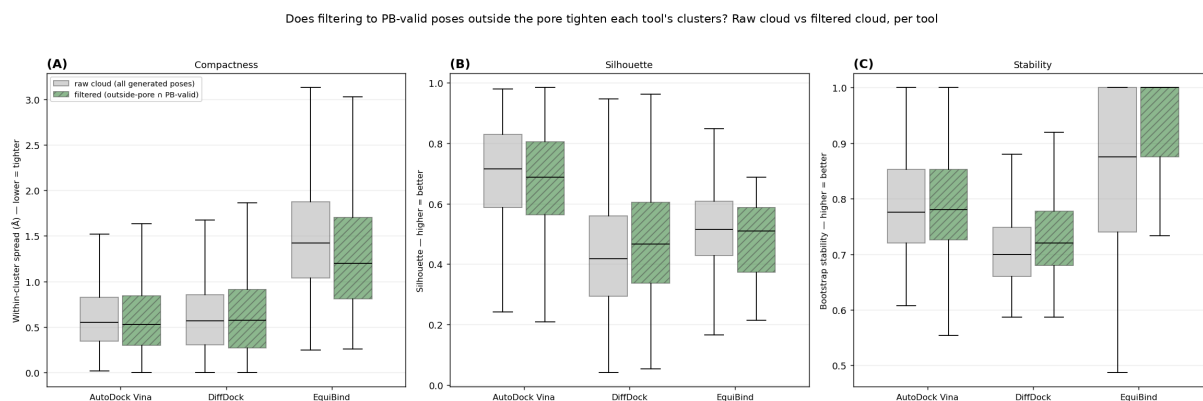


Figure 20: Effect of validity and placement filtering on cluster quality for the benchmark ligands. Each panel contrasts the raw pose cloud with the filtered cloud of PoseBusters-valid poses outside the transmembrane exclusion slab, for panel (A) within-cluster spread, panel (B) silhouette and panel (C) bootstrap stability. The boxes are the unpaired distributions over every frame-ligand unit that has a cloud of the given kind, whereas the medians and the tests quoted in the text are taken over the units that carry both clouds, so the two readings differ wherever filtering removes most of a tool's units, most visibly for EquiBind

0.43 to 0.47 at +0.30 and p below $1e-4$. The EquiBind rise is degenerate rather than informative, since 27 of its 37 paired filtered clouds collapse to a single cluster, a case in which the bootstrap Jaccard stability is identically 1.00 by construction, and on the ten paired units that keep more than one cluster the medians move only from 0.64 to 0.71. Filtering therefore does not detectably tighten the wide scatter along the pore, so on this evidence it is not an artefact of invalid or mislocated poses that filtering could clean away. No equivalence test was run, so this bounds the effect of filtering rather than establishing that the scatter is intrinsic to how the tools sample, and the EquiBind contrast in particular rests on few surviving units. The surviving AutoDock and DiffDock poses do partition more reproducibly.

In summary, the tools do not provide a reliable consensus site on Orail. Pairwise sites differ by a pooled median of roughly 30 Å on both ligand sets, rising to 42 Å on the experimental ligands once each ligand is collapsed to the median over its frames, and agreement within 5 Å reaches at most 15%. AutoDock Vina forms the best-separated pose clouds, and the tightest on the benchmark, but precision is not evidence of accuracy without a reference. The three largest experimental-versus-control gaps among these clustering and agreement proxies are DiffDock's higher physical-validity yield, its larger consensus-site distance from EquiBind and its lower silhouette, and none of them is established on all three ligands once the ligand rather than the frame-ligand pair is taken as the unit, the validity and silhouette gaps falling well short of correction and the consensus-site gap surviving but resting on two experimental ligands. The one cross-panel contrast that survives that collapse at both units on all three ligands is AutoDock Vina's lower usable yield, which is a placement rather than a clustering proxy and is reported with the placement results above. The largely preserved descriptive ordering supports cautious use of the control panel for these proxies, but it does not validate equivalence.

4.2.5 Interaction and Contact Analysis

All poses entering the analysis are PoseBusters-valid and outside the transmembrane slab, capped at each tool's top three ranked poses per frame-ligand unit, and both ligand sets were docked into the same four frames, so the residue axis is directly comparable. PandaMap typed every contact at residue level into fifteen interaction classes. After filtering, the experimental set contributes 31 poses from 11 complexes for AutoDock and 36 from 12 for DiffDock, against 3,406 and 3,524 benchmark poses, whereas EquiBind retains a single experimental pose against 233, so every EquiBind figure below is non-interpretable and is reported only for completeness. Poses within a complex are correlated, so the figures display pose-level values, but every contact-count and type-profile significance test aggregates to the per-complex mean first, before an unpaired Mann-Whitney U test with Cliff's δ oriented so that a positive value means the benchmark is higher. The prose therefore quotes the pose-level medians shown on the plots as the descriptive summary and the per-complex-mean values alongside each δ as the inferential unit.

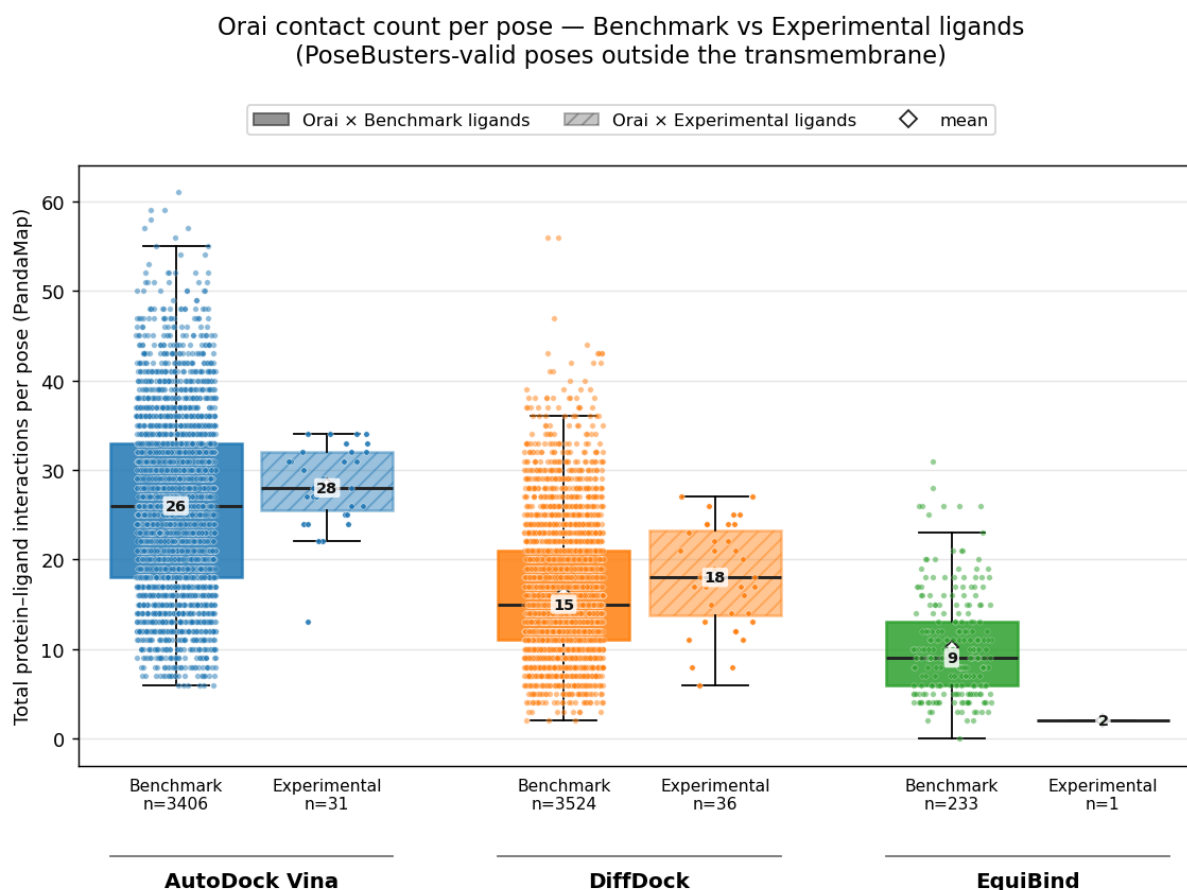


Figure 21: Total protein-ligand contacts per pose on Orai

Figure 21 compares total contacts per pose between the two ligand sets. The pose-level median is printed on each box. AutoDock Vina trends toward denser contacts for the experimental ligands, its median rising from 26 for the benchmark set to 28 for the experimental set, and

DiffDock moves the same way, from 15 to 18. Neither trend survives testing one level up on the per-complex mean, which removes the pose pseudoreplication. AutoDock gives Cliff's $\delta = -0.20$ at a Holm-adjusted $p = 0.254$ at the frame-ligand unit (per-complex-mean medians 25.7 for the benchmark against 29.7 for the experimental set), and DiffDock $\delta = -0.27$ at a Holm-adjusted $p = 0.205$ at the same unit (15.7 against 18.2). Collapsing each ligand to the median over its four frames weakens both further, to $\delta = -0.20$ for AutoDock and -0.19 for DiffDock, each at a Holm-adjusted $p = 1.000$. EquiBind contributes only one usable experimental pose, which is too few to test, so it is left out of the correction family rather than counted as a hypothesis. No difference in total contact count is detected at the available power, which is not evidence that the panels have equivalent contact intensity.

Figure 22 gives the mean count per pose of each interaction type. As for the totals, the plotted bars are pose-level means and the values quoted below are the per-complex means that enter the tests. The same families dominate in both ligand sets for both mature tools, namely alkyl- π , hydrophobic contacts, the π family and hydrogen bonds, but their order is not preserved. Hydrogen bonds fall from second place on the benchmark ligands to fifth on the experimental ones under both tools, which is the one rank change in the leading group, and several differences are chemically coherent and run consistently in one direction. The benchmark ligands form more hydrogen bonds, 3.85 against 2.42 per pose for AutoDock and 2.47 against 1.31 for DiffDock, with $\delta = +0.42$ and $+0.54$. The modulators form more halogen bonds, 0.76 against 0.19 and 1.17 against 0.16. They are also markedly more hydrophobic under both mature tools, 5.48 against 3.50 for AutoDock with $\delta = -0.64$ and 3.42 against 2.36 for DiffDock with $\delta = -0.58$, and richer in alkyl- π contacts under both, 6.36 against 4.75 for AutoDock with $\delta = -0.45$ and 5.06 against 3.31 for DiffDock with $\delta = -0.59$. The charged classes, namely ionic, salt-bridge and attractive-charge contacts, separate the two sets more modestly under AutoDock at $\delta = +0.37$, appearing at 0.44 per pose in its benchmark poses and not at all in its modulator poses, and DiffDock moves the same way on those classes at 0.22 per pose against none. All of these are descriptive patterns and none of them is an established difference. At the frame-ligand unit fifteen of the thirty tool-by-type cells clear Benjamini-Hochberg correction, but at the ligand unit not one of the fifteen AutoDock or fifteen DiffDock contrasts does, the smallest adjusted values being 0.164 and 0.122, which the ligand-level sensitivity analysis reports in full at both units. The direction of the pattern is nonetheless coherent rather than arbitrary, since the real modulators are halogenated lipophilic compounds and the benchmark library is comparatively polar and hydrogen-bond driven.

Figure 23 turns from contact types to their location, reporting for the fifteen most frequently contacted Orai residues the percentage of a tool's poses that touch each one. The fifteen are ranked by pooled contact frequency across the six tool-by-ligand-set groups, weighting each group by its pose count, so no residue reaches the axis on the strength of a group holding very few poses. Both libraries concentrate on the same pocket residues, namely Asp110, Gln108, Tyr115, Lys198, Val107, His113 and Pro146, and no residue separates the two panels once the fifteen tests within a tool are Holm-corrected. Under AutoDock the modulators sit slightly higher

Interaction-type profile on Orai — Benchmark vs Experimental ligands, per tool
(mean count per pose; PoseBusters-valid poses outside the transmembrane)

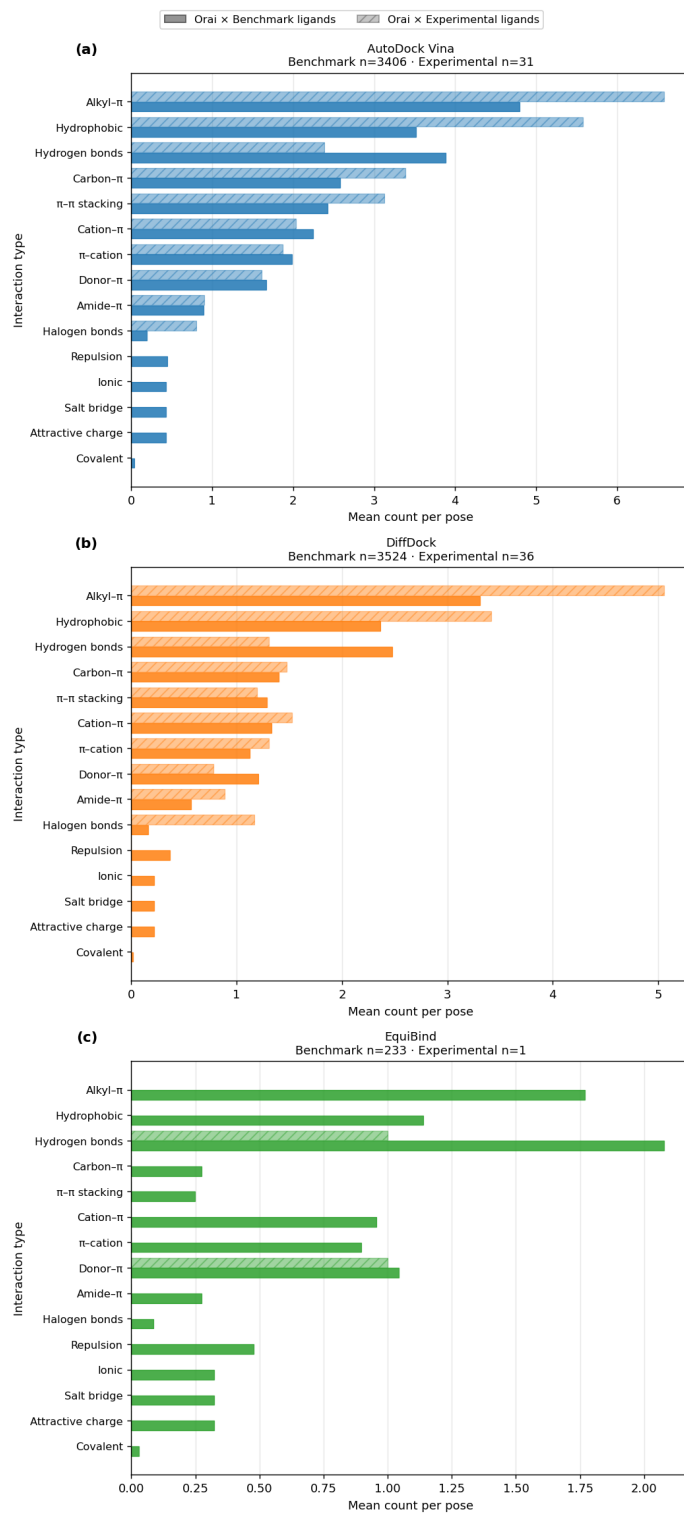


Figure 22: Interaction-type profile on Orai. Mean count per pose of each PandaMap interaction type, with panel (a) AutoDock Vina, panel (b) DiffDock and panel (c) EquiBind. Solid bars are the benchmark ligands and hatched bars the experimental ones. This figure and the two that follow it carry lower-case panel letters, as printed on the panels themselves

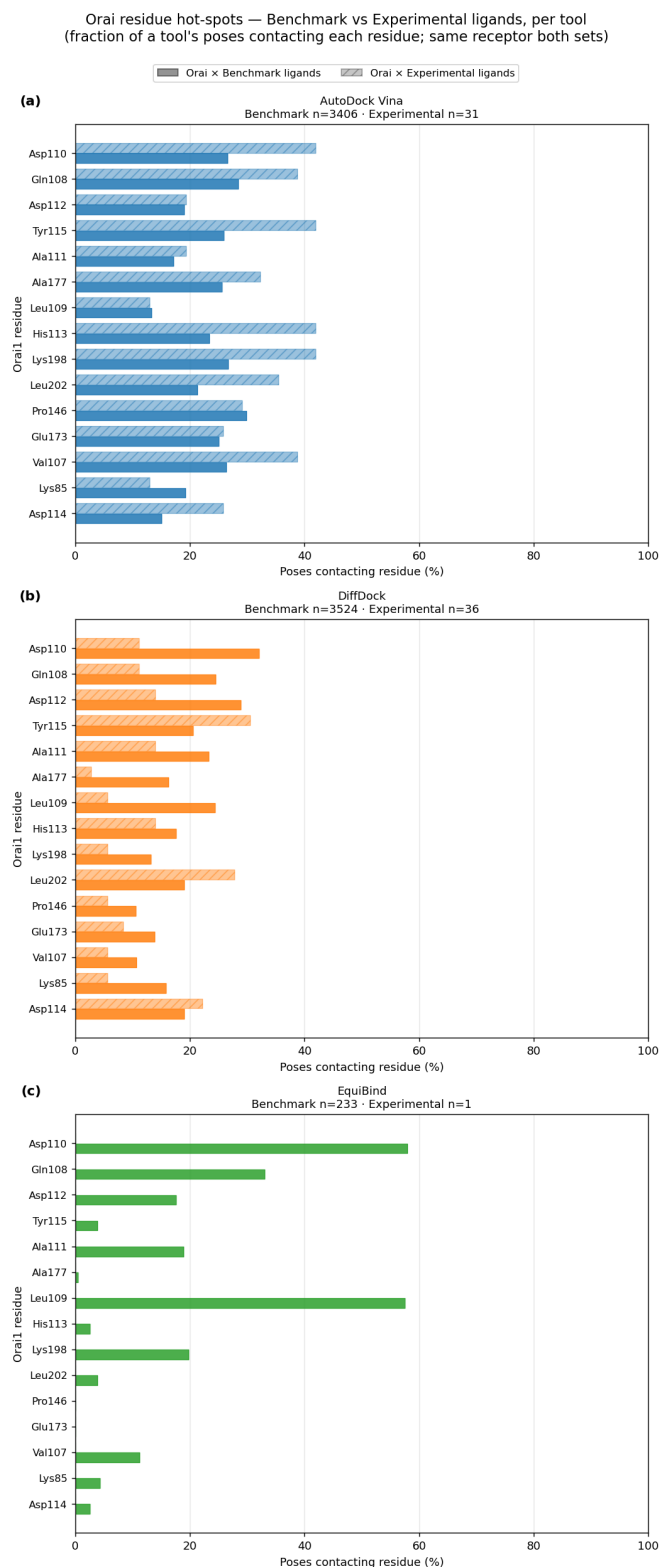


Figure 23: Orai residue hot-spots. Percentage of a tool's poses contacting each of the fifteen most frequently contacted Orai1 residues, with panel (a) AutoDock Vina, panel (b) DiffDock and panel (c) EquiBind. Solid bars are the benchmark ligands and hatched bars the experimental ones. The fifteen are ranked by pooled contact frequency over all 7,231 poses of the six tool-by-ligand-set groups, so each group contributes in proportion to the number of poses it holds and the axis runs from Asp110 at 30.2% of poses down to Asp114 at 16.7%

on almost every frequent residue, the largest raw gaps being His113 at 41.9% of poses against 23.4% of benchmark poses, Tyr115 at 41.9% against 25.9%, Lys198 at 41.9% against 26.6% and Asp110 at 41.9% against 26.5%, but only His113 reaches even an uncorrected p below 0.05 and none survives correction. DiffDock runs the other way on several residues, engaging Asp110 in 32.1% of benchmark poses against 11.1% of modulator poses and Leu109 in 24.4% against 5.6%, and its modulator poses essentially abandon Ala177 and Glu106, at 2.8% each against 16.2% and 17.1%, the Glu106 pair being read from the underlying fingerprint table because that residue falls just outside the fifteen drawn here. These too are uncorrected signals that do not survive Holm. The experimental panel supplies only 31 AutoDock and 36 DiffDock poses, so these pose-level two-proportion tests are both pseudoreplicated and underpowered, and they are descriptive support for the figure rather than inference.

Figure 24 summarises how far the two libraries agree on the residues they engage. Overlap is graded by tool, the residue-level Jaccard being 0.56 for AutoDock and 0.31 for DiffDock, and for AutoDock it falls to 0.49 once the interaction type must match as well, since the two libraries hit the same residues without always forming the same kind of contact there, which is the fingerprint-level echo of the hydrogen-bond against hydrophobic shift above. DiffDock moves the same way but less far, from 0.31 to 0.27. Both Jaccards are taken on the same 10% characteristic-contact threshold, so the residue-to-typed drop measures a change of contact type rather than a change of basis. An earlier reading, on which AutoDock fell to 0.28 and DiffDock rose to 0.35, compared a thresholded residue set against an unthresholded typed one, and most of the apparent AutoDock drop was the long tail of one-off typed contacts that a benchmark set of several thousand poses accumulates. AutoDock shares 33 characteristic residues between the libraries and DiffDock 13, whereas EquiBind shares none, its single experimental pose leaving a Jaccard of zero that carries no meaning. One reading trap deserves flagging. The 10% threshold is an absolute count, roughly four poses for the experimental set against several hundred for the benchmark, and the diverse benchmark ligands spread their contacts across many residues while the three repeatedly docked modulators concentrate theirs, so the experimental-only count exceeding the benchmark-only count, 16 against 10 for AutoDock, is an artefact of set size and homogeneity rather than evidence of promiscuity. The shared set and the per-residue frequencies are the robust readings.

The surviving modulator poses show partial concordance with published pharmacology. Synta-66 has been mapped to the outer TM1/TM3 interface adjacent to the E106 selectivity filter [69]. AutoDock Vina contacts Ala177 in 10 of its 31 modulator poses against 25.6% of its benchmark poses, an uncorrected $p = 0.410$ that is compatible with TM3 involvement without establishing it. DiffDock offers no support on this point, since its modulator poses contact Ala177 in a single pose of 36 against 16.2% of its benchmark poses. Glu106 itself separates neither tool, being touched by one modulator pose under each, so the TM3 side of the filter-adjacent hypothesis rests on the AutoDock arm alone and the filter residue itself is supported by neither. These observations make the poses biologically interpretable, but contacts contributed by different tools cannot be combined to validate one binding mode. The remaining frequent residues define

Orai contact-fingerprint overlap — Benchmark vs Experimental ligands, per tool
(same receptor both sets; characteristic = carried by $\geq 10\%$ of a tool's poses, applied to the residue and typed fingerprints alike)

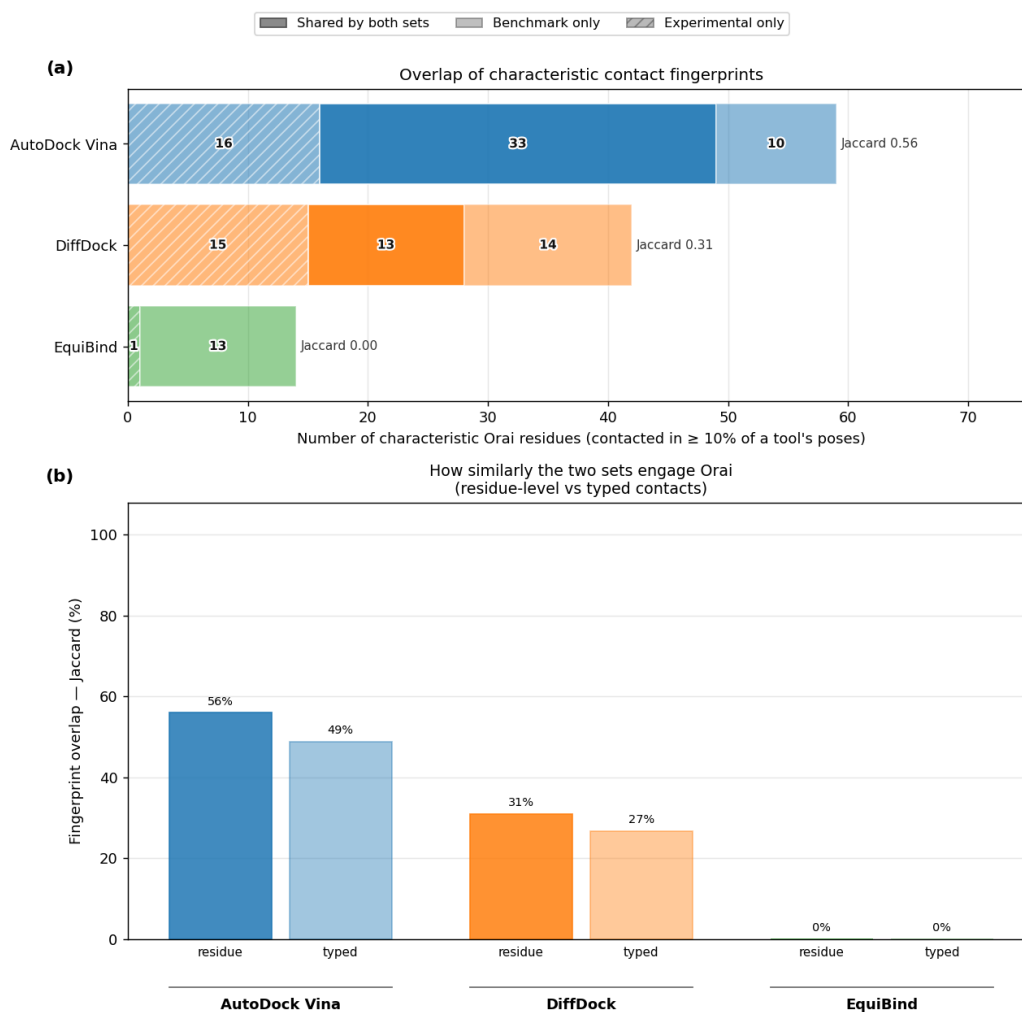


Figure 24: Contact-fingerprint overlap between the two ligand sets. Panel (a) counts the characteristic Orai1 residues, those contacted in at least 10% of a tool's poses, split into shared, benchmark-only and experimental-only. Panel (b) gives the residue-level and typed Jaccard overlap per tool

plausible hypotheses for mutagenesis or ligand-bound simulation.

The comparison is unpaired and confounded by chemistry, so the contact-type split characterises chemotypes rather than tool skill. Without a cognate structure, literature concordance remains partial and some retained poses are still anatomically implausible. Pose-level residue tests are pseudoreplicated, the experimental arm contains three ligands, and EquiBind contributes only one usable experimental pose. Effect sizes and qualitative agreement with pharmacology therefore carry more weight than nominal p-values.

Taken together, the Orai1 analysis inverts the ordering of its own control panel on net usable yield. AutoDock Vina leads that control panel, but on the experimental ligands DiffDock + smina is ahead, which is also the ordering calibration accuracy gives, where DiffDock + smina has the higher recovery point estimate. AutoDock Vina is physically clean yet retains only 50 to 60% of poses per ligand under the operational placement rule, with a median usable yield of 60% across the twelve frame–ligand units. DiffDock leads at a median frame–ligand-unit validity of 100% and usable yield of 65%. Conditional on already valid poses, it also has a lower transmembrane-loss rate than AutoDock Vina on both panels. EquiBind has a median experimental frame–ligand-unit validity of 5%, and 28 of its 29 valid poses fail the transmembrane exclusion rule, leaving one usable pose out of 120.

Three methodological conclusions follow. First, chemical validity alone overstates usefulness on a channel target, since a tool can be geometrically flawless and still dock into the ion-conduction pathway, so usable yield rather than raw validity is the defensible way to summarise performance on this target and the placement filter is what exposes EquiBind. Second, cross-tool consensus is not available as a confidence signal here, because the three tools disagree about where these ligands bind by a pooled median of roughly 30 Å in both ligand sets, agree within 5 Å in at best 15% of pairs, with all three agreeing at once in a single benchmark unit of 467 and in none of the experimental ones, and filtering to valid poses that satisfy the placement rule outside the transmembrane exclusion slab does not detectably tighten that scatter, so it is not obviously an artefact of invalid or mislocated poses, although no equivalence test was run, so that filtering result bounds the effect rather than establishing that the scatter is intrinsic to how the tools sample an elongated pore. Third, the surviving poses do carry chemical signal. Both libraries engage the same pocket residues, yet the real modulators are descriptively more hydrophobic and halogen-rich and less hydrogen-bonded than the benchmark ligands. Both tools place the modulators on the extracellular TM1 and loop residues His113, Tyr115 and Leu202, which both retain in their shared fingerprints. The TM3 sub-pocket lined by Arg170, Leu174 and Ala177 is an AutoDock observation alone, since none of those three residues survives into DiffDock's much smaller shared set, and it is the AutoDock arm that places the modulators in the outer TM1 and TM3 region adjacent to the selectivity filter where the Synta-66 site has been mapped experimentally. Because contacts contributed by different tools cannot be combined to validate one binding mode, that concordance is a single-tool observation and not a two-tool consensus. AutoDock also puts a separate cluster of contacts on the cytosolic N-terminal segment below TM1, at Lys85 and its neighbours, which is the fourteenth of the

fifteen residues drawn in Figure 23. That cluster lies on the opposite face of the membrane from the extracellular site, so it supports no part of the Synta-66 concordance and is noted only for completeness.

The cross-panel comparison is reassuring only in a limited descriptive sense. Three experimental-versus-control contrasts survive correction once the ligand rather than the frame-ligand pair is taken as the unit, and two of them also survive at the frame-ligand unit, of which only AutoDock Vina's lower usable yield on the experimental modulators rests on all three experimental ligands, at ligand-collapsed medians of 60.0% against 95.0% with Cliff's δ of +0.90 and a Holm-corrected p of 0.016. Being significant at both units and resting on all three experimental ligands together makes it the most robust cross-panel observation in this chapter. The remaining large gaps do not survive. DiffDock's higher physical-validity yield and its lower silhouette on the experimental ligands both fall well short of correction under the collapse, the DiffDock against EquiBind inter-tool distance survives but rests on two experimental ligands, and the interaction-type family behaves the same way, with none of its thirty tool-by-type contrasts surviving Benjamini-Hochberg correction at the ligand unit. Across all sixty contrasts three reach an adjusted p below 0.05 at the ligand unit, against nineteen at the frame-ligand unit, and the ligand-level sensitivity analysis flags one of those three, DiffDock's Calinski-Harabasz index, as an artefact of the collapse, since it does not reach correction at the frame-ligand unit at all. Every comparison in this section is nonetheless reported descriptively, as medians, per-ligand values and Cliff's δ , because three ligands and four snapshots of one receptor cannot support corrected inference in either direction, and that caution applies to the surviving usable-yield contrast as much as to the null ones. Interaction-type differences are consistent with the more hydrophobic, halogen-rich experimental chemotypes, and residue-level fingerprint overlap is moderate for AutoDock Vina at a Jaccard of 0.56 and weaker for DiffDock at 0.31. For Orai1, the practical recommendation is to retain AutoDock Vina and refined DiffDock as separate hypothesis generators, prioritise poses that satisfy the validity and placement rules and agree with independent pharmacology, and omit EquiBind in its present configuration. Sparse overlap between the two viable tools should not itself be treated as validation.

4.3 Dataset Summary

Across the 303 common calibration complexes retained from the 308-complex source benchmark, all three pipelines searched the whole receptor without any crystal-derived binding-site information, so the comparison is matched on search scope. Under those conditions the end-points separate rather than agree. AutoDock Vina + gnina is the strongest pipeline on physical validity, at a pooled 99.5% against 84.5% for DiffDock + smina and 51.8% for unguided EquiBind + gnina. DiffDock + smina has the highest point estimate on the headline metric, recovering a PoseBusters-valid pose within 2Å for 34.0% of complexes at rank-1 against 29.4% for AutoDock Vina + gnina, and retains the higher point estimate at every ranking depth and on conformational form as well. Its rank-1 near-nativeness advantage of +4.6 percentage points carries a 95%

interval of -2.2 to $+11.4$ points, so that contrast is inconclusive rather than settled in either direction, as set out in the Results (Near-Nativeness and Form). DiffDock still carries substantial selection headroom, because acceptable poses often occur within a shallow top-15 pool but are not ranked first, although most of its rank-1 failures remain unrecovered even at that depth. EquiBind remains weakest and is limited chiefly by generation, localisation and raw geometry. On Orai1, DiffDock + smina has the highest net usable yield on the experimental ligands and also loses the smaller conditional share of already valid poses to the transmembrane slab, while AutoDock Vina leads the control panel.

The ordering therefore depends on which property is being asked about, and no single tool is best on all of them. That is a different conclusion from the one a crystal-boxed comparison would support, and it is discussed further below.

Two conclusions cut across the datasets. First, post-hoc minimisation makes many learned poses physically admissible but does little to correct gross placement. Second, the residual bottleneck differs by tool. DiffDock leaves substantial ranking headroom, while EquiBind often fails to sample a usable placement at all. It is therefore not accurate to describe all remaining loss as selection rather than search.

The calibration analysis carries the greater inferential weight because it contains 303 crystal-referenced complexes. The Orai1 arm contains three ligands and twelve correlated frame-ligand units without a cognate pose. Its value lies in reference-free workflow diagnosis and partial concordance with filter-adjacent pharmacology, not in a statistically powered estimate of prospective accuracy.

4.4 Pose Production Costs

Accuracy and validity say nothing about what a result costs. The relevant efficiency measure is not throughput but the compute spent per qualifying result, so cost is expressed per pose that clears the deployment bar, namely PoseBusters-valid and within $2 \text{ \AA}$ of the crystal ligand. Two views follow, the wall-clock a user waits for and the hardware resource a cluster bills. All docking ran on a single workstation. That machine used an Intel Core i9-14900HX processor with 32 hardware threads, an NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB of video memory, and 32 GB of system memory. The AutoDock Vina search ran multi-threaded on the processor at exhaustiveness 32, while the DiffDock and EquiBind inference passes ran on the GPU. The refinement stages split the other way. The gnina rescoring that completes the AutoDock arm ran on the GPU, whereas the refinement that completes the EquiBind arm is accounted as processor work and dominates that arm's bill, so neither of those two pipelines sits on a single device.

Every figure in this section is measured on the whole-protein run, over the 302 timing-complete complexes, which are all but one of the 303 complexes carried through the accuracy analysis, the exception being 7WCF_ACP, and the AutoDock arm is AutoDock Vina + gnina with both stages timed. The two stages are worth separating, because they are not comparable in size. The blind Vina search itself took 2.21 wall-hours across the 302 complexes, against 1.16 hours

for the crystal-ligand-boxed search over the 293 complexes the two runs share. Those two totals are not a like-for-like ratio, because they cover different complex sets, and restricted to the 293 shared complexes they stand at 2.15 against 1.16 hours, a ratio of totals of 1.9. The quoted cost is instead the paired per-complex statistic, whose median is 2.3 with an interquartile range of 1.8 to 3.1. It exceeds the ratio of totals because a total weights every complex by its own cost and the most expensive boxed complexes are the ones on which the two runs differ least, at Spearman's $\rho = -0.72$ between a complex's boxed cost and its blind-to-boxed ratio. Withdrawing the search box therefore costs a median factor of 2.3 per complex, less than the enlargement of the search volume would suggest because exhaustiveness rather than volume sets the sampling budget. The gnina rescoring pass then added a further 12.13 hours, so the learned rescorer is roughly five and a half times more expensive than the docking it re-ranks. The two stages ran consecutively and were timed separately on the same run, so the measured end-to-end cost of the AutoDock arm is 14.34 wall-hours, of which the rescoring pass is 84.6%, and it is that total which every per-pose figure below divides.⁴

Figure 25 gives the wall-clock cost per qualifying pose, one value per complex, over the complexes for which a tool produced at least one such pose.

The cost ordering under a blind search is the reverse of the boxed run. Medians are 119.7 s per qualifying pose for AutoDock Vina + gnina (interquartile range 91.2 to 138.8 s, 151 complexes), 50.9 s for DiffDock (15.8 to 133.8 s, 168 complexes) and 8.5 s for EquiBind (3.2 to 22.1 s, 62 complexes). On the 32 complexes where all three succeed the paired tests confirm that ordering (Friedman $\chi^2 = 39.8$, $p = 2.3\text{e-}09$, Kendall $W = 0.62$, every pair surviving Holm correction), with DiffDock costing 0.12 times as much per usable pose as AutoDock and EquiBind 0.09 times. On the boxed run the same endpoint, that is a PoseBusters-valid pose within 2 Å of the crystal ligand, put AutoDock cheapest at 3.5 s, although the boxed AutoDock arm is the raw Vina search with no gnina pass rather than the Vina + gnina pipeline measured here. Between the two runs the classical engine moves from the cheapest pipeline to the most expensive, by a factor of 34, but the box alone does not produce that move. Three effects compound to produce it, and the largest of them is not the box. The blind search costs 2.3 times more per complex, its gnina pass then multiplies that cost by a further six and a half, and the resulting cost is then amortised over fewer qualifying poses, because the median number of near-native valid poses per qualifying complex falls from two in the boxed run to one in the blind run while the number of complexes yielding any such pose falls from 259 to 151. Each factor is the median of its own distribution, so together they indicate the size and direction of the three contributions rather than an exact accounting that multiplies out to 34.

⁴Rescoring runs one pose at a time on the GPU, whereas Vina spreads each dock over 32 CPU threads, so the two stages consume different hardware currencies, namely 70.61 CPU-core-hours for the search and 12.13 GPU-hours for the rescoring. Those two figures cannot be added, whereas the elapsed times can and do sum to the 14.34 wall-hours quoted above. The CPU-core-hour figure is the search wall multiplied by the 32 threads requested rather than a measurement of thread occupancy. Optimiser time is recovered from the per-complex optimiser logs, where cached entries carry the originally measured elapsed time in their provenance sidecar rather than the cost of the cache lookup.

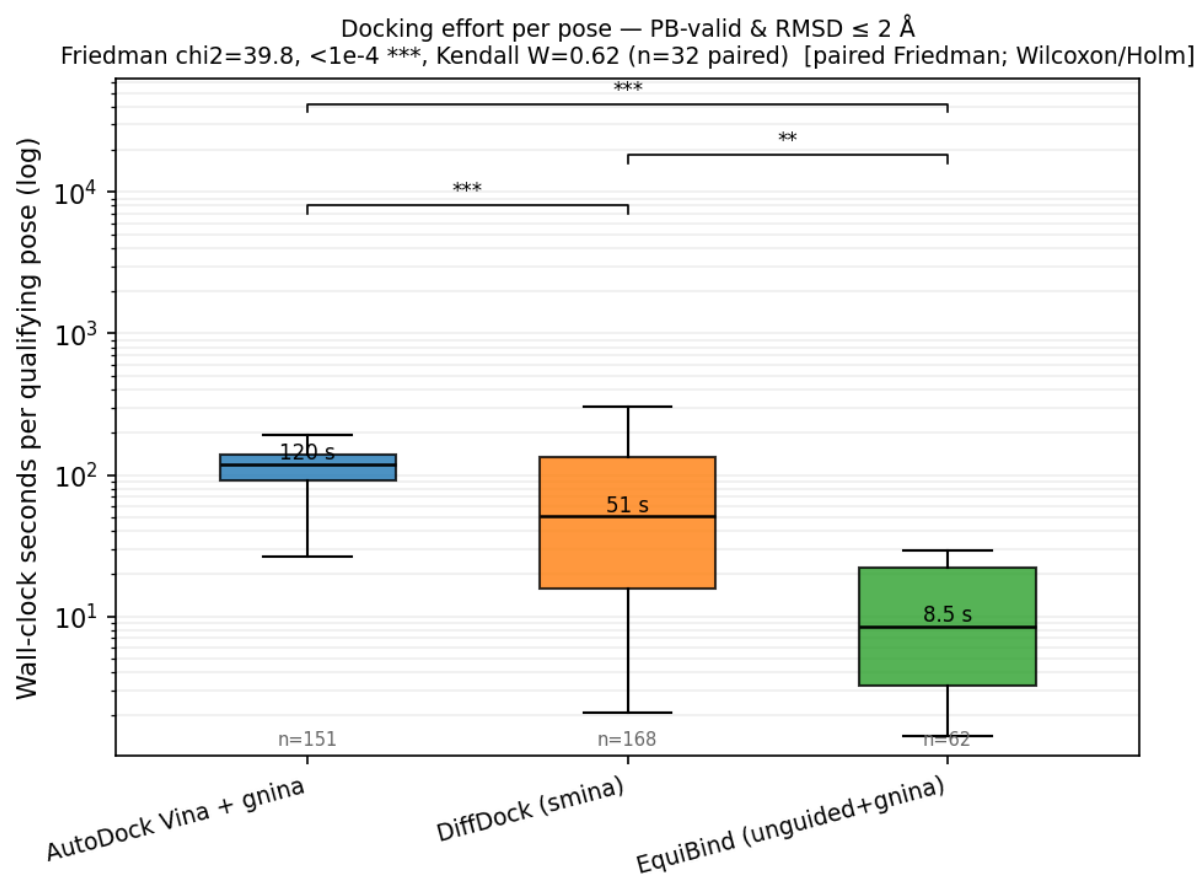


Figure 25: Wall-clock seconds per qualifying pose, where a qualifying pose is PoseBusters-valid and within 2 Å of the crystal ligand. Each box summarises only the complexes for which that tool produced a qualifying pose, namely 151, 168 and 62, whereas the bracketed paired tests use the 32 complexes on which all three succeed

Cost climbs as the quality bar tightens, most steeply for AutoDock. From any generated pose through valid to valid and near-native the median runs 4.3, 4.4 and 119.7 s for AutoDock Vina + gnina, 8.1, 9.8 and 50.9 s for DiffDock and 0.9, 1.2 and 8.5 s for EquiBind, while the contributing complexes fall from 302 to 302 and 151 for AutoDock, from 302 to 299 and 168 for DiffDock, and from 302 to 247 and 62 for EquiBind. Fewer qualifying poses spread the same fixed docking wall further, and many complexes yield none at all. This is where AutoDock's near-perfect validity stops helping, since almost every pose it produces is admissible but only a minority are near-native, so the tightening from valid to valid-and-near-native multiplies its unit cost by a factor of about 27.5 against about 5.2 for DiffDock, whose median rises from 9.8 to 50.9 s. EquiBind's competitive unit cost remains an artefact of its failures, since its 8.5 s is a median over the 62 complexes in which it succeeds and it fails in the remaining 240, and it rests on a wall time modelled as its compute-seconds spread across 32 concurrent workers rather than on a measured elapsed time.

Figure 26 removes parallelism from the comparison, reporting the hardware resource consumed per qualifying pose separately for CPU and GPU, on the same near-native and valid criterion as Figure 25.

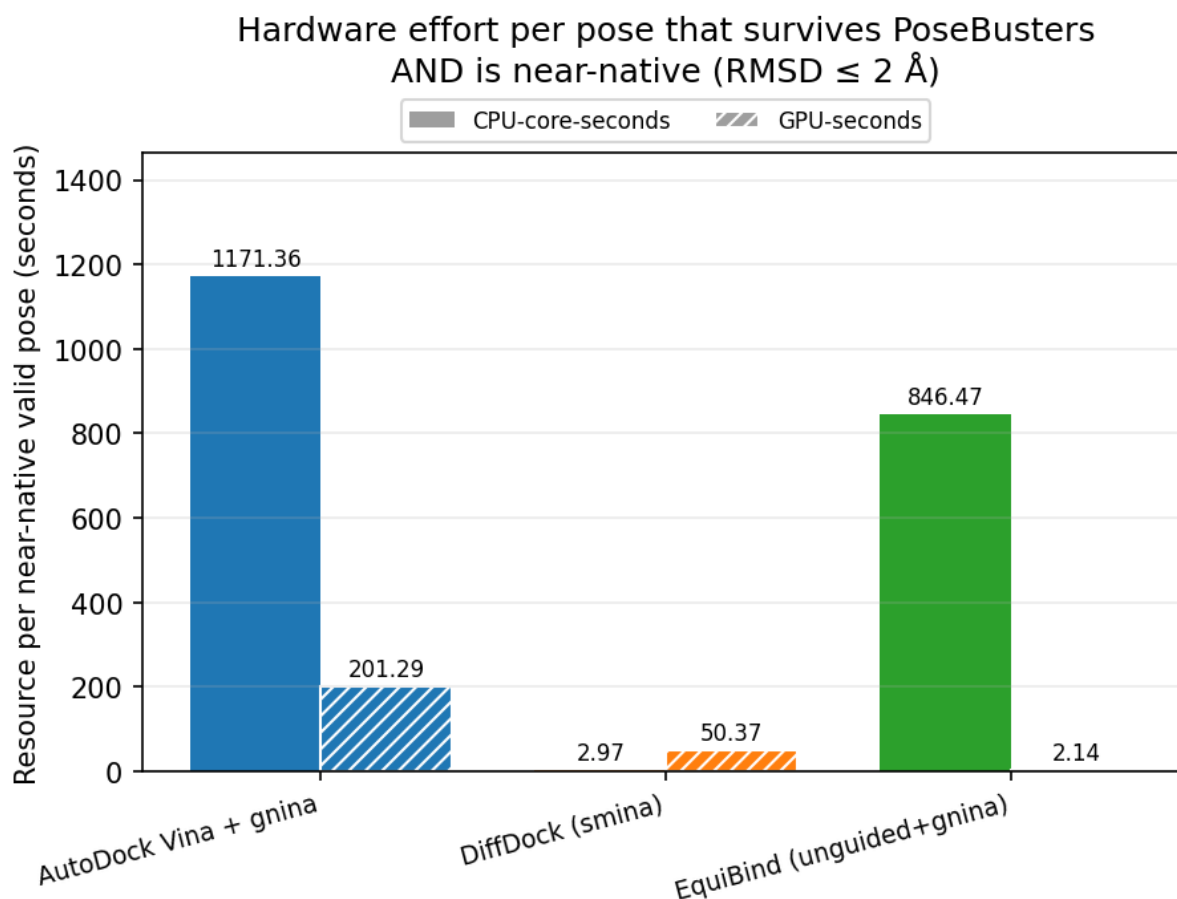


Figure 26: Hardware resource per near-native valid pose, that is per pose that is PoseBusters-valid and within 2 Å of the crystal ligand

On this axis the blind search separates the tools even more sharply. AutoDock Vina + gnina consumes 1,171.4 CPU-core-seconds and 201.3 GPU-seconds per qualifying pose, DiffDock 50.4 GPU-seconds and 2.97 CPU-core-seconds, and EquiBind 846.5 CPU-core-seconds with 2.1 GPU-seconds. AutoDock is now the most expensive pipeline on every axis at once. Its 32-thread parallelism still compresses 70.61 CPU-core-hours of search into 2.21 wall-hours, so the wall-clock advantage was not lost to parallelism ceasing to pay. It was lost to the serial single-GPU rescoring pass appended after the search, which contributes 84.6% of the arm's wall-clock, while the CPU-core-second bill remains what that parallelism always implied. The denominators explain much of the gap, since the three tools convert their compute into 217 (AutoDock Vina + gnina), 1,602 (DiffDock) and 305 (EquiBind) qualifying poses. For both DiffDock and EquiBind the GPU time is the docking itself, whereas AutoDock's GPU time is entirely its gnina rescoring pass, which nonetheless carries the largest per-qualifying-pose GPU bill of the three because its yield of qualifying poses is so small.

Three caveats bound these figures. The wall-clock comparison is cross-hardware, and the device mix differs within pipelines as well as between them, since AutoDock spends 2.21 of its 14.34 wall-hours multi-threaded on the processor and the remaining 12.13 on a single GPU, so it measures elapsed time rather than identical work. The resource values are order-of-magnitude costs, since AutoDock's CPU-core-seconds cover only the Vina search and assume the full 32 threads are occupied, while its gnina pass is accounted separately as GPU occupancy, and EquiBind's wall time is an idealised best-configuration estimate, namely the compute-seconds of its selected configuration divided by the 32 concurrent workers the pipeline was configured for, so it assumes perfect overlap where the AutoDock and DiffDock walls are elapsed times of runs executed as measured. That convention, rather than any difference in the work performed, is what separates EquiBind's two axes. Its CPU-core bill of 71.72 hours sits within 2% of AutoDock's 70.61, while its wall of 2.25 hours is a sixth of AutoDock's 14.34, because Vina already occupies all 32 threads within each docking and so gains nothing from running complexes concurrently, whereas EquiBind is credited with exactly that concurrency. The resource panel carries no inferential test by design, being a pooled aggregate rather than a per-complex distribution. The tools nevertheless differ sharply in whether a timing-complete complex yields any qualifying pose, namely 151 of 302 for AutoDock Vina + gnina, 168 of 302 for DiffDock + smina and 62 of 302 for EquiBind + gnina (Cochran's $Q = 99.2$, $p = 2.9 \times 10^{-22}$).

In summary, under the blind whole-protein configuration that supplies every other result in this chapter, AutoDock Vina + gnina is the most expensive pipeline per usable pose on the wall-clock axis and on both hardware currencies, and it is not the cheapest on any currency per pose generated either. Pooled over the whole campaign, and so as a ratio of totals rather than the per-complex medians quoted earlier, EquiBind is the cheapest per generated pose on wall-clock time at 0.89 seconds against 5.73 for AutoDock and 9.53 for DiffDock, and on GPU time at 0.07 seconds against 4.85 and 9.00, while DiffDock is the cheapest on CPU-core time at 0.53 seconds against 28.20 for AutoDock and 28.50 for EquiBind, so AutoDock sits mid-field on wall-clock and GPU time and is effectively tied with EquiBind at the expensive end

of the CPU-core axis. Its advantage is coverage and admissibility rather than economy, since it returned at least one PoseBusters-valid pose for every one of the 302 complexes and qualifying poses for 151 of them, against 168 for DiffDock and 62 for EquiBind. The ordering also depends on whether poses or docking jobs carry the cost. Per attempted docking the median wall-clock is 129.5 s for AutoDock Vina + gnina against 241.7 s for DiffDock and 25.6 s for EquiBind, and charging the whole campaign to the complexes that yielded a qualifying pose gives 342, 509 and 130 s, so on that denominator AutoDock Vina + gnina is the cheaper of the two leading pipelines on wall-clock and on GPU-seconds, at 289 s per success against 480, while remaining far more expensive on CPU-core-seconds. Because all three tools produce about thirty poses per complex, this is the per-complex median view rescaled rather than an independent measurement, and it does not alter the per-qualifying-pose result. DiffDock is the most cost-effective route to a near-native valid pose in this configuration, and EquiBind's low median elapsed cost remains conditional on the 62 complexes in which it succeeds at all. Hardware-resource conclusions remain currency-specific. GPU time dominates the cost for DiffDock, which nonetheless also spends 2.97 CPU-core-seconds per qualifying pose, whereas CPU time dominates AutoDock Vina and EquiBind, with the qualification that AutoDock also carries a substantial GPU bill in its gnina rescoring pass. The efficiency ordering reported here therefore reverses the one obtained under a crystal-derived search box, which is itself the practical lesson, since a site-restricted benchmark flatters the classical engine on cost as well as on accuracy.

5 Discussion

5.1 Principal Findings and Scope of Inference

This study compared AutoDock Vina, DiffDock and EquiBind in two deliberately different settings, namely crystallographic re-docking on 303 calibration complexes retained from the 308-complex PoseBusters benchmark, and reference-free docking of three Orai1 modulators into four molecular-dynamics snapshots. The calibration benchmark provides the stronger evidence because every prediction can be tested against an experimental pose. In that setting the endpoints separate rather than electing a single winner. AutoDock Vina was the most reliable configured pipeline on physical validity, whereas refined DiffDock had the highest point estimate on the headline metric and matched or exceeded AutoDock on native-pocket recovery at every ranking depth once the search box was removed. Its near-nativeness advantage over AutoDock was inconclusive at the tested depths, because the paired intervals reported in the Results (Near-Nativeness and Form) exclude neither a null difference nor a lead of up to roughly thirteen percentage points, the widest upper bound among the four depths. Refined EquiBind was limited by low raw physical validity and weak in-place accuracy and pocket localisation. The Orai1 case

preserved that split on physical validity but reversed it on usable yield, where refined DiffDock retained more poses than AutoDock Vina under the study's operational validity and placement rules, and the absence of a ligand-bound Orai1 structure means that it cannot establish which predicted binding mode is correct.

The appropriate interpretation is therefore narrower than a general contest between artificial intelligence and physics. The results show that under this study's preparation, search, ranking, refinement and evaluation settings the endpoints split between the two leading pipelines, with AutoDock Vina ahead on physical validity and refined DiffDock-L, the larger retrained DiffDock release used throughout this work [89], having higher point estimates for in-place accuracy, recovery and pocket localisation, although that in-place advantage over AutoDock is inconclusive at the tested depths on the interval evidence given in the Results (Near-Nativeness and Form). Both remain ahead of the tested EquiBind pipeline throughout. They do not show that classical docking universally outperforms current AI docking. EquiBind and the evaluated DiffDock version represent earlier learned architectures, while more recent diffusion, co-folding and hybrid systems report better performance when pockets are specified, receptor flexibility is represented and structural relaxation is included [12], [13], [31], [62], [67]. Conversely, current similarity-controlled analyses continue to show that aggregate accuracy can conceal reliance on training-set analogues and weak extrapolation to new pockets [34], [61], [63], [67]. The principal contribution of this thesis is thus its separation of failure modes, namely sampling, ranking, physical validity and localisation, rather than a permanent ranking of method classes.

The term physics-based also requires care. AutoDock Vina uses an empirical scoring function and an explicit search-and-optimisation procedure. It is not a full molecular-mechanics or free-energy simulation. In this thesis, physics-based is therefore an operational contrast with end-to-end learned pose generation. That distinction matters when interpreting the strong Vina result. It supports this classical search-and-score workflow, not a claim that a complete physical model of binding was evaluated.

5.2 Physical Validity and Post-Docking Optimisation

The largest initial separation between the tools occurred before RMSD was considered. AutoDock Vina + gnina produced 99.5% PoseBusters-valid poses, against 98.2% for the unrescored search, compared with 24.4% for raw DiffDock and 2.9% for raw unguided EquiBind. Refinement substantially changed the usability of the learned outputs, because smina raised DiffDock validity to 84.5%, while gnina raised unguided EquiBind validity to 51.8%. These gains support the central PoseBusters argument that RMSD alone is insufficient, because a geometrically close ligand can still contain unrealistic bond geometry, steric overlap or intermolecular contacts [34]. They are also consistent with broader benchmarking in which learned methods can lead on pose accuracy while classical methods retain an advantage in structural rationality, making combined accuracy-and-validity criteria more discriminating than RMSD alone [13], [32].

Refinement did not have the same practical effect for the two learned methods. For DiffDock,

the principal gain was conversion of already sampled hypotheses into physically admissible structures. Holding each pose at its ranked position, optimisation changed the per-rank near-native rate by only about 0.1 to 0.6 percentage points with smina and 0.1 to 0.7 with gnina, while substantially increasing PoseBusters validity. For EquiBind, the same rate rose more, though still modestly, by approximately 2.5 to 2.6 percentage points with smina and 2.8 to 3.1 with gnina, alongside a much larger validity gain. Table 23 lists these fixed-rank changes by pose-rank band. Local minimisation therefore repaired geometry far more effectively than it corrected blind placement. It could not recover a binding mode that had not been sampled or move a grossly mislocated ligand into the correct pocket.

The selected refiner should be treated as an empirical pipeline choice rather than a demonstrated mechanism. On the variant-selection endpoint the two DiffDock refiners are separated by at most two complexes, at 76 against 74 at rank-1 and 133 against 132 at the top-15 depth, and are effectively tied, whereas EquiBind + gnina clearly outperformed EquiBind + smina. Descriptions such as gentle preservation or aggressive repair are plausible interpretations of these observations, but the study did not directly measure optimisation trajectories or validate refiner selection on an independent dataset. Selecting and evaluating a variant on the same benchmark can overstate the portability of that choice.

Physical validity is also not equivalent to binding stability. Passing all PoseBusters checks (Table 15) removes important chemical, conformational and steric failures, but it does not establish favourable binding free energy, correct protonation and tautomer states, appropriate water or ion mediation, or persistence in a membrane environment. The thesis therefore supports a two-stage interpretation. Physical validity is a necessary admissibility filter, while native-like placement, interaction recovery and prospective experimental evidence provide separate tests of accuracy.

5.3 Sampling, Ranking and Interaction Recovery

The ranking-depth analysis separates pose generation from pose selection. Under the headline metric, a PoseBusters-valid pose within 2 Å of the crystal ligand, rank-1 recovery was approximately 34.0% for DiffDock + smina, 29.4% for AutoDock Vina + gnina and 15.2% for EquiBind + gnina. Retrospective best-of-top-15 recovery rose to approximately 55.1%, 49.8% and 20.1%, respectively. The large DiffDock gain shows that useful poses were often generated but not assigned the highest confidence. AutoDock Vina gains almost as much in absolute terms, at 62 complexes against 64, so under blind conditions the classical engine is selection-limited to a comparable degree rather than being a reliable self-ranker. EquiBind gained only about five percentage points, indicating a predominantly sampling or placement limitation. No re-ranker can recover a binding mode that is absent from the pool.

The form-versus-in-place decomposition supports this diagnosis. At the prespecified 1 Å Kabsch threshold, rank-1 form recovery was 52.1% for DiffDock + smina, 46.9% for AutoDock Vina + gnina and 30.7% for EquiBind + gnina. At the best-of-top-30 depth it reached 74.6%,

71.0% and 49.8%, respectively. DiffDock therefore has the highest point estimate for internal conformational credibility as well as for near-nativeness, although the latter advantage over AutoDock remains inconclusive at the tested depths on the interval evidence given in the Results (Near-Nativeness and Form). What it fails to do is rank the credible pose first rather than generate it. EquiBind was weaker on both axes. The distinction is chemically important because a pose can preserve ligand shape yet present its pharmacophore to the wrong residues.

Pocket recovery was easier than exact pose recovery, but under the whole-protein search it is no longer a settled contest. At rank-1, DiffDock reached the crystallographic pocket cluster for 55% of complexes, AutoDock Vina for 53% and EquiBind for 41%, a spread narrow enough that the AutoDock and DiffDock confidence intervals overlap. By top-15 the corresponding rates were 82%, 76% and 49%, so DiffDock has the higher point estimate wherever ranking depth is allowed to grow, and at that depth the reach advantage over AutoDock is separable on the paired test given in the Results (Cross-Tool Clustering). Pooling the three tools placed the correct pocket somewhere in the combined pose cloud for 93% of complexes, yet the best inexpensive cluster-ranking rule selected it first in only 55%. That shortfall is far larger than the equivalent gap under the superseded crystal-boxed search, where an already localised AutoDock made the pooled site obvious. It is evidence of a site-selection gap in the pooled ensemble, not evidence that exact pose prediction is 93% solved. Cluster membership is a coarser endpoint than a physically valid pose within 2 Å.

The interaction analysis provides an orthogonal check, and it now rests on the same whole-protein search as the rest of the chapter. On the top-5 union of typed contacts DiffDock + smina has the highest mean fingerprint overlap with the crystal ligand (0.45, compared with 0.37 for EquiBind + gnina and 0.36 for AutoDock Vina + gnina), reversing the boxed run on which the site-restricted AutoDock search reached 0.67. That reversal is a property of the AutoDock protocol rather than of the contact chemistry, since the two pipelines that docked blindly in both runs barely move between them, EquiBind + gnina returning 0.37 in each and DiffDock + smina shifting only from 0.44 to 0.45, against the halving of the AutoDock value. The search box is not isolated within that protocol, because the boxed AutoDock arm was raw AutoDock Vina while the blind one is AutoDock Vina + gnina. On the pose a user would actually take the tools are not separable at all, with a rank-1 F1 of 0.53 for both DiffDock and AutoDock Vina against 0.46 for EquiBind, which matches the placement result, where the two pipelines could likewise not be ordered, although only that placement contrast was tested against an equivalence margin and neither the interaction nor the pocket-cluster endpoint supports a claim that the two perform alike. Neither endpoint shows that they perform alike, because in both cases the data are inconclusive rather than positively equivalent. The separation appears with depth instead, where AutoDock decays to 0.36 by rank 5 while DiffDock holds near 0.51. DiffDock's confidence ranking shows no detected association with interaction recovery, so that flat and comparatively good pose pool is left unexploited, whereas AutoDock's gnina ranking orders its poses well but has genuinely worse ones to order. EquiBind missed more native contacts than it recovered at every rank. Ranking quality is therefore endpoint- and distribution-dependent. The improved cross-docking

ranker in gnina 1.3, together with its less uniform redocking behaviour, makes the same point. An external score should be validated for the intended task rather than assumed to improve every setting [65].

5.4 Generalisation in the Context of Current Literature

The post-2021 construction of the PoseBusters benchmark reduces direct temporal overlap with the training data of early learned models, but it does not eliminate similarity leakage. A later crystal structure may contain a homologous receptor, a familiar pocket geometry or a close ligand analogue. PoseBusters reported a marked decline for receptors with low sequence identity to training proteins [34], and the later multidimensional analysis by Li et al. identified binding-pocket novelty as a central generalisation bottleneck [13]. TEMPL further illustrates why a time split alone is not decisive. A simple template-transfer baseline achieved meaningful performance on a conventional PDBbind split, with substantially better results when related protein-ligand templates were available [63].

The 2026 PoseBench study sharpens this qualification. Modern co-folding systems generally outperformed comparable conventional and learned docking baselines overall, but they remained challenged by new protein-ligand binding poses and by the simultaneous demands of structural accuracy and chemical specificity [67]. PoseX similarly reported that recent AI and co-folding systems can outperform physics-based baselines on self- and cross-docking after relaxation, and that pocket information and physics-inspired post-processing materially improve their outputs [12]. These results do not contradict the present thesis. They show that conclusions depend on model generation, benchmark novelty, receptor information, flexibility and relaxation. The poor EquiBind result is consistent with current benchmarks that place early one-shot models below newer diffusion and co-folding systems. The DiffDock result identifies limitations of this configured version's blind placement, chemical validity and confidence ranking rather than an intrinsic ceiling on generative docking.

The thesis therefore provides a useful bridge between two apparently opposing strands of literature. Studies such as PoseBusters, TEMPL and PoseBench warn that physical plausibility and generalisation can lag behind headline RMSD [34], [63], [67]. PoseX and newer hybrid systems show that these deficiencies are not immutable when models receive pocket information and physics-informed relaxation [12], [62]. The present results support both propositions. Raw learned output was often unusable, while post-processing rescued many structures but did not remove sampling and ranking errors.

5.5 Interpretation of the Reference-Free Orai1 Study

The Orai1 analysis addresses a more realistic but less identifiable problem. AutoDock Vina produced PoseBusters-valid poses for essentially all three experimental ligands but retained

only 50 to 60% of generated poses after the study's combined screen for on-receptor placement outside the transmembrane exclusion slab. Refined DiffDock was less consistent on validity yet retained more usable poses, with a median frame–ligand–unit physical-validity yield of 100% and a median usable yield of 65% on the same unit against 60% for AutoDock Vina. EquiBind retained one pose out of 120 after validity and placement screening. These figures show how often each workflow satisfies the predefined admissibility criteria. They do not measure near-nativeness because no experimental ligand pose exists.

This distinction is especially important for the R91-E106 transmembrane exclusion slab. GSK-7975A sensitivity is linked directly or allosterically to selectivity-filter geometry [68], while Synta-66 has been mapped to an extracellular TM1/TM3 and loop region adjacent to that filter [69]. Excluding the central conduction pathway is therefore a defensible operational choice for focusing on an outer-pore hypothesis, but it is not ground truth and cannot prove that every excluded pose is biologically wrong. The correct claim is that EquiBind produced almost no poses satisfying this study's criterion of a valid, on-receptor pose outside the transmembrane exclusion slab.

The low cross-tool agreement reinforces the uncertainty. Consensus sites differed by a pooled median of 30.1 Å on the experimental ligands, against 29.5 Å on the control panel, but the experimental value rises to 42.2 Å once each ligand is collapsed to the median over its four frames, so it is not a settled central estimate and the two libraries cannot be called equally scattered on this evidence. AutoDock Vina and DiffDock agreed within 5 Å in none of the eleven evaluable experimental pairs, against 15% of control-panel pairs, and no three-way experimental consensus occurred. AutoDock Vina formed the best-separated clouds in both ligand sets and the most compact ones on the control panel, whereas DiffDock was the tighter of the two on the experimental ligands at a median within-cluster spread of 0.34 against 0.62 Å, but compactness measures precision rather than accuracy. A method can reproducibly sample the wrong site. Partial recovery of literature-associated residues near the selectivity filter and TM1/TM3 is therefore hypothesis-generating evidence, not validation of a bound pose. Combining one AutoDock contact with a different DiffDock contact likewise does not establish that either complete pose is correct.

One contrast between the three experimental ligands and the larger Orai control panel is established once the ligand rather than the frame–ligand pair is taken as the unit, namely AutoDock Vina's lower usable yield on the experimental modulators, which survives correction at both units and on all three ligands. The three largest of the remaining gaps among the reference-free placement and clustering proxies, namely DiffDock's higher physical-validity yield on the experimental modulators, its larger consensus-site distance from EquiBind and its lower silhouette on the experimental ligands, either lose their corrected significance under that collapse or rest on two experimental ligands, and the interaction-type family, chiefly the halogen-bond, hydrophobic, alkyl- π and hydrogen-bond differences between the two libraries, behaves the same way, with none of its contrasts surviving Benjamini-Hochberg correction at the ligand unit. The both-unit table of the ligand-level sensitivity analysis records this for every contrast, three of

sixty reaching an adjusted p below 0.05 at the ligand unit against nineteen at the frame-ligand unit, with one of those three flagged as an artefact of the collapse. All of these comparisons are nonetheless reported descriptively, and the absence of a corrected difference elsewhere should not be described as proof of transfer or equivalence either. With three chemotypes, four correlated frames of one receptor and almost no surviving EquiBind output, the tests have limited power in both directions, and three ligands cannot support corrected inference either way. The preservation of tool ordering on physical validity is reassuring, but it does not extend to usable yield, where the experimental panel reverses the control-panel ordering, so it supports only cautious use of the control panel for these reference-free proxies. The four receptor frames also provide limited flexibility because three late snapshots are closely related. A stronger Orai study would add independent trajectories, systematic frame selection, explicit membrane and ion conditions where feasible, ligand-bound simulations and experimental tests such as mutagenesis or competition assays. Such experimental validation is the decisive test these reference-free proxies cannot supply, and it falls outside the scope of a computational thesis. It is being taken forward in continuing Orai1 research at the Johannes Kepler University Linz (JKU), to which the present work contributes the physically valid, on-receptor pose hypotheses generated here for the three modulators.

5.6 Computational Efficiency and Practical Tool Selection

Within the measured whole-protein configuration, EquiBind provided the shortest median elapsed cost per qualifying pose at 8.5 s, ahead of DiffDock at 50.9 s and AutoDock Vina + gnina at 119.7 s. EquiBind's apparent advantage is conditional on the 62 of 302 timing-complete complexes where it succeeded, since it failed to produce a qualifying pose for the remaining 240, so DiffDock is the more realistic cost-effective choice at 168 successful complexes. Hardware-resource comparisons use different currencies. CPU-core-seconds cannot be ranked directly against GPU-seconds, and refinement is part of every pipeline's cost here, the AutoDock arm included. The justified conclusion is therefore that, within the measured whole-protein protocol, the classical pipeline is the most expensive route to a qualifying pose on this hardware on both the wall-clock and hardware axes. Its apparent efficiency advantage in the earlier site-restricted configuration is associated with the different crystal-boxed protocol, not established as an intrinsic property of the search box or engine. What AutoDock Vina + gnina buys for that cost is coverage and physical admissibility, not speed.

Current work points toward hybrid rather than exclusive workflows. DiffDock-Glide combines pocket-conditioned diffusion with physics-based post-docking minimisation and reports improved sampling, particularly for targets without training-set homologues [62]. PoseX similarly found that relaxation removed many AI steric failures [12]. gnina 1.3 demonstrates that learned scoring can improve ranking within a classical sampler [65], while the prospective CACHE study shows that such a hybrid open-source workflow can produce experimental hits without establishing universal superiority [66]. These studies align with the strongest finding of this thesis. Learned generation

becomes more useful when followed by explicit physical validation and task-appropriate ranking.

5.7 Implications for Docking Workflows

The results support a staged workflow rather than a single universal engine, namely to generate multiple hypotheses, minimise them, reject physically invalid or biologically inadmissible structures, rank the survivors with a selector validated for the intended endpoint, and retain uncertainty when methods disagree. For the tools studied here, refined DiffDock and AutoDock Vina are best treated as complementary primary engines rather than a baseline and a supplement. DiffDock has higher point estimates for in-place accuracy and pocket localisation, although its near-nativeness advantage over AutoDock Vina is inconclusive at the tested ranking depths on the interval evidence given in the Results (Near-Nativeness and Form), while AutoDock Vina supplies the physically safer geometry. EquiBind contributes too little usable coverage in this configuration to justify routine inclusion. Its single usable Orai1 pose comes from the one arm that applied the in-protein Universal Force Field relaxation described in the Appendix, which deforms ligand geometry, so that figure understates what the method would deliver without that stage. The recommendation does not rest on it, since the relaxation-free Orai1 control panel still returns a usable yield of 2.9% for EquiBind against 79.2% for AutoDock Vina + gnina and 71.1% for DiffDock + smina.

The next comparison should test whether this conclusion persists with current models under matched information and compute budgets. Flexible diffusion and packing methods [31], multi-pocket conditioning [64], hybrid refinement [62], modern co-folding systems [12], [67] and strong template controls [63] should be evaluated on protein-, pocket- and ligand-novel splits. The negative fpocket/P2Rank result for EquiBind also cautions that pocket detection alone is insufficient. The downstream docking model must use that information effectively. Overall, the evidence favours hybrid, flexibility-aware and leakage-controlled evaluation of the kind a standing blind community challenge provides [45], while preserving a strict distinction between benchmark accuracy and reference-free biological hypotheses.

6 Conclusions

This thesis compared three configured docking pipelines on 303 crystallographic re-docking cases and on three Orai1 modulators docked against four receptor snapshots. The findings answer the four research questions as follows. The fourth, on the opportunities and limitations of the tested tools, is summarised here and developed in full in the Limitations and Future Outlook chapter.

6.1 RQ1: Quality and Stability

Under the headline metric, a PoseBusters-valid pose within 2 Å of the crystal ligand, and with every pipeline searching the whole receptor blindly, DiffDock + smina performed best on point estimates, followed by AutoDock Vina + gnina and EquiBind + gnina. Approximate rank-1 recovery was 34.0%, 29.4% and 15.2%, respectively, and the retrospective best-of-top-15 ceiling rose to 55.1%, 49.8% and 20.1%. The DiffDock lead over AutoDock is not statistically separable at any depth on near-nativeness, the paired rank-1 difference being +4.6 percentage points with a 95% interval of -2.2 to $+11.4$ points and the top-15 difference +5.3 points with an interval of -2.1 to $+12.5$. Those intervals order neither pipeline on this endpoint, and two one-sided tests bound the residual difference at about twelve percentage points in either direction, so the contrast is reported here as bounded rather than merely inconclusive. Both remain decisively ahead of EquiBind. The ordering inverts on physical validity, where pooled rates across the 303-complex accuracy set were 99.5% for AutoDock Vina + gnina, 84.5% for DiffDock + smina and 51.8% for unguided EquiBind + gnina. Native-interaction recovery similarly leaves DiffDock + smina and AutoDock Vina + gnina not separable at a rank-1 F1 of 0.53 against 0.46 for EquiBind, a contrast with no detected paired difference on the 278 complexes the two share (Wilcoxon $p = 0.69$) and no equivalence test applied, so reported as an absence of detected difference rather than as equivalence, with DiffDock ahead only once the comparison is widened to the union of the top five poses (0.45 against 0.36 and 0.37 mean fingerprint overlap). On Orai1, the physical-validity ordering appeared again under reference-free proxies. Nearly every AutoDock pose was PoseBusters-valid, while the median frame–ligand–unit validity was 100% for DiffDock and 5% for EquiBind across the twelve experimental units. Usable yield ordered the two leading pipelines the other way. Median usable yields on the same unit under the study's criterion of on-receptor placement outside the transmembrane exclusion slab were 65% for DiffDock + smina, 60% for AutoDock Vina + gnina and 0% for EquiBind, with only one of 120 EquiBind poses surviving both validity and placement screening, on the one arm that carried the in-protein Universal Force Field relaxation. The relaxation-free control panel puts EquiBind at 2.9% usable against 79.2% and 71.1%, so the ordering does not depend on that stage.

Stability was assessed only through static physical validity, clustering reproducibility and sensitivity to receptor frames. Thermodynamic or time-dependent stability of the docked complexes was not tested and cannot be inferred from docking alone.

6.2 RQ2: Effect of Post-Docking Optimisation

Local optimisation mainly repaired physical geometry rather than placement. DiffDock validity increased from 24.4% in the raw output to 84.5% with smina and 87.1% with gnina. Unguided EquiBind increased from 2.9% to 39.3% with smina and 51.8% with gnina. By contrast, optimisation changed the per-rank near-native rate (each pose held at its own rank) by only about 0.1 to 0.7 percentage points for DiffDock and 2.8 to 3.1 points for EquiBind with gnina (0.1 to 0.6 and

2.5 to 2.6 points, respectively, with smina). The larger cumulative top-*k* gains in Table 22 arise because the best-of-top-*k* criterion accumulates these per-rank repairs across the ranked pool, and for EquiBind additionally because gnina re-ranks its otherwise unordered poses. Refinement is therefore necessary to make many AI-generated poses chemically and sterically usable, but it cannot recover an unsampled binding mode or relocate a grossly misplaced ligand.

6.3 RQ3: Computational Efficiency

The efficiency comparison is measured on the whole-protein search across 302 timing-complete complexes, with both stages of the AutoDock arm timed. EquiBind gave the shortest median wall-clock cost per qualifying pose at 8.5 s, compared with 50.9 s for DiffDock and 119.7 s for AutoDock Vina + gnina, and qualifying poses were obtained for 168 complexes with DiffDock, 151 with AutoDock Vina + gnina and only 62 with EquiBind. This reverses the ordering obtained under the crystal-boxed search, on which raw AutoDock was cheapest at 3.5 s. Three measured changes contribute, namely that the blind Vina search costs 2.3 times more per complex, the additional gnina rescoring pass costs a further five and a half times the search, and fewer complexes and poses meet the qualifying endpoint, enlarging the per-qualifying-pose denominator. Because the compared AutoDock pipelines and receptor preparations differ, the reversal is associated with the whole-protein protocol rather than attributable to the box alone. Hardware-resource accounting does not yield one universal ordering because CPU-core-seconds and GPU-seconds are different currencies, though on the whole-protein run AutoDock is the most expensive pipeline per qualifying pose on every axis at once, namely wall-clock, CPU-core-seconds and GPU-seconds. The efficiency result applies to these implementations and hardware settings rather than intrinsically to physics-based and AI-based docking as classes. The AutoDock cost is nonetheless a measurement rather than a model, because both of its stages were timed on the same run and ran consecutively, so the 2.21-hour blind search and the 12.13-hour rescoring pass together give a measured end-to-end cost of 14.34 wall-hours, of which the rescoring pass is 84.6%. Part of that pass is an implementation artefact, because it invoked gnina once per pose 9,013 times and each invocation paid the start-up of a separate process. The fastest invocation took 2.05 s and the median 3.79 s, so at most 42% of the 12.13 hours is attributable to per-invocation overhead that batching could amortise. The remaining 58% is not a lower bound on what a batched pass would cost, because batching would also raise the utilisation of a GPU that a single-pose evaluation leaves substantially idle. Batching would therefore reduce the cost of the learned rescorer by an amount this run does not determine, and no batched implementation was measured.

6.4 RQ4: Opportunities and Limitations

The opportunity lies in the fact that the three pipelines fail in different ways, so a staged workflow that generates candidates with a diffusion model, repairs their geometry by local minimisation, screens for physical validity and admissible placement and then ranks the survivors with a selector validated for the intended endpoint uses each component where it is strongest. The limitations are equally specific. Every result rests on configured toolchains at single settings without multi-seed sweeps, on a benchmark whose post-2021 cut-off narrows but does not remove similarity leakage, and on an Orai1 arm of three chemotypes and four correlated frames of one receptor without a cognate structure, so the reference-free proxies used there cannot establish which predicted pose is correct. Both sides of this answer are developed in the following chapter, which sets out the controls that would sharpen the comparison and the experimental evidence the Orai1 hypotheses still require.

Across these four questions, the overall picture is one of a workflow rather than a winner. For the datasets, tool versions, search settings and refiners tested here, AutoDock Vina produced the highest PoseBusters pass rate while DiffDock had the highest in-place accuracy point estimates, although the paired intervals leave its near-nativeness advantage over AutoDock Vina inconclusive at the tested ranking depths. Both were limited by pose selection to a similar degree, whereas EquiBind was limited chiefly by generation and localisation. The limitations that bound every one of these conclusions are set out in full in the following chapter.

7 Limitations and Future Outlook

The overall picture is narrower than a claim that physics beats AI, and narrower still than the reverse. Under a matched blind search AutoDock Vina gave the highest PoseBusters pass rate while DiffDock had higher in-place accuracy point estimates, without a statistically established near-nativeness advantage over AutoDock Vina at the tested ranking depths. Neither class therefore dominates once the classical engine loses its binding-site advantage. AutoDock Vina remains bound by rigid-receptor docking and empirical scoring. DiffDock was selection-limited, as its confidence ranking failed to surface many credible poses. EquiBind was limited in generation, localisation and geometry. For Orai1, Vina and refined DiffDock can support hypotheses but establish no native pose. The most defensible workflow is hybrid, namely to generate candidates, filter for physical validity and biological placement, rank with a crystal-blind selector, and test the final hypotheses against independent evidence.

DiffDock stochasticity also limits exact regeneration. The 303-complex calibration run predates the local inference-seed patch and is a single unseeded draw, so rerunning that arm may produce a different pose sample even under the same documented settings. The Orai1 experimental

panel is distinct, because it was generated after the seed support was added and uses seed 42, so that panel can be repeated exactly. The Orai1 control panel carries no recorded seed (Table 14) and is therefore no more exactly reproducible than the calibration run, which matters because the control panel supplies most of the Orai1 DiffDock numbers reported here.

These conclusions are bounded. The results describe configured toolchains, thirty EquiBind conformers with gnina refining and ranking, at single settings without multi-seed sweeps, so physics-versus-AI labels oversimplify Vina's empirical and gnina's learned scoring. The dominant variant of each toolchain was also chosen on the same 303 complexes on which it is then scored, with no held-out split between selection and evaluation, so every calibration rate quoted in this chapter and in the Results is optimistic as an estimate of performance on unseen targets. The Orai1 DiffDock arm carries a further caveat of its own, namely the seeding asymmetry set out above, under which the experimental panel can be repeated exactly while the control panel carries no recorded seed. Two defects were also corrected before those results were generated. Graphics-memory was not released between complexes, which made the later complexes of a batch fail without a readable error and left both GSK-7975A forms without poses, and the output directories accumulated poses across runs, so earlier analyses drew on unequal and partly stale pose sets. Every Orai1 DiffDock number on the experimental panel comes from a single run in which each frame-ligand unit contributes thirty poses, capped at ten for analysis. The control panel is not on that footing. It was sampled at ten poses per unit rather than thirty and fell short of even that cap on 178 of its 1,215 units, returning 11,673 poses in all, so its per-unit yields rest on a unit-varying denominator. A post-2021 release does not remove leakage. The search is now matched, because all three pipelines were given the whole receptor, so the earlier crystal-centred-box asymmetry no longer qualifies the calibration result, but a blind search is also a harder task than the cognate re-docking regime in which AutoDock Vina is usually benchmarked, and its absolute numbers here should not be compared with published re-docking rates. Poses from one complex, and four correlated frames spanning three chemotypes, are not independent, so the complex or trajectory is the inferential unit, and non-significant contrasts are not evidence of equivalence. Best-of-top-k and crystal-nearest-cluster results are oracle ceilings, not deployable rank-1 accuracy, and because gnina both optimised and ranked EquiBind, the DiffDock + smina and gnina variants must stay named. PoseBusters checks only gross geometry, not free energy, protonation, solvent or membrane stability, and no ligand-bound dynamics were run. Orai1 offers one model, four frames and three cognate-free ligands, and its transmembrane slab is an operational hypothesis. The Orai1 AutoDock arm is gnina-rescored to match the calibration benchmark, and that rescoring covers 1,201 of the 1,215 analysable control-panel units, the fourteen missing units being macrocyclic ligands whose rescored poses could not be rebuilt to a topology matching their input, so the control panel is marginally smaller than the raw search on that arm. The DiffDock arm covers all 1,215 units but does not always fill the ten-pose cap, returning 11,673 control-panel poses against 12,010 for AutoDock Vina over its 1,201 units and 12,150 for EquiBind, so per-unit yields on that arm rest on a slightly smaller and unit-varying pose count. Because GSK-7975A and Synta-66 act near the pore [68], [69], a usable pose is

not necessarily correct. CPU-core-seconds and GPU-seconds are never summed, and cost per qualifying pose is conditional on success, which flatters EquiBind.

The asymmetric search was the most tractable of these limitations, and it has been removed. Because AutoDock Vina requires a search box while DiffDock and EquiBind localise the ligand and themselves, an equal-information comparison means withdrawing Vina's crystal-centred advantage rather than granting the learned tools a pocket. Benchmarks that evaluate docking blindly withhold the crystallographic pocket but generally supply a predicted one in its place rather than none at all, as PoseBench does when it pairs Vina with P2Rank [67], [59]. The whole-protein box adopted here is therefore the stricter setting, and it matches the pocket-free search baselines of the DiffDock line of work, in which the box is built around the entire receptor with a small fixed buffer [24], [89]. The result is informative because it did not preserve the earlier ordering. In the like-for-like raw Vina arms, validity-aware rank-1 recovery falls from 55.4% (168 of 303) in the superseded crystal-centred run to 25.7% (78 of 303) in the blind run. The selected blind AutoDock Vina + gnina arm reaches 29.4% (89 of 303). DiffDock, which was blind in both analyses, moves from 32.3% to 34.0% on the reported endpoint. These cross-run contrasts associate the whole-protein protocol with the reversal, but they do not isolate the search box itself, because receptor preparation also differs between the runs and the AutoDock treatment must be held fixed for a like-for-like comparison. The physical-validity advantage does persist in the blind analysis, but attributing it to a particular stage remains an interpretation because the design does not separate search, scoring and local optimisation.

Two further controls would sharpen this. The first is an intermediate arm in which AutoDock Vina receives a predicted rather than a crystallographic pocket, for instance the end-to-end P2Rank-Vina pipeline used by PoseBench [67], [59], which would measure how much of the lost ground a realistic site predictor recovers and is the configuration a practitioner would actually deploy. The second is a search-effort control, because exhaustiveness was held at 32 while the search volume grew by orders of magnitude, and box volume and search effort are known to interact, with larger boxes benefiting most from a more exhaustive search [79]. That interaction was characterised on boxes centred on the crystal ligand and no larger than 30 Å per side, so its quantitative thresholds do not transfer directly to a whole-protein search, and the reported gains flatten between exhaustiveness 25 and 50, which bounds the headroom available above the setting used here. On a genuine whole-protein box, by contrast, raising exhaustiveness from 8 to 64 on CASF-2016 reduced the median RMSD by more than one Å [80]. Part of the measured drop is therefore attributable to under-sampling a larger space rather than to the scoring function, and separating the two requires re-running the whole-protein arm at higher exhaustiveness. Until that is done the blind AutoDock numbers should be read as a lower bound on what the engine can do without site information.

Several directions already exist and should be benchmarked directly, not treated as speculation, namely flexible side-chain diffusion [31], co-folding evaluated in PoseX and PoseBench [12], [67], DiffDock-Glide [62], PocketVina [64], updated gnina scoring [65] and template controls such as TEMPL [63]. Because pose accuracy alone is insufficient [32], the strongest near-term workflow

is that hybrid pipeline, evaluated against current models, not the versions tested here.

For Orai1 in particular, the poses reported here are computational hypotheses rather than validated complexes, and the reference-free proxies used to rank them cannot establish which, if any, is correct. Settling that question requires wet-lab evidence, namely site-directed mutagenesis, competition or displacement assays and, ideally, a ligand-bound structure, which no docking study can provide on its own. This experimental validation is being pursued in continuing research at JKU, whose Orai1 modulators motivated the reference-free case study, and the present thesis contributes to that programme the physically screened, on-receptor pose hypotheses together with the tool-selection evidence on which such follow-up can build. The lasting contribution of the reference-free arm is therefore not a bound pose but a documented and physically screened workflow for generating and filtering testable Orai1 hypotheses, and a defensible ordering of the tools that produce them.

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Documentation table of AI-based tools

AI-Based Tool	Intended Use	Prompt, Source, Page, Paragraph. . .
OpenAI Codex	Consistency review	Prompt: "Please conduct a very detailed review of the thesis. Compare the numbers reported in the tables and figures and make sure they are in line with the text."
Claude Code	Code review and code documentation of the analysis and docking scripts, verification of the reported numbers against the underlying data, checking and correction of citations, and application of the resulting corrections to the manuscript text	Representative prompts: "Review the code for errors and bugs"; "Please add a function description and comments."; "Please perform an adversarial review of the thesis. Do all the numbers in the graphs, tables and prose align? Are the citations valid?"; "Can you please correct the citations?"

A Source Code and Data Availability

The source code underlying this thesis is held in the project GitHub repository, which is available from the author on request:

https://github.com/dthemann/master_dev

The repository contains the docking drivers for the three tools (AutoDock Vina, DiffDock and EquiBind), the post-hoc refinement (smina / gnina) and PoseBusters validity pipeline, and the pose-comparison, transmembrane-exclusion, clustering and PandaMap interaction-fingerprint analyses for both the calibration benchmark and the Orai1 application. The docking and analysis outputs themselves run to several hundred thousand files and are not distributed through the repository, so they are held by the author and provided on request. A regeneration guide at Scripts/Analysis/REGENERATE.md maps the figures and tables reported in this document to the output file each was taken from and to the exact command that rebuilds that output, and states which conda environment each command needs and in which order the dependent steps must run. That guide also marks explicitly the items it could not reconstruct, namely the two hand-composed molecular renders of the Orai1 receptor, one reference-free cluster-quality figure that matches its generating output in content but not byte for byte, and the tables whose provenance it did not audit.

B Tools and Functions

B.1 AutoDock Vina

AutoDock Vina is a physics-based docking tool designed to enhance speed and pose prediction accuracy in comparison to earlier AutoDock versions [22]. AutoDock Vina combines an empirical scoring function with an efficient global optimisation strategy, sampling many candidate ligand poses in the search space and ranking them according to an analytically defined estimate of binding free energy.

The docking process in AutoDock Vina is divided into five computational steps. First, receptors and ligands are prepared by removing crystallographic water molecules, adding hydrogen atoms, assigning atom types, calculating Gasteiger partial charges and converting SDF or MOL2 files into PDBQT format. Defining rotatable bonds for ligands allows the exploration of conformational flexibility during docking. Second, a three-dimensional search box restricts the spatial region in which the ligand may be placed. Instead of including the entire protein surface, the search can be limited to known binding pockets for computational efficiency.

Third, Vina begins the actual sampling stage by generating random initial ligand poses within the search box, where each pose is represented through translational, rotational, and torsional degrees of freedom. These candidate poses are then refined by an iterated local search procedure in which random perturbations are followed by local optimisation, enabling the algorithm to move efficiently through the conformational landscape while avoiding complete reliance on brute-force enumeration. A central reason why this is computationally feasible is that Vina uses a smooth scoring function and precomputed interaction grids, which allow fast energy evaluation and support gradient-based local improvement using a BFGS-like quasi-Newton optimiser.

Fourth, after repeated sampling and local optimisation, Vina clusters similar poses by RMSD so that highly redundant solutions are not overrepresented in the output. Fifth, the representative poses of these clusters are ranked by predicted binding free energy, and the lowest-energy poses are returned as the most likely docking solutions. This search-and-score design makes Vina highly attractive as a baseline method because its docking behaviour is explicit, reproducible, and easy to interpret in terms of chemical interactions and search heuristics [23].

The Vina scoring function (Equation 1, [22, p. 2]) determines how candidate poses are ranked. In simplified form, the scoring function combines pairwise intermolecular interaction terms with a penalty for ligand flexibility, so that the total predicted affinity is the weighted sum of steric attraction, steric repulsion, hydrophobic contacts, hydrogen bonding and a torsional penalty proportional to the number of rotatable bonds. More specifically, the function includes two

Gaussian steric terms, one repulsion term, one hydrophobic term, one hydrogen-bond term and one conformation-dependent penalty term [23].

In the formulation of Trott and Olson [22], the predicted binding free energy is written as a sum of distance-dependent interaction terms evaluated over every ligand–receptor atom pair. Each contribution depends not on the raw interatomic distance but on the surface distance d , the interatomic separation minus the sum of the two atoms' van der Waals radii, so that $d = 0$ corresponds to atoms in van der Waals contact, negative values to steric overlap, and positive values to a gap between the surfaces.

The steric contribution shared by all atom pairs is built from three of these terms, namely two attractive Gaussians and one repulsive parabola. The first Gaussian (gauss1) rewards contacts close to van der Waals distance, the second (gauss2) captures a broader, longer-range attraction centred about 3 Å beyond contact, and the repulsion term penalises interpenetration by growing quadratically once the surfaces overlap ($d < 0$). Combining the negative Gaussian weights with the positive repulsion weight reproduces the overall shape of a Lennard-Jones potential without its hard singularity. The remaining two terms apply only to suitable atom types, namely a piecewise-linear hydrophobic term that rewards contact between non-polar atoms, and a piecewise-linear hydrogen-bond term that rewards donor–acceptor pairs at short surface distances [22].

Each term enters the total with a fixed weight obtained once by non-linear regression against the measured binding affinities and native poses of a large set of protein–ligand complexes, which gives Vina its hybrid empirical and knowledge-based character [22]. The weighted per-pair sum is separated into intermolecular and intramolecular contributions, and the reported affinity is divided by a factor that increases with the number of active rotatable bonds ($1 + w \cdot N_{\text{rot}}$). This normalisation acts as the flexibility penalty, down-weighting highly flexible ligands that incur a larger conformational-entropy cost on binding. Because the score is evaluated analytically together with its gradient, each candidate pose can be locally minimised during the search, which is why no separate post-docking optimisation step is required [23].

The scoring function is defined formally as follows, using the notation and weights of Trott and Olson [22]. The conformation-dependent score c is summed over all pairs of atoms that can move relative to one another (normally excluding atoms separated by three or fewer consecutive covalent bonds):

$$c = \sum_{i < j} f_{t_i t_j}(r_{ij}) \quad (1)$$

where atom i carries a type t_i and $f_{t_i t_j}$ is a symmetric interaction function of the interatomic distance r_{ij} . The score separates into an intermolecular and an intramolecular part,

$$c = c_{\text{inter}} + c_{\text{intra}} \quad (2)$$

and the predicted free energy of binding is read from the intermolecular part of the lowest-scoring conformation (denoted 1), while the remaining conformations are ranked with the

intramolecular term of that best binding mode held fixed, so that the ranking is preserved:

$$s_1 = g(c_1 - c_{intra,1}) = g(c_{inter,1}) \quad (3)$$

$$s_i = g(c_i - c_{intra,1}) \quad (4)$$

Each interaction function is written in terms of the surface distance d_{ij} , obtained from the interatomic distance by subtracting the van der Waals radii R of the two atom types (hydrogen atoms are treated implicitly, and all interactions are truncated at $r_{ij} = 8\text{\AA}$):

$$f_{t_it_j}(r_{ij}) \equiv h_{t_it_j}(d_{ij}), \quad d_{ij} = r_{ij} - R_{t_i} - R_{t_j} \quad (5)$$

The function h is a weighted sum of three steric terms common to every atom pair, plus a hydrophobic term and a hydrogen-bond term that act only on the appropriate atom types. The three steric terms are two attractive Gaussians and a repulsive parabola:

$$gauss_1(d) = \exp \left[- \left(\frac{d}{0.5\text{\AA}} \right)^2 \right] \quad (6)$$

$$gauss_2(d) = \exp \left[- \left(\frac{(d - 3\text{\AA})}{2\text{\AA}} \right)^2 \right] \quad (7)$$

$$repulsion(d) = d^2 \quad \text{for } d < 0, \quad 0 \quad \text{for } d \geq 0 \quad (8)$$

The hydrophobic term equals 1 for $d < 0.5\text{\AA}$ and 0 for $d > 1.5\text{\AA}$, with linear interpolation in between. The hydrogen-bond term equals 1 for $d < -0.7\text{\AA}$ and 0 for $d > 0$, again interpolated linearly. The conformation-independent factor g then rescales the intermolecular score by the number of active rotatable bonds N_{rot} between heavy atoms in the ligand:

$$g(c_{inter}) = \frac{c_{inter}}{(1 + w \cdot N_{rot})} \quad (9)$$

The six weights w that fix the relative contribution of each term were obtained by tuning the function against the PDBbind dataset, going beyond simple linear regression. Table 12 reproduces their values from Table 1 of [22].

Table 12: AutoDock Vina Scoring-Function Weights Reproduced from Trott and Olson

Weight	Term
−0.0356	gauss ₁
−0.00516	gauss ₂
0.840	repulsion
−0.0351	hydrophobic

Table 12: AutoDock Vina Scoring-Function Weights Reproduced from Trott and Olson (continued)

Weight	Term
−0.587	hydrogen bonding
0.0585	N_rot

The two Gaussian terms were introduced because they provide a smooth approximation of favourable steric complementarity between ligand and receptor atoms across relevant interatomic distances. One Gaussian term rewards close but non-overlapping contacts, while the second captures broader distance-dependent attraction, together approximating dispersive complementarity without requiring a computationally expensive all-atom physical force field. The repulsion term penalises atomic overlap at very short distances and therefore discourages physically impossible steric clashes.

The hydrophobic term rewards favourable contacts between nonpolar atoms because burial of hydrophobic surface commonly contributes to ligand stabilisation in protein pockets. The hydrogen-bond term captures favourable donor-acceptor interactions within a distance range compatible with stabilising hydrogen bonds, which is essential because many correctly docked poses are distinguished by chemically plausible polar interactions rather than by shape alone. Finally, the torsional penalty reflects the entropic cost of binding a flexible ligand, since ligands with more rotatable bonds generally lose more conformational freedom upon binding and should therefore not be ranked as favourably as equally complementary but more rigid ligands.

The weights were chosen by Trott and Olson [22] by empirical calibration against experimental binding affinity data, while at the same time imposing the practical requirement that the scoring function remain smooth enough for efficient local optimisation. Vina was not designed to reproduce a full physical free-energy decomposition, but rather to provide a fast and well-behaved surrogate objective that correlates reasonably with both correct pose selection and experimental affinity across a broad range of complexes. The parameter values were chosen since they offered the best compromise between speed, robustness, and predictive usefulness on benchmark datasets available to the authors at the time.

Later work built on this design philosophy and produced Vina-related variants such as smina and later Vina releases, which extended atom typing, exposed additional scoring parameters, improved local minimisation behaviour, and enabled easier rescoring or custom scoring-function development for specific benchmark settings. These later variants did not abandon the Vina paradigm but rather preserved the same central idea of a smooth empirical score combined with efficient search, making AutoDock Vina a strong and still widely used reference method in comparisons with modern AI-based docking approaches [13].

A closely related reparametrisation of this scoring function is Vinardo, short for Vina RaDii Optimized [71]. It was originally implemented as an optional scoring function in smina and is now selectable directly in AutoDock Vina through its Vinardo scoring option [23], [71]. Vinardo

retains the overall structure of the Vina score, namely a weighted sum of steric, hydrophobic and hydrogen-bond terms with a rotatable-bond flexibility penalty, but differs from Vina in three main respects. First, it removes the second Gaussian steric term, whose broad minimum about 3Å beyond atomic contact was found to dominate the Vina steric energy and to draw docked ligands too far into the binding pocket, leaving one attractive Gaussian, a repulsion term, a hydrophobic term and a hydrogen-bond term. Second, it re-optimises the atomic radii that fix the surface distance between atoms, enlarging the radius of aliphatic carbon and reducing those of nitrogen and oxygen, a change the authors identify as one of the most important for docking accuracy. Third, it re-fits the remaining weights and interaction ranges, for instance widening the first Gaussian, fixing the hydrophobic onset parameter at zero so that this term reduces to a single linear ramp, and extending the hydrophobic range to a distance close to the diameter of a water molecule. Rather than fitting the weights by regressing predicted against measured affinities, as was done for Vina, the authors developed Vinardo by first screening candidate functions on the average RMSD of crystallographic poses after local minimisation, a quantity they found to track docking ability where affinity correlation does not, and then confirming the best candidates by actual redocking [71]. On the CASF-2013 benchmark they report higher scoring, docking and screening power for Vinardo than for Vina and an improved high-level ranking success, although Vina remains marginally ahead on the low-level ranking criterion. Because CASF-2013 is built on the PDBbind Core 2013 structures that also informed Vinardo's parametrisation, the two overlap in the complexes used, though the authors argue that their docking-oriented optimisation is distinct from the rescoring-oriented CASF tests, so these results are best read as evidence of a well-behaved docking objective rather than as an independent generalisation test. In this work the same AutoDock Vina engine was additionally run with the Vinardo scoring function, but because Vinardo did not improve on the default Vina function for the targets studied here, Vina scoring underlies the reported results and the Vinardo variant is retained only as a minor supporting comparison. That outcome is less surprising than the published CASF-2013 comparison alone would suggest, for two reasons. First, Vinardo was developed and validated under pocket-localised conditions, with every docking run in a box centred on the bound ligand pose, so its reported advantage was established in a regime in which the search never had to locate the site. The whole-protein search used here is the opposite regime, and the term Vinardo removes is precisely the long-range attractive Gaussian that dominates the Vina steric energy at distances beyond contact [71]. Second, the advantage has already been shown not to transfer across benchmarks. On DUD-E, Vinardo outperforms Vina on both area under the curve and early enrichment, yet on the harder LIT-PCBA set the same comparison by the smina and gnina developers finds the two indistinguishable, with Vinardo marginally behind [81]. Which empirical function wins is therefore a property of the target and the protocol rather than a fixed ordering, a point the scoring-function assessment literature makes generally by separating scoring power from docking power and measuring them independently [82].

B.2 AI-Based Molecular Docking

AI-based molecular docking tools employ diverse computational strategies. Common strategies include deep-learning neural networks that predict binding poses directly from protein and ligand structures, diffusion models like DiffDock [24] that treat docking as a generative sampling problem, geometric deep learning approaches such as EquiBind [9], [10], which predicts the bound pose directly with an SE(3)-equivariant network, and TankBind [25], [26], which encodes the protein with an equivariant network but predicts an inter-molecular distance map through a trigonometry module and then derives coordinates from it by optimisation, and regression-based models like KarmaDock [27] and GAABind [28] that predict binding affinities or poses with learned scoring functions. Other notable AI-based algorithms include SurfDock, DiffBindFR, DynamicBind, QuickBind, and hybrid methods like Interformer, which integrate traditional conformational searches with AI-driven scoring functions. These tools also differ in the search space they require. DiffDock, EquiBind and TankBind need no user-specified binding site and localise the ligand themselves, whereas DeepDock and Uni-Mol must be given a pocket definition [12], [34]. Co-folding models likewise require no predefined site, and they predict protein and ligand structure jointly rather than placing a ligand into a fixed receptor structure [12], [29].

In contrast to traditional exhaustive conformational search algorithms and empirical scoring functions, AI-based tools learn complex interaction patterns from large datasets of protein-ligand complexes to directly predict optimal binding conformations. Moreover, AI-based methods tend to be computationally efficient, which improves scalability for large-scale virtual screening.

B.2.1 DiffDock

DiffDock is a diffusion generative model (DGM). It treats protein-ligand pose prediction as a generative task rather than as the optimisation of a hand-crafted energy function. Instead of searching a scoring landscape fixed in advance by empirical interaction terms, DiffDock learns a probability distribution over correct binding poses from structural training data. It then samples candidate dockings through a reverse diffusion process [24].

In DiffDock, the protein and the ligand are first represented as graphs. Nodes encode chemical identity and geometry. Edges encode connectivity or spatial neighbourhood. The ligand pose is parameterised by translation, rotation, and internal torsion angles, so docking becomes a prediction over the full space of rigid-body placement and conformational flexibility. During training, a forward diffusion process gradually corrupts known experimental poses by adding noise to these variables, turning correct conformations into increasingly disordered ones.

At inference time, the reverse process runs. DiffDock starts from random noisy ligand placements and denoises them over several diffusion steps. At each step, an SE(3)-equivariant neural network predicts how to update the translation, rotation, and torsion variables so that the pose moves toward a high-probability bound conformation. Because the network is SE(3)-equivariant, its predictions transform consistently under rigid motions of the input coordinates.

This equivariance is essential for reasoning over three-dimensional geometry.

A practical consequence is that DiffDock produces several plausible poses rather than one deterministic answer. This helps because real docking landscapes are often multimodal. Several poses can be chemically plausible, and near-native conformations may sit far apart in geometry. Direct regression methods struggle to capture this pattern. Finally, a separate confidence model evaluates the generated poses and ranks them by predicted reliability rather than by estimated binding free energy.

DiffDock's scoring concept has two components. The first is the diffusion score model itself. It predicts the gradient of the log-probability density over poses at each noise level and thereby steers the reverse denoising trajectory toward high-likelihood regions of pose space. This is not an affinity score in the classical sense, but a learned geometric-statistical signal that tells the model how to turn a noisy pose into a more plausible one.

The second component, the confidence model, is the one directly relevant to pose ranking. Once DiffDock has generated candidate poses, the confidence network examines each predicted complex and outputs a scalar value meant to correlate with docking correctness. In particular, it reflects whether the pose falls below a conventional RMSD threshold such as 2Å from the reference. The authors added this model because generative sampling alone does not guarantee that the top sampled pose is the best, and because practical docking needs a way to prioritise the most trustworthy candidates among many.

This ranking strategy follows from the multimodal nature of docking. Regressing a single pose can collapse several valid solutions into a physically meaningless average. Diffusion modelling was chosen because it can represent complex pose distributions, and the confidence model because it offers a calibrated way to pick the pose most likely to be near-native, even though the reverse process stays stochastic.

DiffDock's ranking behaviour is learned from large structural datasets. The effective "parameters" that judge poses are therefore encoded implicitly in the network weights and the training objective, not exposed as chemically interpretable constants. This lets DiffDock learn subtle interaction patterns that are hard to represent analytically, but it also makes performance depend heavily on how representative the training data are and on how well the confidence model is calibrated [24].

Later work extended the DiffDock framework toward flexible-pocket docking and hybrid rescoring strategies. Follow-up studies such as DiffDock-Pocket incorporated side-chain flexibility in the receptor pocket [30], [31], while newer hybrid approaches pair diffusion-based pose generation with classical physics-based rescoring [62]. This reflects a growing view. Diffusion models generate candidate poses well but can still benefit from physically motivated ranking criteria [62].

B.2.2 EquiBind

EquiBind represents a third methodological class. It is a direct geometric deep-learning approach to blind docking. Vina searches and scores many poses, and DiffDock samples and ranks many

denoised ones. EquiBind instead predicts the bound pose directly, in a single forward pass of an SE(3)-equivariant neural network [10].

EquiBind first encodes both protein and ligand as graphs whose nodes carry chemical and geometric features. It then runs interaction-aware equivariant message passing between the two graphs. Ligand atoms gather information about the most relevant protein regions, while protein nodes encode ligand-dependent context. This matters because EquiBind uses no predefined binding site. It learns to infer both where the ligand should bind and how it should be oriented [10].

A central innovation is that EquiBind predicts matching sets of key points on the ligand and on the protein surface. It then computes the rigid-body transformation that best aligns the ligand key points to the protein key points, typically via a singular-value-decomposition (Kabsch-style) step [10], [60]. This transformation places the ligand directly into the predicted pocket. A final conformer-fitting stage adjusts internal torsions to improve local geometry while keeping the ligand chemically valid [9], [10].

The full workflow therefore comprises graph construction, joint equivariant message passing, key-point prediction, rigid alignment, and torsional refinement, and it is that sequence that the released implementation executes [9]. Compared with Vina, it drops the iterative stochastic search. Compared with DiffDock, it drops the repeated denoising steps. This runtime advantage is a main reason EquiBind is attractive for large-scale screening and for settings that need rapid blind docking [10].

EquiBind's scoring concept departs from classical docking even more than DiffDock's. In its original form, it uses no separate scoring function to rank alternative poses at inference time. Instead, it is trained with losses that directly penalise the geometric error between the predicted pose and the experimental reference. The network thus internalises an implicit scoring rule during training and outputs a single pose that reflects that learned criterion [10].

The main training signals are ligand RMSD-type coordinate losses, rigid-body alignment losses, and auxiliary terms that push the predicted placement toward the correct region of the protein [10]. These objectives were chosen for two reasons. They match the metrics used in docking evaluation, especially ligand RMSD and centroid distance, and they give stable gradients for end-to-end learning of geometric correspondences. In effect, EquiBind learns which placements are “good” because they minimise structural discrepancy to known complexes in the training data, not because they minimise an explicit analytic energy.

This design shapes how EquiBind should be placed in a comparative review. It is not a classical scoring-based method. It assigns no thermodynamic, affinity-like number comparable to Vina's predicted ΔG . Nor is it a generative, uncertainty-aware sampler like DiffDock. It is a direct pose-prediction model whose ranking criteria are embedded in a learned neural representation of protein-ligand complementarity.

The authors justified this design by arguing that a sufficiently expressive SE(3)-equivariant model can learn the interaction patterns needed for docking directly from structural examples, avoiding expensive sampling and hand-crafted scoring altogether [10]. This was one of the first

influential demonstrations that blind docking can be framed as a geometric correspondence problem rather than a search over an engineered energy landscape [10]. Later analyses, however, show that EquiBind often benefits from optional physics-based refinement when highly flexible ligands or subtle local interactions must be resolved precisely, which is why hybrid pipelines remain important in practice [34].

B.3 Optimisation Tools

B.3.1 smina

Unlike the tools mentioned above, smina and gnina are used here as lightweight refinement tools that re-score and locally optimise generated poses. smina is a fork of AutoDock Vina developed by Koes et al. [53]. It inherits Vina's empirical scoring function and search machinery unchanged, but re-implements them so that each individual energy term can be reweighted or swapped out by the user and so that custom atom types can be defined. The energy smina optimises is therefore the Vina objective itself, an empirical, knowledge-based estimate of the binding free energy formed as a weighted sum of pairwise atom-type interaction terms, namely two attractive Gaussian steric terms, a steric-repulsion penalty, a hydrophobic contact term and a hydrogen-bond term, divided by a factor growing with the number of rotatable bonds so as to penalise ligand flexibility (the same function detailed in the Physics-Based Molecular Docking section above). Beyond full docking, smina exposes a fast local-minimisation mode that relaxes a supplied ligand pose into the nearest minimum of this energy by gradient-based optimisation, without repeating a conformational search. This makes it well suited to post-docking optimisation, since a pose seated in roughly the correct pocket but clashing sterically with the receptor can be minimised against the Vina function, removing the clash while leaving the placement essentially unchanged. Its source code is maintained on SourceForge and mirrored on GitHub [55].

B.3.2 gnina

Because gnina is in turn a fork of smina, both tools share the docking algorithm inherited from Vina [54], [65], [66]. Conformational sampling is carried out by a set of Markov chain Monte Carlo (MCMC) chains that repeatedly apply random perturbations, namely rigid-body translation and rotation of the whole ligand together with rotations about its rotatable bonds, to positions inside the user-defined search box. Each candidate a chain generates is driven to a local energy minimum against the Vina scoring function by gradient-based (quasi-Newton) minimisation, and a Metropolis criterion then decides whether the minimised pose is accepted as the next state of the chain. The Vina empirical energy is thus the quantity optimised during ordinary search and empirical minimisation or rescoring in both smina and gnina. The explicitly evaluated CNN-refinement mode is an exception, where CNNscore supplies the objective for the final local

minimisation. What further distinguishes the tools is how the resulting poses are scored and ranked.

gnina extends this lineage by replacing the hand-crafted empirical score used for ranking with a learned one [54], [65], [66]. Once the MCMC search and Vina minimisation have produced candidate poses, gnina rescores and ranks them with an ensemble of convolutional neural networks (CNNs). Each CNN reads a three-dimensional grid of Gaussian atom-type densities, a voxelised representation of the protein-ligand complex generated by the libmolgrid library, and is trained jointly on two objectives. The first is pose scoring, a classifier, trained with a cross-entropy loss, that predicts the probability a pose lies within 2Å RMSD of the native binding mode, reported as CNNscore. The second is binding-affinity prediction, a regression head, trained with a mean-squared-error loss that is hinged so inaccurate poses are not penalised for underestimating affinity, that predicts a binding constant on a pK (negative-log) scale, reported as CNNaffinity. Their product, $\text{CNNscore} \times \text{CNNaffinity}$, is exposed as CNN_VS and is the preferred metric for ranking compounds in virtual screening [66]. Unlike Vina's ΔG -like number, these CNN outputs are learned criteria rather than analytic energies. In the default rescoring path they rank already empirically minimised poses. In the CNN-refinement mode evaluated here, CNNscore is differentiated and used as the final local-minimisation objective.

The scoring CNNs come in two architectures, a linear five-convolutional-layer network (“Default2018”) and a twelve-layer densely connected network (“Dense”), and are combined into ensembles, since ensembling consistently yields better-ranked poses than any single network. gnina 1.3, the version used here, retrains these models on the updated CrossDocked2020 v1.3 dataset, ports the deep-learning backend from Caffe to PyTorch for faster CPU inference, and applies knowledge distillation to compress a five-member ensemble into a single “fast” student model for high-throughput screening. It also adds covalent-docking support [65]. By default, only the pose score is used to rank docked poses.

It thus occupies a hybrid position between the physics-based and AI-based approaches, namely a physics-based sampler coupled to a deep-learning scoring function, which can serve either as a standalone docking tool or, as in this work, purely as a rescoring and refinement step applied to poses produced elsewhere. It is openly available on GitHub [56].

The gnina CNNs are trained on the CrossDocked2020 set [57], millions of poses obtained by cross-docking ligands into structurally similar Protein Data Bank pockets available up to 2020 [54]. The calibration benchmark used here is restricted to complexes released from 2021 onwards and curated so that none of the evaluated methods, e.g. DiffDock, had seen or been trained on them [34]. The receptor-ligand pairs of the benchmark are therefore absent from gnina's training data as direct entries, making the re-scoring a direct-entry temporal holdout rather than a similarity-controlled test. A later release date does not exclude homologous proteins, familiar pockets, related ligands or templates (see Section 5.4).

Because both tools relax and rescore poses efficiently and use no crystal information, they provide model-agnostic ways to turn the geometrically plausible but physically strained output of the deep-learning methods into physically valid poses, making them suitable tools for post-

docking optimisation.

C Docking Protocols

The three end-to-end pipelines use the same downstream validity definitions, but their receptor preparation and conformer generation are tool-specific. AutoDock Vina and DiffDock receive the same available RDKit [84] ETKDGv3 + UFF start conformer, whereas EquiBind generates thirty ETKDGv3 + MMFF conformers, as detailed below. In the primary benchmark arms, AutoDock Vina, DiffDock and unguided EquiBind all search or process the whole receptor without a crystal-derived restriction, so none is told where the ligand belongs. Separately reported EquiBind variants receive an fpocket- or P2Rank-predicted pocket and crop locally around it. The benchmark is a comparison of deployed workflows rather than a controlled estimate of algorithm class alone. Tool-specific settings and pose budgets are summarised in Table 13, and the corresponding settings for the two Orai1 panels are summarised in Table 14. A fourth engine, Uni-Dock2 0.6.3, was also run over the 303 benchmark complexes, reusing the ligand start conformers and the numeric whole-protein boxes of the AutoDock arm while performing its own receptor preparation under engine-specific search settings, and the 8,605 poses it produced were screened for physical validity together with the rest and remain in the master comparison file. No result reported in this document draws on those poses, which is why Uni-Dock2 appears in no protocol table, and the only trace they leave is in the pose totals quoted from that file, where 8,605 of the 169,993 screened poses are Uni-Dock2 and the reported comparison rests on the remaining 161,388.

All software versions were verified by invoking each binary. Docking used AutoDock Vina v1.2.7-20-g93cdc3d-mod, a locally patched build rather than stock 1.2.7. It also used DiffDock-L (PyTorch 2.4.0 / CUDA 12.4) and a custom EquiBind pipeline (February 2025, PyTorch 2.4.1 / CUDA 12.4). Local optimisation used gnina 1.3.2 and smina (9 November 2017 build, on the Vina 1.1.2 scoring base). AutoDock PDBQT preparation used Meeko 0.7.1 [85] for ligands together with MGLTools 1.5.7 [47] for receptors. Receptor cleaning ahead of that conversion used PDBFixer 1.12.0 on OpenMM 8.3.1. The pipeline also supports ADFRsuite-1.0 `prepare_receptor` [87], but no reported run used it. Open Babel [86] was installed as package version 3.1.1, although the binary itself reports 3.1.0. Blind pocket detection for EquiBind used fpocket 4.0 [58] and P2Rank 2.5 [59]. Physical-validity and interaction screening used PoseBusters 0.6.3 [34] and PandaMap 4.1.0 [88]. The PoseBusters installation is not stock. One statement in `posebusters/modules/energy_ratio.py` was patched locally so that the exception handler guarding the UFF parameter check logs the caught exception itself rather than indexing its second argument. As shipped, that handler also catches the assertion raised when UFF has no parameters for the ligand, and because that assertion carries only one

argument the handler then raises an `IndexError` of its own and aborts the entire `bust()` call instead of returning the empty result the module defines for a failed check. Every pose of a ligand that UFF cannot parameterise, for example one carrying hypervalent sulfur, a phosphate group or a coordinated metal, is therefore discarded without an error being reported. The patch adds no check, changes no threshold and alters no pass criterion, and it only restores the intended failure behaviour of that check for such ligands. A second, purely operational override forces the conformer generation of the same check onto one RDKit thread so that parallel workers do not oversubscribe the processor, which changes scheduling and not verdicts. All validity screening, statistics and figure generation ran in one conda environment holding Python 3.12.12, RDKit 2025.09.1 (distributed as package version 2025.9.1), NumPy 2.3.4, SciPy 1.16.2, pandas 2.3.3 and scikit-learn 1.8.0. The RDKit release matters for interpretation, because the distance-geometry bounds it supplies define the bond-length and bond-angle tolerances and its UFF implementation supplies both the pose energy and the reference ensemble of the internal-energy ratio, so most PoseBusters verdicts are tied to this exact version. All docking ran on a single laptop workstation with an Intel Core i9-14900HX processor (32 hardware threads) and 32 GB of system memory. AutoDock Vina ran multi-threaded on that processor at exhaustiveness 32. All GPU steps ran on a single RTX 4070 Laptop GPU (8 GB, driver 581.80, CUDA 12.4).

C.1 AutoDock Vina

AutoDock Vina [22], [23] provides the classical reference. Receptors are first cleaned with PDBFixer, which strips waters, ions, metals and cofactors, and are then protonated and converted to PDBQT with the MGLTools preparation named above, while ligands use the shared start conformer. The whole-protein receptors are therefore apo-like rather than cofactor-retaining. Heteroatom records survive in only 28 of the 303 prepared receptors and are almost entirely covalently modified amino acids such as phosphoserine, phosphothreonine, phosphotyrosine and carboxylated lysine, together with a small number of other heteroresidues including a bound uridine diphosphate in two structures, so no haem, flavin, metal or ion is present in the docked structure even where the deposited entry carried one. The search box spans the whole receptor, so no crystal-ligand restriction is applied and AutoDock searches blindly, exactly as DiffDock does. Vina runs at exhaustiveness 32, energy range 6 kcal/mol and random seed 42, retaining up to thirty ranked modes per complex.

Removing the box is a deliberate design decision and not a convenience. A box centred on the crystal ligand supplies the answer to part of the question being asked, because it tells the sampler where the ligand belongs before the search begins. The practice is widespread and is acknowledged as such even in the benchmark this study calibrates against, where AutoDock Vina received a 25 Å cube centred on the crystal-ligand heavy atoms so that its search space would be more comparable with the blind deep-learning methods, while remaining centred on the answer [34]. The size of the resulting advantage is substantial. Docking accuracy is known to depend strongly on box dimensions, with the best results obtained when the box is

scaled to roughly three times the ligand radius of gyration and progressively worse results as the box grows [77], and on CASF-2016 a whole-protein box has been reported to reduce Vina pose-prediction success to roughly a third to a half of the rate reported for site-specific docking [80]. The mechanism is that a fixed number of Monte Carlo search threads is diluted over a much larger volume, so sampling becomes correspondingly less dense [78], which is also why search effort has to be raised together with box volume [79]. Because the comparison in this work is against methods that localise the ligand themselves, keeping the crystal-centred box would have made the classical arm structurally incomparable with the learned arms on precisely the quantity being measured. The whole-protein box removes that asymmetry at the cost of a much harder search, and the AutoDock numbers reported here should be read as blind-docking numbers rather than as re-docking numbers.

Six AutoDock variants are produced on this set, arranged as two scoring families crossed with three treatments. The two families differ only in the empirical scoring function driving the shared search engine, namely the default Vina function and the Vinardo reparametrisation selected through the same program [23], [71]. Each is run as an independent authoritative docking pass with identical search settings, so the two are directly comparable, and each supplies the raw variant of its family. The remaining four variants are not separate searches. They are post-hoc passes over the poses of a completed pass, two per scoring family, using gnina 1.3.2 [54], [65] in its two convolutional scoring modes, namely rescoring and refinement. Both passes invoke the same local-minimisation mode of the program, so neither leaves the input pose untouched, and what separates them is which objective drives that minimisation. Each pose is rebuilt to SDF with Meeko and processed individually in a box extended 4 Å around that pose, with a fixed seed of 42 and no crystallographic input.

In the rescoring variant the pose is locally minimised against the empirical function and the convolutional ensemble then rescores the result, so the network reorders the poses without determining their geometry. In the refinement variant the network supplies the objective of that local minimisation, so the network determines the geometry. Two protocol details matter for interpreting the results. The gnina-side empirical objective is the Vina function in all four variants, so in the Vinardo family the poses are searched under Vinardo but minimised under Vina. Ranking after both gnina variants is taken from CNNaffinity rather than from gnina's own default CNNscore ordering, and the original Vina or Vinardo rank is retained alongside it so the two orderings can be compared. The convolutional model was pinned to the `crossdock_default2018` ensemble rather than left at the gnina 1.3 default ensemble.

Refinement was applied in all AutoDock runs despite the recommendation of the gnina authors, who report that it costs about an order of magnitude more computation than rescoring without performing better [54]. It is included here precisely for that reason. Poses produced by a physics-based search are already almost entirely valid, so passing them through a learned optimiser isolates the effect of the network on geometry from the geometric repair that any empirical minimisation would have supplied. Because all variants within a scoring family derive from one docking pass, they share a pose set exactly and can be compared as paired observations.

C.2 DiffDock

DiffDock [24], [8], specifically DiffDock-L [89], is the diffusion-based generative method. It docks blindly over the whole receptor without a supplied search box. ESM2-650M embeddings represent the protein and the shared start conformer provides the ligand graph. Reverse diffusion uses twenty denoising steps and generates up to thirty candidates per complex, each ranked by native confidence. Refinement was attempted independently with both smina and gnina for every raw pose, and each refined variant contains 8,993 outputs from 9,013 raw poses, with failed outputs excluded. The main Results designate the smina-refined variant as DiffDock + smina, while Appendix optimisation and ranking diagnostics label raw and gnina-optimised variants explicitly.

C.3 EquiBind

EquiBind [10] is a direct-regression method that natively emits one pose. To create a comparable retrospective pool, this study docks thirty fixed RDKit ETKDGv3 conformers, minimised with MMFF, in unguided mode against the full uncropped receptor, while fpocket- and P2Rank-guided variants, in which the receptor is cropped to an 11 Å shell around the predicted site (an 8 Å residue-selection radius plus a 3 Å margin), are evaluated separately [58], [59]. This expansion is a custom pipeline rather than native EquiBind behaviour. Refinement was attempted independently with smina and gnina for every output, and the unguided variants contain 9,082 smina outputs and 9,074 gnina outputs from 9,089 raw poses, with failed outputs excluded. The main Results use the unguided EquiBind + gnina variant. Pocket-guided results test whether supplying an external site predictor improves this configured workflow, but they leave EquiBind with binding-site information that neither AutoDock Vina nor DiffDock receives.

The pipeline also implements an intermediate ligand-relaxation stage between pose prediction and refinement, in which each EquiBind pose is minimised with the Universal Force Field against the receptor atoms lying within 6Å of the ligand, for at most two hundred iterations and with every receptor atom held rigid. The stage was switched off for the calibration-benchmark runs reported in this chapter, and all 308 committed EquiBind run records for that set confirm that it was inactive. It was switched off because a minimisation that may move only the ligand cannot lift a buried pose out of a steric overlap and relieves the overlap by deforming the ligand instead. In a superseded benchmark run of the same pipeline with the stage enabled, only 19.3% of EquiBind poses passed the PoseBusters bond-angle check and 22.5% passed the bond-length check, whereas every pose of the reported run passes both. Neither refiner repairs that damage, because smina and gnina move a pose by rigid-body translation, rigid-body rotation and rotation about its rotatable bonds and so leave its bond lengths and bond angles as they were, which is why the bond-angle pass rate of the superseded run was identical for its raw, smina-refined and gnina-refined variants. Because the refiners relieve the steric overlap on their own, switching the stage off roughly doubled the share of physically valid EquiBind poses. The one arm of this

study that still carries the relaxation is the experimental Orai1 panel, as recorded in the Orai1 Panels section below.

C.4 Post-Docking Optimisation and Validity Screening

This section covers the deep-learning pipelines. The corresponding step for AutoDock, where gnina is applied in two distinct modes to each scoring family, is described in the AutoDock Vina section above. Because deep-learning outputs can make invalid physical contact with the receptor even when placed near the intended site, each raw DiffDock and EquiBind pose was independently submitted to smina and to gnina rescoring and local minimisation, yielding two parallel variants from the same search after failed outputs were excluded. smina is a fork of AutoDock Vina tuned for fast local minimisation of the Vina scoring function [53]. gnina adds an ensemble of convolutional neural networks as a learned scoring function [54]. Both are confined to a box extended 4Å beyond the pose, use no crystallographic information and cost only seconds per pose. The two arms are not, however, given the same instruction inside that box, and the difference is worth recording because it cannot be recovered from the pose files. The DiffDock poses were refined in the energy-minimisation mode of the program (`--minimize`), the same mode used for the AutoDock Vina optimisation described above, so each pose is relaxed into the nearest minimum of the objective and no further search is performed. The EquiBind poses were refined in the local-search mode (`--local_only`) instead, which the program documents as a local search rather than a plain energy minimisation, so the pose may be adjusted by a bounded search around its input placement. All 307 committed EquiBind benchmark run records that log this setting record the local-search mode. The refined EquiBind poses are therefore not the product of exactly the same operation as the refined DiffDock and AutoDock Vina poses, which should be borne in mind wherever the refined arms are compared. This step mirrors the post-prediction energy minimisation reported alongside the PoseBusters benchmark [34]. Its quantitative effect on physical validity and geometric accuracy is reported in the Results. All retained poses were finally screened with the PoseBusters validity suite, and geometric accuracy was scored as the symmetry-corrected heavy-atom RMSD to the crystal ligand.

Table 13: Docking-Protocol Parameters for the Three Pipelines on the Calibration Benchmark

Parameter	AutoDock Vina	DiffDock	EquiBind
Method class	Physics-based (Vina / Vinardo)	Diffusion generative	E(3)-equivariant regression

Table 13: Docking-Protocol Parameters for the Three Pipelines on the Calibration Benchmark (continued)

Parameter	AutoDock Vina	DiffDock	EquiBind
Search space	Blind (whole receptor), 0.375Å grid	Blind (whole receptor)	Blind (whole receptor); 11Å crop only in the pocket-guided variants
Start conformer	RDKit ETKDGV3 + UFF	RDKit ETKDGV3 + UFF	30 × ETKDGV3 + MMFF
Sampling	Exhaustiveness 32; ΔE 6 kcal/mol	20 diffusion steps; no final noise	30 unguided poses
Poses per complex	≤ 30	≤ 30	≤ 30
Analysis ranking	gnina CNNAffinity in the gnina arms, Vina or Vinardo score in the raw arms	Model confidence	gnina affinity
Random seed	42	—	30 fixed seeds
Post-optimisation	gnina rescoring and gnina refinement, each evaluated per scoring family	both refiners evaluated; primary = DiffDock + smina	both refiners evaluated; primary = EquiBind + gnina
Refiner mode	Energy minimisation	Energy minimisation	Local search
In-protein UFF relaxation	Not applicable	Not applicable	Implemented, switched off
Total poses (303 pairs)	9,043 (Vina), 9,062 (Vinardo)	9,013	$\approx 9,080$

C.5 Orai1 Panels

The Orai1 chapter runs two ligand panels through the same three pipelines against the same four receptor frames, and the two panels were not sampled at the same settings. The experimental panel, which pairs the Orai1 modulators with the four frames, inherited the sampling budget of the calibration benchmark, namely thirty AutoDock Vina modes inside a 6 kcal/mol energy window, thirty DiffDock samples and thirty unguided EquiBind conformers. The control panel pairs all 308 calibration ligands with the same four frames, giving 1,232 complexes, and was cut back to ten Vina modes inside a 3 kcal/mol window, ten DiffDock samples and ten unguided

EquiBind conformers. Both settings are given side by side in Table 14. The two panels share the four receptor frames, the whole-receptor search boxes, the exhaustiveness of 32, the AutoDock seed and the post-docking variant carried forward for each tool. They are therefore matched on search settings rather than on every input, and the three further differences that remain are stated in the paragraphs below, namely the receptor converter, the DiffDock sampler seed and the in-protein UFF relaxation on the EquiBind arm.

The Orai1 yield, transmembrane-placement and pose-clustering analyses are then taken over the top ten ranked poses of each frame-ligand unit, and that cap removes most of the asymmetry rather than concealing it. For AutoDock Vina the two panels searched on the same settings, because the box, the exhaustiveness and the seed are the same and the mode count and the energy window govern only how many of the resulting modes are written out. The receptor files themselves were not prepared by the same converter, since the control receptors were written by MGLTools and the experimental receptors that enter the analysis by Meeko, so the two panels are matched on search settings rather than on every input file. The first ten modes of a thirty-mode run are therefore the same ranked prefix that a ten-mode run reports, and the twenty deeper experimental modes are discarded before analysis. The narrower control window was not binding either, because all 1,232 control complexes wrote the full ten modes, so no complex was truncated by the 3 kcal/mol limit before the cap applied. The DiffDock seed is a further recorded difference, since the reproducibility hook that fixes the sampler seed appears in the experimental run logs and is absent from the control ones.

For DiffDock and EquiBind the prefix argument is weaker, so the residual difference is stated rather than argued away. Both tools rank a pool of independently generated samples, so the ten retained experimental poses are the best ten of thirty whereas the retained control poses are simply all that were written. For EquiBind that is ten of ten on every unit. For DiffDock it is ten of ten on 1,037 of its 1,215 control units and fewer than ten on the remaining 178, ten of which return a single pose, so its control denominator varies by unit. The experimental panel therefore enters every comparison with the more heavily selected pose set. That bias runs against the direction of the reported result, because the control panel returns the higher usable yield for all three tools, so a deeper experimental pool would if anything have narrowed the gap rather than opened it. One further difference is confined to EquiBind, namely that the experimental poses passed through the in-protein UFF relaxation while the control poses did not, the relaxation having been switched off after the calibration runs showed it distorting ligand geometry to relieve clash. The EquiBind arm of the two-panel comparison is therefore not a clean per-panel match and is read as such.

Table 14: Docking-Protocol Parameters for the Three Pipelines on the Two Orai1 Panels

Parameter	AutoDock Vina	DiffDock	EquiBind
Search space	Whole receptor, per-frame box 91.8 to 102.0 Å per edge, 0.375Å grid	Blind (whole receptor)	Blind (whole receptor), unguided only
Exhaustiveness	32, both panels	Not applicable	Not applicable
Denoising steps	Not applicable	20, both panels	Not applicable
Modes or samples requested	Exp. 30, Ctrl. 10	Exp. 30, Ctrl. 10	Exp. 30, Ctrl. 10
Energy window	Exp. 6 kcal/mol, Ctrl. 3 kcal/mol	Not applicable	Not applicable
Random seed	42, both panels	Exp. 42, Ctrl. none recorded	Exp. 30 fixed ETKDGV3 seeds, Ctrl. the first ten of the same list
Poses written per unit	Exp. 30 on every unit, Ctrl. 10 on each of the 1,201 units it covers	Exp. 30 on every unit, Ctrl. 10 requested but returned in full on only 1,037 of 1,215 units	Exp. 30 on every unit, Ctrl. 10 on each of the 1,215 processed units
In-protein UFF relaxation	Not applicable	Not applicable	Exp. applied, Ctrl. not applied
Analysis cap	Top ten ranked poses per unit	Top ten ranked poses per unit	Top ten ranked poses per unit
Analysis ranking	gnina CNNAffinity order for both the top-ten cap and the top-three interaction cap	Model confidence	gnina refinement affinity
Post-optimisation	gnina rescoring, crossdock_default2018 ensemble, 4Å autobox, seed 42	smi minimisation, 4Å autobox, seed 0	gnina local refinement, 4Å autobox, seed 0

Exp. denotes the experimental panel of Orai1 modulators and Ctrl. the control panel of calibration-benchmark ligands. The analysis cap applies to every Orai1 analysis except the interaction and contact analysis, which uses the top three ranked poses per unit as stated in the Interaction Fingerprints section below. Box edges are the per-frame whole-receptor extents,

namely $98.3 \times 91.8 \times 92.6 \text{ \AA}$ for Fr0, $97.0 \times 97.0 \times 102.0 \text{ \AA}$ for Fr300, $97.7 \times 96.6 \times 101.9 \text{ \AA}$ for Fr400 and $97.8 \times 96.9 \times 101.6 \text{ \AA}$ for Fr499.

C.6 PoseBusters Validity Checks

Table 15 lists the individual checks of the PoseBusters validity battery referenced in the Methods (Physical and Chemical Validity) [34]. Twenty-two of the twenty-six listed entries were applied in this work. The four remaining entries compare the docked pose against the co-crystallised reference ligand, and none of them ran in any dataset reported here, because every pose was screened without a reference molecule supplied. Their columns are consequently absent from the results tables, and a pose is PB-valid only if it passes every applied check. The verdict set also covers two entries that a bare listing of the geometric tests omits, namely that the receptor file itself loads and that the ligand has not been placed beyond the maximum permitted distance from the protein.

Table 15: PoseBusters Validity Checks

Check	Description
Chemical validity and consistency	
File loads	The docked pose can be parsed into an RDKit molecule object.
Receptor file loads	The receptor structure, together with any retained cofactors, can be parsed so that the intermolecular tests have something to test against.
Sanitisation	The molecule passes RDKit's chemical sanitisation checks.
InChI convertible	The molecule can be converted to a standard InChI string.
All atoms connected	All atoms form a single connected component (no disconnected fragments).
No radicals	The molecule contains no unpaired electrons (radical atoms).
Molecular formula ¹	The molecular formula is identical to that of the reference ligand. Not applicable in this work and not run.
Bonds	The bond graph (atom connectivity) is identical to the reference ligand. Not applicable in this work and not run.

¹This entry and the three that follow it (bonds, tetrahedral chirality and double-bond stereochemistry) are the reference-based members of the battery. Each needs the co-crystallised ligand as a second input, which was supplied in none of the runs reported here, so all four were not applicable and were not run, and they contribute no columns to the results tables.

Table 15: PoseBusters Validity Checks (continued)

Check	Description
Tetrahedral chirality	Tetrahedral stereocentres match the reference configuration. Not applicable in this work and not run.
Double-bond stereochemistry	Double-bond (cis/trans) configuration matches the reference. Not applicable in this work and not run.
Intramolecular validity (ligand geometry and strain)	
Bond lengths	Every bond length lies within 0.75–1.25 of the RDKit distance-geometry bounds (25% tolerance).
Bond angles	Every bond angle lies within 0.75–1.25 of the distance-geometry bounds.
Planar aromatic rings	All atoms of 5- and 6-membered aromatic rings lie within 0.25Å of their mean plane.
Non-aromatic ring non-flatness	Non-aromatic 6-membered rings deviate from their mean plane by at least 0.05Å (they are not spuriously flattened).
Planar double bonds	The two carbons of aliphatic C=C bonds and their four substituents lie within 0.25Å of their mean plane.
Internal steric clash	Non-covalently bound atom pairs are no closer than 0.7 of the distance-geometry lower bound, the value implied by the clash tolerance of 0.3 set in the docking configuration used throughout this work.
Internal energy ratio	The UFF energy of the pose is at most 100× the mean energy of a 50-conformer ETKDG ensemble (no gross strain).
Intermolecular validity (ligand versus receptor)	
Protein–ligand maximum distance	The ligand lies within the maximum permitted separation from the receptor, so it has not been placed away from the protein altogether.
Minimum protein–ligand distance	Protein–ligand atom pairs are farther apart than 0.75× the sum of their van der Waals radii.
Minimum distance to organic cofactors	Ligand–organic-cofactor atom pairs exceed 0.75× the sum of their van der Waals radii.
Minimum distance to inorganic cofactors	Ligand–inorganic-cofactor atom pairs exceed 0.75× the sum of their covalent radii.

Table 15: PoseBusters Validity Checks (continued)

Check	Description
Minimum distance to waters ²	Ligand–water atom pairs exceed $0.75 \times$ the sum of their van der Waals radii.
Volume overlap with protein	Less than 7.5% of the ligand volume intersects the protein (van der Waals radii scaled by 0.8).
Volume overlap with organic cofactors	Less than 7.5% of the ligand volume intersects organic cofactors (radii scaled 0.8).
Volume overlap with inorganic cofactors	Less than 7.5% of the ligand volume intersects inorganic cofactors (radii scaled 0.5).
Volume overlap with waters	Less than 7.5% of the ligand volume intersects water molecules (radii scaled 0.5).

C.7 Interaction Fingerprints

Protein-ligand interaction fingerprints are used in the Results chapter to compare where and how each tool contacts the receptor, and the procedure behind them is documented here because it first appears there. Every docked pose is written into a single coordinate file together with its receptor, with the ligand forced to heteroatom records under one reserved residue name, and the resulting complex is profiled with PandaMap 4.1.0 [88] through its hybrid protein-ligand mapper. Detection is run in the package's standard mode rather than in its stricter experimental variant, which is not functional in this release, so the classification below is the one actually executed.

The contact classes and their thresholds follow the conventions established by PLIP [52] and are applied as heavy-atom distance criteria between the ligand and the receptor. A hydrogen bond is recorded between a ligand nitrogen or oxygen and a receptor nitrogen or oxygen separated by 2.4 to 3.5 Å. A hydrophobic contact is recorded between a ligand carbon and a carbon of alanine, valine, leucine, isoleucine, methionine, phenylalanine, tryptophan, proline or tyrosine within 4.0 Å. Ionic contacts, salt bridges and attractive-charge contacts between a

²The two water-related checks (minimum distance to waters and volume overlap with waters) were run together with the rest of the battery and are among the twenty-two applied entries, so both contribute verdict columns to the results tables. Receptor preparation left no crystallographic water molecules in any receptor file, so nothing the two checks were designed to catch can be present. The residual failures they report, namely 5,203 poses on minimum distance to waters and 270 on volume overlap with waters, come from the full 169,993-pose whole-protein comparison file, which includes 8,605 UniDock2 poses not displayed in the Results; the displayed rows account for the other 161,388 poses. The package attributes these failures to non-water receptor atoms. Every affected pose already fails at least one non-water check, so the set passing all twenty-two applied checks is identical to the set passing the twenty non-water checks and PB-validity is unaffected.

charged ligand atom and an aspartate or glutamate oxygen or an arginine, lysine or histidine nitrogen use 4.0 Å, and same-sign repulsion uses 6.0 Å. The aromatic family, namely pi-stacking, cation-pi, pi-cation, carbon-pi, donor-pi, amide-pi and alkyl-pi, uses 5.5 Å. Halogen bonds use 3.5 Å to an oxygen, nitrogen or sulfur acceptor, metal coordination uses 2.8 Å and covalent contacts use 2.1 Å to the cysteine, serine, lysine or histidine side-chain atoms that can form them. Atom pairs closer than 1.5 Å are discarded as clashes or coordinate errors. Angular criteria are not applied in this execution path, so a hydrogen bond is a distance-and-atom-type assignment and carries a nominal rather than a measured donor angle, and the aromatic classes are likewise geometric proximity assignments rather than ring-normal tests. Every fingerprint comparison in this work is therefore a comparison of contact patterns under one fixed and uniformly applied definition, not a claim about hydrogen-bond geometry.

Each detected contact is reduced to the key formed by its interaction class together with the residue name, residue number and chain identifier of the receptor residue involved, and the fingerprint of a pose is the set of those keys. Because the keys carry residue numbering, fingerprints only compare across tools if every tool is profiled against one numbering scheme. The whole-protein preparation used by some engines rennumbers residues sequentially from one, and one prepared receptor carries no chain identifiers at all, so left alone their keys would never match the author-numbered crystal reference and their measured native recovery would collapse towards zero. All poses of the calibration benchmark are therefore profiled against a canonical receptor set that carries author numbering and was verified heavy-atom coordinate-compatible with each tool's own docking receptor, so this relabels residues without moving any atom. A pose that yields no contact at all is retained with an empty fingerprint and counts as a zero, while a pose whose complex cannot be assembled or parsed is written to a separate error table and excluded from every figure, so that a mapping failure is never silently scored as an absence of contacts.

Only physically valid poses enter the analysis, and the pose set is capped by ranking depth before mapping. On the calibration benchmark the top five ranked poses of each tool per complex are retained, and on Orai1 the top three, with EquiBind ranked by its gnina refinement affinity because it emits no confidence score of its own. The per-complex fingerprint of a tool is the union of the fingerprints of its retained poses for that complex, so a contact counts once whether it appears in one retained pose or in all of them. The inferential unit throughout is the complex rather than the pose. Each tool contributes one value per complex, either that union or a per-complex mean count, and tools are compared on the complexes they share, with a Friedman omnibus across tools followed by Wilcoxon signed-rank pairwise tests under Holm or Benjamini-Hochberg correction. Reporting per pose would treat the several retained poses of one complex as independent observations, which they are not.

D Orai1 Receptor Model

The docking receptor is a homology model of the human Orai1 hexamer, built on the sixfold architecture first resolved in the *Drosophila melanogaster* Orai crystal structure [36]. Each of the six identical subunits contributes residues 66 to 288 (223 residues), spanning the tail of the cytoplasmic N-terminus, the four transmembrane helices M1 to M4 and the M4 C-terminal extension, giving 1,338 residues and 10,362 heavy atoms across the assembly. The model is apo, carrying no ligand, no Ca^{2+} and no other heteroatoms, and no pH-based protonation was applied to it at any stage. The supplied molecular-dynamics coordinates carry CHARMM hydrogens and CHARMM histidine states, and the preparation step removes every hydrogen and renames HSD, HSE and HSP to plain HIS, so the delivered receptor is a purely heavy-atom structure of 10,362 atoms. Polar hydrogens are then re-added by the PDBQT converter under its own geometric and atom-type rules rather than by a titration model. One qualification attaches to the Fr0 snapshot. Its AutoDock search receptor was the only one that required the permissive residue filter of the PDBQT converter, which dropped five residues carrying 41 heavy atoms, namely Ser93 in chains B and D, Leu248 in chain C, Phe253 in chain A and His256 in chain C, so that search ran against 1,333 residues and 10,321 heavy atoms rather than the full assembly. Every other frame carried the complete 1,338 residues, and PoseBusters screened all frames including Fr0 against the intact heavy-atom structure. Forty-two of the 240 Fr0 AutoDock poses lie within 8Å of a dropped residue, and two Synta-66 poses fail only the protein-distance and protein-overlap checks against an atom of the deleted Phe253, so the reported Fr0 AutoDock validity is marginally conservative for those poses.

The pore-lining signature residues are quoted in the human Orai1 numbering carried by the supplied homology model itself, which is offset from the *Drosophila* crystal numbering and is therefore not taken from that structure. R91 is the basic cytoplasmic residue associated with the R91W severe combined immunodeficiency phenotype, V102 is the hydrophobic gate and E106 forms the selectivity filter. Two of the three tested ligands act at or near the extracellular pore mouth and the selectivity filter, with GSK-7975A linked to filter geometry [68] and Synta-66 mapped to an extracellular TM1/TM3 and loop region adjacent to the filter [69]. The site of 2-APB is not experimentally resolved and its documented STIM1 arm is cytosolic, so the same expectation cannot be asserted for it on structural grounds. The three should not be described as sharing one proven binding site.

The receptor ensemble contains an equilibrated starting structure (Fr0) and three later molecular-dynamics snapshots (Fr300, Fr400 and Fr499). The files carry membrane-simulation features, including a periodic box and CHARMM-style residue and atom naming, which together indicate that the frames were drawn from a membrane-embedded simulation. The representation follows the structural and molecular-dynamics work of the contributing Orai group [37], [70]. Ensemble docking presents each ligand with several thermally sampled conformations and partly mitigates, but does not eliminate, the rigid-receptor approximation.

The provenance of the model stops at that point, and the gap is stated here rather than left

silent. The coordinates were supplied without the modelling program or the template Protein Data Bank identifiers used to build the homology model, without the force field name or version, without the lipid composition of the bilayer, without the solvent and ion treatment, and without the trajectory length, the identity of the replicate or the simulation times at which the four frames were taken. None of that metadata accompanied the files, so no part of it is reconstructed or inferred here beyond the CHARMM-style naming and periodic box just noted. Every statement in this work that depends on the receptor therefore rests on the coordinates as delivered, and the four frames are treated as four unlabelled conformational samples of one membrane simulation rather than as timed points along a characterised trajectory.

The four frames occupy separate molecular-dynamics coordinate systems, so raw $C\alpha$ displacement before receptor superposition mixes conformational change with rigid-body translation and rotation and is not interpreted as structural spread. That aligned comparison was carried out for the frame-spread statement reported in the Results chapter, where a Kabsch superposition on the 1,338 $C\alpha$ atoms still leaves the starting frame 5.90, 5.88 and 6.02 Å from Fr300, Fr400 and Fr499. The cross-frame pose analysis itself continues to use alignment-free residue-contact fingerprints folded onto residue number across the symmetric hexamer.

Pocket detection identifies the elongated ion pore as the top-ranked cavity in each frame, with M1 residues 66 to 110 and additional transmembrane residues in the more open frame. This region overlaps the functionally implicated pore and filter environment [68], [69], but it is not a conventional buried hydrophobic pocket. The complementary geometric finder returns hundreds of low-druggability cavities, illustrating why blind site selection is unusually difficult on this membrane channel.

E Experimental Ligand Pharmacology

The three experimental compounds introduced with the experimental dataset in Section 3.1.2 are described pharmacologically here in turn, in the order GSK-7975A, Synta-66 and 2-APB (docked in its neutral form, 2abp-NH₂).

GSK-7975A is a pyrazole-derived Orai blocker and the most selective of the three, though not a clean Orai1 tool in native tissue. Derler et al. report a half-maximal inhibitory concentration near 4 μ M for both Orai1 and Orai3 currents, with roughly 8 μ M against L-type voltage-gated calcium channels and above 10 μ M against the transient receptor potential channels tested, and that micromolar figure is the value the field works from for concentration selection [68]. Its sensitivity is tied to selectivity-filter geometry, either directly or allosterically, since the pore mutant Orai1-E106D is markedly less sensitive while the outer-pore triple mutant Orai1-D110/112/114A leaves potency intact, and 2-APB-gated Orai3 is similarly resistant [68]. That is a mutagenesis-derived link to the filter environment and not a resolved ligand pocket, since no ligand-bound Orai

structure places the compound. Extracellular application blocks the current, and the compound does not disturb STIM1 puncta formation or STIM1 to Orai1 coupling, so it acts on the channel rather than as an upstream inhibitor, although the cited evidence does not establish that the extracellular face is its only route of access [68]. Its block develops more slowly than that of trivalent lanthanum [68] but far faster than BTP2, whose block of the calcium release-activated calcium current takes more than two hours to reach its full effect [92]. Selectivity liabilities are real and not confined to TRPV6, which it blocks at 10 μM as completely and as rapidly as lanthanum [68]. In mouse aorta its most sensitive action is on non-selective cation channel-mediated contraction, near 1.2 μM against roughly 12 μM for the voltage-gated component, so it behaves there functionally as a repolarising agent and leaves contractile store refilling essentially untouched [39]. In a preparation with appreciable non-selective or voltage-gated conductance a GSK-7975A effect is therefore not automatically an Orai1 effect.

Synta-66 is an aryl-amide Orai1 blocker with a markedly different isoform fingerprint. It abolishes Orai1 function while potentiating Orai2 and leaving Orai3 untouched, and concatemeric Orai1–Orai1 dimers are inhibited to the same extent as Orai1 itself whereas Orai1–Orai2 and Orai1–Orai3 heterodimers are only partially inhibited and Orai2–Orai3 dimers are not inhibited at all, which makes it a probe of Orai1 dependence rather than a general Orai blocker [91]. Structurally it retains an aryl amide core flanked by a fluoropyridine and a 2,3-dimethoxyaryl moiety. Waldherr et al. localise its action to an extracellular site formed by the outer parts of TM1 and TM3 together with extracellular loops 1 and 3, immediately adjacent to the selectivity filter, with docked hydrogen bonds to Asp110 and His113, reduced inhibition in the loop1 mutants L109D, H113G and Y115G and the loop3 mutants F199G, P201G and L202G, and molecular dynamics showing the compound anchoring progressively deeper towards the filter over about 100 ns [69]. Additional sites are not excluded, but the peer-reviewed evidence points to the extracellular face rather than to both faces of the hexamer. Consistent with a filter-proximal site, E106D largely abolishes the block, STIM1 puncta formation and STIM1 to Orai1 coupling are unaffected, and the block depends on calcium-selective permeation, since in divalent-free sodium solution 10 μM inhibits wild-type Orai1 only partially whereas the same protocol in calcium gives complete block [69]. Potency should therefore not be quantified under divalent-free conditions or in non-selective mutants. In glioblastoma lines the compound blocks store-operated entry almost completely at 1 and 10 μM in A172, LN-18 and U-87 MG without impairing viability or migration, and at 10 μM it modestly increased growth of LN-18 and U-87 MG, significantly for LN-18 at 48 h, while slightly increasing temozolomide sensitivity in U-87 MG [69].

The docking input 2abp-NH2 is native 2-APB (2-aminoethoxydiphenyl borate) in its neutral protonation state, so the pharmacology below characterises the docked molecule directly. 2-APB is the oldest and by far the least selective of the three and is better described as a bimodal modulator with two points of attack than as an inhibitor. On the channel its sign reverses with concentration, overexpressed Orai1 being potentiated around 5 μM and inhibited above roughly 40 μM , Orai2 being less sensitive to that block, and Orai3 being potentiated even at high concentration [68]. On Orai3 it does more than gate the channel, widening the minimum

pore from about 3.8 Å in the store-operated state to more than 5.34 Å and destroying calcium selectivity so that monovalent ions permeate in both directions [90]. A smaller dilation of the STIM1-activated Orai1 pore has also been described, but the numerical value usually quoted for it does not originate in the source cited here, so it is left unquantified. It also acts on the sensor. Wei et al. show that it leaves STIM1 to STIM1 interactions unchanged while enhancing the intramolecular coupling between CC1 and SOAR, locking STIM1 towards its autoinhibited resting state, an effect abolished by L258G and L416G, and that it directly inhibits the constitutively active Orai1 mutants P245T and V102A whereas the constitutively active V102C mutant, when unbound from STIM1, is not inhibited at all, and that it indirectly impairs functional STIM1 to Orai1 coupling through the ETON-region residue K85 [38]. Two consequences matter for interpretation. Inhibition of puncta and inhibition of store-operated entry are dissociable, since co-expression of Orai1 abolishes the puncta effect while the entry block persists, so a 2-APB-sensitive signal identifies neither module [38]. Application timing also changes the result, pre-incubation before store depletion inhibiting about 77% against about 57% for application afterwards [38].

Read as a set, the three ligands therefore span a micromolar Orai1 and Orai3 pore block, an Orai1-selective extracellular block with an internal Orai2 contrast, and a bimodal modulator that acts both on the pore and on the STIM1 sensor. The evidence for GSK-7975A and Synta-66 converges on the extracellular pore mouth and the selectivity filter rather than on the conduction pathway, with Synta-66 mapped to a specific extracellular region and GSK-7975A linked to filter geometry without a resolved pocket, which is precisely what the placement criterion used in the results chapter encodes, whereas the site of 2-APB is not experimentally resolved and its STIM1 arm is cytosolic, so for that compound the criterion is a uniform study assumption rather than an established fact. Their differing mechanisms are also why the biologically defensible region is not identical for the three compounds. For the docking comparison this matters twice over, once because a chemically valid pose inside the radially unbounded transmembrane exclusion slab is implausible for the two compounds whose evidence points to the extracellular face and only assumed implausible for 2-APB, and once because the size and flexibility of the three scaffolds differ enough to load the tools unevenly. Should an orthogonal chemotype be needed to cross-check an Orai1-dependence conclusion in future work, the pyrazole-core compound C79413 and the indole-like C63368, which block the CRAC current at 2.2 and 7.0 μ M rapidly and reversibly and are unaffected by the mutations that weaken Synta-66, occupy a different region of the pore and are better characterised for that purpose than 2-APB [40]. The docked structure represents the neutral protonation state of 2-APB. Its aminoethyl nitrogen is basic enough that a protonated fraction is expected at physiological pH, which is an inference from the chemistry rather than a measured value taken from a cited source, so poses for the cationic form are not sampled here and remain for separate validation.

F Ranking Quality Under the Superseded Site-Restricted Search

Figures 27 to 30 turn from whether a near-native pose exists to whether a tool can surface it. For every complex, poses are sorted by a ground-truth quality criterion and assigned an oracle rank, where rank 1 is the best pose in that tool's own pool. Each panel then plots the median oracle rank at every tool-assigned rank position. Perfect ordering follows the diagonal, while random ordering centres near $(N+1)/2$, which is 15.5 for the thirty poses DiffDock and EquiBind supply and 15.0 for AutoDock Vina, whose pools have a median size of twenty-nine.

Because oracle rank is defined within each tool's own pose pool, these panels measure ranking fidelity rather than absolute accuracy and must be read with the preceding recovery analyses. Figure 29 is different. It reports physical-validity frequency by rank. This appendix deliberately compares native Vina ranking, raw DiffDock confidence ranking and gnina-ranked, gnina-optimised EquiBind output. Only the AutoDock and DiffDock arms differ from the main Results, which use AutoDock Vina + gnina and DiffDock + smina, whereas the gnina-ranked, gnina-optimised EquiBind arm used here is the same variant the main Results report. The AutoDock and DiffDock ranking values in this appendix should therefore not replace the main headline estimates.

A second and larger scope limit applies to the whole chapter. Every figure, table and statistic in it was computed on the earlier crystal-ligand-boxed run and none has been regenerated for the whole-protein search that supplies every headline estimate in the Results chapter, so the supersession covers all three tools and not only AutoDock. It bites unevenly. DiffDock and EquiBind do search blindly and their pose pools, and with them every purely geometric quantity, are identical in the two runs, which is why their median oracle ranks and their Kendall τ values on as-placed crystal RMSD and on best-fit Kabsch RMSD are the same numbers under either configuration. Physical validity is not carried over in the same way, because PoseBusters re-evaluates each pose against the receptor preparation of its own run, so every validity-dependent quantity here, namely the validity-by-rank panel and its table, the validity concordance row and the combined criterion that ranks validity first, belongs to the boxed run for DiffDock and EquiBind as much as for AutoDock Vina. On the whole-protein receptor those validity numbers move, for example raw DiffDock rank-1 validity from about 39% to about 41%. For AutoDock Vina the supersession is deeper still, since the search itself differed. The difference is not cosmetic. Recomputed on the blind whole-receptor pool the AutoDock placement advantage does not survive on the raw Vina arm shown here, since DiffDock is then ahead of raw AutoDock Vina on both as-placed crystal concordance, at a median Kendall τ of 0.154 against 0.140, and best-fit Kabsch concordance, at 0.122 against 0.053, and raw AutoDock Vina is no longer separable from EquiBind on either criterion. That reversal is specific to the raw arm and must not be read as applying to the published one, because the gnina-rescored AutoDock variant that the main Results designate as dominant leads DiffDock on as-placed crystal concordance at 0.177 against

0.154 on the same pool, and only on best-fit Kabsch concordance, at 0.048 against 0.122, does DiffDock stay ahead of it. This chapter is therefore retained as a diagnostic of ranking behaviour under a site-restricted search, and neither its tool orderings involving AutoDock nor any of its validity-based values for any tool may be carried over to the blind configuration reported in the main text.

By as-placed crystal RMSD, the criterion that rewards putting the ligand in the right place, AutoDock Vina was the only tool whose ordering carried appreciable signal in this boxed diagnostic. Its rank-1 pose was the second best of its roughly thirty, its rank-2 pose the fourth best and its rank-3 pose the sixth best, so its curve lay closest to the diagonal at early ranks before rising towards the mid-range around $k \approx 15$. That advantage belongs to the site-restricted search and does not carry over to the blind whole-protein configuration reported in the main text. DiffDock and EquiBind both start far above the diagonal, with their most confident pose landing only at median oracle rank 9 and 10 respectively, so their ordering is weak, although the distance from the mid-range value shows it is not entirely absent. A tool's single most trusted pose is therefore rarely its own best-placed pose, and the ranking signal that does exist is concentrated at the top of each list.

Figure 28 repeats the construction with best-fit Kabsch RMSD as the oracle criterion, which measures the residual conformational difference remaining after the pose is optimally superposed on the crystal ligand, that is its internal form with placement removed. Here every curve moves towards the mid-range and the advantage Vina held on placement in the boxed run disappears. Its rank-1 pose was then only the fifth best of its set by form, while DiffDock and EquiBind placed their top pose at oracle rank 10 and 12 respectively. On form the physics-based and learned orderings were statistically indistinguishable, and DiffDock's median concordance was in fact nominally the higher of the two ($\tau = 0.122$ against Vina's 0.108, $p = 0.504$), while EquiBind's gnina order carried essentially no form signal ($\tau = 0.048$). No tool ordered its own poses well by pure internal shape, which shows that the ranking signal Vina possessed under a site-restricted search concerned where the ligand sits rather than how its conformation is resolved.

Figure 29 changes the question, because PoseBusters validity is binary and admits no oracle rank. Instead, the panel plots, at each rank position, the percentage of rank- k poses that pass the PoseBusters checks, with each tool's overall valid rate drawn as a dotted random-order reference. A useful validity ranker starts above its dotted line and decays towards it as rank worsens. AutoDock Vina is essentially flat at about 94 to 97% across the fifteen ranks plotted, sitting on its own high baseline because nearly every Vina pose is already valid and there is little left to sort. DiffDock front-loads validity in the right direction but from a low level, falling from about 39% at rank 1 to its 23% baseline, so that even its most confident pose is more likely invalid than valid. The gnina-optimised, gnina-ranked EquiBind variant shows the clearest ranking signal on the plot, declining broadly, if not strictly monotonically, from about 77% at rank 1 towards its 52% baseline, which makes gnina affinity an effective validity sorter for these EquiBind poses even though it is a weak geometry sorter. Every percentage in this paragraph belongs to the boxed run, for the learned tools as well as for Vina, since validity was rescored

Rank vs oracle — Crystal RMSD

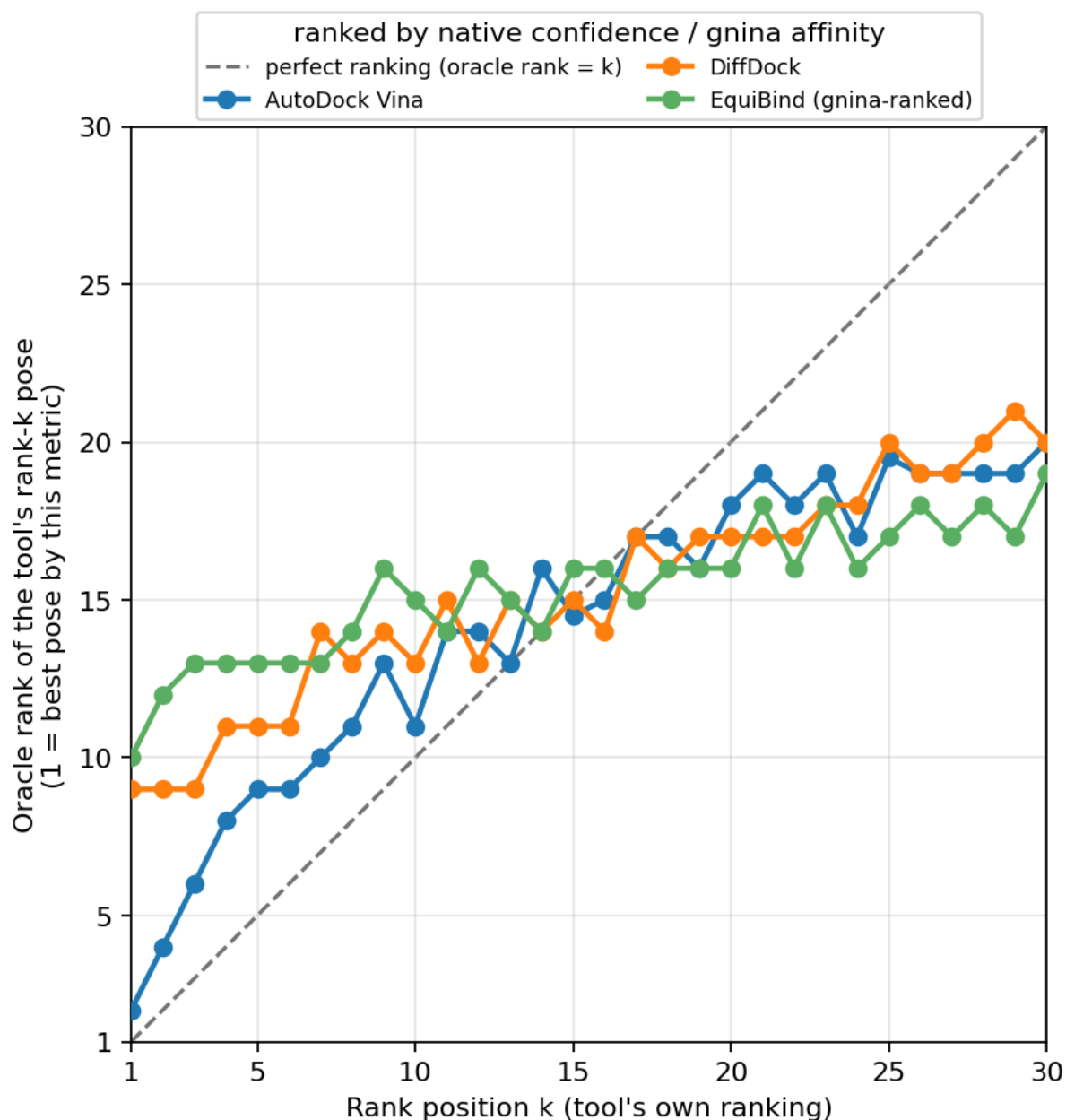


Figure 27: Rank versus oracle for as-placed crystal RMSD. The AutoDock Vina curve comes from the superseded crystal-ligand-boxed search and not from the blind whole-protein configuration of the Results chapter. The horizontal axis is each tool's own rank position k and the vertical axis the median oracle rank of the pose it placed at k , where oracle rank 1 is the pose closest to the crystal ligand. The dashed diagonal is perfect ranking and the random-ordering reference is about 15, at $(N+1)/2$ for a pool of N poses

Rank vs oracle — Kabsch RMSD

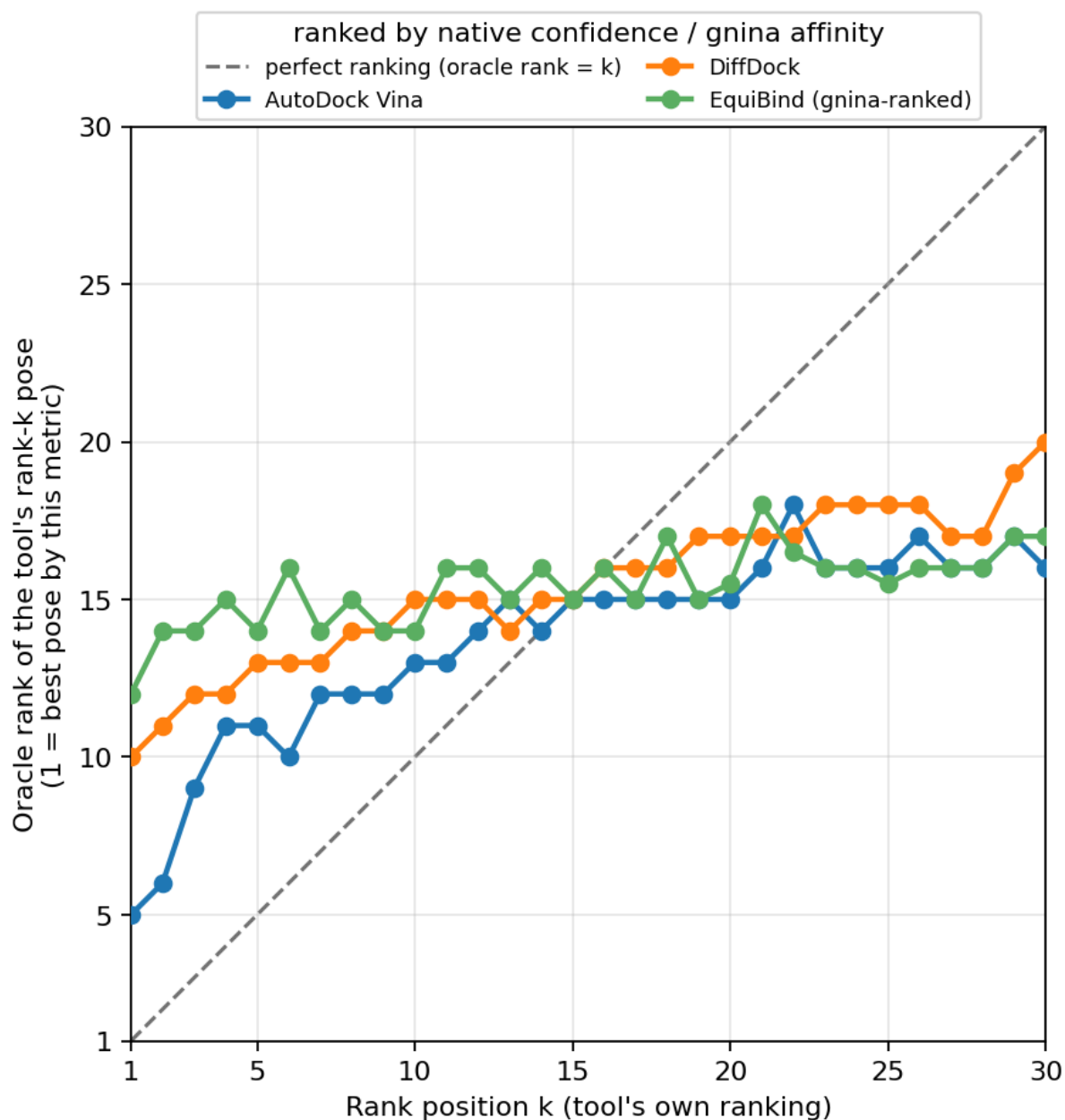


Figure 28: Rank versus oracle for best-fit (Kabsch) RMSD, representing internal form after optimal superposition. The AutoDock Vina curve is superseded, coming from the crystal-ligand-boxed search rather than the blind whole-protein one. Axes as in Figure 27

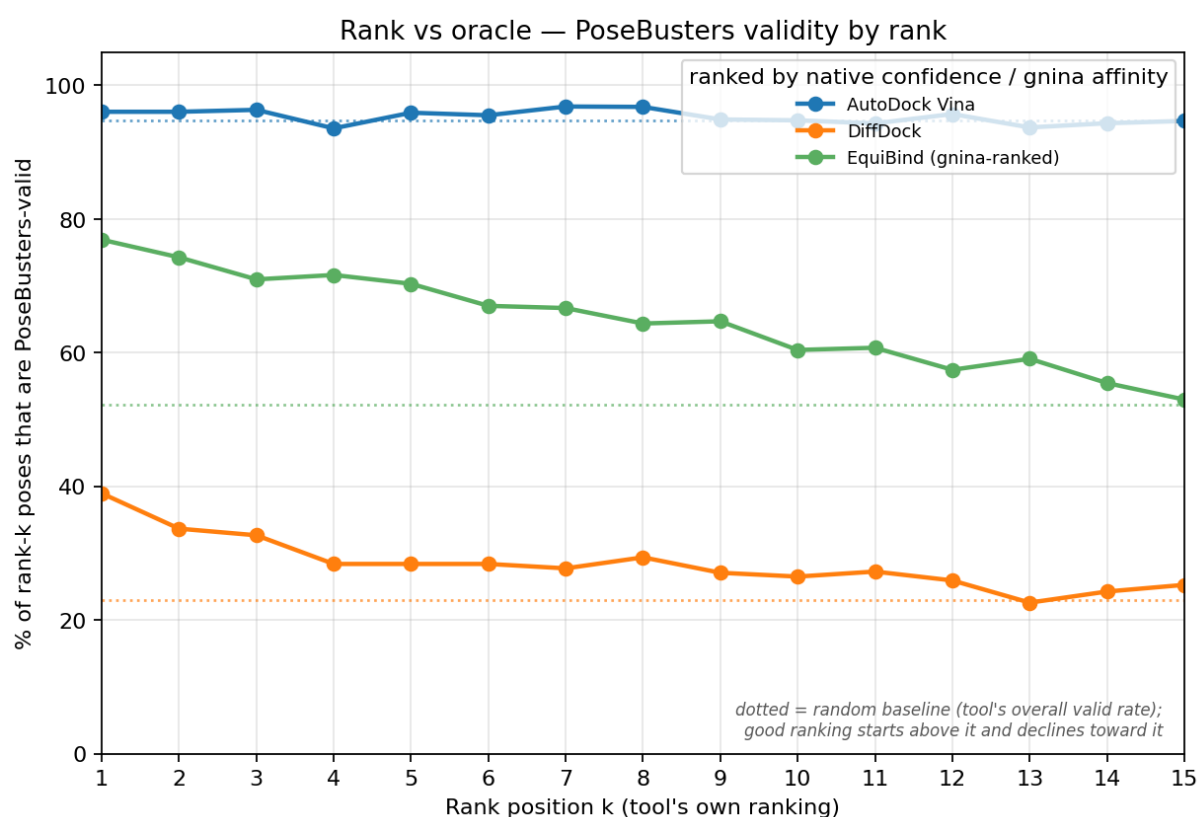


Figure 29: PoseBusters validity by rank. All three curves are superseded, coming from the crystal-ligand-boxed run rather than the blind whole-protein one, because PoseBusters validity was re-evaluated against each run's receptor preparation and so differs for DiffDock and EquiBind as well as for AutoDock Vina. The solid curve is the percentage of rank-k poses that are PoseBusters-valid and the dotted line is each tool's overall valid rate, the random-order reference. Good ranking starts above the dotted line and declines towards it. This panel spans ranks 1 to 15 only, unlike the three rank-versus-oracle panels of this chapter, which run to rank 30

against that run's receptor. The shape of each curve survives the change of receptor, but its level does not, so these values are read as a ranking diagnostic and never as validity rates for the whole-protein configuration reported in Table 4.

Rank vs oracle — Combined (valid + close + form)

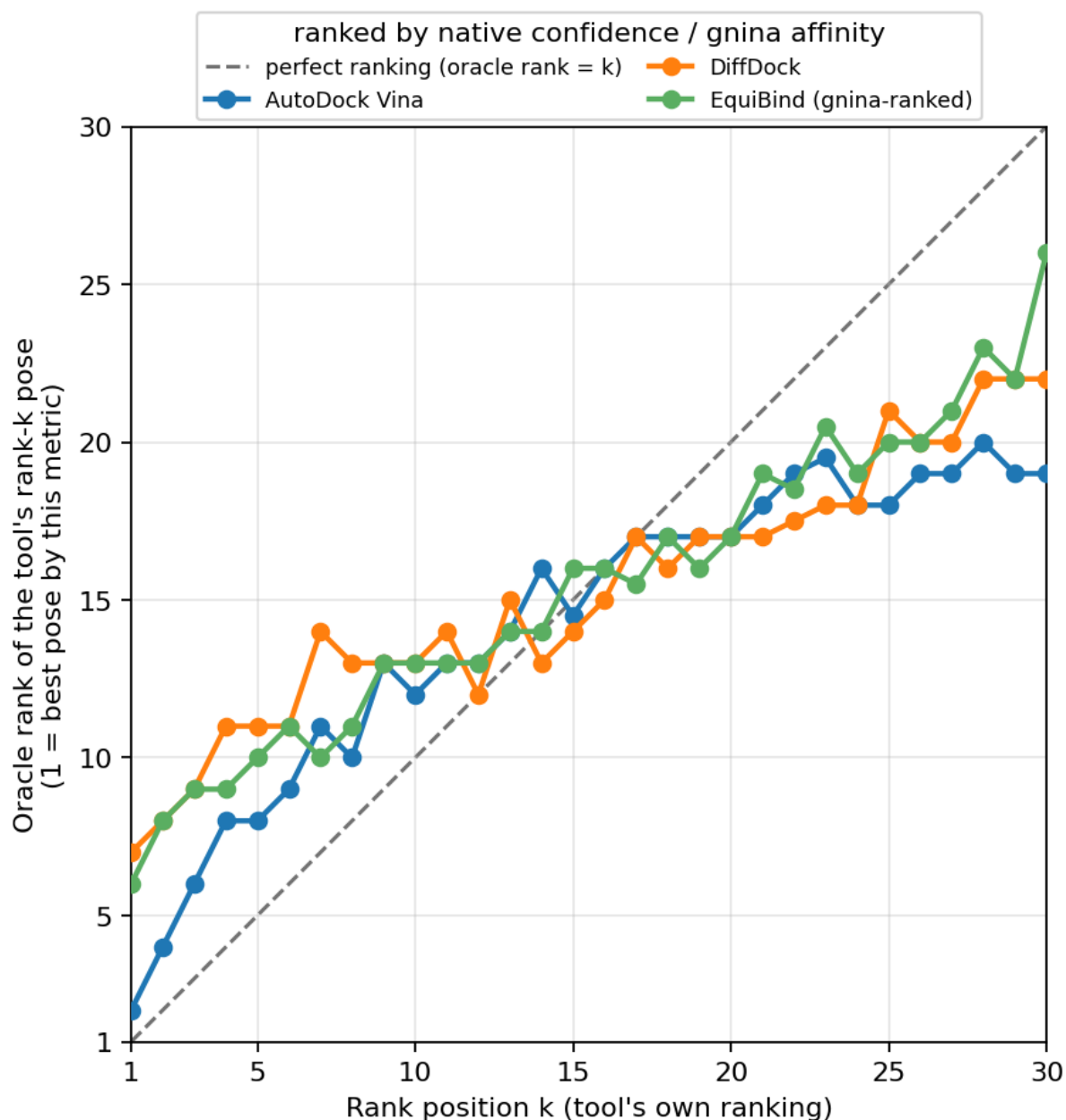


Figure 30: Rank versus oracle for the combined deployment criterion (validity, then as-placed RMSD, then form). All three curves are superseded, coming from the crystal-ligand-boxed run rather than the blind whole-protein one, because the criterion ranks physical validity first and validity was re-evaluated against each run's receptor preparation. PoseBusters-invalid poses are penalised below valid poses before valid poses are ordered by geometry. Axes as in Figure 27

Figure 30 applies a deployment-relevant oracle criterion, in which a global penalty forces every PoseBusters-invalid pose below every valid one within a complex and the surviving valid poses are then ordered by placement and form together. It is an evaluation target rather than a selector usable in practice, since it relies on crystal RMSD, which is unavailable at real deployment time. The composite is also structurally opinionated, prioritising validity lexicographically and then summing crystal and Kabsch RMSD with equal weight, so that form is partly double-counted because as-placed RMSD already contains conformational error, and every conclusion drawn from the panel is conditional on that prespecified choice. In the boxed diagnostic Vina's rank-1 pose sat at oracle rank 2, almost unchanged from its crystal curve because it was already about 95% valid and the validity-first key barely reshuffled it. DiffDock improved slightly to oracle rank 7, helped by the validity front-loading. The clearest movement was EquiBind, whose rank-1 pose rose to oracle rank 6, better than its crystal figure of 10 or Kabsch figure of 12 and ahead of DiffDock. Because the composite rewards validity first, EquiBind's strong validity ordering lifted its whole-list concordance to a level not detectably different from Vina's, although Vina retained the better descriptive rank-1 median. Every comparison in this paragraph belongs to the boxed run for all three tools, because the composite is keyed on validity and validity is run-specific.

The four line charts are deliberately free of inference marks, so the formal comparison rests on three layers of distribution-free statistics computed per complex. First, a Kendall τ -b correlation between each tool's rank order and the oracle badness order reduces every complex to a single concordance score, where +1 is perfect agreement, 0 is chance and -1 is a reversed ordering. A complex contributes only when it has at least three poses with non-constant rank and badness, which is why the validity τ is defined solely on the mixed-validity complexes that contain both valid and invalid poses, namely 94 for Vina, 229 for DiffDock and 215 for EquiBind in this boxed run, against 57, 236 and 216 on the whole-protein receptor. The resulting per-tool medians are collected in Table 16.

Table 16: Median Per-Complex Kendall Tau-b between Tool Ranking and Oracle Badness. Every entry is computed on the crystal-ligand-boxed run rather than on the blind whole-protein configuration reported in the Results chapter. The whole AutoDock Vina column is superseded, and so are the validity and combined rows for all three tools, since both depend on physical validity, which was re-evaluated against each run's receptor. The DiffDock and EquiBind entries on crystal RMSD and on Kabsch RMSD are the only ones that carry over unchanged

Criterion	AutoDock Vina	DiffDock	EquiBind (gnina-ranked)
Crystal RMSD (as placed)	0.271	0.154	0.140
Kabsch RMSD (form)	0.108	0.122	0.048
PoseBusters validity	0.064	0.202	0.461

Table 16: Median Per-Complex Kendall Tau-b between Tool Ranking and Oracle Badness (continued)

Criterion	AutoDock Vina	DiffDock	EquiBind (gnina-ranked)
Combined (valid + close + form)	0.273	0.214	0.333

A Friedman omnibus with Kendall's W as its effect size then asks whether the three tools' per-complex τ differ, and a Wilcoxon signed-rank test with Holm correction and a matched-pairs rank-biserial effect size resolves the individual pairs. Three of the four omnibus tests are highly significant ($p < 0.0001$) and the fourth, the combined criterion, reaches $p = 0.00037$, but the large sample simply confers high power, and it is Kendall's W that measures how large the systematic between-tool effect actually is. For the geometric criteria Kendall's W is only 0.050 for crystal, 0.035 for Kabsch and 0.026 for the combined score, which indicates a small between-tool effect against considerable complex-to-complex variation, so the population ordering does not translate into a reliable per-complex ranking. Only physical validity produces a substantial and stable ordering, with $W = 0.329$, and even that rests on just the 41 complexes that are mixed-validity for all three tools at once. On the pairwise tests, which refer throughout to the boxed run for AutoDock and, on the validity and combined criteria, for the learned tools as well, Vina beats DiffDock and DiffDock beats EquiBind on crystal RMSD (rank-biserial +0.25 and +0.24, with Vina over EquiBind at +0.44), Vina and DiffDock tie on Kabsch while both beat EquiBind, EquiBind dominates both on validity (rank-biserial -0.80 against Vina and -0.86 against DiffDock), and on the combined criterion Vina beats DiffDock ($p = 0.012$) and EquiBind beats DiffDock ($p = 0.020$) while Vina and EquiBind show no significant difference ($p = 0.432$), which is an absence of a detected difference rather than demonstrated equivalence. One further caution applies to the tails of the curves, where the number of contributing complexes falls unequally across tools, from 303 at rank 1 to 149 for Vina, 285 for DiffDock and 302 for EquiBind at rank 30, so deep-rank comparisons rest on different subsets and are read only qualitatively. The persistence of signal into the tail is nonetheless visible, for example in EquiBind's combined median of 26 at rank 30, which shows its lowest-ranked poses being pushed towards the bottom of the pool.

Three findings follow, each requiring careful framing. AutoDock Vina is the strongest self-ranker for as-placed RMSD in this diagnostic, but the effect is a population tendency rather than a per-complex guarantee. Its use of the same empirical function during search and final ordering is a plausible mechanism for self-consistency [22], [50], but that mechanism is not directly tested here.

Second, both learned ranking pipelines retain a large selection gap. DiffDock's native confidence and EquiBind's gnina re-ranking place their top pose only around the ninth or tenth best of thirty by as-placed crystal RMSD, better than chance but far from the oracle. This is consistent with reported confidence-calibration and generalisation limitations for early learned docking

systems [24], [34]. Third, internal form is difficult for every score. All three rank Kabsch form less effectively than as-placed RMSD, and gnina ranking carries little form signal for EquiBind.

The single largest number in the analysis, gnina-ranked EquiBind's validity τ of 0.461, is genuine but must not be read as evidence that EquiBind is intrinsically the better-ranking tool, because it is confounded in three ways. The validity τ is defined only on mixed-validity complexes, and EquiBind, valid barely half the time overall, contributes 215 such information-rich complexes whereas Vina, valid about 95% of the time, contributes only 94, and those 94 are Vina's hardest cases, so the three tools are scored on non-comparable subpopulations with only 41 in common. The ranker doing the work is gnina, a convolutional network trained to recognise realistic protein–ligand geometry, so its validity enrichment is an empirical property rather than something it was explicitly trained for, and the result largely restates that a geometry-aware model orders validity better than scores never aimed at it. And because the gnina poses are energy-minimised, gnina both optimises and ranks the EquiBind poses, so this is not a clean rescoring-only comparison nor a measurement of intrinsic EquiBind behaviour. The honest reading is that a validity-aware rescorer recovers usable ordering from a model that emits none, not that EquiBind holds an intrinsic ranking advantage, and this is single-model rescoring rather than any consensus effect.

The combined-criterion result for Vina and EquiBind is likewise structural rather than evidence of equivalence. The validity-first composite favours the gnina-ranked EquiBind subset because gnina both optimises and orders those poses, while Vina remains descriptively stronger at rank 1. The appropriate conclusion is limited to this diagnostic, namely that gnina can enrich valid EquiBind poses within mixed-validity complexes, but that does not overcome EquiBind's low absolute near-native recovery or establish prospective parity with Vina.

Together, these panels illustrate the established separation between sampling and scoring [48], [50]. The main recovery curves show that acceptable poses are sometimes present below rank 1, while the oracle-rank curves show that the tested scores do not reliably surface them. Under the site-restricted search diagnosed here Vina retained the strongest as-placed-RMSD ordering, a standing that the blind whole-protein configuration of the Results chapter does not reproduce, raw learned output exhibits the physical-validity failures emphasised by PoseBusters [34], and gnina demonstrates that an external learned score can recover some useful ordering [65]. Every continuous ranking result remains a population tendency, and the EquiBind validity result describes gnina's behaviour on a favourable mixed-validity subset rather than an intrinsic EquiBind capability.

Tables 17 and 18 provide the numerical data for Figures 27 to 30. Table 17 lists median oracle rank for the crystal, Kabsch and combined criteria of Figures 27, 28 and 30. Table 18 lists PoseBusters validity by rank for Figure 29.

Table 17: Median Oracle Rank at Positions 1–15. Every entry is computed on the crystal-ligand-boxed run rather than on the blind whole-protein configuration reported in the Results chapter. All three Vina columns are superseded, and so are the three combined columns (Cb) for the learned tools, since that criterion ranks physical validity first and validity was re-evaluated against each run’s receptor. The DiffDock and EquiBind crystal (C) and Kabsch (K) columns are identical in the two runs. Entries are exact medians and are not rounded, so a half-integer appears wherever the number of contributing complexes is even

Rank k	Vina (C)	DiffDock (C)	EquiBind (C)	Vina (K)	DiffDock (K)	EquiBind (K)	Vina (Cb)	DiffDock (Cb)	EquiBind (Cb)
1	2	9	10	5	10	12	2	7	6
2	4	9	12	6	11	14	4	8	8
3	6	9	13	9	12	14	6	9	9
4	8	11	13	11	12	15	8	11	9
5	9	11	13	11	13	14	8	11	10
6	9	11	13	10	13	16	9	11	11
7	10	14	13	12	13	14	11	14	10
8	11	13	14	12	14	15	10	13	11
9	13	14	16	12	14	14	13	13	13
10	11	13	15	13	15	14	12	13	13
11	14	15	14	13	15	16	13	14	13
12	14	13	16	14	15	16	13	12	13
13	13	15	15	15	14	15	14	15	14

Table 17: Median Oracle Rank at Positions 1–15 (continued)

Rank k	Vina (C)	DiffDock (C)	EquiBind (C)	Vina (K)	DiffDock (K)	EquiBind (K)	Vina (Cb)	DiffDock (Cb)	EquiBind (Cb)
14	16	14	14	14	15	16	16	13	14
15	14.5	15	16	15	15	15	14.5	14	16

Table 18: PoseBusters Validity by Rank. Every column is superseded, being computed on the crystal-ligand-boxed run rather than on the blind whole-protein configuration reported in the Results chapter. This holds for DiffDock and EquiBind as well as for AutoDock Vina, because PoseBusters validity was re-evaluated against each run's receptor preparation. On the whole-protein receptor the raw DiffDock column reads 41, 36, 34, 30, 29, 32, 30, 31, 28, 28, 29, 28, 25, 26 and 27 across ranks 1 to 15

Rank k	AutoDock Vina	DiffDock	EquiBind (gnina-ranked)
1	96	39	77
2	96	34	74
3	96	33	71
4	94	28	72
5	96	28	70
6	96	28	67
7	97	28	67
8	97	29	64
9	95	27	65
10	95	26	60
11	94	27	61
12	96	26	57
13	94	23	59

Table 18: PoseBusters Validity by Rank (continued)

Rank k	AutoDock Vina	DiffDock	EquiBind (gnina-ranked)
14	94	24	55
15	95	25	53

G Dataset

The PoseBusters benchmark provides 308 high-quality protein-ligand crystal structures released after 2021 for testing generated poses [34]. Every figure, table and count in this chapter describes those 308 entries, which are the ones enumerated in `posebusters_pdb_ccd_ids.txt`, published by the PoseBusters authors through the project repository. The copy of the distribution held for this work also ships a larger list of 428 entries in `posebusters_benchmark_set_ids.txt`, the set as it stood in the earlier preprint release, from which the benchmark authors withdrew 120 entries whose ligands lie within 5.0 Å of a protein symmetry mate and therefore carry crystal contacts. The release window reduces exact-entry overlap with the nominal training cutoff of early learned docking models, but it does not exclude related proteins, ligands, pockets or templates. The present study maps the chemical and structural breadth of the source set and compares it with the three Orai1-related docking inputs.

G.1 Composition and Provenance

Each of the 308 entries pairs one receptor with one cognate ligand, every complex carrying a distinct PDB identifier and a distinct ligand (308 unique PDB IDs and 308 unique chemical-component codes)¹. The crystallographic pose is the experimentally validated reference for scoring every docked pose, while the generated start conformer mimics the ligand information a docking engine sees at runtime.

The experimentally resolved ligand conformations provide a reference for symmetry-corrected pose comparison. However, a later PDB release date is not proof that the biological or chemical pattern was absent from training data. Homology-, pocket-, scaffold- and template-similarity analyses are needed before making a strong no-leakage claim. Multiple crystallographic copies of the same ligand within a structure are handled by the stated symmetry-corrected comparison.

¹For every entry the benchmark provides four files, namely the receptor stripped of solvent but retaining its cofactors (`*_protein.pdb`), all crystallographic instances of the ligand of interest (`*_ligands.sdf`), the single chosen crystal instance that serves as the ground-truth answer pose (`*_ligand.sdf`), and one computer-generated starting conformer produced with RDKit's ETKDGV3 algorithm followed by UFF minimisation (`*_ligand_start_conf.sdf`).

G.2 Ligand Physicochemical Profile

The 308 source ligands cover a wide region of chemical space. The median ligand has 24 heavy atoms, with an interquartile range of 16 to 32 and a full range of 6 to 64, and a median molecular weight of 359 Da. The distributions are right-skewed and extend beyond 600 Da, reflecting peptides, nucleotides, sugars and cofactor-like molecules alongside conventional small molecules (Figure 31). Table 19 summarises the corresponding descriptors. The median ligand has five rotatable bonds, six hydrogen-bond acceptors, three donors and a topological polar surface area of 107 Å².

Ligand attribute distributions — PoseBusters Benchmark (n=308)

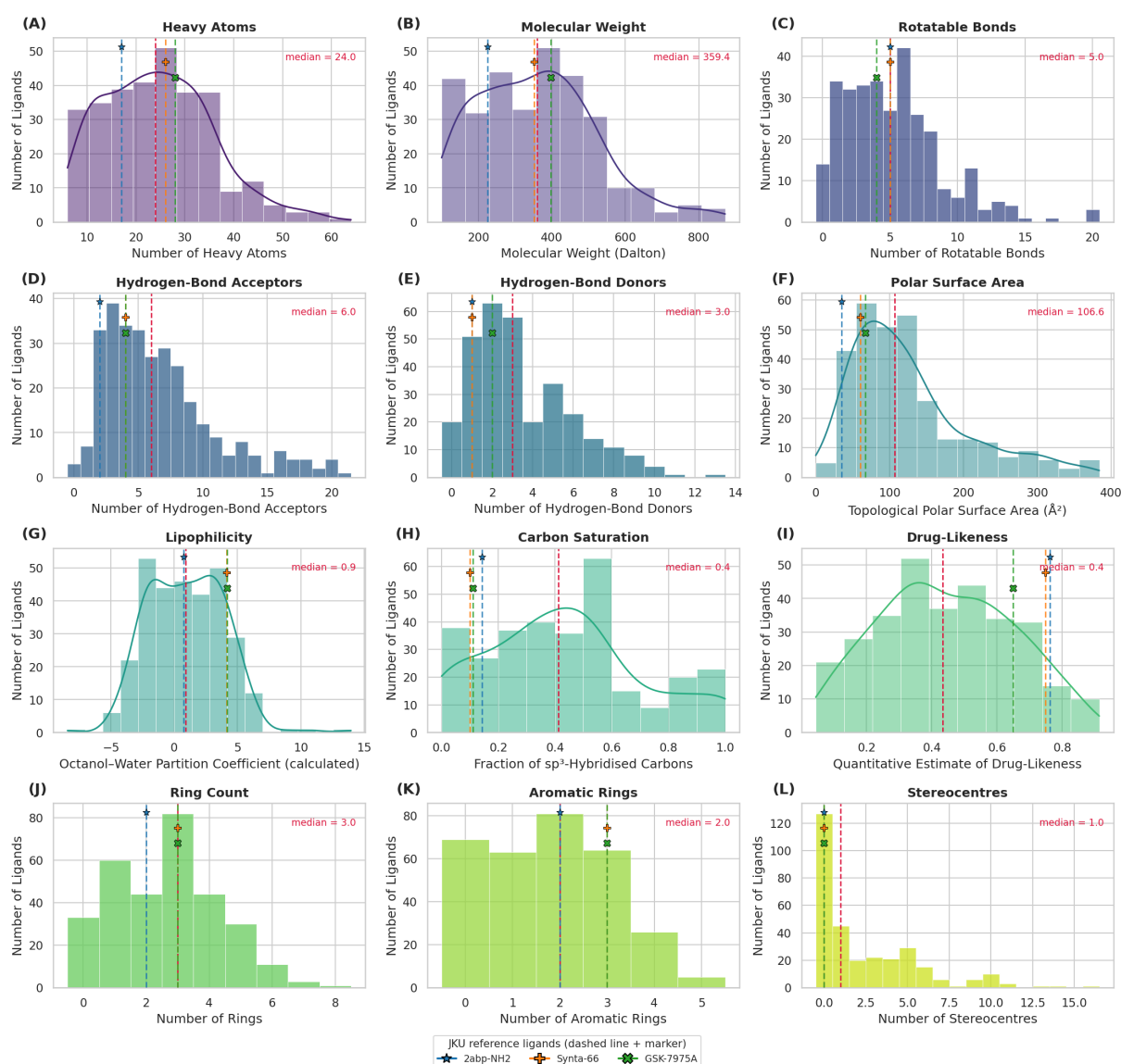


Figure 31: Univariate distributions of twelve physicochemical attributes across the 308 benchmark ligands

Table 19: Distribution of Principal Ligand Descriptors in the PoseBusters Benchmark

Descriptor	Median	IQR (25–75 %)	Range (min–max)
Heavy atoms	24	16–32	6–64
Molecular weight (Da)	359	236–462	100–872
Rotatable bonds	5	2–7	0–20
H-bond acceptors	6	3–9	0–21
H-bond donors	3	2–5	0–13
TPSA (Å ²)	107	66–160	0–384
cLogP	0.9	–1.6–3.2	–8.4–14.0
Fraction sp ³ C	0.41	0.23–0.57	0.00–1.00
QED	0.43	0.29–0.60	0.04–0.91
Ring count	3	1–4	0–8

G.3 Drug-Likeness

Drug-likeness metrics confirm that the set deliberately reaches past classical small-molecule drug space. Under Lipinski's rule of five,² 83% of ligands incur at most one violation and 65% none at all, yet a substantial 17% violate two or more criteria (Figure 32). The quantitative estimate of drug-likeness (QED) sits at a median of just 0.43, and only 67% of ligands clear the 0.34 threshold usually called “attractive”. This spread is intentional. Keeping polar, charged and oversized ligands stresses each method's ability to generalise rather than rewarding performance on an easy, lead-like subset.

G.4 Elemental and Charge Composition

The elemental make-up widens the challenge further (Figure 33). Roughly a quarter of the ligands contain phosphorus (25%) or sulfur (24%), 22% carry at least one halogen, and metal atoms are essentially absent from the ligand records. Net formal charges are uncommon, since only 3% of ligands are charged and the vast majority are neutral, but the heteroatom count is high and broadly spread (median 8, reaching nearly 30), matching the many nucleotide- and

²Lipinski's “rule of five” is a heuristic for oral drug-likeness. Poor absorption or permeation is more likely when a compound has more than five hydrogen-bond donors, more than ten hydrogen-bond acceptors, a molecular weight above 500 Da, or a calculated logP (cLogP) above 5, each threshold being a multiple of five. See C. A. Lipinski, F. Lombardo, B. W. Dominy and P. J. Feeney, “Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings”, *Advanced Drug Delivery Reviews*, 1997, 23 (1–3), 3–25 (DOI: 10.1016/S0169-409X(96)00423-1).

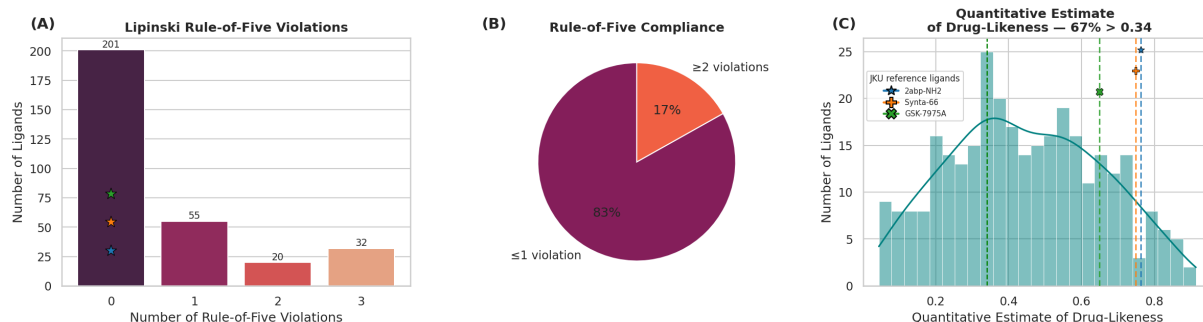


Figure 32: Drug-likeness of the benchmark ligands. Lipinski rule-of-five violation counts (A), overall rule-of-five compliance (B) and the QED distribution (C)

phosphate-containing species. These are exactly the features that the physical and chemical validity checks, namely protonation, geometry and steric clashes, probe most sharply.

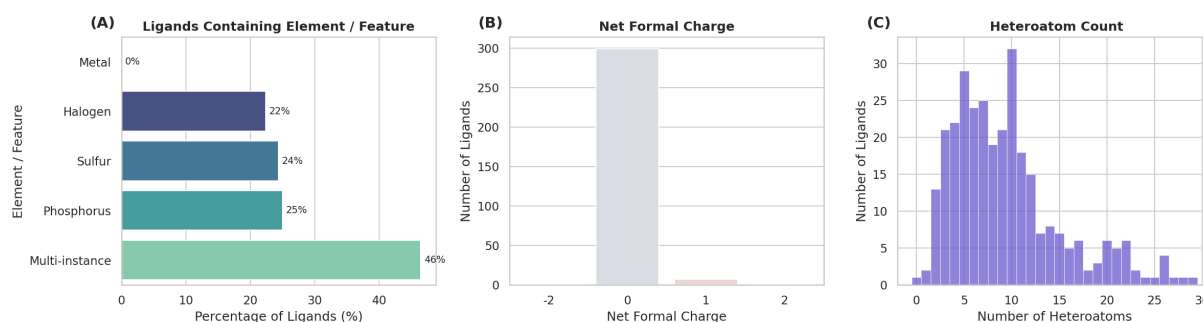


Figure 33: Elemental and charge composition. Prevalence of selected elements and features (A), formal-charge distribution (B) and heteroatom counts (C)

G.5 Joint Distributions and Correlation Structure

The descriptors are considerably interdependent (Figure 34). The size-related quantities, namely heavy-atom count, molecular weight, hydrogen-bond acceptor count, polar surface area and heteroatom count, correlate strongly and positively, while QED falls as molecules grow larger and more polar. Lipophilicity runs opposite to polar surface area, and the sp^3 -carbon fraction opposes aromatic-ring count, separating flat aromatic scaffolds from saturated three-dimensional ones. The two-dimensional mappings make this explicit. Heavy-atom and rotatable-bond counts rise together ($r = +0.70$) (Figure 35 (A)), mass and polar surface area co-vary ($r = +0.66$), lipophilicity and polarity trade off ($r = -0.61$), and drug-likeness declines with size ($r = -0.52$) (Figure 35). The pairwise joint distributions of the six core descriptors (Figure 36) confirm that the benchmark forms one broad cloud rather than distinct clusters.



Figure 34: Pairwise Pearson correlation of the sixteen ligand descriptors

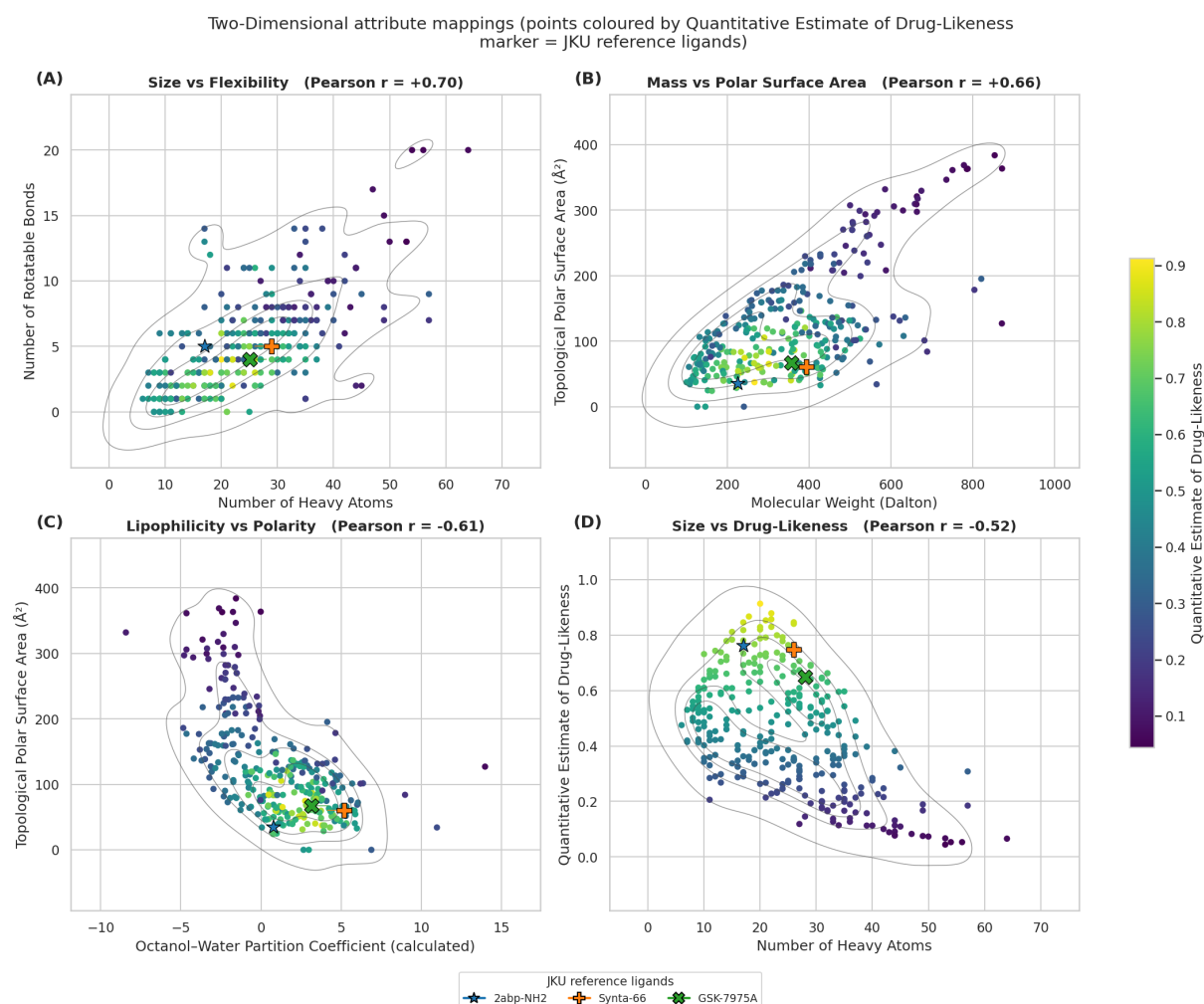


Figure 35: Two-dimensional attribute mappings with points coloured by QED, with panels (A) size against flexibility, (B) mass against polar surface area, (C) lipophilicity against polarity and (D) size against drug-likeness. The three experimental Orai1 ligands are marked. Where two of them almost coincide, the plotting routine fans their markers apart along the horizontal axis so that neither is hidden, which displaces Synta-66 and GSK-7975A in panels (A), (B) and (C) and reverses their apparent left-to-right order there. A thin connector runs from each displaced marker to its true position. Their true values are given in Table 20, and the markers in panel (D) are drawn at their exact coordinates. r denotes the Pearson correlation

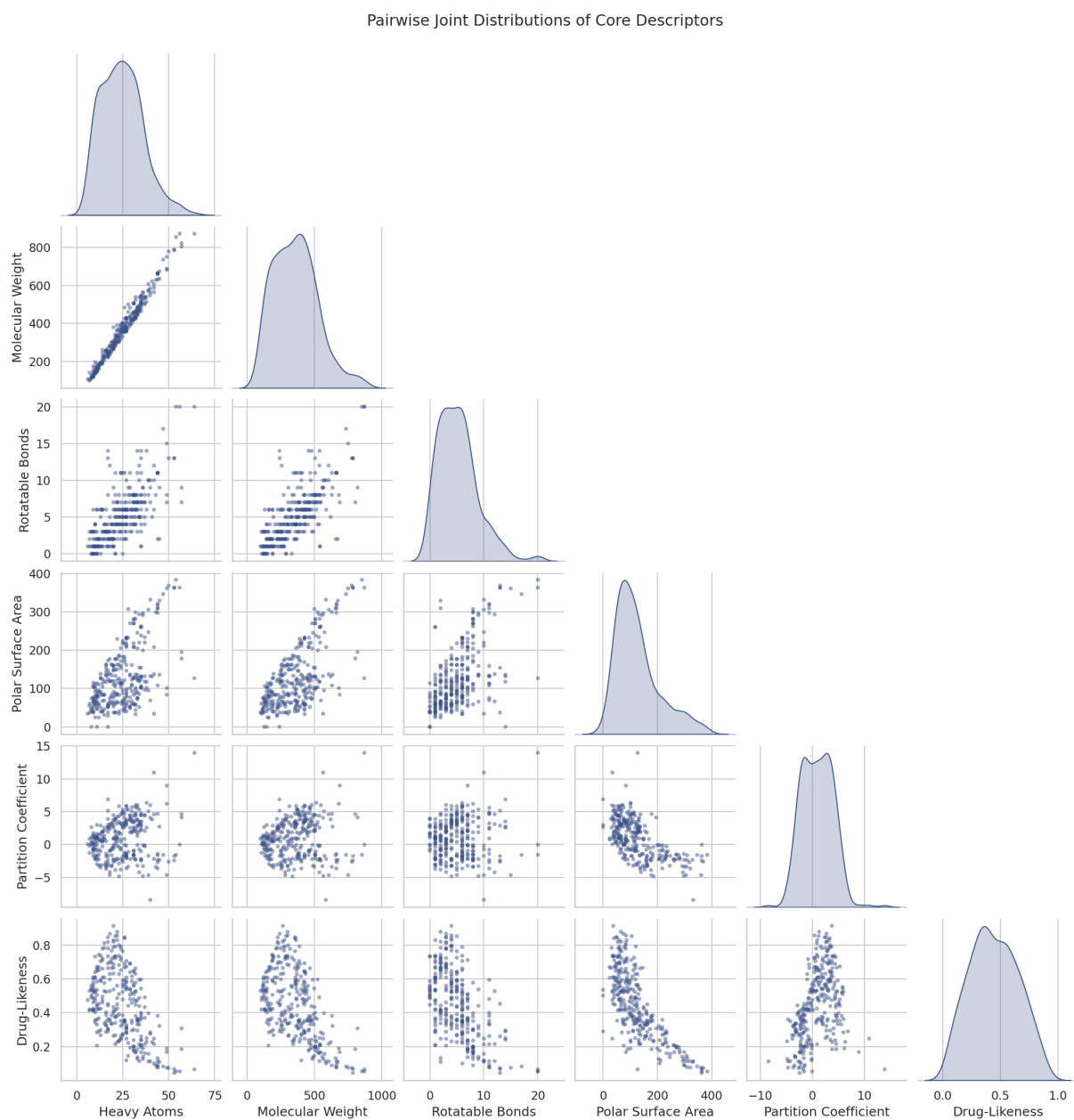


Figure 36: Pairwise joint distributions of six core descriptors with kernel-density estimates on the diagonal

G.6 Principal-Component Analysis — Chemical Space

Summarising this correlated space, a principal-component analysis was run on the sixteen standardised ligand descriptors. Two components already capture roughly two-thirds of the variance (PC1 $\approx$ 45%, PC2 $\approx$ 23%) and four reach about 81% (Figure 37), showing that only a few underlying factors govern the set's diversity. The first component is a size-and-polarity axis. Heavy-atom count, molecular weight, hydrogen-bond acceptors and donors, polar surface area, heteroatom and stereocentre counts all load strongly and positively, while QED loads strongly negatively, so moving along PC1 trades rising size, polarity and complexity against falling drug-likeness. The second component separates flat, lipophilic, aromatic, halogenated scaffolds (positive loadings on aromatic rings, cLogP, ring count and halogen count) from saturated, sp³-rich, hydrogen-bond-donating ones. The minor components capture halogenation and element diversity (PC3) and net formal charge (PC4). Overlaying the per-ligand difficulty of reproducing the crystal pose on the score plot, it rises towards positive PC1 (Figure 38), identifying molecular size and flexibility, not aromatic or lipophilic character, as the dominant driver of docking difficulty.

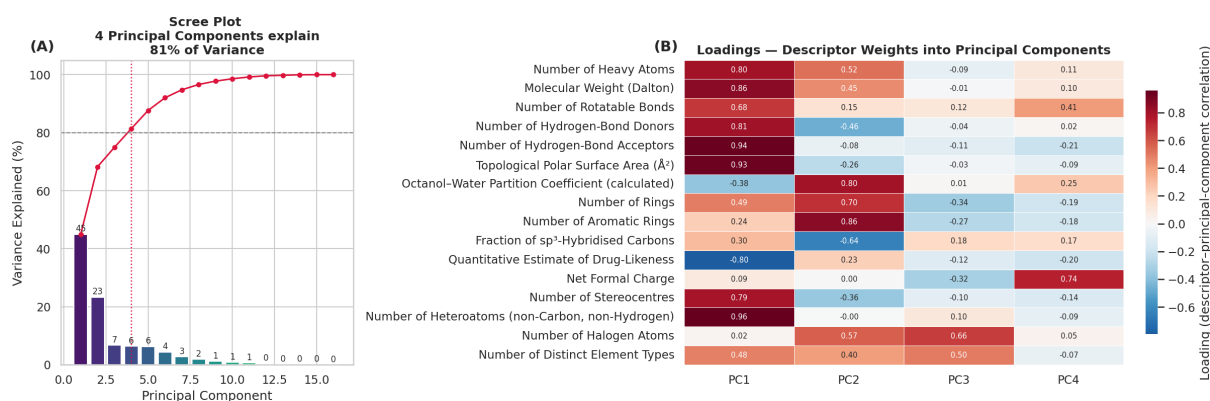


Figure 37: Principal component analysis of the standardised ligand descriptors. Scree plot of explained variance (A) and component loadings (B), where each cell is the correlation between a descriptor and a component

G.7 Receptor Profile

The receptor side is equally varied (Figure 39). A typical complex resolves 472 residues (interquartile range 306 to 778) and is a monomer or dimer, but the distribution has a long tail of large multi-chain assemblies running to forty chains and several thousand residues. In the deposited source structures, 89% of the complexes retain at least one cofactor or hetero-component, although receptor preparation removed most cofactors before docking. Ligand size is essentially independent of receptor size, meaning small ligands are docked into both small and very large pockets and receptor dimensions cannot act as a confounding shortcut.

Ligand chemical space (Principal Component Analysis) — benchmark cloud and JKU reference ligands

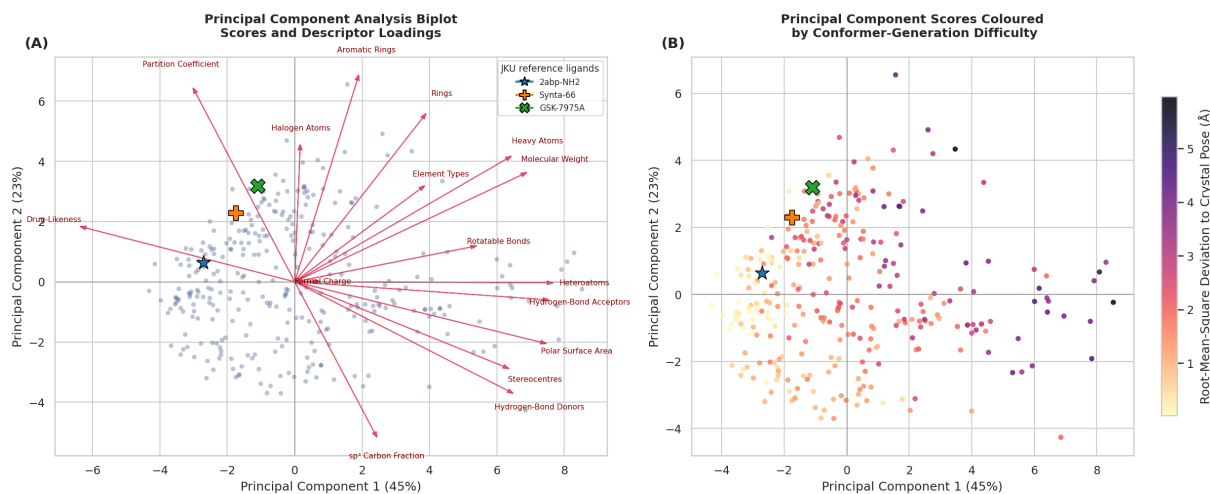


Figure 38: PCA biplot with descriptor loadings (A) and PC scores coloured by conformer-generation difficulty (B). The three experimental Orai1 ligands are projected into the same space

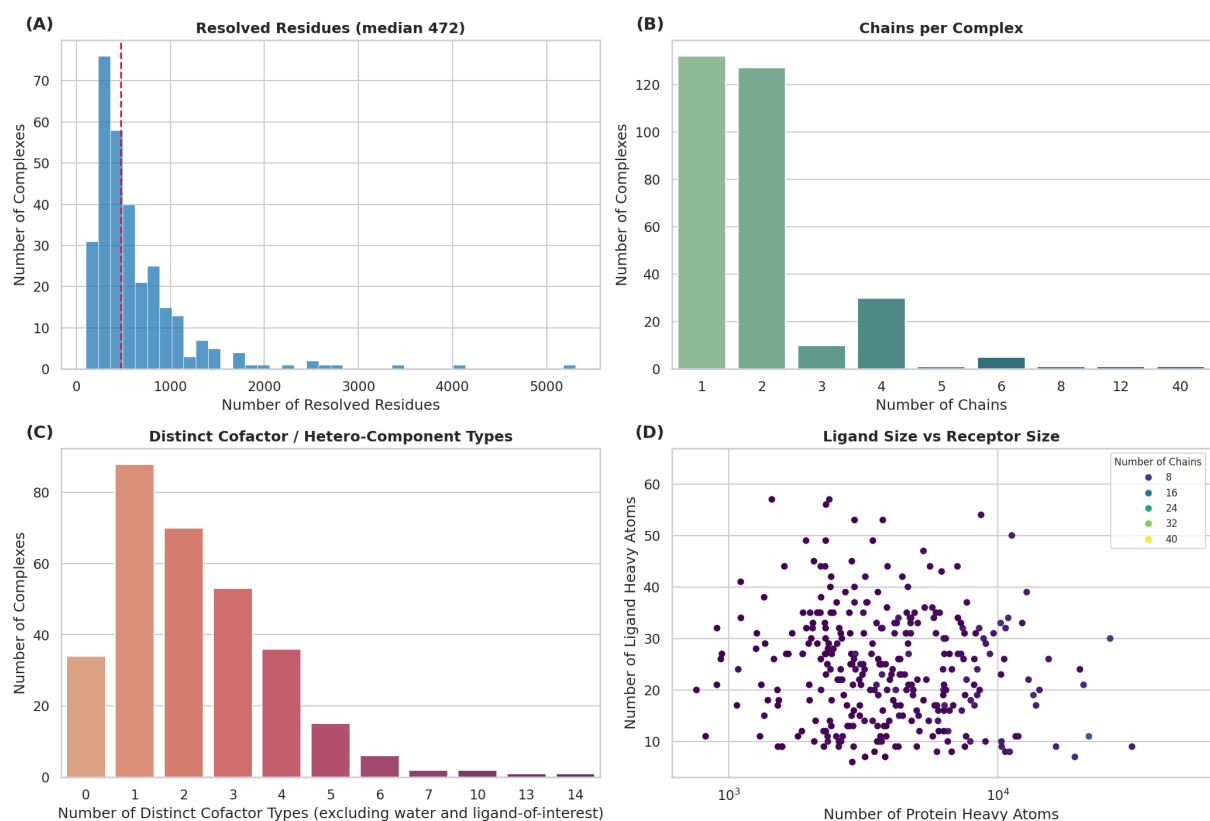


Figure 39: Receptor profile. Resolved residues, chains per complex, distinct cofactor types and ligand-versus-receptor size

G.8 Conformer-Generation Difficulty

The single ETKDGv3 + UFF starting conformer supplied with each ligand gives a lower-bound estimate of how far a docking engine must move the ligand to recover the crystal pose. The symmetry-corrected RMSD between this start conformer and the crystallographic pose has a median of 1.56Å, and only 65% of conformers fall within 2Å of the answer (Figure 40). This deviation grows steadily with rotatable-bond count ($r = +0.75$) and heavy-atom count ($r = +0.80$), quantifying the search burden the set imposes and confirming that ligand size and flexibility are its principal sources of difficulty.

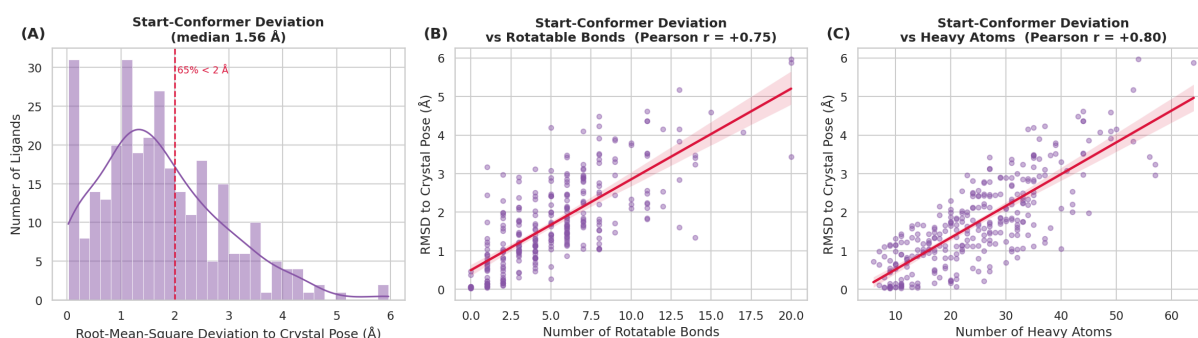


Figure 40: Conformer-generation difficulty. The start-conformer-to-crystal RMSD distribution (A) and its dependence on rotatable bonds (B) and heavy-atom count (C)

G.9 Contrast with the Functionally Tested Ligands

The geometry-optimised Orai1 ligands were characterised with the identical descriptor pipeline and projected onto the same chemical-space maps for comparison. Unlike the benchmark complexes, these compounds are experimental and carry no crystallographic reference pose, so their binding modes must be predicted. Their physicochemical profile is markedly narrower and more lead-like than the benchmark set (Table 20). With molecular weights of 225 to 397 Da, at most 28 heavy atoms, four to five rotatable bonds, one or two hydrogen-bond donors and polar surface areas below 68Å², all three pass Lipinski's rule of five with zero violations and show QED values of roughly 0.65 to 0.76 (above the benchmark median of 0.43). Across the distribution figures the three experimental ligands consistently land on the small, drug-like, aromatic-lipophilic side of every attribute, and in the principal-component projection they cluster at negative PC1 and positive PC2, the compact, lead-like, comparatively easy-to-reproduce corner of the benchmark's chemical space.

Table 20: Physicochemical Descriptors of the Geometry-Optimised Experimental Orai1 Ligands, as Docked and Analysed

Ligand	MW (Da)	Heavy atoms	Rot. bonds	HBD / HBA	TPSA (Å ²)	cLogP	Ro5 viol.
2abp-NH2	225	17	5	1 / 2	35.3	0.77	0
Synta-66	352	26	5	1 / 4	60.5	4.16	0
GSK-7975A (neutral)	397	28	4	2 / 4	67.2	4.19	0

G.10 Suitability of the Dataset

This contrast is what makes the PoseBusters benchmark a sound calibration anchor. The benchmark brackets and clearly exceeds the chemical and structural complexity of the experimental Orai1 ligands. It is larger, more polar, more flexible and more diverse, and it sets a harder conformer-search problem. Evaluating AutoDock Vina, DiffDock and EquiBind across this demanding, experimentally grounded set establishes the configured pipelines' performance within the tested range before they are applied to the smaller, lead-like experimental Orai1 compounds. Testing the three tools across far more chemical diversity than the experimental targets demand, and grounding that test in experimentally resolved poses, supplies the objective reference that the experimental dataset itself cannot.

H Ranking-Recovery Statistics

The following tables give depth-resolved statistics for the variants displayed below. Recovery is reported for $k = 1, 5, 10$ and 15 over the $N = 303$ calibration complexes. Percentages are complex-level recovery rates and bracketed intervals are Wilson 95% confidence intervals. Paired comparisons use McNemar's exact test with Holm correction across the full 24-test family, which is the six prespecified between-variant comparisons taken at each of the four depths, where significance is marked as * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$, and ns denotes not significant. The family is computed on the gina-rescored AutoDock Vina arm that the recovery tables below display, ranked by that arm's own rescored order, so every corrected p value in this chapter refers to a variant printed here. The six prespecified comparisons, each taken on the near-native endpoint, are DiffDock (raw) against DiffDock (gina-opt), EquiBind (raw) against

EquiBind (gnina-opt), AutoDock Vina + gnina against DiffDock (raw), AutoDock Vina + gnina against DiffDock (gnina-opt), DiffDock (gnina-opt) against EquiBind (gnina-opt), and AutoDock Vina + gnina against EquiBind (gnina-opt). The validity-aware endpoint is not part of the family, because the validity-aware set is a strict subset of the near-native set, so its gap is reported as an effect size in Table 25 rather than tested. Every table in this chapter is computed on the blind whole-protein run that supplies the Results chapter, unlike the explicitly superseded site-restricted Ranking Quality chapter.

Table 21: Top-k Recovery with Wilson 95% Confidence Intervals

Variant	k	Near-native% [95% CI]	Validity-aware% [95% CI]
AutoDock Vina + gnina	1	29.7 [24.8, 35.1]	29.4 [24.5, 34.7]
	5	45.2 [39.7, 50.8]	44.9 [39.4, 50.5]
	10	48.5 [42.9, 54.1]	48.5 [42.9, 54.1]
	15	49.8 [44.2, 55.4]	49.8 [44.2, 55.4]
DiffDock (raw)	1	33.7 [28.6, 39.2]	18.2 [14.2, 22.9]
	5	44.6 [39.1, 50.2]	25.4 [20.8, 30.6]
	10	48.2 [42.6, 53.8]	28.7 [23.9, 34.0]
	15	49.8 [44.2, 55.4]	31.0 [26.1, 36.4]
DiffDock (gnina-opt)	1	35.6 [30.5, 41.2]	34.3 [29.2, 39.8]
	5	49.2 [43.6, 54.8]	48.2 [42.6, 53.8]
	10	52.8 [47.2, 58.4]	51.8 [46.2, 57.4]
	15	56.1 [50.5, 61.6]	55.1 [49.5, 60.6]
EquiBind (raw)	1	1.3 [0.5, 3.3]	0.7 [0.2, 2.4]
	5	3.0 [1.6, 5.5]	0.7 [0.2, 2.4]
	10	4.0 [2.3, 6.8]	0.7 [0.2, 2.4]
	15	5.9 [3.8, 9.2]	0.7 [0.2, 2.4]
EquiBind (gnina-opt)	1	15.5 [11.9, 20.0]	15.2 [11.6, 19.7]
	5	19.8 [15.7, 24.7]	18.8 [14.8, 23.6]
	10	20.8 [16.6, 25.7]	19.8 [15.7, 24.7]
	15	21.1 [16.9, 26.1]	20.1 [16.0, 25.0]

Table 22: Raw-to-gnina Refinement Effect on Cumulative Top-k Near-Native Recovery

Tool	k	raw% → gnina-opt%	discordant (raw-only / opt-only)	p (raw)	p (Holm)	sig
DiffDock	1	33.7 → 35.6	2 / 8	0.109	0.984	ns
	5	44.6 → 49.2	5 / 19	0.007	0.073	ns
	10	48.2 → 52.8	6 / 20	0.009	0.094	ns
	15	49.8 → 56.1	6 / 25	0.00088	0.011	*
EquiBind	1	1.3 → 15.5	1 / 44	<1e-4	<1e-4	***
	5	3.0 → 19.8	1 / 52	<1e-4	<1e-4	***
	10	4.0 → 20.8	1 / 52	<1e-4	<1e-4	***
	15	5.9 → 21.1	1 / 47	<1e-4	<1e-4	***

Recovery here is the cumulative best-of-top- k complex rate, in which a complex counts when *any* of its top- k poses is near-native. For DiffDock the top- k pool is the same set of confidence-ranked poses before and after refinement, so its raw → gnina gain is geometry repair alone, accumulated across ranks by the best-of-top- k union rather than produced by any re-ordering. EquiBind has no native ranking, so its gnina variant is additionally re-ranked by gnina affinity, and its gain also reflects which pooled pose is drawn into the top k . Both gains are therefore larger than the fixed-rank, per-pose near-native rate change of 0.1–0.7 (DiffDock) and 2.5–3.1 (EquiBind) percentage points reported in the Discussion, which holds each pose at its own rank and measures only the geometry repair of that same pose. Those per-rank changes are broken out by rank band in Table 23. Every entry is taken from the blind whole-protein run. The uncorrected p values are identical in the superseded crystal-ligand-boxed run, because the two runs share the DiffDock and EquiBind pose pools and their crystal RMSD values, but the Holm column is not, since Holm is applied jointly to all six pre-specified comparisons at all four depths and three of those six comparisons take AutoDock Vina as their baseline, which places it in twelve of the twenty-four tests, and its recovery differs between the two searches. On the whole-protein family the DiffDock refinement effect is therefore significant only at $k = 15$.

Table 23: Fixed-Rank Refinement Gain in Near-Native and Validity Rates by Pose-Rank Band

Tool	Criterion	1–10 smina	1–10 gnina	11–20 smina	11–20 gnina	21–30 smina	21–30 gnina
DiffDock	near-native	+0.6	+0.7	+0.1	+0.1	+0.5	+0.4
DiffDock	PoseBusters-valid	+57	+59	+60	+62	+63	+67
DiffDock	near-native and PB-valid	+12.9	+13.0	+7.7	+7.6	+5.2	+5.2
EquiBind	near-native	+2.6	+3.1	+2.5	+2.8	+2.6	+3.0
EquiBind	PoseBusters-valid	+36	+47	+36	+49	+37	+50
EquiBind	near-native and PB-valid	+2.5	+3.0	+2.7	+3.1	+2.8	+3.2

Entries are the change in the per-rank rate (percentage points) averaged within each pose-rank band, with every pose held at its own confidence rank for DiffDock or generation rank for EquiBind, so no re-ranking is involved. Every entry is computed on the blind whole-protein run that supplies the Results chapter, using the pose-count-weighted pooled rate within each band. The near-native gnina bands span 0.1 to 0.7 percentage points for DiffDock and 2.8 to 3.1 for EquiBind, and the smina bands span 0.1 to 0.6 and 2.5 to 2.6, which give the composite 0.1–0.7 and 2.5–3.1 ranges quoted in the Discussion and in the note to Table 22. The three bands partition ranks 1 to 30 with almost equal pose counts, so their mean reproduces the whole-run validity gain implied by Table 4, namely +60 percentage points for DiffDock with smina and +63 with gnina, against 84.5% and 87.1% validity from a raw 24.4%, and +36 and +49 for unguided EquiBind, against 39.3% and 51.8% from a raw 2.9%. The derived near-native and PB-valid column is shown for context and is near-nativeness and validity by construction.

Tables 24 and 25 provide the corresponding cross-tool recovery contrasts and the share of near-native complexes excluded by physical-invalidity screening.

Table 24: Cross-Tool Comparisons of Near-Native Recovery

Comparison (near-native%)	k = 1	k = 5	k = 10	k = 15
AutoDock Vina + gnina versus DiffDock (raw)	29.7 versus 33.7	45.2 versus 44.6	48.5 versus 48.2	49.8 versus 49.8
AutoDock Vina + gnina versus DiffDock (gnina-opt)	29.7 versus 35.6	45.2 versus 49.2	48.5 versus 52.8	49.8 versus 56.1
DiffDock (gnina-opt) versus EquiBind (gnina-opt)	35.6 versus 15.5	49.2 versus 19.8	52.8 versus 20.8	56.1 versus 21.1
AutoDock Vina + gnina versus EquiBind (gnina-opt)	29.7 versus 15.5	45.2 versus 19.8	48.5 versus 20.8	49.8 versus 21.1

Entries are complex-level near-native recovery percentages over the $N = 303$ calibration complexes, computed on the blind whole-protein run, and they match the near-native column of Table 21. These four contrasts are four of the six prespecified comparisons of the 24-test Holm family described at the head of this chapter, the other two being the DiffDock and EquiBind refinement contrasts of Table 22. No significance marks are printed in the cells above, so the corrected results of those sixteen tests are given here. Neither AutoDock contrast against DiffDock is significant at any depth, at a Holm-corrected p of 1.000 at all four depths against DiffDock (raw) and of 0.984, 1.000, 1.000 and 0.984 against DiffDock (gnina-opt). Both contrasts against EquiBind (gnina-opt) are significant at every depth, at a Holm-corrected p below 0.0002 throughout, so all eight of those tests clear the 0.001 threshold.

Table 25: Accurate-but-Invalid Share by Ranking Depth

Variant	k = 1	k = 5	k = 10	k = 15
AutoDock Vina + gnina	0.3 [0.1, 1.8]	0.3 [0.1, 1.8]	0.0 [0.0, 1.3]	0.0 [0.0, 1.3]
DiffDock (raw)	15.5 [11.9, 20.0]	19.1 [15.1, 23.9]	19.5 [15.4, 24.3]	18.8 [14.8, 23.6]
DiffDock (gnina-opt)	1.3 [0.5, 3.3]	1.0 [0.3, 2.9]	1.0 [0.3, 2.9]	1.0 [0.3, 2.9]
EquiBind (raw)	0.7 [0.2, 2.4]	2.3 [1.1, 4.7]	3.3 [1.8, 6.0]	5.3 [3.3, 8.4]
EquiBind (gnina-opt)	0.3 [0.1, 1.8]	1.0 [0.3, 2.9]	1.0 [0.3, 2.9]	1.0 [0.3, 2.9]

Entries are percentages of the N = 303 calibration complexes with bracketed Wilson 95% confidence intervals, computed on the blind whole-protein run. The share is the complex-level gap between the two columns of Table 21, that is the near-native rate minus the validity-aware rate at the same depth, and counts complexes whose best top-*k* pose is within 2 Å of the crystal ligand but never physically valid at that depth. The validity-aware set is a strict subset of the near-native set, so the gap is one-directional and is reported as an effect size rather than tested. The raw AutoDock Vina arm, which Table 21 does not carry, sits at 0.7 [0.2, 2.4] at all four depths.